

Hypercom Device Support Manual

BASE24-pos[®]



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Contents

Preface	xi
Conventions Used in This Manual	xv
1: Introduction	1-1
BASE24-pos Hypercom Device Handler Module Overview	1-2
Network Access Controller (NAC)	1-2
Network Management System (NMS)	1-2
Supported Terminals	1-3
Supported Protocols	1-4
Supported Features	1-5
Expanded Transaction Set	1-6
Multiple Terminal Support	1-6
Multiple Language Support	1-6
Hypercom External Message Format	1-6
CVV2 Data Processing	1-7
Download Support	1-7
American Express Data Collection	1-7
Address Verification	1-8
Draft Capture Options	1-8
BASE24-pos Settlement and Cutover	1-9
Terminal Balancing	1-9
BASE24 Transaction Security	1-10
Sequence Number Checking	1-10
Event Message Generation	1-11
Electronic Benefits Transfer	1-11
Context Data Removal	1-12
Hypercom Triple DES Master/Session	1-12
Derived Unique Key per Transaction (DUKPT)	1-12

Introduction *continued*

HPDH Transaction Support	1-14
Financial Transactions	1-14
Administrative Transactions	1-16
Hypercom-Specific Messages	1-18
Integration with BASE24-pos	1-19
Transaction Flow	1-19
Tying the HPDH Module to BASE24-pos	1-20
HPDH Module Versions	1-22
 2: Terminal Configuration	2-1
HPDH Configuration Files	2-2
Hypercom Configuration File (HPCF)	2-2
Hypercom T7 Configuration File (HPTC)	2-2
Acquirer Processing Code File	2-2
Institution Definition File (IDF)	2-3
Log Message Configuration File (LMCF)	2-3
POS Retailer Definition File (PRDF)	2-3
BASE24-pos Terminal Data Files (PTD)	2-4
POS Transaction Log File	2-5
Logical Network Configuration File (LCONF)	2-6
Assigns	2-6
Params	2-7
 3: Hypercom Device Handler (HPDH) Module	3-1
HPDH Module Overview	3-2
Enforcing Transaction Sequence	3-2
Allowed Transaction Checking	3-2
Cutting Over Terminals	3-4
Decrypting PINs	3-4
Authenticating Messages	3-4
Downloading	3-5
Controlling Response Timers	3-5
Reversal Processing	3-5
File Usage	3-7

Hypercom Device Handler (HPDH) Module *continued*

HPDH Module Initialization and Warmboot	3-9
Initializing the HPDH Module	3-9
Warmbooting the HPDH Module	3-10
Warmbooting Static POS Terminal Data	3-10
Network Text Commands	3-11
Event Message Generation	3-14
4: Hypercom External Message	4-1
Message Overview	4-2
Transport Protocol Data Unit (TPDU) Message Header	4-3
Application Messages	4-3
Network Management System (NMS) Messages	4-5
Message Text Format	4-10
Message Type Identifier	4-10
Bit Map	4-12
Primary Data Elements	4-13
Primary Data Element Descriptions	4-14
Message Structures	4-27
Hypercom Authorization Transaction	4-29
Hypercom Preauthorization Transaction	4-31
Hypercom Balance Inquiry Transaction	4-33
Hypercom Card Verification Transaction	4-35
Hypercom Sale Transaction	4-37
Hypercom Cash Transaction	4-39
Hypercom Sale and Cash Back Transaction	4-41
Hypercom Void Sale Transaction	4-43
Hypercom Refund Transaction	4-45
Hypercom Guarantee Late Arrival Transaction	4-47
Hypercom Offline Sale Transaction	4-49
Hypercom Offline Cash Transaction	4-51
Hypercom Sale Completion Transaction	4-53
Hypercom Adjustment Sale Transaction	4-55
Hypercom Check Guarantee Transaction	4-57
Hypercom Check Verification Transaction	4-59
Hypercom Batch Upload Transaction	4-60

Message Structures *continued*

Hypercom Reversal Transaction	4-63
Hypercom Settlement Transaction	4-64
Hypercom Test Transaction	4-66
Hypercom Logon Transaction	4-67
Hypercom Parameter Download Transaction—T4, T5, T6 Terminals	4-69
Hypercom Parameter Download Transaction—T7 Terminals	4-70
Table IDs	4-72
Lodging Details	4-73
Lodging Program Indicator	4-73
Level II Commercial Card Indicator	4-73
CVV2 Processing Data	4-74
Payment Services (PS2000) Data	4-74
PS2000 Terminal Data	4-75
CPS 2000 Data	4-75
Tax Amount #1	4-76
Cash Back Amount	4-76
Surcharge Amount	4-77
Payment Plan	4-77
EBT Terminal Data	4-78
EBT Response Data	4-78
Miscellaneous Amount	4-79
Address Verification Request Data	4-79
Address Verification Response Data	4-80
Customer ID or Purchase Order Number	4-82
Level II Commercial Card Data Subfield	4-82
American Express Data Collection	4-83
Restaurant Format	4-83
General Retail Format	4-84
Lodging Format	4-85
Check Guarantee and Verification Methods	4-86
Account Number	4-87
Drivers License	4-87
Check MICR Data	4-88
Full MICR Data	4-88

5: Download Data	5-1
Hypercom ISO Download Response Messages—T4, T5, and T6 Terminals	5-2
Card Acceptor Name—Data Element P-43	5-2
Configuration Parameters—Data Element P-60	5-2
Configuration Parameters—Data Element P-61 (Card Definition Table)	5-4
Configuration Parameters—Data Element P-62 (Product Code Table)	5-5
Configuration Parameters—Data Element P-63 (Range Definition Table)	5-6
Hypercom Download Examples—T4, T5, and T6 Terminals	5-7
Download Completes Successfully—18 or More Entries	5-8
Download Completes Successfully—Fewer Than 18 Entries	5-11
Hypercom ISO Download Response Message—T7 Terminals	5-14
Message Format	5-14
ICC Table Entry	5-16
Chip Data Table	5-18
Key Data Table	5-19
Revoked Certificate Key Table	5-20
Terminal Configuration Table	5-23
Card Range Table	5-34
Issuer Table	5-35
Acquirer Table	5-43
Descriptor Table	5-47
Additional Data Prompt Table	5-48
Hypercom T7 Download Example	5-51
6: HPDH Configuration Considerations	6-1
PIN Encryption	6-2
Derived Unique Key per Transaction (DUKPT)	6-3
Sequence Number Checking	6-4
Message Authentication Codes (MACs)	6-5
Setting Up MAC Use	6-6
Generating and Verifying a MAC	6-6
Failed MAC Procedure	6-7
Draft Capture	6-8
Draft Capture Options	6-8
Authorization Only with Paper Follow-up	6-9

Draft Capture <i>continued</i>	
Authorization and Draft Capture	6-9
Setting the Draft Capture Flag in the PSTM.	6-9
Transaction Accumulation Totals	6-11
PTDD1 Totals	6-11
Terminal Totals	6-11
Maintaining Totals	6-12
Handshaking	6-15
Passthrough Configuration Example	6-16
HPDH Module Configuration	6-16
Terminal Configuration	6-17
Network Management System (NMS) Configuration	6-18
7: Transaction Message Flows	7-1
Normal Transaction with Response	7-2
Normal Transaction without Response—Terminal Times Out.	7-3
Normal Transaction—Unable to Deliver Response to Terminal	7-5
Normal Transaction - NAC Unable to Deliver Response to Terminal	7-7
NMS Passthrough Completes Successfully	7-9
NMS Unable to Deliver Message to NAC	7-11
Settlement Transaction—Batch in Balance	7-13
Settlement Transaction—Draft Capture Totals In Balance and	
Total Business Out of Balance	7-15
Settlement Transaction—Draft Capture Totals Out of Balance	7-17
8: Hypercom Configuration File (HPCF)	8-1
Hypercom T4, T5, and T6 Terminal Configuration Data	8-2
Configuration Parameters Screen	8-3
Product Data Screen	8-13
Card Definition Table Screen Function Keys	8-15
Card Definition Table Screen	8-16
Range Definition Table Screen Function Keys	8-26
Range Definition Table Screen	8-27

9: Hypercom T7 Configuration File (HPTC).....	9-1
Hypercom T7 Terminal Configuration Data.....	9-2
Screen 1 Function Keys	9-3
Screen 1.....	9-4
Configuration Parameters Table Screen 2.....	9-5
Configuration Parameters Table Screen 3.....	9-12
Configuration Parameters Table Screen 4.....	9-25
Card Range Table Screen 2 Function Keys.....	9-30
Card Range Table Screen 2.....	9-31
Card Range Table Screen 3 Function Keys.....	9-34
Card Range Table Screen 3.....	9-35
Issuer Table Screen 2 Function Keys	9-38
Issuer Table Screen 2	9-39
Issuer Table Screen 3 Function Keys	9-48
Issuer Table Screen 3	9-49
Acquirer Table Screen 2 Function Keys	9-61
Acquirer Table Screen 2	9-62
Acquirer Table Screen 3 Function Keys	9-70
Acquirer Table Screen 3	9-71
Product Data Screen 2.....	9-76
Additional Prompt Screen 2	9-78
A: American Express Standard Industry Formats	A-1
Lodging.....	A-2
Restaurant	A-4
General Retail.....	A-6
B: Odd-Parity Adjustment Algorithm	B-1
Index	Index-1
Field Name Index	Index-5
Data Name Index.....	Index-9

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Preface

The *BASE24-pos Hypercom Device Support Manual* is designed to provide readers with a thorough understanding of the BASE24-pos Hypercom Device Handler (HPDH) module. It presents readers with an extensive overview of the HPDH module and its options, along with providing information on how to integrate the HPDH module for use with BASE24-pos. In addition, this manual describes, in detail, the processing logic and configuration issues associated with the HPDH module.

Audience

There are three targeted audiences of this manual. One audience reads this manual to get a general overview of the versatility of the HPDH module and terminals that meet its specifications. Another audience consists of terminal vendors who want to manufacture terminals that are compatible with the HPDH module. The third audience of this manual is customers who are purchasing the HPDH module and contracting with terminal vendors.

Prerequisites

Although a knowledge of BASE24-pos is helpful for the customer who is reading this manual, the manual is designed to stand alone. Terminal vendors and personnel who seek an overview of the HPDH module should be able to find as much information about BASE24-pos as they require to perform their functions. However, if more information about BASE24-pos is desired, readers should refer to the *BASE24-pos Transaction Processing Manual*.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for

readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Base Files Maintenance Manual*** discusses the base files used with the HPDH module.
- The ***BASE24 CRT Access Manual*** contains general information on logging on to the BASE24 system, using the default function keys, and creating screen access security.
- The ***BASE24 Logical Network Configuration File Manual*** discusses in detail the assigns and params used in configuring the HPDH module.
- The ***BASE24 Event Message Reference Manual*** discusses in detail how to access event messages generated by the HPDH module.
- The ***BASE24 Tokens Manual*** discusses how BASE24 products use tokens, including field descriptions for all available tokens.
- The ***BASE24 Text Command Reference Manual*** contains detailed descriptions and syntax of all text commands.
- The ***BASE24 Transaction Security Manual*** discusses in detail PINs, message authentication codes (MACs), and dynamic key management used to set up transaction security for the HPDH module.
- The ***BASE24-pos Settlement and Reporting Manual*** discusses in detail BASE24-pos settlement, cutover, balancing, and reporting.
- The ***BASE24-pos Transaction Processing Manual*** discusses in detail transaction processing used in configuring the HPDH module.
- The ***BASE24-pos Files Maintenance Manual*** discusses the BASE24-pos files used with the HPDH module.

In addition, this manual contains references to the following Hypercom publications:

- The ***Hypercom ECB Message Authentication Code Generation*** document describes the Hypercom proprietary MAC algorithm.
- The ***Hypercom T7 Standard Software Initialization Parameters*** document describes modem modes used with Hypercom terminals.

Software Release

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

Section 1, “Introduction,” provides an overview of the HPDH module and its functionality. This section includes information explaining how the HPDH module can be integrated with BASE24-pos. This section also discusses the role of the HPDH module during transaction processing.

Section 2, “Terminal Configuration,” explains the options available to customers configuring their POS terminals. This section explains the files that must be set up in order to tailor HPDH module configuration options to meet the needs of a particular customer. This section includes descriptions of the Logical Network Configuration File (LCONF) assigns and params used by the HPDH module.

Section 3, “Hypercom Device Handler (HPDH) Process,” contains information about the HPDH process. It includes an overview of the process, a description of each file used by the HPDH module during processing, a description of the steps the HPDH module performs during initialization and warmbooting, and a list of the network control commands recognized by the HPDH module.

Section 4, “Hypercom External Message,” contains information about the Hypercom External Message and how the HPDH module formats requests and responses to send to terminals.

Section 5, “Download Data” contains information about the role of the HPDH module when downloading configuration data to terminals for use during processing. It includes information about the records that can be fully downloaded to terminals from the Hypercom Configuration File (HPCF), the Hypercom T7 Configuration File (HPTC), and the BASE24-pos Terminal Data files (PTD).

Section 6, “HPDH Configuration Considerations,” presents information on topics the customer must consider when deciding how to configure the HPDH module. It includes sequence number checking options, message authentication code (MAC) information, draft capture options, and transaction accumulation totals information.

Section 7, “Transaction Message Flows,” contains transaction flows between the terminal and the HPDH module.

Section 8, “Hypercom Configuration File (HPCF),” contains the records used in configuring Hypercom T4, T5, and T6 terminals. BASE24-pos file maintenance manipulates these records through the use of formatted screens. This section shows the formatted screens used by the HPDH module and describes each field shown on these screens.

Section 9, “Hypercom T7 Configuration File (HPTC),” contains the records used in configuring Hypercom T7 terminals. BASE24-pos file maintenance manipulates these records through the use of formatted screens. This section shows the formatted screens used by the HPDH module and describes each field shown on these screens.

Appendix A, “American Express Standard Industry Formats,” defines the American Express standard industry formats used with the AMEX Data Collection option. The AMEX Data Collection option enables customers to perform draft capture on transactions originating from American Express cardholders. Generally, American Express has defined categories under which transactions are placed. For each of these categories, American Express requires different data to be sent from the device. Therefore, American Express has set forth standard industry formats for each category to ensure the data they require is captured by the device and sent to BASE24.

Appendix B, “Odd-Parity Adjustment Algorithm,” describes the algorithm used to adjust the PIN Master key.

Readers can use the index by field name to locate information about a particular screen field and the index by data name to locate information about a particular field from a file or record structure.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this PRODUCT_NAME publication. The publication number (for example, PS-DP390-01 for the *BASE24-pos Hypercom DH Manual*) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Dec-2009 for December, 2009) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in This Manual

This section explains how field descriptions, blank characters, and unlabeled fields on screens are documented in this manual.

Field Descriptions

Each field appearing on a file maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the Hypercom T7 Configuration File (HPTC).

Example

TERMINAL GROUP — The group to which the terminals belong. This value corresponds to the TERM GROUP field on BASE24-pos Terminal Data files (PTD) screen 1. The TERMINAL GROUP field combined with the RECORD TYPE field are the key to the HPTC.

Field Length: 1–4 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: HPTC.KEY-S.TERM-GRP

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term <i>alphanumeric</i> includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term <i>hexadecimal</i> includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is <i>System Protected</i>.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the F8 key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

Blank Characters

Field descriptions for internal and external messages, tokens, files maintenance screens, and reports often include lists of valid values. When the value contains a blank character, this manual uses the symbol *b* to indicate the blank character.

Unlabeled Fields on Screens

Angle brackets (< >) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

Unlabeled fields are not included in the index by field name; they appear by subject in the main index.

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1: Introduction

This section contains introductory-level information describing the BASE24-pos Hypercom Device Handler (HPDH) module and the functions it can perform. The main purpose of this section is to provide readers with the background information they require as a basis to build a solid foundation for understanding the HPDH module. This section introduces the main topics important when using the HPDH module and provides general overviews for each specific topic identified.

This section is intended for readers who require only an overview of the HPDH module and its functionality. It provides information on topics important in gaining a basic understanding of the HPDH module, such as supported features and terminals. However, this section is also helpful to readers who require a detailed explanation of the HPDH module and the functions it can perform. This section lays the groundwork to prepare these readers for information covered in the rest of the manual by providing them with previews of topics discussed in greater detail in subsequent sections.

This section also includes the following topics concerning the HPDH module:

- Supported terminal models
- Supported protocols
- Supported features
- Supported transactions
- Integration with BASE24-pos
- HPDH module version information

BASE24-pos Hypercom Device Handler Module Overview

The Hypercom Device Handler (HPDH) module operates in conjunction with the BASE24-pos product. BASE24-pos is a point-of-sale transaction processing system. The HPDH module provides flexible transaction support, configuration options, and allows retailers to select the functionality to best meet their particular business requirements.

The HPDH module can handle processing for a total of 19 POS transactions sent from a terminal. A complete list of transaction types supported by the HPDH module is included later in this section. The HPDH module also offers flexible security options to ensure that transactions are processed only once, a technique to verify whether data has been accidentally or fraudulently altered, and various PIN encryption techniques.

In addition, two major components that allow the HPDH module to perform its functions are the network access controller (NAC) and the network management system (NMS). The NAC provides end-to-end network optimization and concentration to free the host of processing that is better handled elsewhere. The NMS provides a method to control NACs and terminals throughout the network. The NAC is required in order to use the HPDH module with BASE24-pos, whereas the NMS is optional.

Network Access Controller (NAC)

The network access controller (NAC) provides an interface between terminals and the HPDH module. It provides a uniform interface to the terminals supported by the HPDH module, performing the translation required to convert the native mode messages of a terminal into the ISO 8583 message format supported by the HPDH module.

Network Management System (NMS)

The network management system (NMS) is an optional diagnostic and configuration tool that can be purchased from Hypercom to control NACs and Hypercom T4, T5, T6, and T7 terminals. Software maintenance can be performed through downloading from the NMS. Along with the application facilities, the NMS also provides a range of diagnostic and statistical functions for communications and terminal hardware.

Supported Terminals

The Hypercom Device Handler (HPDH) module provides support for the following terminals:

- Hypercom T4
- Hypercom T5
- Hypercom T6
- Hypercom T7

The HPDH module also can communicate with any POS terminal having the capability of sending messages to and receiving messages from the HPDH module using the message format supported by the HPDH module. Terminals must be able to communicate with the HPDH module and allow the HPDH module to communicate with them. Terminals can communicate with the HPDH module over dial lines, leased lines, or a communications protocol network.

Supported Protocols

The HPDH module can be configured to use the following communication protocols:

- Bisynchronous 3650 (EBCDIC)
- SDLC/HDLC
- SNA 3650
- X.25

The protocol selected by the customer depends on the line communications environment established between the HPDH module and its terminals. Line communications environments for the HPDH module can be dial, leased, or X.25. For example, the Bisynchronous 3650 protocol can be used to accommodate leased bisynchronous communications with the HPDH module, whereas the X.25 protocol can be used to accommodate X.25 communications with the HPDH module. In general, any protocol that supports the transmission of binary data can be used with the HPDH module. Any protocol that requires protocol-specific processing on the part of the HPDH module is not supported.

ACI personnel assist customers at installation or when a system is configured in determining the type of protocol best suited for the customer's particular requirements.

Supported Features

An enhanced set of features is available for use with the HPDH module. Customers can select and configure the features necessary to meet their processing requirements. This subsection provides a list of these features, followed by a summary of each feature. The following features are supported by the HPDH module:

- Expanded transaction set
- Multiple terminal support
- Multiple language support
- Hypercom external message format
- CVV2 Data Processing
- Download support
- American Express data collection support
- Address verification support
- Draft capture options
- BASE24-pos settlement and cutover support
- Terminal balancing
- BASE24 transaction security support
- Sequence number checking
- Event message generation
- Event Benefits Transfer transactions
- Context Data Removal

Also, the following tables are now supported:

- **Level II Commercial Card Indicator:** Level II Commercial Card Table (Table ID 15, private use field 63). This also includes support for the corresponding Response Table (Table ID 73).
- **CPS 2000 Data:** CPS 2000 Data Table (Table ID 30, private use field 63).
- **Surcharge Amount:** Surcharge Amount Table (Table ID 44, private use field 63).
- **Payment Plan:** Payment Plan Table (Table ID 45, private use field 63).

Expanded Transaction Set

The HPDH module can handle processing for a total of 19 different POS transactions sent from the terminal, including both financial and administrative transactions. A complete list of the transaction set supported by the HPDH module is included later in this section.

Multiple Terminal Support

The HPDH module supports several different terminal types. The Hypercom T4, T5, T6, and T7 devices are the most common type of POS terminal used with the HPDH module. However, customers are not limited only to the Hypercom devices. The HPDH module is flexible enough to communicate with any third generation POS terminal having the capability of sending messages to and receiving messages from the HPDH module using the message format supported by the HPDH module. Customers can use any POS terminal that conforms to the HPDH module requirements set forth in this manual.

Multiple Language Support

Customers have the option of sending responses to a terminal in one of three different languages. Terminals display responses in the language set up in the LANGUAGE-IND field in the BASE24-pos Terminal Data files (PTD). The LANGUAGE-IND field identifies the language table used by the HPDH module when sending responses to the terminal. For more information on the BASE24-pos Terminal Data files, refer to the ***BASE24-pos Files Maintenance Manual***.

Hypercom External Message Format

The Hypercom terminal uses its own external message when processing transactions. The Hypercom External Message is a variable-content, variable-length message, based on the ISO 8583 message format developed by the International Organization for Standardization (ISO). The HPDH module configures each message based on the type of message being sent. All HPDH module messages consist of a transport protocol data unit (TPDU) header and data elements.

CVV2 Data Processing

Hypercom has added a new table for requests, the CVV2. This represents the card authentication value printed on the card signature line. The CVV2 is a three or four character field used to authenticate the card when the card is not present at the time of the transaction. The CVV2 element arrives from the terminal as a 6 byte field. The device handler will forward the CVV2 value to the authorizer and the result code logged.

The CVV2 data table is described in the Hypercom Message Specification document, v3.42, section 6.28.4.7

Download Support

The HPDH module supports full terminal parameter downloads. This enables terminals to receive full configuration file information. In addition, the download data can pertain to as many terminals as the customer selects. Terminal parameters are defined for terminal groups; one download record is **not** required for each terminal.

American Express Data Collection

The HPDH module provides customers with the option of performing draft capture on transactions originating from American Express cardholders. Generally, American Express has defined categories into which transactions being processed are placed. For each of these categories, American Express requires different data to be sent from the device. To ensure the data they require is captured by the device and sent to BASE24, American Express has set forth standard industry formats for each category.

In order to perform draft capture on transactions originating from an American Express cardholder, customers must use a Device Handler module flexible enough to accept and recognize the American Express standard industry formats. The HPDH module is able to recognize these standard industry formats and, subsequently, capture and process transactions using them. In conjunction with this option, the customer must set up Standard Industrial Classification (SIC) codes or Merchant Category codes in the BASE24-pos Terminal Data files (PTD). SIC and Merchant Category codes identify the merchant's line of business.

For more information on the PTD, refer to the ***BASE24-pos Files Maintenance Manual***. For the American Express standard industry formats, refer to appendix A.

Address Verification

The HPDH module is compatible with the BASE24-pos add-on Address Verification module. Address verification assists merchants in controlling losses that can result from credit card fraud during transactions where the cardholder cannot be readily identified. The HPDH module places address verification information in the PSTM. See section 4 for information about the data fields used to format address verification data in the PSTM. For more information on the BASE24-pos add-on Address Verification module, refer to the ***BASE24-pos Address Verification Manual***.

Draft Capture Options

The HPDH module provides an efficient method of draft capture with third generation terminals. It does this by performing draft capture in a single transaction.

As a comparison, the draft capture option common for first-generation terminals consists of authorizing the transaction online and then sending paper follow-up. This method requires merchants to send the paper follow-up before receiving payment.

The draft capture option common for second-generation terminals consists of authorizing the transaction online and then sending a separate transaction at the end of a batch to verify that the transaction was completed correctly. In effect, this consists of two transactions for every purchase. Since most of the transactions that are authorized are completed, this requires almost twice the amount of disk space and communication costs. Second-generation terminals also can use the draft capture option offered with first-generation terminals.

In the process used by the HPDH module with third-generation terminals, the HPDH module oversees authorization and capture of the data required to pay the merchant in a single step. If a transaction is not completed correctly, the operator who entered the transaction sends an adjustment transaction. This procedure compensates for the second-generation terminals' system of ensuring that transactions are completed correctly. The HPDH module also supports transactions

that require paper follow-up. This method reduces the amount of disk space required and cuts communication costs normally associated with POS service providers.

BASE24-pos Settlement and Cutover

BASE24-pos provides an efficient method of settlement and cutover. *Settlement*, in terms of BASE24-pos, is actually a reconciliation effort whereby transaction activity for the current day is closed out and reported, and the system is shown to be in balance. The term *cutover* refers to the point in time where transactions stop being logged to the business for the current day and start being logged to business for the next day. It refers to the closing of the current posting day and the opening of the next.

BASE24-pos settlement differs from the settlement carried out by participating BASE24-pos financial entities (for example, retailers, retailer sponsors, cardholders, card issuers, interchanges) in that BASE24-pos does not move funds to settle accounts. It does, however, provide the support that enables these financial entities to initiate the movement of funds. An extensive set of reports provides financial entities with the information they require to settle their accounts. BASE24-pos also provides the POS Transaction Log File (PTLF), which keeps a record of each transaction processed by BASE24-pos during its defined reporting day. In addition, BASE24-pos provides an automated method to cut over terminals, retailers, institutions, interchanges, and the BASE24-pos network to a new posting day. For more information on BASE24-pos settlement, cutover, and reporting, refer to the ***BASE24-pos Settlement and Reporting Manual***.

Terminal Balancing

The HPDH module enables merchants to balance their terminals with BASE24-pos. The terminal retains totals independent of BASE24-pos and therefore terminal totals can be compared to the BASE24-pos totals in the BASE24-pos Terminal Data files (PTD) to determine if the totals balance.

The HPDH module provides a method for merchants to balance terminals with BASE24-pos. Merchants can perform a close batch transaction to determine if BASE24-pos and the terminal have accumulated the same totals. Merchants can also enter adjustment transactions, if necessary, before closing the batch. When a close batch transaction is received by the HPDH module, the HPDH module creates a batch totals record from the terminal totals in the PTD and writes it to the PTLF. It then clears the batch totals in the PTD. If the terminal can send its own

batch totals, the HPDH module compares the terminal totals to the PTD totals, and, if the debits, credits, and adjustments do not match exactly, writes the terminal totals to the PTLF immediately following the PTD totals.

Merchants also have the option of receiving a report a day later, or they can contact the service provider to receive terminal balancing information. The BASE24-pos Retailer Reconciliation Report (B24POS35) contains terminal balancing information. Retailers use this report to balance transaction activity at the terminal batch level and to determine whether their terminals balance with BASE24-pos.

BASE24 Transaction Security

The HPDH module offers various ways of ensuring that transactions are transmitted and secured correctly. These include options that ensure every transaction is being processed only once and options that allow customers to select from various PIN encryption methods. See section 6 for more information about PIN encryption options.

In addition, the HPDH module supports the use of message authentication codes (MACS) for T4, T5, T6 and T7 terminals. MACs are used to verify whether data has been accidentally or fraudulently altered. Any data sent between the terminal and the HPDH module can be verified through the use of the MACs. See section 6 for more information about message authentication options.

For more information on BASE24 transaction security, refer to the ***BASE24 Transaction Security Manual***.

Sequence Number Checking

The HPDH module supports sequence number checking, which is a method of ensuring that every message being sent from the terminal is received only once. The HPDH module uses the Systems Trace Number (P-11) data element in the incoming ISO message to detect duplicate transactions. If the sequence number is included in the message, the HPDH module can ensure that no messages have been duplicated.

Event Message Generation

The HPDH module generates event messages that keep operators aware of activity occurring in the system. Event messages can be used to assist operators with their daily tasks or to notify operators of problems occurring in the system. Event messages provide operators with insight as to how the system is running.

In addition, ACI has developed a method that enables customers to view event message documentation electronically. This event message documentation provides customers with greater flexibility by allowing them to print out event message documentation as needed at their location or to electronically manipulate the data to suit their needs. Event message documentation provides the message number, severity, and text for each event message generated by the HPDH module. In addition, it also provides the cause of each message and the recommended action to take.

Refer to section 3 for a discussion of accessing the documentation for event messages generated by the HPDH module. Refer to the ***BASE24 Event Message Reference Manual*** for a complete explanation of event message documentation.

Electronic Benefits Transfer

Hypercom now supports Electronic Benefits Transfer (EBT) transactions through the following features.

- Two tables have been added to pass response data from the host to the terminal: the EBT Terminal Data Table and the EBT Response Data Table.
- Two values have been added to the account selection positions of the processing code: Value “96” (EBT Cash Benefit) and Value “98” (EBT Food Stamp Benefit).
- A new transaction format has been added to support EBT processing that will apply to all three EBT transaction types: EBT Cash Only, EBT Sale, and EBT Sale and Cash.

An EBT transaction is an authorization request. When approved, the transaction is stored in the terminal’s batch for later settlement. The 0200 message type is sent when the capture flag is set to “Y.”

Thus authorization is obtained in the same manner as any other financial transaction for each of the EBT transaction types.

Context Data Removal

The Hypercom device handler suppresses Track 2 and Track 1 discretionary data in the HPDH device dependent data that is written to the POS Terminal Data files (PTD). The device handler ensures that Track 2 and Track 1 discretionary data does not reside in the temporary context area of the PTD once the transaction response has been received from the authorizing entity. The manual card verification digits (CVD2) are also suppressed in the temporary context area of the PTD.

Track data and the CVD2 data are suppressed at the time of response processing. The suppression of the CVD2 data and Track 1 discretionary data occurs before the token data is logged to the PTDD2. The Track 2 discretionary data suppression takes place prior to updating the PTDD1 record.

Suppression occurs for all message types, including authorizations, financial, advices, and reversals.

Hypercom Triple DES Master/Session

The Hypercom device handler (HPDH) for BASE24-pos supports triple DES PIN encryption in coordination with the Transaction Security Services (TSS) module. The HPDH supports double-length PIN encryption keys of 32 hex characters (16 bytes) was added, enabling it to generate either single or double-length PIN encryption keys. When the HPDH receives an initialization request, it generates and sends a double-length PIN encryption key (if so configured) during the download of initialization parameters to the terminal.

Derived Unique Key per Transaction (DUKPT)

The HPDH module provides support for POS devices that derive a unique key for each transaction processed. Each unique key is derived from a base derivation key loaded into the terminal's security module (TSM) when it is initialized. The same base derivation key is loaded into a database at the host. When a POS device derives a unique key for each transaction, the PIN block in the request message is encrypted under the derived key and the request message includes a key serial number field in the message. The key serial number consists of a host derivation key identifier, a host derivation key subordinate identifier, the TSM identifier, and a transaction counter. When the host receives a PIN block encrypted under a

derived key, it uses the key serial number passed in the message and the terminal's base derivation key stored in the host database to derive the same unique transaction key, decrypt the PIN block, and verify the PIN.

HPDH Transaction Support

All of the transactions supported by the HPDH module must originate with a request from the terminal. A response is sent from the HPDH module for each request received unless the HPDH module is unable to read the BASE24-pos Terminal Data files (PTD) records that correspond to the terminal sending the request. In this case, no response is sent and the terminal times out. Message requests and responses are explained in more detail later.

The HPDH module supports both financial and administrative transactions. Financial transactions are monetary transactions performed on behalf of a cardholder, and adjustments to these transactions. Administrative transactions are monetary and nonmonetary transactions performed in support of financial transactions, typically by supervisory personnel.

The tables below provide lists of the BASE24-pos financial and administrative transactions supported by the HPDH module. Each transaction is followed by a corresponding description.

Financial Transactions

The following table describes the financial transactions supported by the HPDH module. Each transaction contained in the table is followed by a short description.

Financial Transactions	
Transaction	Description
Balance Inquiry	Online inquiry into the balance of a customer's account. This transaction can be initiated using a credit or debit card.
Card Verification	Online check to see if a customer's card is valid. This transaction can be initiated using a credit card.
Cash Advance	Cash disbursement from a customer's credit card account at the point of sale. This transaction can be initiated using a credit or debit card.

Financial Transactions	
Transaction	Description
Check Guarantee	Electronically reserves funds in a customer's checking account to cover a check written by the customer.
Check Verification	Online check to see if a customer's check is tied to a valid checking account.
Mail or Telephone Order	Purchase of goods or services where the cardholder is not present at the point of sale, in this case through the mail or by telephone.
Merchandise Return	Return of merchandise to a retailer for refund. The refund is electronically credited to the customer account.
Merchandise Return Adjustment	Corrects an error in a previously completed merchandise return transaction.
Normal Purchase	Purchase of goods or services at the point of sale using a credit or a debit card.
Online Void (credit)*	Voids (adjusts to 0) a credit transaction.
Online Void (debit)*	Voids (adjusts to 0) a debit transaction.
Preauthorization Purchase	Online check of a customer's account to determine if funds are available for an intended purchase. The amount preauthorized is placed on hold in the customer's account until the purchase is completed, a timer expires, or the hold is removed.
Preauthorization Purchase Completion	Completion of a purchase that was preauthorized. The actual amount of the purchase is entered using this transaction and the hold for the amount entered in the preauthorization purchase transaction is released. The transaction is then completed using the correct amount.
Purchase Adjustment	Corrects an error in a previously completed purchase transaction.

Financial Transactions	
Transaction	Description
Purchase With Cash Back	Combination purchase and cash withdrawal at the point of sale. The requested transaction amount is higher than the price of the purchase. The customer receives the difference in cash. This transaction is used only with debit cards.

* Specific to Hypercom devices.

Administrative Transactions

The following table describes the administrative transactions supported by the HPDH module. Each transaction contained in the table is followed by a short description.

Administrative Transactions	
Transaction	Description
Close Batch Request	Ends the current batch and opens a new batch to which data is logged and accumulated. When a close batch transaction is received, the system creates a batch totals record from the terminal totals in the BASE24-pos Terminal Data files (PTD) and writes it to the PTLF. It then clears the batch totals in the PTD. If the terminal can send its own batch totals, the system also compares the terminal totals to the PTD totals, and, if the debits, credits, and adjustments do not match exactly, writes the terminal totals to the PTLF immediately following the PTD totals. Note: When a batch is closed, the system also automatically writes a service totals record to the PTLF (prior to the batch totals record) and clears the service totals in the PTD.

Administrative Transactions	
Transaction	Description
Download Request	Causes a full download to be sent to the terminal.
Handshake Request	Verifies the status of the communications link.
Logon	Causes the HPDH module to generate new PIN and MAC keys. The module then returns the new keys in a separate response to the device. The Logon request is entered by the operator at the Hypercom terminal.

Hypercom-Specific Messages

The HPDH module supports message types that are specific to Hypercom terminals.

A batch upload message is a 0320 message from the terminal to the host that occurs for each transaction in a batch if the batch is out of balance. During a close batch request transaction, the HPDH module compares the BASE24 totals and the terminal totals. When the terminal and BASE24 do not have matching totals, BASE24 sends a response with a response code of 95, indicating that the totals are out of balance. This causes the terminal to start a batch upload. The HPDH module logs only the 0320 messages to the POS Transaction Log File (PTLF).

A terminal statistics message is an 0800 message sent from the T7 terminal after a batch close sequence has been processed. The terminal statistics message is the last message in a batch close sequence. The 0800 message is specific to Hypercom T7 terminals.

Refer to the *BASE24-pos Transaction Processing Manual* for definitions of the 0320 and 0800 messages.

Integration with BASE24-pos

The HPDH module can be integrated with BASE24-pos. BASE24-pos is a complete point-of-sale authorization and switching system that uses the technology of the Compaq NonStop system to provide reliable, fault-tolerant service 24 hours a day, seven days a week.

BASE24-pos offers routing of transactions based on card type and retailer floor and ceiling limits. If the primary destination for authorization is not available, BASE24-pos allows the transaction to be sent to up to two additional authorization centers before a decision is made about approving, denying, or referring the transaction. Destinations to which transactions can be routed include host computers and interchange networks.

In addition, BASE24-pos offers a comprehensive set of authorization options on the Compaq NonStop system. These include local and national negative identification, positive identification, and positive identification with account balances. Another authorization option for credit card accounts, parametric authorization, enables institutions to define additional criteria on which to base the authorization of transaction requests that exceed standard cardholder limits.

BASE24-pos also includes online control of all network communications elements and POS terminals, as well as a comprehensive set of daily and periodic reports for service providers and individual retailers.

Transaction Flow

When a Hypercom terminal sends a message to BASE24, the XPNET process receives the message and sends it to the bound Device Handler/Router/Authorization process.

The HPDH module is bound together with a Router module and Authorization module into a Device Handler/Router/Authorization process. The Router module receives any message coming into the bound process from a POS device, then routes it to the Device Handler module. The Router module also receives messages coming into the bound process from any other destination, but routes them instead to the Authorization module.

Tying the HPDH Module to BASE24-pos

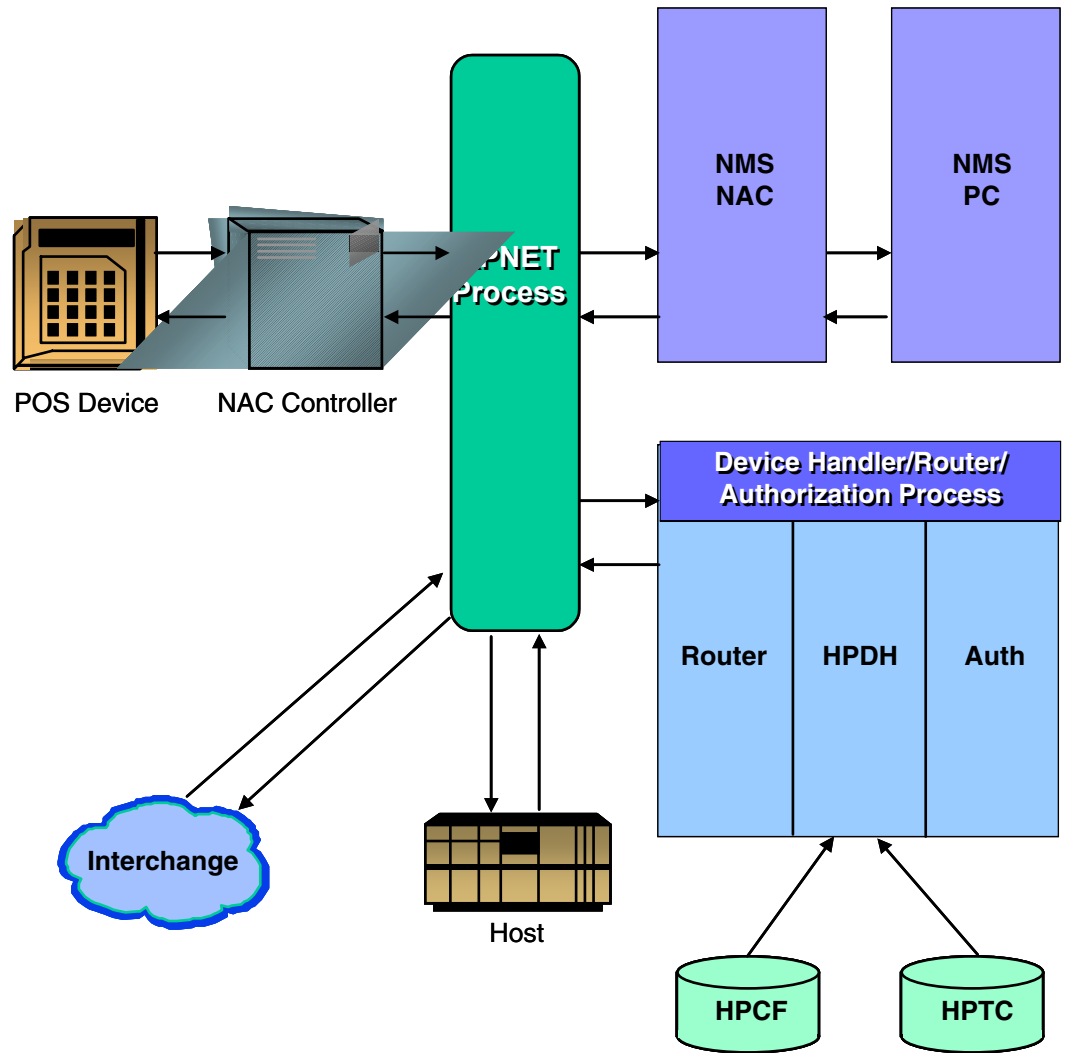
Within a BASE24-pos system, the HPDH module provides the link between the POS terminal and BASE24. Messages coming from Hypercom terminals must be sent to the HPDH module before proceeding to an authorizer. The HPDH module is the last BASE24 process to handle any messages leaving BASE24 on their way to the terminal.

The HPDH module translates messages from the Hypercom External Message format into BASE24-pos Standard Internal Message Format (PSTM), which is understood by all of the other BASE24-pos processes. When BASE24-pos is sending a message to the terminal, the HPDH module translates the message from the PSTM format to the Hypercom External Message format.

The network access controller (NAC) is another component that plays an important role in linking terminals to BASE24. The NAC is an interface between terminals and BASE24. It is responsible for collecting messages being sent to BASE24 and providing a more efficient means of distributing them.

Two types of NACs exist. When BASE24-pos sends a message to the terminal, the terminal NAC receives the message from the HPDH module in the Hypercom External Message format and translates it into the terminal's native mode. The network management system (NMS) NAC provides an interface between NMS PCs and BASE24. The NMS is a diagnostic tool used with the HPDH module.

The following diagram illustrates the relationships between the terminal, the terminal NAC, the NMS NAC, the NMS PC, the HPDH module, and BASE24 when a message is sent by a POS terminal to BASE24. The diagram also shows the configuration files for the HPDH module. The Hypercom Configuration File (HPCF) contains configuration parameters for Hypercom T4, T5, and T6 terminals and the Hypercom T7 Configuration File (HPTC) contains configuration parameters for Hypercom T7 terminals.



HPDH Module Versions

Release 6.0 of the HPDH module, which is documented in this manual, accommodates terminals that have firmware for Hypercom Program release P09. Older releases of the HPDH module accommodate terminals that have firmware for Hypercom Program release P03.

2: Terminal Configuration

The Hypercom Device Handler (HPDH) module allows customers the flexibility to configure their POS terminals to meet their exact specifications. For example, customers can determine the information that is downloaded to terminals and the configuration options supported by specific card prefix ranges.

Options supported by the HPDH module are based on configuration information in the following files:

- Acquirer Processing Code File (APCF)
- BASE24-pos Terminal Data files (PTD)
- Hypercom Configuration File (HPCF)
- Hypercom T7 Configuration File (HPTC)
- Institution Definition File (IDF)
- Logical Network Configuration File (LCONF)
- POS Retailer Definition File (PRDF)
- POS Transaction Log File (PTLF)

HPDH Configuration Files

This section describes the files used to configure the HPDH module. The descriptions explain the role that each file plays in determining the options supported by the HPDH module.

Hypercom Configuration File (HPCF)

The Hypercom Configuration File (HPCF) contains the configuration information for the HPDH module when Hypercom T4, T5, or T6 terminals are being used. It contains configuration data that is downloaded to a terminal. The HPCF also contains the configuration options allowed for cards within specific card prefix ranges. This file is accessed through BASE24-pos file maintenance. The HPCF screens are described in section 8.

Hypercom T7 Configuration File (HPTC)

The Hypercom T7 Configuration File (HPTC) contains the configuration information for the HPDH module when Hypercom T7 terminals are being used. It contains configuration data that is downloaded to a terminal. The HPTC also contains the configuration options allowed for cards within specific card prefix ranges. This file is accessed through BASE24-pos file maintenance. The HPTC screens are described in section 9.

Acquirer Processing Code File

The Acquirer Processing Code File (APCF) contains one record for each combination of acquirer transaction profile, message category, and ISO processing code that is supported by BASE24-pos acquirer endpoint in the system. The APCF also supports the use of wildcard characters for the acquirer transaction profile and message category, so that one APCF record can cover many combinations. Each acquirer transaction profile defines a set of transactions supported for an individual acquiring terminal or group of terminals. For BASE24-pos, retailer and administrative card transaction profiles are also used to determine whether an administrative card is required to perform a transaction and whether a transaction is allowed for a specific administrative card.

When processing a transaction from a Hypercom terminal, the HPDH module accesses the APCF extended memory table to determine whether a transaction is allowed from that device. The HPDH module accesses the APCF directly when downloading the transactions allowed table to a Hypercom device.

Institution Definition File (IDF)

The Institution Definition File (IDF) contains one record for each institution participating in the logical network and defines processing for each institution. The IDF contains routing tables for transaction routing within BASE24 and each institution's parameters for cards, dates, and processing control. For more information on the IDF, refer to the *BASE24 Base Files Maintenance Manual*.

The HPDH module uses the IDF to obtain the token retrieval option, if it is not set at the terminal level in the BASE24-pos Terminal Data files (PTD), and to obtain the default acquirer transaction profile, if it is not set at the retailer level in the PRDF or at the terminal level in the PTD. The token retrieval option indicates whether tokens are included in reversal messages, and if so, from where the token data is retrieved. The acquirer transaction profile specifies a set of allowed transaction processing codes and is used to determine whether an acquired transaction is allowed at a terminal.

The HPDH module accesses information in the IDF using the IDF extended memory table.

Log Message Configuration File (LMCF)

The Log Message Configuration File (LMCF) contains event message information such as text, routing code, and severity, that has been modified by the user. If the event message has been modified, the Device Handler/Router/Authorization process logs the modified information instead of the standard event message.

POS Retailer Definition File (PRDF)

The POS Retailer Definition File (PRDF) contains one record for each retailer in the logical network. These records are used in validating POS transactions and defining authorization and routing parameters. For more information on the PRDF, refer to the *BASE24-pos Files Maintenance Manual*.

BASE24-pos Terminal Data Files (PTD)

The BASE24-pos Terminal Data files (PTD) contain one record for each POS terminal in the logical network. The TERMINAL TYPE field located on screen 1 of the PTD identifies the type of terminal defined in specific PTD records, and must be set to a value of D0 for all terminals using the HPDH module (that is, Hypercom T4, T5, T6, and T7 terminals). The PTD also contains fields that apply only to Hypercom terminals. These fields are located in a PTD device dependent area. The PTD consists of the following three files:

- The BASE24-pos Terminal Data Dynamic File—general data (PTDD1) contains dynamic data such as terminal totals that changes during transaction processing.
- The BASE24-pos Terminal Data Static File—general data (PTDS1) contain static terminal that rarely, if ever, changes during the course of transaction processing.
- The BASE24-pos Terminal Data Dynamic File—scratch pad (PTDD2) is an optional file used to store last message token data for reversal processing. If this file is not used, last message token data is retrieved from the PTLF for reversal processing.

The HPDH module uses the TERMINAL ID field contained in request messages to access the appropriate records in the PTD for Hypercom terminals.

The PTD also contains information about the logging of transactions processed by BASE24-pos. All approved or denied transactions are logged to the POS Transaction Log File (PTLF). Values in the TERMINAL CUTOVER field and the five LOG TOTALS fields on PTD screen 3 determine if and when totals are logged to the PTLF.

For more information about the PTD and PTLF, refer to the ***BASE24-pos Files Maintenance Manual***.

POS Transaction Log File

The POS Transaction Log File (PTLF) contains information about the transactions processed by BASE24-pos. Approved or denied transactions are logged to the PTLF only if they are sent by a terminal. When a close batch request, a close shift request, or a close day request transaction is performed, totals are logged to the PTLF if logging PTLF totals is configured in the LOG TOTALS field on screen 3 of the PTD. If logging PTLF totals is not configured, no totals are logged to the PTLF.

Last message token data can be retrieved from the PTLF for reversal processing if a POS terminal is configured to do so.

Logical Network Configuration File (LCONF)

One Logical Network Configuration File (LCONF) exists for each logical network. It contains the assigns and params associated with the network.

Assigns associate the physical names of objects in the logical network to names used in BASE24 code. Assign information must be present; otherwise BASE24 is unable to process. Most assigns link the actual physical file names to logical file names as they are referenced in BASE24 program code. For example, in order to access the HPCF, a process or server must be able to identify the fully qualified name of the file as it exists in a logical network. The LCONF must contain an assign linking the HPCF name to the actual physical file. Other assigns link the physical names for different hardware to the names as they are referenced in the BASE24 program code. For example, the types of hardware referenced in BASE24 program code includes spooler locations, terminal locations, and security devices.

Params set processing values at the logical network level. In some instances when param information is not present, a process abends during initialization; however, some params are not critical for processing.

See the *BASE24 Logical Network Configuration File Manual* for additional information about LCONF assign and param descriptions and maintenance.

Assigns

The HPDH module requires several assigns in the LCONF in order to perform processing. These assigns are as follows and are in addition to any assigns required by the Router and Authorization modules bound to the HPDH module.

APCF — The fully qualified file name of the Acquirer Processing Code File (APCF).

KEYD — The fully qualified file name of the Derivation Key File (KEYD).

POS-HPDH-HPCF — Contains the name of the Hypercom Configuration File (HPCF) used to configure Hypercom T4, T5, and T6 terminals.

POS-HPDH-HPTC — Contains the name of the Hypercom T7 Configuration File (HPTC) used to configure Hypercom T7 terminals.

POS-PTD-DYN-GNRL — Contains the filename of the general BASE24-pos Terminal Data Dynamic File—general data (PTDD1). This file contains dynamic core terminal data (i.e., data common to all types of devices that is subject to change), device information (i.e., data specific to a device type), and device-dependent data (i.e., data specific to each terminal defined).

POS-PTD-DYN-SCRATCH-PAD — Contains the filename of the BASE24-pos Terminal Data Dynamic File—scratch pad (PTDD2). This file contains dynamic token data for reversal processing. This assign is optional. If this assign is used, the Standard POS Device Handler module stores and retrieves token data from the PTDD2. If this assign is not used, data for reversal processing is obtained from last transaction data in the POS Transaction Log File (PTLF).

POS-PTD-STATIC-GNRL — Contains the filename of the BASE24-pos Terminal Data Static File—general data (PTDS1). This file contains static data, which is information that does not change during the course of transaction processing.

POS-DH-PTD-EXTMEM-SWAPVOL — Contains the name of the POS Terminal Data File extended memory swap volume.

Params

The HPDH module uses the following params during processing. These params are in addition to any params required by the Router and Authorization modules bound to the HPDH module.

POS-DH-DUKPT-UPDATE-METHOD — Specifies whether the HPDH module can automatically update the PIN ENCRYPTION TYPE field on BASE24-pos Terminal Data files (PTD) screen 7 to a value of 07 (DUKPT) when the Key Serial Number (KSN) is received in a message for the first time and a Derivation Key File (KEYD) record exists for the HPDH terminal.

POS-DH-KEYD-READ-FROM-DISK — Specifies whether the HPDH module reads the Derivation Key File (KEYD) from disk or from memory.

POS-DH-KSN-APPRV-OPT — Controls the action taken by the HPDH if the static data received in the Key Serial Number (KSN) of a request message does not match the KSN static data maintained in PTD device dependent data.

POS-DH-KSN-CNTR-APPRV-OPT — Controls the action taken by the SPDH based upon the the transaction counter for two transactions from the same terminal.

POS-DH-KSN-DESCR — Specifies the default key serial number (KSN) descriptor required by the Thales e-Security hardware security module (HSM) to perform a translate command.

POS-DH-KSN-MAX-DIFFERENCE — Indicates the largest allowable difference in the transaction counter between two transactions from the same terminal.

POS-DH-MAX-TERMS — The maximum number of POS Terminal Data file records that each HPDH module can hold in extended memory at one time. Thus, this param indicates the maximum number of terminals each HPDH module can service.

When an HPDH module initializes or re-initializes, space is allocated in extended memory for the BASE24-pos Terminal Data file records the module maintains. By accessing this data in extended memory rather than from a disk file, these modules can process transactions more quickly.

If the value in this param is set too low, the HPDH module can run out of extended memory space and be unable to process for all of its assigned devices. If the value in this param is set too high, the excess extended memory space is wasted. The value of this param should be equal to the total number of devices that are associated with each HPDH module. This param can be set to any numeric value greater than zero. However, the value in this param should be greater than the values in the POS-DH-MAX-TERMS-IN-DYN-TBL and the POS-DH-MAX-TERMS-IN-STATIC-TBL params. The default value is 50.

POS-DH-MAX-TERMS-IN-DYN-TBL — The maximum number of BASE24-pos Terminal Data Dynamic File (PTDD1 and PTDD2) records that an HPDH module can store in extended memory at one time. Thus, this param indicates the amount of extended memory each HPDH module must reserve for PTDD1 and PTDD2 records and the maximum number of transactions each HPDH module can process at one time.

The HPDH module stores a PTDD1 record, and optionally a PTDD2 record, in memory each time it receives a transaction from a POS device. The HPDH module retains these PTDD1 and PTDD2 records in memory until it receives an authorization response from the Authorization module and, if required, a completion message from the POS device.

If the value in this param is too low, the HPDH module can run out of extended memory during peak processing times and transactions are declined until more extended memory becomes available. If the value in this param is set too high, the excess extended memory is wasted.

The value in this param is an estimate of the maximum number of devices that can have transactions in progress at one time. This param can be set to any numeric value greater than zero. However, the value in this param should not be greater than the value in the POS-DH-MAX-TERMS param. The default value is 25.

POS-DH-MAX-TERMS-IN-STATIC-TBL — The maximum number of BASE24-pos Terminal Data Static File—general data (PTDS1) records that an HPDH module can store in extended memory at one time. Thus, this param indicates the amount of extended memory each HPDH module must reserve for PTDS1 records and the maximum number of transactions each HPDH module can process at one time.

The HPDH module stores a PTDS1 record in memory the first time it receives a transaction from a POS device. The module continues to dynamically add PTDS1 records to memory until it runs out of space, the static data is re-initialized, or the process is warmbooted. When the module runs out of space, the oldest record is deleted from memory and the new record is added. When the static data is re-initialized or the process is warmbooted, all of the PTDS1 records are deleted from memory and the module begins adding them again dynamically as transactions are processed.

If the value in this param is too low, the HPDH module can run out of extended memory during peak processing times and transactions are declined until more extended memory becomes available. If the value in this param is set too high, the excess extended memory is wasted.

The value in this param is an estimate of the maximum number of devices that can have transactions in progress at one time. This param can be set to any numeric value greater than zero. However, the value in this param should not be greater than the value in the POS-DH-MAX-TERMS param. The default value is 25.

POS-HPDH-AUTH-RESP-TIMEOUT — Indicates the number of seconds an HPDH module waits for a response from the Authorization module.

POS-HPDH-LOG-TERM-STATS — Indicates whether to log Hypercom terminal statistics to the PTLF.

POS-HPDH-MAX-HPCFS — Indicates the maximum number of Hypercom Configuration File (HPCF) records to be stored in memory by the HPDH module at one time.

POS-HPDH-MAX-HPTCS — Indicates the maximum number of Hypercom T7 Configuration File (HPTC) records to be stored in memory by the HPDH module at one time.

POS-HPDH-NAC-ID*identifier* — A Hypercom network access controller (NAC) ID and its corresponding station name. The *identifier* specifies the NAC ID. This param is optional and needs to be set up only if NMS processing is being used.

POS-HPDH-NMS-ID*identifier* — A Hypercom network management system (NMS) ID and its corresponding station name. The *identifier* specifies the NMS ID. This param is optional and needs to be set up only if NMS processing is being used.

POS-HPDH-OLDMSG-TIMEOUT — Indicates the number of seconds a message is allowed to remain in the HPDH queue.

3: Hypercom Device Handler (HPDH) Module

In BASE24-pos, the Device Handler module is bound together with the Router and Authorization modules to create the Device Handler/Router/Authorization process. Each module has distinct functions, but they operate together and are defined as one bound process. Because the three modules are bound, transactions passed between them do not travel through the XPNET process.

The Router and Authorization modules are the same for each Device Handler/Router/Authorization process. However, the Device Handler module differs between Device Handler/Router/Authorization processes depending on the type of POS device attached to the system. For example, a different Device Handler module is required for BASE24-pos Standard POS devices than is required for Hypercom devices.

This section describes the processing associated with the Device Handler module required for Hypercom devices: the HPDH module.

This section describes the processing performed by the HPDH module. It includes the following information about the HPDH module:

- An overview of the HPDH module
- A description of each file used by the HPDH module during processing
- A description of the steps the HPDH module performs during initialization and warmbooting
- Network text commands recognized by the HPDH module
- A description of event message generation and documentation

HPDH Module Overview

The HPDH module is the primary interface between a Hypercom device and BASE24-pos. Each type of POS device has its own message format and method of processing transactions. The HPDH module supports Hypercom devices. The HPDH module reformats messages sent from Hypercom devices into the POS Standard Internal Message (PSTM). The PSTM enables the other processes associated with BASE24-pos to process the transaction.

All messages that BASE24-pos sends to Hypercom devices must be reformatted into *native mode*, a message format the Hypercom device can understand. The HPDH module performs this translation before the transaction message is forwarded to the Hypercom device.

The HPDH module also controls several other functions in the BASE24-pos system. These functions are listed and briefly described in the following paragraphs.

Enforcing Transaction Sequence

The HPDH module can enforce the sequence in which transactions are processed in the system.

Allowed Transaction Checking

For each transaction acquired from a Hypercom device, the HPDH module uses an acquirer transaction profile to determine whether the transaction is allowed from that terminal. Acquirer transaction profiles can be set at the terminal level in the BASE24-pos Terminal Data files (PTD), at the retailer level in the PRDF, or at the institution level in the IDF. The HPDH module always attempts to retrieve the acquirer transaction profile from the PTD record first. If the acquirer transaction profile is found in the PTD, it is used to read the Acquirer Processing Code File (APCF) extended memory table as described below. If the acquirer transaction profile is blank in the PTD record, the module attempts to retrieve the acquirer transaction profile from the retailer's PRDF record. If the acquirer transaction profile is found in the PRDF, it is used to read the APCF extended memory table as described below. If the acquirer transaction profile is blank in the retailer's PRDF record, the module attempts to retrieve the default acquirer transaction profile from the terminal owner's IDF record. If the acquirer transaction profile is found in the

IDF, it is used to read the APCF extended memory table as described below. Finally, if the acquirer transaction profile is blank in the IDF, a value of all asterisks is used to read the APCF extended memory table.

Once the acquirer transaction profile value used to read the APCF extended memory table has been determined, the HPDH module attempts to read the APCF extended memory table using the acquirer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The APCF supports the use of wildcard characters for the acquirer transaction profile and message category, so that one APCF record can cover many combinations. The HPDH module calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the APCF extended memory table.

Using the acquirer transaction profile, message category, and ISO processing code, the HPDH module steps through the records in the APCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Acquirer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed. If the transaction is not allowed, the transaction is declined with response code 055 (invalid transaction). If the transaction is allowed, the HPDH module performs the following:

- Adds the Transaction Profile token with the acquirer transaction profile used.
- Adds the Acquirer Routing token if the transaction is allowed and an authorization destination is specified in the APCF record.

Note: The APCF also enables you to specify an authorization destination for a transaction other than the Router/Authorization module bound in with the HPDH module. This mechanism allows you to route transactions received from a POS device to an application process in the XPNET system. When a

transaction is routed in this way, you can also specify whether the transaction response returned from the authorization destination is logged to the POS Transaction Log File (PTLF).

- Adds the Transaction Description token with the transaction description from the APCF record. If adding this token causes the maximum message length to be exceeded, it is not added.

Cutting Over Terminals

The HPDH module uses a cutover transaction to cut over terminals for daily settlement. The institution designates a timeframe in which cutover must take place and designates a transaction to start cutover. When the Device Handler/Router/Authorization process receives that transaction during the designated timeframe, the terminal is cut over. If the HPDH module does not perform this cutover, the Settlement Initiator process performs a force cutover.

Decrypting PINs

The HPDH module uses software decryption to decrypt PINs for incoming transactions. However, if decryption is done with a hardware security module, the Device Handler module passes the transaction to the Authorization module. The Authorization module then passes the transaction to the hardware security module. Refer to the ***BASE24 Transaction Security Manual*** for more information on PIN decryption.

Authenticating Messages

The HPDH module can authenticate messages passed between BASE24-pos and a POS device by using message authentication codes (MACs). The HPDH module supports the Hypercom proprietary algorithm for software verification of message authentication codes (MACs). The Hypercom proprietary algorithm uses a form of the DES encryption algorithm to generate the MAC. Refer to section 6 of this manual for a detailed discussion of how MACs are used specifically with the HPDH module.

Downloading

The HPDH module performs full downloads consisting of terminal configuration information. The information is downloaded from configuration files. The HPDH module downloads information from the Hypercom Configuration File (HPCF) to Hypercom T4, T5, and T6 devices, and from the Hypercom T7 Configuration File (HPTC) to Hypercom T7 terminals.

Controlling Response Timers

The HPDH module sets and stops transaction response timers as transactions pass in and out of the HPDH module. When the HPDH module receives a transaction from a device, it sets a response timer and then passes the transaction on for authorization. When processing completes, the HPDH module deletes the timer. If processing does not complete before the timer expires, the HPDH module declines the transaction and sends a timeout response to the terminal.

Reversal Processing

The TOKEN RETRIEVAL OPTION field on Institution Definition File (IDF) screen 19 indicates whether BASE24-pos Device Handler modules include tokens in reversal messages, and if so, whether the token data is retrieved from the second BASE24-pos Terminal Data Dynamic File—scratch pad (PTDD2) or from the POS Transaction Log File (PTLF). This option can be overridden at the terminal level using the TOKEN RETRIEVAL OPTION field on POS Terminal Data files (PTD) screen 2.

When the HPDH module receives a request from a POS device, it creates the TLF token with a field indicating the token reversal processing option. If the option is set to a value of 2 in the TLF token, the Router/Authorization module updates the token with POS Transaction Log File (PTLF) key data when processing the transaction response. The HPDH module updates device-dependent PTD extended memory with the TLF Token data and writes this data to the BASE24-pos Terminal Data Dynamic File—general data (PTDD1). It can then use this information to retrieve the record from the PTLF during reversal processing. If the token retrieval option is set to a value of 0 or 1, the Router/Authorization module does not update the TLF token.

When the HPDH module receives a response from the Router/Authorization module, it checks the token retrieval option in the TLF token. If the option is set to a value of 1, the HPDH module stores token data from the response message in the

PTDD2. If no record exists in the PTDD2 for the terminal from which the transaction originated, a new record is created using the terminal ID as the key. If a record already exists for the terminal, the record is updated with the token data from the current message. The HPDH module can then retrieve the record from the PTDD2 during reversal processing.

File Usage

The HPDH module uses the following files during processing. Each file listed is followed by a brief description explaining how it is used by the HPDH module.

Acquirer Processing Code File (APCF) — The HPDH module accesses the APCF extended memory table to determine whether a transaction is allowed from a POS device. The HPDH module accesses the APCF directly when downloading the transactions allowed table to a POS device.

Hypercom Configuration File (HPCF) — The HPCF contains the configuration information for the HPDH module when Hypercom T4, T5, or T6 terminals are being used. It contains configuration data that is downloaded to a terminal. The HPCF also contains the configuration options allowed for cards within specific card prefix ranges.

Hypercom T7 Configuration File (HPTC) — The HPTC contains the configuration information for the HPDH module when Hypercom T7 terminals are being used. It contains configuration data that is downloaded to a terminal. The HPTC also contains the configuration options allowed for cards within specific card prefix ranges.

Institution Definition File (IDF) — The HPDH module accesses the IDF extended memory table to obtain the acquirer transaction profile (if it is not defined in the POS Terminal Data files or PRDF) and token retrieval option (if it is not defined in the POS Terminal Data files).

Logical Network Configuration File (LCONF) — The LCONF contains the assigns and params used during processing. The HPDH module reads the LCONF at initialization and re-initialization, and maintains the assigns and params in memory.

Log Message Configuration File (LMCF) — The LMCF contains event message information such as text, routing code, and severity, that has been modified by the user. At initialization, the Device Handler/Router/Authorization process builds a table in memory from the LMCF records that contain modified information. When an event message is generated, the Device Handler/Router/Authorization process accesses the table to verify whether that event message has been modified. If the event message has been modified, the Device Handler/Router/Authorization process logs the modified information instead of the standard event message.

POS Retailer Definition File (PRDF) — The HPDH module accesses the PRDF to obtain the acquirer transaction profile if it is not defined in the BASE24-pos Terminal Data files (PTD).

BASE24-pos Terminal Data files (PTD) — The PTD contains the characteristics of the POS terminal, including the transaction security methods, terminal owner, logging and transaction data, floor limits, card type, and the POS retail transaction set. It is also used to accumulate transaction and settlement counts and amounts and preserve transaction context. The PTD is read and updated every time a transaction is processed for a POS terminal. The HPDH module uses records in the PTD to build a table of PTD key positions in extended memory. This table enables the HPDH module to access and update the PTD records as transactions are processed.

POS Transaction Log File (PTLF) — The PTLF provides an audit trail for activity occurring at POS terminals. The HPDH module logs all financial transactions occurring at Hypercom terminals throughout the logical network to the PTLF. The HPDH module also logs services totals and settlement totals for Hypercom terminals to the PTLF.

HPDH Module Initialization and Warmboot

This section describes the steps taken by the HPDH module when it is initialized or warmbooted.

Initializing the HPDH Module

When the Device Handler/Router/Authorization process is initialized, the Router module begins initialization first. Part of the initialization process required by the Device Handler module is within the Router module. The steps described in this section assume the Router module has already initialized.

The HPDH module performs as follows to initialize:

1. Initiates direct communication with an EMS Collector process.
2. Reads the LMCF for modified event messages.
3. Allocates an extended memory segment for the network access controller (NAC) and network management system (NMS) tables.
4. Retrieves the following assigns and params from the LCONF.

Assigns

APCF
POS-DH-PTD-EXTMEM-SWAPVOL
POS-HPDH-HPTC
POS-HPDH-HPCF
POS-PTD-POS-PTD-DYN-GNRL
POS-PTD-DYN-SCRATCH-PAD
POS-PTD-STATIC-GNRL

Params

POS-DH-MAX-TERMS
POS-DH-MAX-TERMS-IN-DYN-TBL
POS-DH-MAX-TERMS-IN-STATIC-TBL
POS-HPDH-AUTH-RESP-TIMEOUT
POS-HPDH-MAX-HPCFS
POS-HPDH-MAX-HPTCS
POS-HPDH-OLDMSG-TIMEOUT
POS-HPDH-LOG-TERM-STATS
POS-HPDH-NMS-ID*Identifier*
POS-HPDH-NAC-ID*Identifier*

5. Opens the BASE24-pos Terminal Data files (PTD) and sets the file mode so any process attempting to lock or read a PTD record while the HPDH module has the record locked receives an error 73. This error indicates the file or record is locked. If the HPDH module is unable to open the BASE24-pos Terminal Data files (PTD), the HPDH module abends.
6. Allocates and initializes extended memory space for the number of BASE24-pos Terminal Data file records it needs at one time. The number of records that can be in memory at one time is specified in the POS-DH-MAX-TERMS param.
7. Opens the Hypercom Configuration File (HPCF) and the Hypercom T7 Configuration File (HPTC). These files are used in the configuration of the terminal. If the HPDH module is unable to open the HPCF or the HPTC, the HPDH module abends.
8. Allocates extended memory for the HPCF based on the setting of the POS-HPDH-MAX-HPCFS param.
9. Allocates extended memory for the HPTC based on the setting of the POS-HPDH-MAX-HPTCS param.

Warmbooting the HPDH Module

When the HPDH module receives a 9501 (WARMBOOT) command, it performs steps 3 through 9 of the above initialization procedures. The HPDH module keeps the EMS process communication established in step 1 and the modified event messages obtained from the LMCF in step 2 of the initialization procedures.

Warmbooting Static POS Terminal Data

When the HPDH module receives a WARMBOOT PTD command, it warmboots static terminal data held in extended memory. The module re-reads its BASE24-pos Terminal Data Static File—general data (PTDS1) into extended memory. Any time static data is changed for a Hypercom device on the PTD screens, it is not used in processing until the associated HPDH module receives this command or is re-initialized.

Network Text Commands

Network text commands are used to send information from a network control facility to a process. These commands can be entered by an operator using a network control facility and serve as a tool that an operator can use to perform network management functions such as changing certain processing parameters, warmbooting a specific process, monitoring a specific process, and resolving problems encountered by a specific process. The following text commands can be used with the HPDH module. For detailed descriptions and the syntax of all text commands, refer to the *BASE24 Text Command Reference Manual*.

9501 (WARMBOOT) — Directs the HPDH module process to warmboot. For a list of the steps the HPDH module follows when a WARMBOOT command is issued, refer to the “HPDH Module Initialization and Warmboot” topic presented earlier in this section.

9525 (DEACTIVATE FLAG) — Sets the status of the terminal to activated, deactivated, or pending deactivation. If the terminal is deactivated, the HPDH module declines transactions originated by that terminal. The default (initialized value) is deactivated. If the deactivate flag is in the command message set to 1 (deactivate) and all totals (batch, shift, day, and network) are zero, the terminal is deactivated. If any totals are not zero, the terminal is identified as pending deactivation and only balancing transactions are accepted from the terminal. If the deactivate flag in the command message is not 1, the terminal is activated.

This command activates or deactivates a single terminal, a range of terminals, or all of the terminals in the network whose terminal type is the same as the Device Handler process (that is, all the Hypercom terminals in the network).

9529 (WARMBOOT PTD) — Directs the HPDH module to warmboot static terminal data held in extended memory. This command should be issued any time an operator performs file maintenance on a BASE24-pos Terminal Data files (PTD) field that is stored in the BASE24-pos Terminal Data Static File—general data (PTDS1). Static data that is modified in the PTDS1 is not used in processing until the PTDS1 is warmbooted or the HPDH module is re-initialized.

ALTERATTRIBUTE — Directs the HPDH module to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical

Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. The EMT warmboot notify list params are used only when issuing the command from the DCT. For more information on the EMT Control Command screen, refer to the *BASE24 Device Control Manual*.

CLOSEBATCH — Causes the HPDH module to close its current batch. Batch and service totals are logged to the PTLF and the batch and service totals in the PTD are cleared. This command can be used when the terminal has lost its batch information and you need to clear the PTD totals so that the batch balances.

CLOSEFLAGON — Causes a flag to be set in all responses to the terminal indicating that the terminal should perform a batch close transaction. This function is supported only by Hypercom T4, T5, T6, and T7 terminals. The flag is set in all responses to the terminal until it is cleared by a settlement transaction or by a CLOSEFLAGOFF command.

CLOSEFLAGOFF — Causes a flag to be deactivated in all responses to the terminal indicating that the terminal should not perform a batch close transaction. This function is supported only by Hypercom T4, T5, T6, and T7 terminals.

COMMKEY — Takes the key that was injected into the terminal and enters it into the HPDH module. The HPDH module can then store this key in the PTD. This command is used when the terminal does not support BASE24 parameter downloading. If the terminal does not receive the PIN key from the HPDH module (that is, if the PIN key is not downloaded from the HPDH module), this command makes the HPDH module aware of the new key.

LOADFLAGON — Indicates that download data is waiting for the terminal. Upon receipt of the LOADFLAGON command, the HPDH module sets the REQUEST-LOAD flag in the device dependent area of the BASE24-pos Terminal Data Dynamic File—general data (PTDD1) to on. When the next response is being formatted to the terminal, the HPDH module sets a bit in the Processing Code (P-2) data element to indicate that a download is waiting.

LOADFLAGOFF — Indicates that download data is not waiting for the terminal. The LOADFLAGON command sets the REQUEST-LOAD flag in the device dependent area of the PTDD1 to off. When the next response is being formatted to the terminal, the HPDH module sets a bit in the Processing Code (P-2) data element of the message to indicate that a download is not waiting.

RESETSTATUS — Resets the transaction status in the TRAN-STATUS field in the device dependent area of the PTDD1 to NO-TRAN-IN-PROGRESS. This command is intended for use if a transaction abends or some other uncontrollable event leaves the transaction status corrupted in the PTDD1.

STATUSEXTMEM — Directs the HPDH module to display statistics about its BASE24-pos Terminal Data files (PTD) extended memory segment on the network log. This command enables network operators to monitor extended memory usage and determine whether adjustments are needed.

Event Message Generation

The Hypercom Device Handler (HPDH) module generates event messages to assist network operators in performing their daily tasks. Event messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Event messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

Customers can access event message documentation electronically. Event messages are contained in files on a designated Compact NonStop system volume and subvolume. Each file contains event messages for a specific BASE24 module. Files on the designated Compaq NonStop system volume and subvolume are organized according to product module ID (PMID). One PMID exists for each process in BASE24. PMIDs identify a specific process in the BASE24 system (for example, PSHYPRCM identifies the HPDH module). Providing event message documentation in files gives customers greater flexibility by allowing them to print out event message documentation as needed at their location or to electronically manipulate the data to suit their needs.

Documentation is available in file form for the event messages generated by the HPDH module. It is located on the BAxxLOGM subvolume in a file named PSHYPRCM, where xx is the number of the current release. For a complete explanation of event message documentation, refer to the ***BASE24 Event Message Reference Manual***.

4: Hypercom External Message

The Hypercom External Message is a variable-content, variable-length message, based on the ISO 8583 message format developed by the International Organization for Standardization (ISO). The message is configured differently, based on the type of message being sent. This section contains the following information concerning the Hypercom External Message:

- Message overview
- Transport protocol data unit (TPDU) message header
- Application messages
- Network management system (NMS) messages
- Message text format
- Primary data elements
- Transaction codes

Message Overview

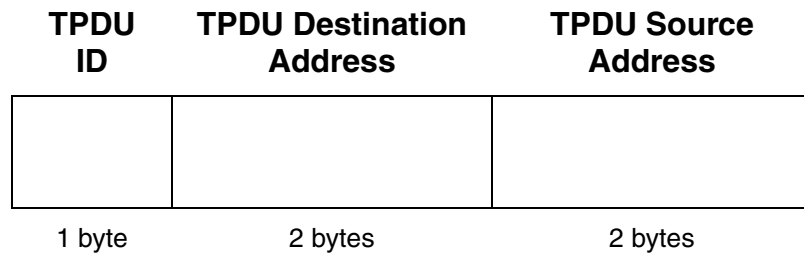
Every message being sent to or from a Hypercom Device Handler (HPDH) module must be preceded by a transport protocol data unit (TPDU) message header. The TPDU message header identifies the type of message being sent, the destination of the message, and the port from which the message originated.

The TPDU message header is followed by the message text. Two types of messages exist: application messages and network management system (NMS) messages. For application messages, the Hypercom External Message is a variable-length, variable-content message, based on the ISO 8583 message format. Every message associated with the HPDH module consists of a message type identifier, a 64-bit primary bit map, and up to 64 primary data elements. Not all primary data elements are currently supported. For NMS messages, the Hypercom External Message contains data that is not used by BASE24, but is used by the network access controller (NAC) and the NMS NAC. BASE24 serves as a passthrough to route message information to the appropriate destination (that is, a NAC or an NMS NAC) for NMS messages.

Any supported primary data elements received in requests are mapped to their corresponding fields in the POS Standard Internal Message (PSTM) before being sent for authorization. Any supported primary data elements required in responses are filled in from their corresponding fields in the PSTM response. Unsupported primary data elements required in responses are echoed from the request, if possible, or assigned initialized values. Refer to the ***BASE24-pos Transaction Processing Manual*** for more information concerning the PSTM.

Transport Protocol Data Unit (TPDU) Message Header

The HPDH module uses three parts of the TPDU, as shown in the following illustration.



The TPDU ID is the first TPDU message header field used by the HPDH module. The HPDH module reads the TPDU ID to determine the type of message being sent. The TPDU ID contains one of two values. If the TPDU ID contains a value of 60 (hexadecimal), the message is a transaction message and is processed by the HPDH module. A TPDU ID value of 68 (hexadecimal) indicates the message should be sent to a network access controller (NAC) hooked up to terminals or an NMS NAC hooked up to NMS PCs for further processing.

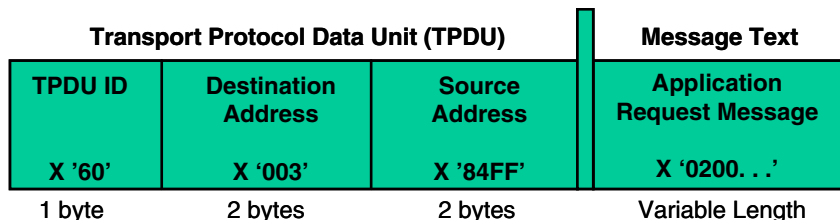
Data contained in the TPDU destination address and TPDU source address fields depends on the TPDU ID and the direction of the message. The following pages contain examples to illustrate the various possibilities.

Application Messages

This section describes the components contained in an application message. The following examples show the components of an application message being sent from a network access controller (NAC) to BASE24 and an application message being sent from BASE24 to a NAC. The components for each are explained after the example.

NAC to BASE24

This example illustrates an application message being sent from a NAC to BASE24.



The first field in this message contains the transport protocol data unit (TPDU) ID. A value of 60 (hexadecimal) in this field indicates the message is an application message.

The second field in this message contains the TPDU destination address, which breaks down into an application type and the destination international ID. The destination address is a 2-byte field and can be defined as follows:

Number of Bits	0	Appl. Type	Destination International ID	
	1	3	4	8

The value of the application type represents the type of terminal from which the transaction originated. The valid values are as follows:

- 0 = Hypercom
- 5 = Visa I
- 6 = Visa II

The third field in this message contains the TPDU source address, which identifies the port from which the message originated.

BASE24 to NAC

This example illustrates an application message being sent from BASE24 to a NAC.

Transport Protocol Data Unit (TPDU)			Message Text
TPDU ID	Destination Address	Source Address	Application Response Message
X '60'	X '84FF'	X '003'	X '0210. . .'
1 byte	2 bytes	2 bytes	Variable Length

The first field in this message contains the TPDU ID. A value of 60 (hexadecimal) in this field indicates the message is an application message.

The HPDH module obtains the addresses used in the second and third fields of this message from the second and third fields of the message received from the NAC. The HPDH module uses the source address in the message received from the NAC as the destination address in this message and uses the destination address in the message received from the NAC as the source address in this message.

Network Management System (NMS) Messages

This section describes the components contained in a network management system (NMS) request and response message. The HPDH module acts as a passthrough for NMS messages between the terminal NAC and the NMS PC, routing NMS messages to and from the NMS PC and the Hypercom NACs. The HPDH module provides the NMS PC and NACs with the source and destination of the messages. When an NMS message from a NAC is routed through the BASE24 system, the HPDH module must inform the NMS PC of the NAC from which the message originated. The following examples show the components of NMS messages. The components for each are explained after the example.

NMS NAC to BASE24

This example illustrates an NMS message being sent from an NMS NAC to BASE24.

Transport Protocol Data Unit (TPDU)			Message Text
TPDU ID	Destination Address	Source Address	NMS Request/Response Message
X '68'	X '84FF'	X '0221'	X '...'
1 byte	2 bytes	2 bytes	Variable Length

The first field in this message is the TPDU ID. A value of 68 (hexadecimal) in this field indicates the message is an NMS message.

The second field in this message contains the TPDU destination address used by the NAC.

The third field in this message contains the TPDU source address, which is a 2-byte field that contains the NMS ID and NAC ID. The following table illustrates the composition of the source address and the corresponding values from the NMS message example.

	Source Address		
	0	NMS ID	NAC ID
Number of Bits	4	4	8
Example Value	0	2	21

In the NMS message example, the HPDH module receives the message from the station associated with NMS ID 2 (LCONF param POS-HPDH-NMS-ID2) and routes the message to the station associated with NAC ID 21 (LCONF param POS-HPDH-NAC-ID21).

BASE24 to NAC

This example illustrates an NMS message being sent from BASE24 to a NAC.

Transport Protocol Data Unit (TPDU)			Message Text
TPDU ID	Destination Address	Source Address	NMS Request/Response Message
X '68'	X '84FF'	X '0221'	X '...'
1 byte	2 bytes	2 bytes	Variable Length

The first field in this message contains the TPDU ID. A value of 68 (hexadecimal) in this field indicates the message is an NMS message.

The second field in this message contains the TPDU destination address used by the NAC.

The third field in this message contains the TPDU source address, which is a 2-byte field that contains the NMS ID and NAC ID. The following table illustrates the composition of the source address and the corresponding values from the NMS message example.

Number of Bits	Source Address		
	0	NMS ID	NAC ID
	4	4	8
	0	2	21

When the NMS sends a message to BASE24 for routing to a NAC, the HPDH module scans through an internal NMS ID table to verify that the NMS message it received for routing to a NAC was sent from a defined NMS. BASE24 then uses the value in the NAC ID field to route the message to the correct station. In the example, the message is received from the station associated with NMS ID 2 (LCONF param POS-HPDH-NMS-ID2). The message is then routed to the station associated with NAC ID 21 (LCONF param POS-HPDH-NAC-ID21).

NAC to BASE24

This example illustrates an NMS message being sent from a network access controller (NAC) to BASE24. The NAC swaps the destination and source address in its response to BASE24.

Transport Protocol Data Unit (TPDU)			Message Text
TPDU ID	Destination Address	Source Address	NMS Request/ Response Message
X '68'	X '0221'	X '84FF'	X '...'
1 byte	2 bytes	2 bytes	Variable Length

The first field in this message contains the TPDU ID. A value of 68 (hexadecimal) in this field indicates the message is an NMS message.

The second field in this message contains the TPDU destination address, which is a 2-byte field that contains the NMS ID and NAC ID. The following table illustrates the composition of the destination address and the corresponding values from the NMS message example.

Number of Bits	Destination Address		
	0	NMS ID	NAC ID
	4	4	8
	0	2	21

When the NAC sends a message to BASE24 for routing to an NMS NAC, the HPDH module scans through an internal NAC ID table to verify that the NMS message was sent from a defined NAC for routing to an NMS NAC.

The third field in this message contains the TPDU source address, which identifies the port from which the message originated.

BASE24 to NMS NAC

This example illustrates an NMS message being sent from BASE24 to an NMS NAC.

Transport Protocol Data Unit (TPDU)			Message Text
TPDU ID	Destination Address	Source Address	NMS Request/Response Message
X '68'	X '0221'	X '84FF'	X '...'
1 byte	2 bytes	2 bytes	Variable Length

The first field in this message contains the TPDU ID. A value of 68 (hexadecimal) in this field indicates the message is an NMS message.

The second field in this message contains the TPDU destination address, which is a 2-byte field that contains the NMS ID and NAC ID. The following table illustrates the composition of the destination address and the corresponding values from the NMS message example.

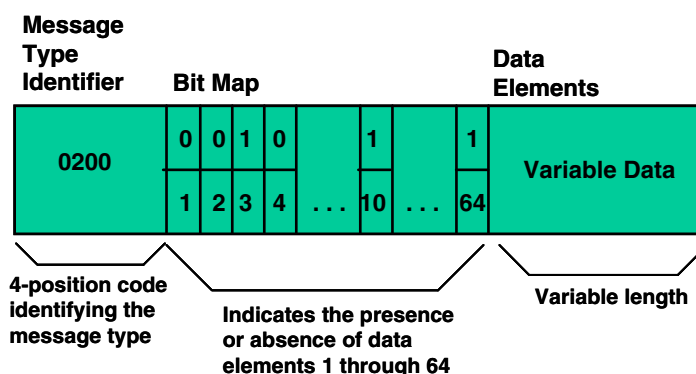
Destination Address		
	NMS ID	NAC ID
Number of Bits	4	8
Example Value	2	21

The HPDH module routes this message to the station that corresponds to NMS ID 2 (LCONF param POS-HPDH-NMS-ID2). The NMS PC uses the NAC ID to determine the NAC from which the original message originated.

The third field in this message contains the TPDU source address, which identifies the port from which the message originated.

Message Text Format

The message text immediately follows TPDU information in a message. It consists of the three parts illustrated in the following diagram and described below. The diagram is an example for an application type message. The message text for network management system (NMS) processing is informational only and the HPDH module serves as a passthrough to route message information to the appropriate destination (that is, a NAC or an NMS NAC).



Message Type Identifier

The Hypercom External Message has a standard format, regardless of whether the message is an authorization request, response, completion, reversal, or adjustment. As messages are routed between processes, each process determines the type of message it is receiving, and hence how it is to be handled, by the transaction code in the message.

As the message changes character, from request to response to completion, the message type is updated in the message. All transaction codes presume ISO-based message type identifiers are being sent to the host.

The message type identifier consists of the first four digits of the message text and is used to identify the message type, function, and transmission mode. The first and second digits identify the class of message. Valid values are as follows:

- 01 = Authorization message
- 02 = Financial transaction message
- 03 = File update or transfer message

- 04 = Reversal message
- 05 = Reconciliation message
- 08 = Network management message (download, logon, and handshake transactions)

The third and fourth digits identify the message function and transmission mode. Valid values are as follows:

- 00 = Interactive request
- 10 = Interactive request response
- 20 = Noninteractive advice
- 30 = Noninteractive advice response

The four bytes described above are combined to form the message type identifier that is contained in the message. The following message type identifiers are available.

Authorization Messages

The following message type identifiers indicate authorization messages:

- 0100 = Authorization request
- 0110 = Authorization request response

Financial Transaction Messages

The following message type identifiers indicate financial transactions:

- 0200 = Financial transaction request
- 0210 = Financial transaction request response
- 0220 = Financial transaction advice
- 0230 = Financial transaction advice response

File Update or Transfer Messages

The following message type identifiers indicate file updates or transfers:

- 0320 = File update or transfer advice
- 0330 = File update or transfer advice response

Reversal Messages

The following message type identifiers indicate reversal transactions:

- 0400 = Reversal request
- 0410 = Reversal request response

Reconciliation Messages

The following message type identifiers indicate reconciliation messages:

- 0500 = Card acceptor reconciliation request
- 0510 = Card acceptor reconciliation request response

Network Management Messages

The following message type identifiers are used in network management messages. These include download, logon, and handshake transactions.

- 0800 = Network management request
- 0810 = Network management request response

Note: The messages described here should not be confused with NMS PC processing. The messages described are the transactions identified above. NMS PC processing is a diagnostic tool that can be used with the HPDH module.

Bit Map

The ISO 8583 message format uses a concept called a *bit map*. The primary bit map is the second component of the message text. The primary bit map consists of two rows containing 64 bits each. The second row consists of 64 bits numbered from the left from 1 through 64. Each of the 64 bits contained in the primary bit map corresponds to one of the 64 primary data elements. The first row contains a value of 0 or 1 in each of the 64 bits. The values in the first row correspond to values in the second row to determine whether a particular primary data element is present in a message. A value of 1 in the bit map indicates that the corresponding primary data element is present in the message text. A value of 0 in the bit map indicates that the corresponding primary data element is absent from the message text. For example, if bit map position 4 is set to 1, the Transaction Amount (P-4)

data element is present in the particular message. If bit map position 64 is set to 0, the Message Authentication Code (P-64) data element is not present in the message.

Primary Data Elements

The primary data elements associated with the message contain characteristics such as names, attributes, and conditional code values. The primary data elements required for a particular message depends on the type of transaction being performed. The primary data elements associated with the Hypercom External Message are listed below and described in numeric order on the following pages.

Data Element	Data Element Name	Data Element Length
P-2	Primary Account Number	19 numeric characters
P-3	Processing Code	6 numeric characters
P-4	Transaction Amount	12 numeric characters
P-11	Systems Trace Number	6 numeric characters
P-12	Transaction Time	6 numeric characters
P-13	Transaction Date	4 numeric characters
P-14	Expiration Date	4 numeric characters
P-22	POS Entry Mode	3 numeric characters
P-24	Network International ID	3 numeric characters
P-25	POS Condition Code	2 numeric characters
P-32	Acquiring Institution ID	11 numeric characters
P-35	Track 2 Data	37 alphanumeric characters
P-37	Retrieval Reference Number	12 alphabetic and numeric characters
P-38	Authorization ID Response	6 alphabetic and numeric characters

Data Element	Data Element Name	Data Element Length
P-39	Response Code	2 alphabetic and numeric characters
P-41	Card Acceptor Terminal ID	8 alphanumeric characters
P-42	Card Acceptor ID	15 alphanumeric characters
P-43	Card Acceptor Name	40 alphanumeric characters
P-44	Additional Response Data	20 alphanumeric characters
P-45	Track 1 Data	76 alphanumeric characters
P-47	Supervisor Card Data	37 alphanumeric characters
P-52	Cardholder PIN Block	64 binary characters
P-53	Supervisor PIN Block	64 binary characters
P-54	Additional Amount	12 alphanumeric characters
P-55	EMV Data	259 alphanumeric and binary characters
P-60	Private Use	22 alphanumeric characters
P-61	Private Use	8 alphanumeric characters
P-62	Private Use	40 alphanumeric characters
P-63	Private Use	256 alphanumeric characters
P-64	Message Authentication Code	64 binary characters

Primary Data Element Descriptions

Descriptions of the primary data elements are included on the following pages in the format listed below. Explanations of the headings are included in the format.

Data Element	Format	Description and Contents (associated BASE24-pos field)
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The fields contained in the box shown above are described below.

Data Element — The identifier assigned to the primary data element in the Hypercom External Message. This identifier corresponds to the bit that is turned on when a Hypercom External Message contains the primary data element.

Format — The primary data element attributes and length specifications. Valid values are as follows:

N = Numeric characters
 B = Binary characters
 AN = Alphabetic and numeric characters
 ANS = Alphabetic, numeric, and special characters

A value of N6 indicates the primary data element is a fixed-length, 6 numeric character field. A value of N...11 indicates the primary data element is a variable-length, 11 numeric character field.

Description and Contents — A description of the primary data element and its use, when appropriate. If the primary data element relates directly to another BASE24-pos field, the field is listed in parentheses.

P-2	N...19	Primary Account Number (PSTM.TRAN. TRACK2)
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The Primary Account Number data element contains the primary account number (PAN) when the card number is manually entered at the device. The PAN and the expiration date (found in the Expiration Date (P-14) data element) are used to build the Track 2 data in the PSTM. The format of the Track 2 data is as follows:

Position	Format	Description
01	X(1)	Entry ID (M)
02–20	X(19)	PAN (variable-length, 19 bytes maximum)
21	X(1)	Separator character (=)
22–25	9(4)	Expiration Date (YYMM)
26	X(1)	End marker (?)

P-3	N6	Processing Code (PSTM.TRAN.TRAN-CDE)
------------	-----------	---

The Processing Code data element contains six digits used to describe the type of transaction and the account affected. This data element is mandatory for all messages. The first two digits of the processing code are used to identify the transaction type. Valid values are as follows:

- 00 = Sale/authorization
- 01 = Cash
- 02 = Online void or adjustment (debit)
- 04 = Check verification
- 09 = Sale and cash back
- 20 = Refund
- 22 = Online void or adjustment (credit)
- 30 = Preauthorization
- 31 = Balance inquiry
- 38 = Card verification
- 90 = Guaranteed late arrival or initialization
- 91 = Statistics
- 92 = Settlement request or logon
- 96 = Settlement after upload
- 99 = Test transaction

The third and fourth digits define account selection. Valid values are as follows:

- 00 = Default
- 10 = Savings
- 20 = Checking
- 30 = Credit
- 40 = Universal
- 96 = EBT Cash Benefit
- 98 = EBT Food Stamp Benefit

If the account selection option is used (that is, the CUSTOMER ACCT field on the Hypercom Card Definition screen or the CUSTOMER ACCT SELECTION field on Hypercom T7 Issuer Table screen 2 is set to 1), the code selected by the cardholder is included in the third and fourth digits of the processing code. If the account selection option is not used, the third and fourth digits of the processing code are set to the default account type from the card definition table (the DEFAULT ACCOUNT field on the Hypercom Card Definition Table screen or the DEFAULT ACCOUNT TYPE on Hypercom T7 Issuer Table screen 2).

Digit five in the processing code is not used and default to zero.

The host uses digit six for conveying extra information to the terminal in the response message. When converted to binary format, digit six contains four bytes representing an unassigned bit, an initialize after transaction value, a force close batch request value, and a more messages indicator.

The first bit is unassigned and is always set to 0.

The second bit indicates whether to instruct the terminal to perform an Initialization transaction at the first possible opportunity. The host can use this bit to force the terminal to request a new parameter load when updates have been made to the database. Valid values are as follows:

- 0 = No, do not instruct the terminal to perform an Initialization transaction.
- 1 = Yes, instruct the terminal to perform an Initialization transaction.

The third bit indicates whether the terminal is to display a message to the merchant stating that the batch should be closed. Valid values are as follows:

- 0 = No, do not have the terminal display a message stating that the batch should be closed.
- 1 = Yes, have the terminal display a message stating that the batch should be closed.

After the terminal sends the message indicating that the batch should be closed, the merchant is not allowed to perform any transactions other than a Close Batch Request. Failure to close the batch can cause discrepancies between host batch totals and terminal batch totals.

The fourth bit indicates whether there are more messages to be sent. Valid values are as follows:

- 0 = No, there are no more messages to be sent.
- 1 = Yes, there are more messages to be sent.

A value of 1 is used in the initialization response to the terminal to tell the terminal to send another initialization request message to the host. This bit should be set to 0 in the last block from the host. During a batch upload, the terminal sets this bit to 1 in all upload transactions except the last one, which should have the bit set to 0.

P-4	N12	Transaction Amount
------------	------------	---------------------------

The Transaction Amount data element contains the amount of funds requested either for debit or credit in the currency of the source location of the transaction.

- For transactions that involve one amount, this is the transaction amount and is moved into the PSTM.TRAN.AMT-1 field. This value is right-justified and zero filled. Negative balances are not supported.
- For transactions that involve two amounts, this is the revised amount and is moved into the PSTM.TRAN.AMT-2 field. This value is right-justified and zero filled. The original amount is in the Private Use (P-60) data element.

P-11	N6	Systems Trace Number (PSTM.SEQ-NUM, PSTM.INVOICE-NUM, PSTM.PRE-AUTH-SEQ-NUM)
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The Systems Trace Number data element contains a number that must be sent by a message sender and echoed by a message receiver. It is used for matching responses to original messages and is not intended to remain the same throughout the life of a transaction (for example, a reversal could have a number that differs from the original transaction).

P-12	N6	Transaction Time (PSTM.TRAN-TIM)
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The Transaction Time data element contains the time (hhmmss) the transaction entered the BASE24-pos system from a device. The transaction time is the local time at which the transaction began at the card acceptor location.

P-13	N4	Transaction Date (PSTM.TRAN-DAT)
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The Transaction Date data element contains the local month and day that the transaction was initiated by the card acceptor.

P-14	N4	Expiration Date (PSTM.TRAN.TRACK2)
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The Expiration Date data element contains the expiration date (YYMM) of the card that initiated the transaction if the card is manually entered at the device. The primary account number (PAN) and expiration date are used to build Track 2 data in the PSTM. Refer to the description of the Primary Account Number (P-2) data element for more information on building Track 2 data.

P-22	N3	POS Entry Mode (PSTM.TRAN.TRACK2, PSTM.PT-SRV-ENTRY-MDE)
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The POS Entry Mode data element contains codes indicating the manner in which the transaction entered BASE24-pos. The point of service entry mode consists of two codes. The first code is two positions in length and indicates the method by which Track 2 data was entered into the system. The second code is one position in length and indicates the PIN entry capabilities available at the point of service.

Valid values for the first and second bytes are as follows:

- 00 = Unspecified
- 01 = Manual
- 02 = Magnetic stripe
- 05 = Integrated circuit card (ICC)
- 08–60 = Reserved for ISO use
- 61–79 = Reserved for national use
- 80 = Magnetic stripe read from ICC-capable card
- 81–90 = Reserved for private use
- 91 = Contactless magnetic stripe transaction ¹
- 92–99 = Reserved for private use

¹ Hypercom terminals use the presence of data in data element 55 to distinguish contactless EMV transactions from contactless magnetic stripe transactions. Contactless EMV transactions contain data in data element 55 while contactless magnetic stripe transactions do not.

Valid values for the third byte are as follows:

- 0 = Unspecified
- 1 = PIN entry capability
- 2 = No PIN entry capability
- 3–5 = Reserved for ISO use
- 6–7 = Reserved for national use
- 8–9 = Reserved for private use

P-24	N3	Network International ID
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This data element is read from the native request message but is not used.

P-25	N2	POS Condition Code (PSTM.TRAN.TRAN-CDE, PSTM.PT-SRV-COND-CDE)
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The POS Condition Code data element contains a code indicating a special condition that exists at the time a transaction is initiated. Valid values are as follows:

00 = Normal presentment
01 = Customer not present
06 = Preauthorization request
08 = Mail order or telephone order
27 = Unattended terminal unable to retain card

P-32	N...11	Acquiring Institution ID (PTDD1.PTDD1-CORE.RTTN)
-------------	---------------	---

The Acquiring Institution ID data element contains the institution's routing and transit number or issuer identification number of the terminal owner. This data element contains one digit followed by 10 digits of identification data. The identification data is right-justified with high-order zero fill.

P-35	ANS...37	Track 2 Data (PSTM.TRAN.TRACK2)
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The Track 2 Data element consists of a group of fields containing information encoded on the Track 2 magnetic stripe of the customer's plastic card. This data can be obtained by entering it manually or by passing the card through the terminal's card reader. The Track 2 Data element is present on a request only if the card was read by the terminal. For manually entered cards, the TRACK2 field in the PSTM is built using the Primary Account Number (P-2) data element and the Expiration Date (P-14) data element.

This data element is mandatory for all authorization, financial transaction, and reversal messages.

The format indicates which of the two methods was used. The format is as follows for data entered manually:

Position	Format	Description
01	X(1)	Entry ID (M)

Position	Format	Description
02–20	X(19)	PAN (variable-length, 19 bytes maximum)
21	X(1)	Separator character (=)
22–25	9(4)	Expiration Date (YYMM)
26	X(1)	End marker (?)

The format is as follows for data read from the magnetic stripe:

Position	Format	Description
01	X(1)	Start marker (;)
02–20	X(19)	PAN (variable-length, 19 bytes maximum)
21	X(1)	Separator character (=)
22–25	9(4)	Expiration Date (YYMM)
26	X(1)	End marker (?)

P-37	AN12	Retrieval Reference Number (PSTM.PRE-AUTH-SEQ-NUM)
-------------	-------------	---

The Retrieval Reference Number data element contains the number assigned by BASE24-pos to uniquely identify a transaction. This number remains unchanged throughout the life of the transaction. The retrieval reference number is the system trace number of the original transaction. The system trace number is sent back in the response.

P-38	AN6	Authorization ID Response (PSTM.TRAN.APPRV-CDE)
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The Authorization ID Response data element contains the transaction approval code typically received by referral and then submitted in a purchase completion transaction. It represents a unique number generated by the authorizer for the transaction. It can be generated by BASE24-pos, a host, or an interchange.

P-39	AN2	Response Code (PSTM.TRAN.RESP-CDE)
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The Response Code data element contains the response code as determined by the authorizer for this transaction. The value set initially in the PSTM by the Router module, Authorization module, Host Interface process, or Interchange Interface process is mapped to the native response code.

P-41	ANS8	Card Acceptor Terminal ID (PSTM.TERM-ID)
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The Card Acceptor Terminal ID data element contains a unique identifier of the terminal where the transaction originated. This value is set initially in the PSTM by the Device Handler module or Host Interface process. The Device Handler module retrieves the value it places in this data element from the TERMINAL ID field on PTD screen 1. The Host Interface process obtains the value it places in this data element from the acquiring host.

This data element is required in all authorization, financial transaction, and reversal messages.

P-42	ANS15	Card Acceptor Identification Code (PSTM.CRD-ACCPT-ID-NUM)
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The Card Acceptor Identification Code data element identifies the card acceptor in a transaction if the card acceptor is different from the acquiring institution.

If this data element is present in incoming BASE24-pos messages, the first 11 bytes are placed in the PSTM.CRD-ACCPT-ID-NUM field.

If the PSTM.CRD-ACCPT-ID-NUM field is not set to blanks or zeros in outgoing BASE24-pos messages, BASE24 moves the first 11 bytes of the CRD-ACCPT-ID-NUM field to the Card Acceptor Identification Code (P-42) data element, setting the rest of the data element to blanks.

P-43	ANS40	Card Acceptor Name/Location
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The Card Acceptor Name/Location data element contains the name and location of the card acceptor. This data element defines the point of service in both local and interchange environments.

The structure of this data element is shown below. Included for each field is its position in the data element, its length, and a description of its contents.

Position	Length	Description
01–22	22	Terminal owner name The name of the institution owning the terminal. This value is taken from the TERM-OWNER-NAME field in the PTDS1.
23–35	13	Terminal city The city in which the transaction-originating terminal is located. This value is taken from the TERM-CITY field in the PTDS1.
36–38	3	Terminal state The state in which the transaction-originating terminal is located. This value is taken from the TERM-ST field in the PTDS1.
39–40	2	Terminal country A code indicating the country in which the transaction-originating terminal is located. This value is taken from the TERM-CNTRY-CDE field in the PTDS1.

P-44	ANS20	Additional Response Data
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This data element is read from the native request message but is not used.

P-45	ANS...76	Track 1 Data
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This data element is read from the native request message but is not used.

P-47	ANS...37	Supervisor Card Data
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This data element is read from the native request message but is not used.

P-52	B64	Cardholder Pin Block (PSTM.PIN.SIZE), (PSTM.PIN)
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The Cardholder PIN Block data element contains the customer-entered PIN in clear or encrypted form, depending on whether the terminal supports PIN encryption.

P-53	B64	Supervisor PIN Block
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This data element is read from the native request message but is not used.

P-54	ANS...12	Additional Amount (PSTM.TRAN. AMT-2)
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The Additional Amount data element is an information-only value that is logged in the user data field in the PTLF. It can contain one of the following amounts:

- The cash back amount for a purchase with cash back transaction. The cash back amount for a purchase with cash back transaction can also be carried in table ID 41 of the Private Use (P-63) data element.
- The balance amount in a balance inquiry. The HPDH module returns a 0 if the balance amount is negative.

The tip amount in a restaurant transaction. The tip amount can also be carried in table ID 53 of the Private Use (P-63) data element.

P-55	AN&B259	EMV Data
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For detailed information on the use of EMV Data (P-55) refer to the ***BASE24-pos EMV Support Manual***.

P-60	ANS...22	Private Use
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This private-use data element can contain one of the following items, depending on the transaction or message:

- Original amount for adjustment transactions. This value is the amount of funds requested either for debit or credit in the currency of the source location of the transaction and is moved to the TRAN.AMT-1 field in the PSTM.
- Hypercom batch number for settlement transactions.
- Original message type, original systems trace number, and retrieval reference number for batch uploads.
- Software identification code for statistics requests.
- Software name, revision level, and download revision number for Hypercom T4, T5, and T6 terminal download requests.
- All initialization download response messages for Hypercom T7 terminals.

P-61	ANS...8	Private Use
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This private-use data element can contain one of the following items, depending on the message:

- Up to four product codes for Hypercom terminals configured for product code entry.
- Card definition tables for Hypercom T4, T5, and T6 terminal download responses.

P-62	ANS...40	Private Use
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This private-use data element can contain one of the following items, depending on the transaction or message:

- Invoice and ECR reference number for Hypercom terminals.
- Response display message for Hypercom settlement transactions.
- Product code table for Hypercom T4, T5, and T6 terminal download responses.

P-63	ANS...256	Private Use
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This private-use data element can contain one of the following items, depending on the transaction or message:

- Transaction information distinguished by selected Hypercom table IDs. See “Table IDs” later in this section for a discussion of table IDs.
- Check data for check guarantee and verification transactions.
- Terminal totals for Hypercom settlement transactions.
- Terminal statistics for terminal statistics messages.
- Range definition tables for Hypercom T4, T5, and T6 terminal download responses.

P-64	B64	Message Authentication Code
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The Message Authentication Code data element contains the message authentication code (MAC), if the terminal is configured to support MACs.

Message Structures

The Hypercom Device Handler (HPDH) module supports Hypercom transactions that are currently supported by BASE24-pos. The HPDH message does not contain a transaction code as such. The HPDH module determines the BASE24-pos transaction code associated with a message using information in the Message Type Identifier field, Processing Code (P-3) data element, and POS Condition Code (P-25) data element.

The BASE24-pos transaction code, in addition to indicating the type of transaction, identifies whether a debit or a credit card initiated the transaction, and the type of application account used when the transaction was initiated by a debit card. In the case of debit cards, the type of application account is required in the transaction code because debit cards can be associated with checking, savings, or credit accounts.

The following pages defines message structures associated with Hypercom transactions and how they are mapped to BASE24. The message structures are organized and described according to Hypercom transaction type.

Data Element	Format	Request	Response	PSTM Mapping
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Data Element — The name and number of the primary data element in the Hypercom External Message.

Format — The primary data element attributes and length specifications. Refer to the format description in the “Primary Data Element Descriptions” presented earlier in this section.

Request — A code or other value that indicates whether a primary data element is included in the request.

Response — A code or other value that indicates whether a primary data element is included in the response. Values echoed from the request are also noted. Valid code values are as follows:

Request/Response Code and Description for Hypercom Messages

The Request and Response columns below define the terminal request and response messages content comprising a transaction.

M = Mandatory data element

O = Optional data element

Cxx The conditional field, where “xx” represents one of the following values

01 = Conditional data element that is present when information is entered from a keyboard

02 = Conditional data element that is present when information is taken from the Track 2 magnetic stripe on the card used to initiate the transaction

03 = Conditional data element that is present when information is taken from the Track 1 magnetic stripe on the card used to initiate the transaction

04 = Conditional data element that stores the primary account number and expiration date from reading the track

05 = Conditional data element that identifies the digits of the POS entry

06 = Conditional data element that includes bit 2 if the unit number is entered from the keyboard

07 = Conditional data element present when a transaction is performed from a terminal with ICC capabilities that processes an EMV-compliant credit or debit transaction

The values in the Code column are the actual values for the following primary data elements:

- The Message Type ID data element contains a code that identifies the message type in the Hypercom External Message.
- The Processing Code (P-3) data element contains a code that identifies the type of transaction and the account affected. Refer to the description of the Processing Code (P-3) data element in the “Message Text Format” topic for a complete description of valid values.
- The POS Entry Mode (P-22) data element contains a code that indicates how the Track 2 data is entered and what PIN entry capabilities are available at the device where the transaction is initiated. Refer to the description of the Processing Code (P-3) data element in the “Message Text Format” topic for a complete description of valid values.

- The POS Condition Code (P-25) data element contains a code that indicates a special condition that exists when the transaction is initiated. Refer to the description of the POS Condition Code (P-25) data element in the “Message Text Format” topic for a complete description of valid values.

PSTM Mapping — Contains the name of the POS Standard Internal Message (PSTM) field associated with the primary data element. If the field relates directly to another BASE24-pos field, the field is listed in parentheses.

BASE24-pos carries the information from some primary data elements in tokens attached to the PSTM instead of fields in the PSTM. This field contains the token information when appropriate. See the *BASE24 Tokens Manual* for an explanation of how BASE24 products use tokens, including field descriptions for all available tokens.

Hypercom Authorization Transaction

Hypercom authorization transactions are mapped to a BASE24 normal purchase transaction if the POS Condition Code (P-25) data element contains a value of 00 or a BASE24 mail or telephone order transaction if the POS Condition Code (P-25) data element contains a value of 08. The following table describes the message structure of a Hypercom authorization transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0100	0110	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	00A00X	00A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	O	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2

Data Element		Format	Request	Response	PSTM Mapping
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00, 08		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom authorization transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Preauthorization Transaction

Hypercom preauthorization transactions are mapped to a BASE24 preauthorization transaction. The following table describes the message structure of a Hypercom preauthorization transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0100	0110	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	30A00X	30A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	O	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.PRE-AUTH-SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM

Data Element		Format	Request	Response	PSTM Mapping
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00, 27		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom preauthorization transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Balance Inquiry Transaction

Hypercom balance inquiry transactions are mapped to a BASE24 balance inquiry transaction. The following table describes the message structure of a Hypercom balance inquiry transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0100	0110	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	31A00X	31A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12		M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT

Data Element		Format	Request	Response	PSTM Mapping
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-54*	Additional Amount	ANS12		O	
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	PSTM.USER DATA
P-64	Message Authentication Code	B64	O	O	

* The Additional Amount (P-54) data element contains the account balance only if it is a positive amount.

Private Use Data Elements

Hypercom balance inquiry transactions use two Private Use data elements, as follows:

Data Element	Description
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Card Verification Transaction

Hypercom card verification transactions are mapped to a BASE24 card verification transaction. The following table describes the message structure of a Hypercom card verification transaction.

	Data Element	Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0100	0110	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	38A00X	38A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	O	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE

Data Element		Format	Request	Response	PSTM Mapping
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00, 027		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom card verification transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.

Data Element	Description
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Sale Transaction

Hypercom sale transactions are mapped to a BASE24 normal purchase if the POS Condition Code (P-25) data element contains a value of 00 or 27. Transactions are mapped to a BASE24 mail or telephone order transaction if the POS Condition Code (P-25) data element contains a value of 08. The following table describes the message structure of a Hypercom sale transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0200	0210	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	00A00X	00A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	

Data Element		Format	Request	Response	PSTM Mapping
P-25	POS Condition Code	N2	00, 08, 27		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-54*	Additional Amount	ANS..12	O		PSTM.TRAN.AMT-2 PSTM.USER-DATA
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

* The Additional Amount (P-54) data element is an information-only field containing the restaurant tip amount if restaurant processing is set on and a tip amount was entered.

Private Use Data Elements

Hypercom sale transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Cash Transaction

Hypercom cash transactions are mapped to a BASE24 cash advance transaction. The following table describes the message structure of a Hypercom cash transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0200	0210	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	01A00X	01A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT

Data Element		Format	Request	Response	PSTM Mapping
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom cash transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Sale and Cash Back Transaction

Hypercom sale and cash back transactions are mapped to a BASE24 purchase with cash back transaction. The following table describes the message structure of a Hypercom sale and cash back transaction. The BASE24 purchase with cash back transaction is only supported with debit cards.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0200	0210	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	09A00X	09A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT

Data Element		Format	Request	Response	PSTM Mapping
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP- CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-54*	Additional Amount	ANS..12	M		PSTM.TRAN.AMT-2 PSTM.USER-DATA
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

* The Additional Amount (P-54) data element is an information-only field containing the cash back amount. If the Additional Amount (P-54) data element is unavailable, the cash back amount can be moved from table ID 41 in the Private Use (P-63) data element to the TRAN.AMT-2 field in the PSTM.

Private Use Data Elements

Hypercom sale and cash back transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Void Sale Transaction

Hypercom void sale transactions can be mapped to two BASE24 transactions, depending on the processing code set by the device. A processing code of 02A00X is mapped to a BASE24 merchandise return transaction, and a processing code of 22A00X is mapped to a BASE24 merchandise return adjustment transaction. The following table describes the message structure of both Hypercom void sale transactions.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0200	0210	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	02A00X 22A00X	02A00X 22A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM

Data Element		Format	Request	Response	PSTM Mapping
P-12	Transaction Time	N6	M*	M	PSTM.TRAN-TIM
P-13	Transaction Date	N4	M [†]	M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12	M	M	PSTM.ORIG-INVOICE-NUM
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-54 [‡]	Additional Amount	ANS..12	O		PSTM.TRAN.AMT-2 PSTM.USER-DATA
P-60	Private Use (see “Private Use Data Elements”)	ANS..12	O		PSTM.TRAN.AMT-1 PSTM.USER-DATA
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	M		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

* The Transaction Time (P-12) data element contains the time of the original transaction.

- † The Transaction Date (P-13) data element contains the date of the original transaction.
- ‡ The Additional Amount (P-54) data element is an information-only field containing the restaurant tip amount if restaurant processing is set on and a tip amount was entered.

Private Use Data Elements

Hypercom void sale transactions use four Private Use data elements, as follows:

Data Element	Description
P-60	Original transaction amount.
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Refund Transaction

Hypercom refund transactions are mapped to a BASE24 merchandise return transaction. The following table describes the message structure of a Hypercom refund transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0200	0210	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	20A00X	20A00X	PSTM.TRAN.TRAN-CDE

Data Element		Format	Request	Response	PSTM Mapping
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00, 08		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12	O	M	PSTM.ORIG-INVOICE-NUM
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-52	Cardholder PIN Block	B64	O		PSTM.PIN.SIZE PSTM.PIN
P-54*	Additional Amount	ANS..12	O		PSTM.TRAN.AMT-2 PSTM.USER-DATA
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

- * The Additional Amount (P-54) data element is an information-only field containing the restaurant tip amount if restaurant processing is set on and a tip amount was entered.

Private Use Data Elements

Hypercom refund transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Guarantee Late Arrival Transaction

Hypercom guarantee late arrival (hotel or motel) transactions are mapped to a BASE24 purchase transaction. The following table describes the message structure of a Hypercom guarantee late arrival transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0220	0230	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	90A00X	90A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1

Data Element		Format	Request	Response	PSTM Mapping
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6	M	M	PSTM.TRAN-TIM
P-13	Transaction Date	N4	M	M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	M		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	01		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6	M		PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-54	Additional Amount	ANS..12	O		PSTM.TRAN.AMT-2
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	M		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	M		PSTM.USER-DATA
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom guarantee late arrival transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Offline Sale Transaction

Hypercom offline sale transactions are mapped to a BASE24 force post purchase transaction if the POS Condition Code (P-25) data element contains a value of 00 or 27. Transactions are mapped to a BASE24 force post mail or telephone order transaction if the POS Condition Code (P-25) data element contains a value of 08. The following table describes the message structure of a Hypercom offline sale transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0220	0230	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	00A00X	00A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM

Data Element		Format	Request	Response	PSTM Mapping
P-12	Transaction Time	N6	M		PSTM.TRAN-TIM
P-13	Transaction Date	N4	M		PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00, 08, 27		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6	M		PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-54*	Additional Amount	ANS..12	O		PSTM.TRAN.AMT-2 PSTM.USER-DATA
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

* The Additional Amount (P-54) data element is an information-only field containing the restaurant tip amount if restaurant processing is set on and a tip amount was entered.

Private Use Data Elements

Hypercom offline sale transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Offline Cash Transaction

Hypercom offline cash transactions are mapped to a BASE24 force post cash advance transaction. The following table describes the message structure of a Hypercom offline cash transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0220	0230	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	01A00X	01A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM PSTM.ORIG-INVOICE-NUM
P-12	Transaction Time	N6	M		PSTM.TRAN-TIM
P-13	Transaction Date	N4	M		PSTM.TRAN-DAT

Data Element		Format	Request	Response	PSTM Mapping
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	
P-38	Authorization ID Response	AN6	M		PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		PSTM.CRD-ACCPT-ID-NUM
P-54*	Additional Amount	ANS..12	O		PSTM.TRAN.AMT-2 PSTM.USER-DATA
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

* The Additional Amount (P-54) data element is an information-only field containing the cash back amount. If the Additional Amount (P-54) data element is unavailable, the cash back amount can be moved from table ID 41 in the Private Use (P-63) data element to the TRAN.AMT-2 field in the PSTM.

Private Use Data Elements

Hypercom offline cash transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Sale Completion Transaction

Hypercom sale completion transactions are mapped to a BASE24 preauthorization completion transaction. The following table describes the message structure of a Hypercom sale completion transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0220	0230	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	00A00X	00A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM
P-12	Transaction Time	N6	M		PSTM.TRAN-TIM
P-13	Transaction Date	N4	M		PSTM.TRAN-DAT

Data Element		Format	Request	Response	PSTM Mapping
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	06		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12	M	M	PSTM.PRE-AUTH-SEQ-NUM PSTM.ORIG-INVOICE-NUM
P-38	Authorization ID Response	AN6	M		PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2	M*	M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-54 [†]	Additional Amount	ANS..12	O		PSTM.TRAN.AMT-2 PSTM.USER-DATA
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

* The Response Code (P-39) data element contains the response code from the original transaction.

† The Additional Amount (P-54) data element is an information-only field containing the cash back amount. If the Additional Amount (P-54) data element is unavailable, the cash back amount can be moved from table ID 41 in the Private Use (P-63) data element to the TRAN.AMT-2 field in the PSTM.

Private Use Data Elements

Hypercom sale completion transactions use three Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Adjustment Sale Transaction

Hypercom adjustment sale transactions can be mapped to two BASE24 transactions, depending on the processing code set by the device. A processing code of 02A00X is mapped to a BASE24 purchase adjustment transaction, and a processing code of 22A00X or 29A00X is mapped to a BASE24 merchandise return adjustment transaction. The following table describes the message structure of all three Hypercom adjustment sale transactions.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0220	0230	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	02A00X, 22A00X, 29A00X	02A00X, 22A00X, 29A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-2

Data Element		Format	Request	Response	PSTM Mapping
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM
P-12	Transaction Time	N6	M		PSTM.TRAN-TIM
P-13	Transaction Date	N4	M		PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	011, 012, 021, 022		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS..37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12	M	M	PSTM.ORIG-INVOICE- NUM
P-38	Authorization ID Response	AN6	M		PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2	M*	M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-54 [†]	Additional Amount	ANS..12	O		PSTM.USER-DATA
P-60	Private Use (see “Private Use Data Elements”)	ANS..12	M		PSTM.TRAN.AMT-1 PSTM.USER-DATA
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	M		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

* The Response Code (P-39) data element contains the response code from the original transaction.

† The Additional Amount (P-54) data element is an information-only field containing the restaurant tip amount if restaurant processing is set on and a tip amount was entered.

Private Use Data Elements

Hypercom adjustment sale transactions use four Private Use data elements, as follows:

Data Element	Description
P-60	Original transaction amount.
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Check Guarantee Transaction

Hypercom check guarantee transactions are mapped to a BASE24 check guarantee transaction. The following table describes the message structure of a Hypercom check guarantee transaction.

	Data Element	Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0100	0110	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	03200X	03200X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M		PSTM.TRAN.AMT-2

	Data Element	Format	Request	Response	PSTM Mapping
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	021		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS..37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	PSTM.ORIG-INVOICE-NUM
P-38	Authorization ID Response	AN6		O	PSTM.TRAN-APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	M		PSTM.CHECK-AUTH
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom check guarantee transactions use the Private Use (P-63) data element for check data in a format determined by the check guarantee method used. See “Check Guarantee and Verification Methods” for format details and the processing performed with each method.

Hypercom Check Verification Transaction

Hypercom check verification transactions are mapped to a BASE24 check verification transaction. The following table describes the message structure of a Hypercom check verification transaction.

	Data Element	Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0100	0110	
	Bit Map	B64	M	M	
P-3	Processing Code	N6	04A00X	04A00X	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	M	PSTM.TRAN.AMT-2
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	C1		PSTM.TRAN.TRACK2
P-22	POS Entry Mode	N3	0110		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	00		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-37	Retrieval Reference Number	AN12		M	PSTM.ORIG-INVOICE-NUM
P-38	Authorization ID Response	AN6		O	PSTM.TRAN-APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	M		PSTM.CHECK-AUTH
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom check verification transactions use two Private Use data elements, as follows:

Data Element	Description
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	Check data in a format determined by the check verification method used. See “Check Guarantee and Verification Methods” for format details and the processing performed with each method.

Hypercom Batch Upload Transaction

BASE24 maps a Hypercom batch upload transaction in the same manner as it would a normal batch upload transaction. If the terminal and BASE24 draft capture totals do not agree at settlement, the batch draft capture transactions are uploaded to BASE24 for variance analysis. The following table describes the message format associated with a BASE24 batch upload transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0320	0330	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	M		PSTM.TRAN.TRACK2
P-3	Processing Code	N6		M	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M		PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM
P-12	Transaction Time	N6	M	M	PSTM.TRAN-TIM
P-13	Transaction Date	N4	M	M	PSTM.TRAN-DAT
P-14	Expiration Date	N4	O		PSTM.TRAN.TRACK2

Data Element		Format	Request	Response	PSTM Mapping
P-22	POS Entry Mode	N3	M		PSTM.TRAN.TRACK2 PSTM.PT-SRV-ENTRY-MDE
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	M		PSTM.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS..64	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12	M	M	PSTM.ORIG-INVOICE-NUM
P-38	Authorization ID Response	AN6	O		PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2	O	M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-54	Additional Amount	ANS..12	O		PSTM.USER-DATA
P-55	EMV Data	AN&B259	O		
P-60	Private Use (see “Private Use Data Elements”)	ANS..22	M		
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O		

Private Use Data Elements

Hypercom batch upload transactions use three Private Use data elements, as follows:

Data Element	Description
P-60	Original message data. See “Original Message Data Format” on the following page for format information.
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Original Message Data Format

The Private Use (P-60) data element contains original message data in the following format:

Field Name	Format
Original Message Type ID	N4
Original Systems Trace Number	N6
Retrieval Reference Number	AN12

Hypercom Reversal Transaction

Hypercom reversal transactions are mapped to a BASE24 normal reversal transaction. The reversal format is identical to the request or advice being reversed, except for the substitution of 0400 for the message type ID. The following table describes the message structure of a Hypercom reversal transaction.

	Data Element	Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0400	0410	
	Bit Map	B64	M	M	
P-2	Primary Account Number	N..19	C1		PSTM.TRAN.TRACK2
P-3	Processing Code	N6	M	M	PSTM.TRAN.TRAN-CDE
P-4	Transaction Amount	N12	M	O	PSTM.TRAN.AMT-1
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM
P-12	Transaction Time	N6		M	
P-13	Transaction Date	N4		M	
P-24	Network International ID	N3	M	M	
P-25	POS Condition Code	N2	M		PTST.TRAN.TRAN-CDE PSTM.PT-SRV-COND-CDE
P-35	Track 2 Data	ANS..37	C2		PSTM.TRAN.TRACK2
P-37	Retrieval Reference Number	AN12		M	PSTM.PRE-AUTH-SEQ-NUM
P-38	Authorization ID Response	AN6		O	PSTM.TRAN.APPRV-CDE
P-39	Response Code	AN2		M	PSTM.TRAN.RESP.CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		PSTM.CRD-ACCPT-ID-NUM
P-61	Private Use (see “Private Use Data Elements”)	ANS..8	O		PSTM.USER-DATA
P-62	Private Use (see “Private Use Data Elements”)	ANS..6	O		PSTM.USER-DATA

Data Element		Format	Request	Response	PSTM Mapping
P-63	Private Use (see “Private Use Data Elements”)	ANS..256	O	O	
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom adjustment sale transactions use four Private Use data elements, as follows:

Data Element	Description
P-61	Up to four 2-digit product codes, determined by an optional setting in the card definition table. If fewer than four product codes are entered in the Private Use (P-61) data element, the unused fields are blank filled.
P-62	Invoice number or electronic cash register reference number, determined by an optional setting in the card definition table.
P-63	One or more of the Hypercom table IDs and related data, if available. See “Table IDs” later in this section for a discussion of table IDs.

Hypercom Settlement Transaction

Hypercom settlement transactions are mapped to a BASE24 normal settlement transaction. The second digit in the processing code is 2 (92000X) for an initial settlement request. If the host response code is 95 (totals do not agree), the terminal uploads all transactions to the host and sends a second 0500 settlement request to the host. The second digit in the processing code for the second settlement request is 6 (96000X). The following table describes the message structure of a Hypercom settlement transaction.

Data Element		Format	Request	Response	PSTM Mapping
	Message Type ID	N4	0500	0510	
	Bit Map	B64	M	M	

Data Element		Format	Request	Response	PSTM Mapping
P-3	Processing Code	N6	92000X 96000X	92000X 96000X	PSTM.TRAN.TRAN-CDE
P-11	Systems Trace Number	N6	M	M	PSTM.SEQ-NUM PSTM.INVOICE-NUM
P-12	Transaction Time	N6		M	PSTM.TRAN-TIM
P-13	Transaction Date	N4		M	PSTM.TRAN-DAT
P-24	Network International ID	N3	M	M	
P-37	Retrieval Reference Number	AN12	M	M	PSTM.ORIG-INVOICE-NUM
P-39	Response Code	AN2		M	PSTM.TRAN.RESP-CDE
P-41	Card Acceptor Terminal ID	ANS8	M	M	PSTM.TERM-ID
P-42	Card Acceptor ID	ANS15	M		
P-60	Private Use (see “Private Use Data Elements”)	ANS..6	M		PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..40		O	PSTM.USER-DATA
P-63	Private Use (see “Private Use Data Elements”)	ANS..90	M		PSTM.USER-DATA
P-64	Message Authentication Code	B64	O	O	

Private Use Data Elements

Hypercom settlement transactions use two Private Use data elements, as follows:

Data Element	Description
P-60	Batch number
P-63	This data element has a batch totals format for request messages and a host response text format for response messages. See “Batch Totals Format” below for the request message format.

Batch Totals Format

Batch totals appear in the Private Use (P-63) data element in the following format:

Field Name	Format
Length attribute	N2
Capture cards—number of debits sales	N3
Capture cards—amount of debits sales	N12
Capture cards—number of credits refunds	N3
Capture cards—amount of credits refunds	N12
Debit cards—number of debits sales	N3
Debit cards—amount of debits sales	N12
Debit cards—number of credits refunds	N3
Debit cards—amount of credits refunds	N12
Authorize cards—number of debits sales	N3
Authorize cards—amount of debits sales	N12
Authorize cards—number of credits refunds	N3
Authorize cards—amount of credits refunds	N12

Totals are accumulated separately for authorization only, draft capture, and debit card transactions. However, only draft capture and debit card transactions are held in the terminal.

Hypercom Test Transaction

Hypercom test transactions are mapped to a BASE24 handshake transaction. The following table describes the message structure of a Hypercom test transaction. PSTM mapping is not applicable to this transaction.

Data Element		Format	Request	Response
	Message Type ID	N4	0800	0810

Data Element		Format	Request	Response
	Bit Map	B64	M	M
P-3	Processing Code	N6	99000X	99000X
P-12	Transaction Time	N6		M
P-13	Transaction Date	N4		M
P-24	Network International ID	N3	M	M
P-41	Card Acceptor Terminal ID	ANS8	M	M
P-42	Card Acceptor ID	ANS15	M	

Hypercom Logon Transaction

Hypercom logon transactions return Regulation E information plus new PIN and message authentication code (MAC) communication keys. The following table describes the message structure of a Hypercom logon transaction. PSTM mapping is not applicable to this transaction.

	Data Element	Format	Request	Response
	Message Type ID	N4	0800	0810
	Bit Map	B64	M	M
P-3	Processing Code	N6	92000X	92000X
P-11	Systems Trace Number	N6	M	M
P-12	Transaction Time	N6		M
P-13	Transaction Date	N4		M
P-24	Network International ID	N3	M	M
P-39	Response Code	AN2		M
P-41	Card Acceptor Terminal ID	ANS8	M	M
P-42	Card Acceptor ID	ANS15		M
P-62	Private Use (see “Private Use Data Elements”)	ANS..85		M

	Data Element	Format	Request	Response
P-63	Private Use (see “Private Use Data Elements”)	ANS..85		M

Private Use Data Elements

Hypercom logon transactions use two Private Use data elements, as follows:

Data Element	Description
P-62	This data element contains single length keys for response messages. See “Single Length Key Format” below for the response message format.
P-63	This data element has a double length PIN key for response messages. See “Double Length Key Format” below for the request message format.

Single Length Key Format

Hypercom logon transactions use the Private Use (P-62) data element for response data in the following format:

Field Name	Format
Length Attribute	N2
PIN working key	8 hexadecimal
MAC working key	8 hexadecimal
Regulation E merchant name and address	ANS69

Double Length Key Format

Hypercom logon transactions use the Private Use (P-63) data element for response data in the following format:

Field Name	Format
Additional Data Length	N2
Table ID	ANS2
eKTM (KPP)	B..192

Hypercom Parameter Download Transaction—T4, T5, T6 Terminals

Hypercom parameter download transactions for Hypercom T4, T5, and T6 terminals are mapped to a BASE24 normal parameter download transaction. The following table describes the message structure of a Hypercom download transaction for T4, T5, and T6 terminals. PSTM mapping is not applicable to this transaction.

	Data Element	Format	Request	Response
	Message Type ID	N4	0800	0810
	Bit Map	B64	M	M
P-3	Processing Code	N6	90000X	90000X
P-11	Systems Trace Number	N6	M	M
P-24	Network International ID	N3	M	M
P-39	Response Code	AN2		M
P-41	Card Acceptor Terminal ID	ANS8	M	M
P-43	Card Acceptor Name	ANS40		O
P-60	Private Use (see “Private Use Data Elements”)	ANS..11	M	
P-60	Private Use (see “Private Use Data Elements”)	ANS..54		M

Data Element		Format	Request	Response
P-61	Private Use (see “Private Use Data Elements”)	ANS..999		M
P-62	Private Use (see “Private Use Data Elements”)	ANS..198		M
P-63	Private Use (see “Private Use Data Elements”)	ANS..999		M

Private Use Data Elements

Hypercom parameter download transactions for Hypercom T4, T5, and T6 terminals use four Private Use data elements, as shown in the following table. See section 5 for a discussion of download responses.

Data Element	Description
P-60	Software name, revision level, and download revision number for requests. Configuration parameters for responses.
P-61	Card definition table.
P-62	Product code table.
P-63	Range definition table.

Hypercom Parameter Download Transaction—T7 Terminals

Hypercom parameter download transactions from Hypercom T7 terminals are mapped to a BASE24 normal parameter download transaction. The following table describes the message structure of a Hypercom download transaction from T7 terminals. PSTM mapping is not applicable to this transaction.

Data Element		Format	Request	Response
	Message Type ID	N4	0800	0810
	Bit Map	B64	M	M
P-3	Processing Code	N6	93000X	93000X

Data Element		Format	Request	Response
P-11	Systems Trace Number	N6	M	M
P-24	Network International ID	N3	M	M
P-25	POS Condition Code	N2	00	
P-39	Response Code	AN2		M
P-41	Card Acceptor Terminal ID	ANS8	M	M
P-60	Private Use (see “Private Use Data Elements”)	ANS..999		O

Private Use Data Elements

Hypercom parameter download transactions for Hypercom T7 terminals use the Private Use (P-60) data element for all download information. See section 5 for a discussion of download responses.

Table IDs

The Private Use (P-63) data element can contain transaction information distinguished by selected Hypercom table IDs, as shown in the following table. A single message can contains any combination of these table IDs with their related data. BASE24-pos recognizes only a subset of the table IDs defined by Hypercom and ignores any other table IDs that appear in this data element.

ID	Description	Request	Response
11	Lodging Details	✓	
13	Lodging Program Indicator	✓	
15	Level II Commerical Card Indicator	✓	✓
16	CVV2 Data Processing	✓	✓
20	Payment Services (PS2000) Data	✓	✓
21	PS2000 Terminal Data	✓	
30	CPS 2000 Data	✓	✓
39	Tax Amount #1	✓	
41	Cash Back Amount	✓	
44	Surcharge Amount	✓	
45	Payment Plan	✓	
47	EBT Terminal Data	✓	
48	EBT Response Data		✓
53	Miscellaneous Amount (Bar Amount)	✓	
54	Address Verification Request Data	✓	
55	Address Verification Response Data		✓
68	Customer ID for Purchase Order	✓	
73	Level II Commercial Card	✓	✓

The table IDs and their related data can appear in any order. The data for each table ID has its own format, as shown in the following tables.

Lodging Details

Table ID 11 contains lodging information in the following format:

Format	Value	Description
N4	0026	Additional Data Length
ANS2	11	Table ID
AN6		Arrival Date
AN6		Departure Date
AN12		Room Rate

Data associated with this table ID is moved to corresponding fields in the lodging format of the American Express token.

Lodging Program Indicator

Table ID 13 contains the lodging program indicator in the following format:

Format	Value	Description
N4	0003	Additional Data Length
ANS2	13	Table ID
AN1		Program Indicator

Data associated with this table ID is moved to the PROGRAM-CDE field in the lodging format of the American Express token.

Level II Commercial Card Indicator

Table 15 is used for Request and Response processing Level II commercial cards. These cards include Business, Corporate, and Purchase card categories.

Format	Value	Description
N4	0003	Additional Data Length
ANS2	15	Table ID
AN1		Commercial Card Request Code
AN1		Commercial Card Response Code

CVV2 Processing Data

Table 16 contains CVV2 values.

Format	Value	Description
N4	0LLL	BCD length of the date to follow
ANS2	16	Table ID
AN6	6	Six character CVV2 request code
AN1	1	One character CVV2 result code

Payment Services (PS2000) Data

Table ID 20 contains payment services (PS2000) data in the following format:

Format	Value	Description
N4	0026	Additional Data Length
ANS2	20	Table ID
ANS1		PS2000 Indicator
ANS15		Transaction Identifier
ANS4		Validation Code

Format	Value	Description
ANS2		Visa Response Code
ANS2		POS Entry Mode

The PS2000 indicator, transaction identifier, and validation code correspond to fields in the Visa Payment Service 2000 token.

PS2000 Terminal Data

Table ID 21 contains the total authorized amount for PS2000 transactions in the following format:

Format	Value	Description
N4	0014	Additional Data Length
ANS2	21	Table ID
AN12		Authorized Amount

Data associated with this table ID is moved to the TOT-AMT-AUTH field in the Visa Payment Service 2000 Offline token.

CPS 2000 Data

Table 30 is used to carry data as required for the Visa CPS 2000 compliance in the Lodging industry. The table is required for Visa cards only.

Format	Value	Description
N4	0046	Additional Data Length
ANS2	30	Table ID
AN1		Authorization Characteristics Indicator (ACI)
AN 15		Transaction ID
AN4		Value returned by Issuer

Format	Value	Description
AN1		Market specific data
AN1		Requested payment service (RPS)
AN12		First authorization amount
AN3		Duration of stay
AN1		Extra charges reason code
AN6		Extra charges

Tax Amount #1

Table ID 39 contains the tax amount associated with a transaction in the following format:

Format	Value	Description
N4	0014	Additional Data Length
ANS2	39	Table ID
ANS12		Tax Amount

Data associated with this table ID is moved to the TAX-AMT field in the Purchasing Card and Fleet Card token, if that token is present. The Purchasing Card and Fleet Card token is added only if table ID 68 contains the customer ID number.

Cash Back Amount

Table ID 41 contains the cash back amount for a purchase with cash back transaction in the following format:

Format	Value	Description
N4	0014	Additional Data Length
ANS2	41	Table ID

Format	Value	Description
ANS12		Cash Back Amount

This cash back amount is used if the Additional Amount (P-54) data element is unavailable, and is moved to the TRAN.AMT-2 field in the PSTM.

Surcharge Amount

Table 44 passes the transaction surcharge amount.

Format	Value	Description
N4	0014	Additional Data Length
ANS2	44	Table ID
ANS12		Surcharge amount

Payment Plan

Table 45 contains the payment plan for the items purchased. The acquirer defines “Number of Payments” and “Pay Plan”.

Format	Value	Description
N4	0006	Additional Data Length
ANS2	45	Table ID
AN2		Number of Payments. Selected at point of purchase
ANS2		Pay Plan. Selected at point of purchase

EBT Terminal Data

Table 47 contains data on EBT processing.

Format	Value	Description
N4	0043	Additional Length Data
ANS2	47	Table ID
ANS10		Clerk ID
ANS10		Supervisor ID
ANS15		Voucher Number
AND6		Generation Number

EBT Response Data

Table 48 passes EBT response data from the host to the terminal.

Format	Value	Description
N4	0026	Additional Data Length
ANS2	48	Table ID
AN8		EBT Food Balance. Can contain “-”. All spaces means benefit not available. Can’t print balance on receipt
AN8		EBT Cash Balance. Can contain “-”. All spaces means benefit not available. Can’t print balance on receipt
AN8		Case Number

Miscellaneous Amount

Table ID 53 contains the bar amount in the following format:

Format	Value	Description
N4	0014	Additional Data Length
ANS2	53	Table ID
AN12		Bar Amount

The bar amount is moved to the RSTRNT.BEVERAGE-AMT field of the American Express token.

Address Verification Request Data

Table ID 54 contains information required for an address verification request in the following format:

Format	Value	Description
N4	0031	Additional Data Length
ANS2	54	Table ID
AN9		ZIP Code
AN20		Address

Address verification is available on any transaction that can be performed by mail, which includes authorization, sale, refund, and offline sale transactions.

The ZIP code is moved to the ADDR-FLDS.ZIP-CDE field in the PSTM and the address is moved to the ADDR-FLDS.ADDR field in the PSTM.

Address Verification Response Data

Table ID 55 contains the response to an address verification request in the following format:

Format	Value	Description
N4	0005	Additional Data Length
ANS2	55	Table ID
AN1		Address Match Valid values are as follows: N = Street address does not match X = Service is unavailable or has not been completed Y = Street address matches ␣ = No response (where ␣ indicates a blank)
AN1		ZIP Code Match Valid values are as follows: N = ZIP Code does not match X = Service is unavailable or has not been completed Y = ZIP Code matches ␣ = No response (where ␣ indicates a blank)
AN1		AVS Response Code Valid values are as follows: A = Address matches, ZIP code does not match E = Error N = Address does not match, ZIP code does not match R = Retry S = Service not supported U = Address information unavailable W = Address does not match, 9-digit ZIP code matches X = Address matches, 9-digit ZIP code matches Y = Address matches, 5-digit ZIP code matches Z = Address does not match, 5-digit ZIP code matches ␣ = Address verification information not included, where ␣ indicates a blank* 0 = Address verification information included, but not verified*

- * The BASE24-pos Authorization module supports this value. However, if this value appears in the ADDR-FLDS.ADDR-VRFY-STAT field in the PSTM, the HPDH module places a value of U in the AVS Response Code field.

Address verification is available on any transaction that can be performed by mail, which includes authorization, sale, refund, and offline sale transactions.

The AVS response code is based on the value in the ADDR-FLDS.ADDR-VRFY-STAT field in the PSTM. The address match and ZIP code match values in table ID 55 are based upon the AVS response code, as shown in the following table.

ADDR-FLDS. ADDR-VRFY- STAT Field	AVS Response Code	Address Match	ZIP Code Match
A	A	Y	N
E	E	X	X
N	N	N	N
R	R	X	X
S	S	X	X
U	U	X	X
W	W	N	Y
X	X	Y	Y
Y	Y	Y	Y
Z	Z	N	Y
<i>b</i>	U	X	X
0	U	X	X

Customer ID or Purchase Order Number

Table ID 68 contains the customer ID or purchase order number associated with a transaction in the following format:

Format	Value	Description
N4	Variable	Additional Data Length
ANS2	68	Table ID
ANS Variable		Customer ID Number

Data associated with this table ID is moved to the CUST-REF-IND field in the Purchasing Card and Fleet Card token.

Note: Hypercom terminals accept 20-character ID numbers. However, the Visa Interchange Interface process accepts only 17 characters. The Hypercom Device Handler module truncates leading digits if the terminal accepts more than 17 characters.

Level II Commercial Card Data Subfield

Table 73 transmits additional data for Table 15. If a card is within a group of commercial cards, the terminal can request additional data for the transaction sent from Table 73.

Format	Value	Description
N4	0032	Additional Data Length
ANS2	73	Table ID
AN1		Optional Amount ID
N12		Optional Amount. Default is zero

American Express Data Collection

American Express Data Collection support enables BASE24-pos to collect information that is not available in the POS Transaction Log File (PTLF) or in the BASE24 database. This information is used to build the Descriptive Billing File (DBF). The HPDH module maps the data to the American Express token. Hypercom terminals include American Express Data Collection support for general retail, restaurant and lodging industry formats.

The industry format is derived from the following settings:

Message Code	Processing Code	Data Elements Used	Industry Format
N/A	A value other than 09	Data element 54 contains a value and data element 61 contains a value	Restaurant
N/A	N/A	Data element 54 is blank and data element 61 contains a value	General retail
0220	90	Data element 63 contains a value	Lodging

The remainder of this subsection describes the fields that the HPDH module can provide for each industry format. For more information on the American Express standard industry formats, refer to appendix A. For more information about the American Express token, refer to the *BASE24 Tokens Manual*.

Restaurant Format

The HPDH module uses data sent in the message to fill four fields in the Restaurant portion of the American Express token. All other fields in the token are set to default values. These fields can also be completed by the host. The

following table shows the data element that contains the information to be placed in each American Express token field, the name of the American Express token field, and the format of the token field:

Data Element	American Express Token Field Name	Format
	RSTRNT.FRMT-CDE*	N2
37	RSTRNT.REFERENCE-CDE	N9
61	RSTRNT.CHARGE-CDE1.BYTE	AN4
54	RSTRNT.TIP1-AMT	N18

* The American Express format code for restaurant is 12. This value is hard-coded.

General Retail Format

The HPDH module uses data sent in the message to fill six fields in the General Retail portion of the American Express token. All other fields in the token are set to default values. These fields can also be completed by the host. The following table shows the data element that contains the information to be placed in each American Express token field, the name of the American Express token field, and the format of the token field:

Data Element	American Express Token Field Name	Format
	GEN-RETL.FRMT-CDE*	N2
37	GEN-RETL.REFERENCE-CDE	N9
61	GEN-RETL.CHARGE-CDE1.BYTE	N1
61	GEN-RETL.CHARGE-CDE2.BYTE	N1

Data Element	American Express Token Field Name	Format
61	GEN-RETL.CHARGE-CDE3.BYTE	N1
61	GEN-RETL.CHARGE-CDE4.BYTE	N1

* The American Express format code for general retail is 20.

Lodging Format

The HPDH module uses data sent in the message to fill six fields in the Lodging portion of the American Express token. All other fields in the token are set to default values. These fields can also be completed by the host. The following table shows the data element that contains the information to be placed in each American Express token field, the name of the American Express token field, and the format of the token field:

Data Element	American Express Token Field Name	Format
	LDG.FRMT-CDE [*]	N2
37	LDG.REFERENCE-CDE	N9
	LDG.CHARGE-TYPE [†]	N1
63	LDG.ARRIVAL-DAT	N6
63	LDG.DEPART-DAT	N6
63	LDG.ROOM-RATE	N18

* The American Express format code for lodging is 11.

† The charge type is hard-coded with a value of 1.

Check Guarantee and Verification Methods

The method used to perform check guarantee or verification determines the format of the information in the Private Use (P-63) data element because the information required varies according to the method. This method is identified by the Verification Method field in the Private Use (P-63) data element, as shown in the following table.

Verification Method	Format
03, 07	Account number
02, 08	Drivers license
01, 04	MICR data
09	Full MICR (MICR data entered with an option for using a MICR reader)

Note: Option 09 is available only with Hypercom T7 devices.

The HPDH module obtains the verification method from one of the following fields and uses the corresponding format.

- For Hypercom T4, T5, and T6 devices, the value is taken from the CHECK GUARANTEE field on the Configuration Parameters screen of the Hypercom Configuration File (HPCF).
- For Hypercom T7 devices, the value is taken from the CHECK GUARANTEE/SERVICE field on Configuration Parameters Table screen 2 of the Hypercom T7 Configuration File (HPTC).
- If no value is contained in the HPCF or HPTC, or if the value is invalid, the value in the DFLT CHECK ID field on BASE24-pos Terminal Data files (PTD) screen 1 is used.

The following format descriptions include the fields in the Check Guarantee/Verification token used to carry check verification information with the PSTM. Any fields in the Check Guarantee/Verification token that are not required are filled with blanks.

Account Number

When the required identification is an account number, the following format is used for the Private Use (P-63) data element.

Private Use (P-63) Data Element Field Name	Format	Check Guarantee/Verification Token Field Name
Length Attribute	N3	
Verification Method	N2	CHK-ID-TYP (set to 03)

The Private Use (P-63) data element contains only the verification method because the remaining information placed in the Check Guarantee/Verification token is obtained from other data elements, as shown below.

Data Element Name	Format	Check Guarantee/Verification Token Field Name
Card Expiration Date (P-14)	ANS6	BIRTH-DAT
Primary Account Number (P-2)	ANS24	CHK-ID-NUM

Drivers License

When the required identification is a drivers license, the following format is used for the Private Use (P-63) data element.

Private Use (P-63) Data Element Field Name	Format	Check Guarantee/Verification Token Field Name
Length Attribute	N3	
Verification Method	N2	CHK-ID-TYP (set to 02)
State Code	ANS2	STATE-CDE
Birth Date	ANS6	BIRTH-DAT
Drivers License Number	ANS24	CHK-ID-NUM

Check MICR Data

When the required identification is check MICR data, the following format is used for the Private Use (P-63) data element.

Private Use (P-63) Data Element Field Name	Format	Check Guarantee/Verification Token Field Name
Length Attribute	N3	
Verification Method	N2	CHK-ID-TYP (set to 11)
MICR Data	ANS15	CHK-ID-NUM

Full MICR Data

When the required identification is full MICR data, the format used for the Private Use (P-63) data element depends on whether the data is received from a MICR reader or is entered manually. The data ID in each of the following formats is a code that identifies the source of the data provided, as follows:

- A = Manually entered account data (right-justified and zero filled in token)
- C = Check reader data
- R = Manually entered routing number data (left-justified in token)

Data Received from a Check Reader

When the data is received from a check reader, the following format is used for the Private Use (P-63) data element.

Private Use (P-63) Data Element Field Name	Format	Check Guarantee/Verification Token Field Name
Length Attribute	N3	
Verification Method	N2	CHK-ID-TYP (set to 11)
Data Length Attribute	N2	
Data ID (set to C)	ANS1	
Data	ANS40	CHK-ID-NUM

Data Entered Manually

When the data is entered manually, the following format is used for the Private Use (P-63) data element.

Private Use (P-63) Data Element Field Name	Format	Check Guarantee/Verification Token Field Name
Length Attribute	N3	
Verification Method	N2	CHK-ID-TYP (set to 11)
Data 1 Length Attribute	N2	
Data 1 ID (set to A)	ANS1	
Data 1 (account number)	ANS19	CHK-ID-NUM (ahead of routing number)
Data 2 Length Attribute	N2	
Data 2 ID (set to R)	ANS1	
Data 2 (routing number)	ANS11	CHK-ID-NUM (following the account number)

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5: Download Data

The HPDH module supports only full downloads for Hypercom terminals. Configuration parameters downloaded to the terminal are set in the Hypercom Configuration File (HPCF), Hypercom T7 Configuration File (HPTC), and the BASE24-pos Terminal Data files (PTD).

This section contains the following information concerning downloading configuration parameters using the HPDH module:

- Description of downloading to Hypercom T4, T5, and T6 terminals
- Description of downloading to Hypercom T7 terminals
- Examples of downloading to Hypercom devices

Hypercom ISO Download Response Messages—T4, T5, and T6 Terminals

The HPDH module maps data from the BASE24 database to Hypercom External Message response fields for use during downloads. The HPDH module uses specific data elements to download data and provide the appropriate configuration information to the Hypercom T4, T5, or T6 terminal. The HPDH module uses data elements P-43, P-60, P-61, P-62, and P-63 when downloading configuration data to the device. The contents of these five data elements is shown in this subsection, along with their corresponding BASE24 fields.

Card Acceptor Name—Data Element P-43

The following table shows the BASE24 mapping for the Card Acceptor Name (P-43) data element. The Card Acceptor Name (P-43) data element contains the card acceptor name and location.

Field Name	Bytes	BASE24 Mapping
Name	22	PTDS1.PTDS1-CORE.TERM-OWNER-NAM
City	18	PTDS1.PTDS1-CORE.TERM-CITY-ST.CITY PTDS1.PTDS1-CORE.CITY-EXT
State	3	PTDS1.PTDS1-CORE.TERM-CITY-ST.ST
Country	2	PTDS1.PTDS1-CORE.CNTRY-CDE

Configuration Parameters—Data Element P-60

The following table shows the BASE24 mapping for the Private Use (P-60) data element. The Private Use (P-60) data element contains configuration parameters used in adjustments, batch uploads, download requests, and requests for statistics.

Field Name	Bytes	BASE24 Mapping
Length Attribute	2	0LLL—Length of the data element in bytes, excluding the length attribute.

Field Name	Bytes	BASE24 Mapping
DLL Revision Number	1	HPCF.CONFIGURATION-PARAMS.DLL-REVISION- NUM
Initialization Control	1	HPCF.CONFIGURATION-PARAMS.INITIALIZATION- CNTL
ECR Baud Rate	1	HPCF.CONFIGURATION-PARAMS.ECR-BAUD-RATE
Control Parameters	3	HPCF.CONFIGURATION-PARAMS.CONTROL- PARAMS
Printer Control	1	HPCF.CONFIGURATION-PARAMS.PRINTER- CONTROL
Date and Time	6	YYMMDDhhmmss—Compaq NonStop system date and time. The time is adjusted according to the value set in the PTDD1.PTDD1-CORE.TIM-OFST field.
Telephone Dial Option	1	HPCF.CONFIGURATION-PARAMS.PHONE-DIAL- OPTION
Reauthorize Margin	1	HPCF.CONFIGURATION-PARAMS.REAUTHORIZE- MARGIN
Password	2	HPCF.CONFIGURATION-PARAMS.PASSWORD
Amount Dual Entry	1	HPCF.CONFIGURATION-PARAMS.AMOUNT-DUAL- ENTRY
Display Mag Stripe	1	HPCF.CONFIGURATION-PARAMS.DISPLAY-MAG- STRIPE
Help Desk Telephone Number	7	HPCF.CONFIGURATION-PARAMS.HELP-DESK- PHONE
Local Terminal Options	1	HPCF.CONFIGURATION-PARAMS.LOCAL-TERM- OPTIONS
Check Guarantee Options	1	HPCF.CONFIGURATION-PARAMS.CHECK- GUARANTEE-OPTIONS
Terminal Options	1	HPCF.CONFIGURATION-PARAMS.TERM-OPTIONS
Reserved for future use	1	HPCF.CONFIGURATION-PARAMS.USER-FLD-1

Field Name	Bytes	BASE24 Mapping
Current Batch Number	3	PTDD1.PTDD1-CORE.BATCH-NUM
Next Batch Number	3	PTDD1.PTDD1-CORE.BATCH-NUM plus 1
Default Transaction	1	HPCF.CONFIGURATION-PARAMS.DEFAULT-TRAN-CODE
Supervisor Password	1	HPCF.CONFIGURATION-PARAMS.SUPERVISOR-PASSWORD
PIN Working Key	8	PTDD1.PTDD1-CORE.ENCN-PIN.ENCN-KEYS.M-KEY
MAC Working Key	8	PTDD1.PTDD1-CORE.MAC-DATA.ENCN-KEYS.MAS-KEY
Regulation E Merchant Data	69	PTDS1.PTDS1-CORE.TERM-NAM-LOC PTDS1.PTDS1-CORE.TERM-CITY-ST.CITY PTDS1.PTDS1-CORE.CITY-EXT PTDS1.PTDS1-CORE.TERM-CITY-ST.ST
PIN Pad Control	1	HPCF.CONFIGURATION-PARAMS.PIN-PAD-CNTL

Configuration Parameters—Data Element P-61 (Card Definition Table)

The following table shows the BASE24 mapping for the Private Use (P-61) data element. The Private Use (P-61) data element contains configuration parameters for the card definition table. The set of fields beginning with the Card Acquirer Table Index field and ending with the Track Sent field occurs twice.

Field Name	Bytes	BASE24 Mapping
Length Attribute	2	0LLL—Length of the data element in bytes, excluding the length attribute.
Card Acquirer Table Index	1	An index identifying the entry being sent. Valid values are 1 through 16.
Primary Card Prefix—Low	4	HPCF.CARD-DEFINITION-TBL.LOW-PREFIX

Field Name	Bytes	BASE24 Mapping
Primary Card Prefix—High	4	HPCF.CARD-DEFINITION-TBL.HIGH-PREFIX
Card Name	10	HPCF.CARD-DEFINITION-TBL.CARD-NAM
Card Processing Options	7	HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS
Primary Telephone Number	7	HPCF.CARD-DEFINITION-TBL.PRIMARY-PHONE- NUM
Secondary Telephone Number	7	HPCF.CARD-DEFINITION-TBL.BACKUP-PHONE- NUM
Referral Telephone Number	7	HPCF.CARD-DEFINITION-TBL.REFERRAL-PHONE- NUM
Network International ID	2	HPCF.CARD-DEFINITION-TBL.NETWORK- INTERNATIONAL-ID
Card Acceptor ID	15	HPCF.CARD-DEFINITION-TBL.CRD-ACCEPTOR-ID
Reserved for future use	2	HPCF.CARD-DEFINITION-TBL.USER-FLD-3
Floor Limit	2	HPCF.CARD-DEFINITION-TBL.FLOOR-LIMIT
Terminal Timeout	1	HPCF.CARD-DEFINITION-TBL.TERMINAL- TIMEOUT
Default Account	1	HPCF.CARD-DEFINITION-TBL.DEFAULT-ACCT
Track Sent	1	HPCF.CARD-DEFINITION-TBL.TRACK-SENT

Configuration Parameters—Data Element P-62 (Product Code Table)

The following table shows the BASE24 mapping for the Private Use (P-62) data element. The Private Use (P-62) data element contains configuration parameters for the product code table. This data element is used when the Hypercom T4, T5,

or T6 terminal is configured for product code entry. The set of fields beginning with the Code Select Key field and ending with the Product Description field occurs 9 times.

Field Name	Bytes	BASE24 Mapping
Length Attribute	2	0LLL—Length of the data element in bytes, excluding the length attribute.
Code Select Key	1	An index identifying the product code being sent. Valid values are 01 through 09.
Product Code	2	HPCF.PRODUCT-CODE.CDE
Product Description	20	HPCF.PRODUCT-CODE.DESCRPTION

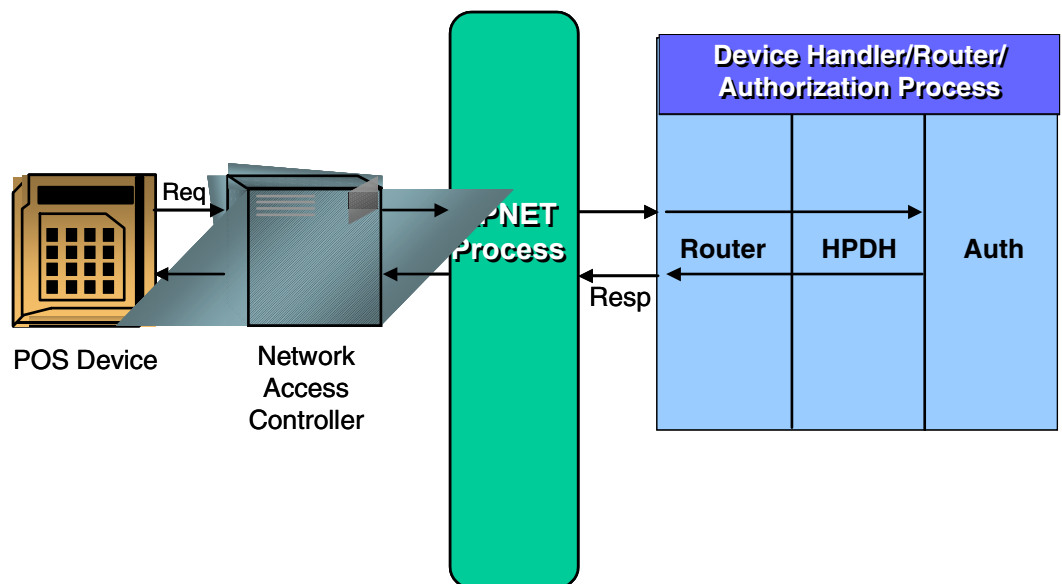
Configuration Parameters—Data Element P-63 (Range Definition Table)

The following table shows the BASE24 mapping for the Private Use (P-63) data element. The Private Use (P-63) data element contains configuration parameters for the range definition table. The set of fields beginning with the Card Number Range Table Index field and ending with the Card Prefix—High field can occur 18 times.

Field Name	Bytes	BASE24 Mapping
Length Attribute	2	0LLL—Length of the data element in bytes, excluding the length attribute.
Card Number Range Table Index	2	An index identifying the entry being sent. Valid values are 0001 through 0100.
Card Acquirer Table Index	1	HPCF.RDT.RANGE-DEFINITION-TBL.CARD-DEFINITION-TBL-INDEX
Card Prefix—Low	4	HPCF.RDT.RANGE-DEFINITION-TBL.LOW-PREFIX
Card Prefix—High	4	HPCF.RDT.RANGE-DEFINITION-TBL.HIGH-PREFIX

Hypercom Download Examples—T4, T5, and T6 Terminals

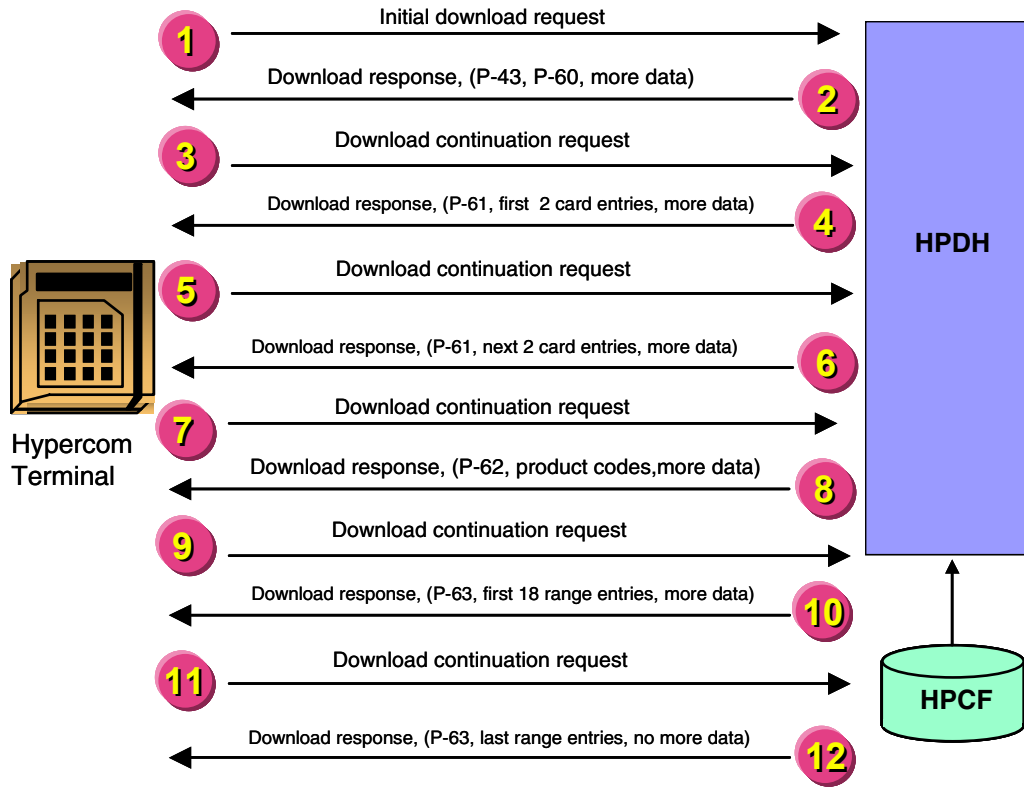
The Hypercom network access controller (NAC) and the HPDH module support full parameter downloads to Hypercom T4, T5, and T6 terminals. The NAC maps the initial download request from a Hypercom device to an ISO message format. The HPDH module checks the terminal type in the Transport Protocol Data Unit (TPDU) message header to determine that the response should be formatted as a Hypercom response. The HPDH module retrieves the required data from the Hypercom Configuration File (HPCF) and formats a native-mode response to a Hypercom device. The diagram below illustrates the basic message flow beginning with a request from the Hypercom terminal and completing with a response from the HPDH module. Note that the request travels through the NAC, the XPNET process, and the Router module, and the response travels through the XPNET process and the NAC.



The following examples describe typical transaction flows that occur when the HPDH module is downloading information to Hypercom terminals. Each example describes each step that is taken during the download, and the order in which the steps are performed.

Download Completes Successfully—18 or More Entries

The following example describes the steps taken during a successful download when between 18 and 100 card prefix range entries are contained in the range definition table on screen 4 of the Hypercom Configuration File (HPCF). For simplicity, message interactions with the NAC, XPNET process, and Router module are not shown.



1. The terminal formats and sends a download request to the HPDH module. The last digit of the processing code contained in the request is 0, indicating that this is an initial request.
2. Since this is an initial request, the HPDH module resets a device dependent data index for accessing the HPCF, and formats and sends a response to the terminal NAC. This response contains the Card Acceptor Name (P-43) data element and contains configuration parameters in the Private Use (P-60) data element. The last digit of the processing code is a 1, indicating more messages are to come.

3. The terminal formats the download information it has received and sends a download continuation request through the terminal NAC to the HPDH module. The last digit of the processing code is a 1, indicating this is a download continuation request.
4. The HPDH module searches its index for accessing the HPCF and determines that the next element to be sent in the download is the card definition table. The card definition table is set up on screen 3 of the HPCF. The card definition table contains up to 16 screens. All 16 occurrences of the card definition table are downloaded to the terminal. The HPDH module updates its index for accessing the HPCF and formats a response in the Private Use (P-61) data element, containing the first two entries of the card definition table. The last digit of the processing code is a 1, indicating more messages are to come.
5. The terminal formats the download information it has received from the HPDH module and sends a download continuation request through the terminal NAC to the HPDH module. The last digit of the processing code is a 1, indicating that this is a download continuation request.
6. The HPDH module updates its index for accessing the HPCF, then formats a response to send the next two card definition table entries in the Private Use (P-61) data element through the terminal NAC to the terminal. The last digit of the processing code is a 1, indicating more messages are to come.
7. The terminal formats the download information it has received from the HPDH module and sends a download continuation request through the terminal NAC to the HPDH module. The last digit of the processing code is a 1, indicating that this is a download continuation request. Steps 6 and 7 are repeated until the entire card definition table is sent.
8. The HPDH module searches its index for accessing the HPCF and determines that the next element to be sent in the download is the product code table. The product code table is set up on screen 2 of the HPCF. The HPDH module updates its index, then formats and sends a response through the terminal NAC to the terminal. This response contains the product code table in the Private Use (P-62) data element. The last digit of the processing code is a 1, indicating more messages are to come.
9. The terminal formats the download information it has received from the HPDH module and sends a download continuation request through the terminal NAC to the HPDH module. The last digit of the processing code is a 1, indicating that this is a download continuation request.
10. The HPDH module searches its index for accessing the HPCF and determines that the next element to be sent in the download is the range definition table. The range definition table is set up on screen 4 of the HPCF. The HPDH module updates its index, then formats a response containing the first 18

entries of the range definition table in the Private Use (P-63) data element. The last digit of the processing code is a 1, indicating more messages are to come.

11. The terminal formats the download information it has received from the HPDH module and sends a download continuation request to the HPDH module. The last digit of the processing code is a 1, indicating that this is a download continuation request. Steps 10 and 11 are repeated until fewer than 18 valid range definition entries remain.
12. The HPDH module searches its index for accessing the HPCF and determines that fewer than 18 valid range definition entries remain to be sent to the terminal in the download. This can be because the HPDH module has reached the end of the range definition table or because the HPDH module has reached an end marker (that is, an entry in the range definition table that has been zero filled). The HPDH module updates its index for accessing the HPCF and formats a response in the Private Use (P-63) data element containing the remaining valid range definition table entries.

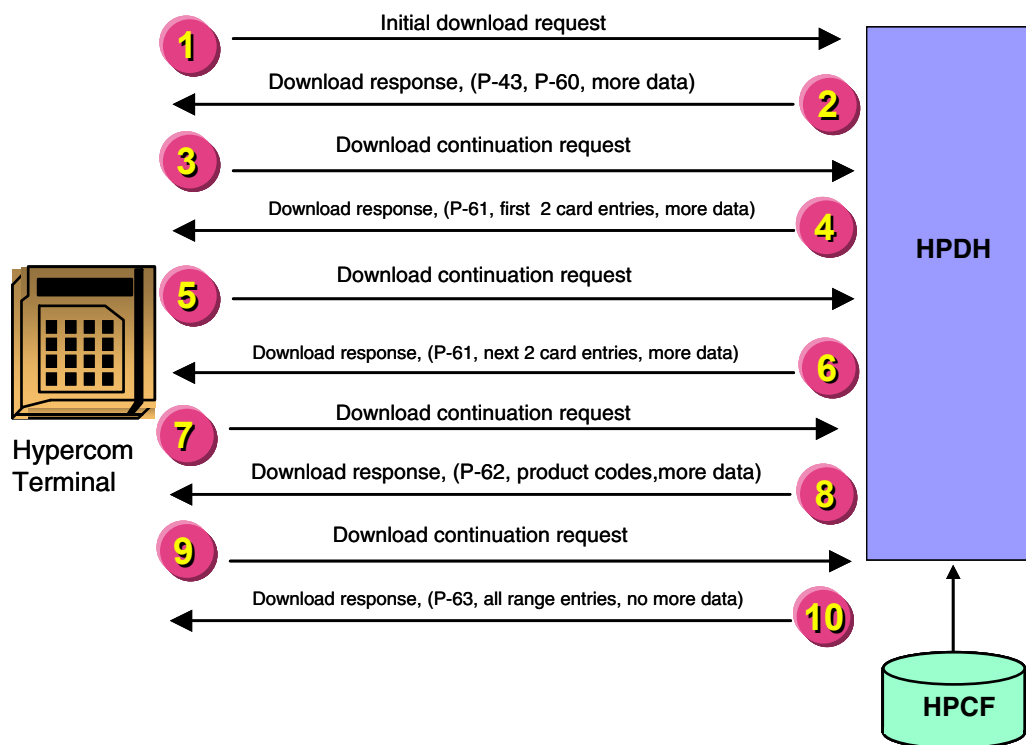
If the last entry being sent is not 100, the HPDH module formats a dummy range definition table entry equal to the next entry that would have been sent. The high and low prefix ranges contained in this dummy entry are zero filled to serve as an end marker. The dummy entry indicates that no more range definition table information exists.

If the last valid entry in the range definition table is a multiple of 18, the dummy entry formatted by the HPDH module is the only entry contained in the Private Use (P-63) data element.

The last digit of the processing code is a 0, indicating no more messages are to come.

Download Completes Successfully—Fewer Than 18 Entries

The following example describes the steps taken during a successful download when less than 18 card prefix range entries are contained in the range definition table on screen 4 of the Hypercom Configuration File (HPCF). For simplicity, message interactions with the NAC, XPNET process, and Router module are not shown.



1. The terminal formats and sends a download request through the terminal NAC to the HPDH module. The last digit of the processing code contained in the request is 0, indicating that this is an initial request.
2. Since this is an initial request, the HPDH module resets a device dependent data index for accessing the HPCF, and formats and sends a response to the terminal NAC. This response contains the Card Acceptor Name (P-43) data element and contains configuration parameters in the Private Use (P-60) data element. The last digit of the processing code is a 1, indicating more messages are to come.
3. The terminal formats the download information it has received and sends a download continuation request through the terminal NAC to the HPDH module. The last digit of the processing code is a 1, indicating that this is a download continuation request.

4. The HPDH module searches its index for accessing the HPCF and determines that the next element to be sent in the download is the card definition table. The card definition table is set up on screen 3 of the HPCF. The card definition table contains up to 16 screens. All 16 occurrences of the card definition table are downloaded to the terminal. The HPDH module updates its index for accessing the HPCF and formats a response in the Private Use (P-61) data element, containing the first two entries of the card definition table. The last digit of the processing code is a 1, indicating more messages are to come.
5. The terminal formats the download information it has received from the HPDH module and sends a download continuation request through the terminal NAC to the HPDH module. The last digit of the processing code is a 1, indicating that this is a download continuation request.
6. The HPDH module updates its index for accessing the HPCF, then formats a response to send the next two card definition table entries in the Private Use (P-61) data element through the terminal NAC to the terminal. The last digit of the processing code is a 1, indicating more messages are to come.
7. The terminal formats the download information it has received from the HPDH module and sends a download continuation request through the terminal NAC to the HPDH module. The last digit of the processing code is a 1, indicating that this is a download continuation request. Steps 6 and 7 are repeated until the entire card definition table is sent.
8. The HPDH module searches its index for accessing the HPCF and determines that the next element to be sent in the download is the product code table. The product code table is set up on screen 2 of the HPCF. The HPDH module updates its index, then formats and sends a response through the terminal NAC to the terminal. This response contains the product code table in the Private Use (P-62) data element. The last digit of the processing code is a 1, indicating more messages are to come.
9. The terminal formats the download information it has received from the HPDH module and sends a download continuation request through the terminal NAC to the HPDH module. The last digit of the processing code is a 1, indicating that this is a download continuation request.
10. The HPDH module searches its index for accessing the HPCF and determines that the next element to be sent in the download is the range definition table. The range definition table is set up on screen 4 of the HPCF. The HPDH module determines that fewer than 18 valid range definition entries are to be sent to the terminal in the download (that is, one of the first 18 entries in the range definition table has been zero filled). The HPDH module updates its index into the HPCF and formats a response in the Private Use (P-63) data element containing the valid range definition table entries.

The HPDH module formats a dummy range definition table entry equal to the next entry that would have been sent. The high and low prefix ranges contained in this dummy entry are zero filled to serve as an end marker. The dummy entry indicates that no more range definition table information exists.

The last digit of the processing code is a 0, indicating no more messages are to come.

Hypercom ISO Download Response Message—T7 Terminals

The HPDH module maps data from the BASE24 database to ISO response subfields for use during downloads to Hypercom T7 terminals. The HPDH module uses subfields of the Private Use (P-60) data element to download information for the following tables to the T7 device:

- Terminal configuration
- Card range
- Issuer
- Acquirer
- Product descriptor
- Additional data prompt

The contents of the Private Use (P-60) data element subfields are described in this subsection along with their corresponding BASE24 fields. The message format is included.

Also, four new tables have been added to this section:

- ICC
- CHIPDATA
- KEYDATA
- REVKEY

Message Format

The HPDH module responds to an 0800 initialization request with an 0810 response. The Private Use (P-60) data element of the 0810 response message contains the total field length and subfield data.

The following is the format of the response message the HPDH module sends to the Hypercom T7 terminal.

Field Name	Bytes	Description
Total Field Length	2	The total length of this field.
Subfield	variable	The initialization data portion of the response.

The response can contain any number of subfields in any order, up to the maximum message size supported by the network and the T7 terminal. Each subfield of the 0810 message contains the following information.

Subfield	Bytes	Description
Table ID	1	A code identifying the subfield. Valid values are as follows: 00 = Clear all initialization tables 01 = Terminal configuration table 02 = Card range table 03 = Issuer table 04 = Acquirer table 05 = Descriptor table 09 = Additional data prompt table 0C = ICC table ID 0D = Chipdata table ID 0E = Keydata table 0F = REV Key table
Subfield Length	2	The total length of the subfield. Note: This field should contain a value of 0000 when the Table ID field is set to 00.
Subfield Data	variable	Table data

ICC Table Entry

The ICC table (ICC) can have a maximum of 8 individual records. The Table ID for the ICC table records is “0C”. Each record will have the following format:

Field Name	Bytes	Type	Description
Table ID	1	N	0C - ICC table
Table Length	2	N	Total length of the following table entry
Tblnmbr	1	N	Position in table Range: 01h-08h
Iccisid	1	N	Issuer ID
ICCOptions 1	1	L	MSB 7 6 5 4 3 2 1 0 Always print and check signature
ICCOptions 2	1	L	MSB 7 6 5 4 3 2 1 0 Always print and check signature

Field Name	Bytes	Type	Description
ICCOptions 3	1	L	MSB 7 6 5 4 3 2 1 0 Always print and check signature
Target percentage for biased random selection	1	N	Target percentage for biased random selection
Max target percentage for biased random select	1	N	Max target percentage for biased random select
Threshold amount for biased random select	2	N	Threshold amount for biased random select
TAC terminal action code Denial	5	N	TAC terminal action code denial
TAC terminal action code online	5	N	TAC terminal action code online
TAC terminal action code Default	5	N	TAC terminal action code Default
Transaction Category Code	1	A	Transaction Category Code default 'R'
	25		Data length does not include table ID or the table length

Chip Data Table

The Chip Data table (CHIPDATA) has a single record entry (similar to the existing Terminal Configuration Parameters table). The Table ID for the CHIPDATA table record is “0D”. The CHIPDATA record will have the following format:

Field Name	Bytes	Type	Description
Table ID	1	N	0D - Chipdata table
Table Length	2	N	Total length of the following table entry
Tableid	1	N	Position in the table
AddTID	1	N	Additional Tags ID
Iccaid Length	1	N	Length of the following field (Iccaid)
Iccaid	16	N	Terminal Application ID
ICCOptions 1	1	L	MSB 7 Unused 6 Skip EMV Processing for this AID 5 Use additional tags 4 Easy Entry allowed 3 Go online if AAR 2 Special processing ARQC --> AAC 1 User issuer floor limit 0 Select this AID if exact match only
Floor	2	N	AID's floor limit
TAIDVER1	2	N	Supported AID's Version Number 1
TAIDVER2	2	N	Supported AID's Version Number 2
TAIDVER3	2	N	Supported AID's Version Number 3

Field Name	Bytes	Type	Description
TAIDTDOL	20	N	Terminal Default TDOL
TAIDTDOL	20	N	Terminal Default DDOL
TAIDOFL_APPR	2	A	Offline Approved Response Code
TAIDOFL_DECL	2	A	Offline Declined Response Code
TAIDUNAB_APPR	2	A	Unable to go online (Offline Approved) Response Code
TAIDUNAB_DECL	2	A	Unable to go online (Offline Declined) Response Code
	76		Length of data does not include the table ID or the table length

Key Data Table

The Key Data table (KEYDATA) is a Certification Authority Public Key table. It is used to specify a Load KEYDATA table entry and assign parameters to it. The table can have a maximum of 60 individual record occurrences.

Field Name	Bytes	Type	Description
Table ID	1	N	OE - Keydata table
Table Length	2	N	Total length of the following table entry
Keydata Table Number	1	N	Position in the table
Public Key - Key Index	1	N	Public Key index
Public Key - Key Length	1	N	Key length
Public Key - Key	248	N	Public Key

Field Name	Bytes	Type	Description
Public Key - RID	5	N	Public key associated with RID
Public Key - Exponent	1	N	Public Key exponent
Public Key - Hash	20	N	Public Key Hash value
	277		Data length does not include table ID or table length

Revoked Certificate Key Table

The Revoked Certificate Key table (REVKEY) is a Certification Authority Public Key table. The REV Key table is used to specify a Revoked Certificate Keys table entry and assign parameters to it. The table has a single record.

Field Name	Bytes	Type	Description
Table ID	1	N	0F - REV Key table
Table Length	2	N	Total length of the following table entry
Key RID 1	5		Revoked certificate key scheme application identifier 1
Key IDX 1	1		Revoked certificate key index 1
Key Cert Num 1	3		Revoked certificate key number 1
Key RID 2	5		Revoked certificate key scheme application identifier 2
Key IDX 2	1		Revoked certificate key index 2

Field Name	Bytes	Type	Description
Key Cert Num 2	3		Revoked certificate key number 2
Key RID 3	5		Revoked certificate key scheme application identifier 3
Key IDX 3	1		Revoked certificate key index 3
Key Cert Num 3	3		Revoked certificate key number 3
Key RID 4	5		Revoked certificate key scheme application identifier 4
Key IDX 4	1	N	Revoked certificate key index 4
Key Cert Num 4	3	N	Revoked certificate key number 4
Key RID 5	5	N	Revoked certificate key scheme application identifier 5
Key IDX 5	1	N	Revoked certificate key index 5
Key Cert Num 5	3	N	Revoked certificate key number 5
Key RID 6	5	N	Revoked certificate key scheme application identifier 6
Key IDX 6	1	N	Revoked certificate key index 6
Key Cert Num 6	3	N	Revoked certificate key number 6

Field Name	Bytes	Type	Description
Key RID 7	5	N	Revoked certificate key scheme application identifier 7
Key IDX 7	1	N	Revoked certificate key index 7
Key Cert Num 7	3	N	Revoked certificate key number 7
Key RID 8	5	N	Revoked certificate key scheme application identifier 8
Key IDX 8	1	N	Revoked certificate key index 8
Key Cert Num 8	3	N	Revoked certificate key number 8
Key RID 9	5	N	Revoked certificate key scheme application identifier 9
Key IDX 9	1	N	Revoked certificate key index 9
Key Cert Num 9	3	N	Revoked certificate key number 9
Key RID 10	5	N	Revoked certificate key scheme application identifier 10
Key IDX 10	1	N	Revoked certificate key index 10
Key Cert Num 10	3	N	Revoked certificate key number 10
	90		Data length. Does not include table ID or table length

Terminal Configuration Table

The terminal configuration table is used to set terminal-wide parameters that control terminal operation. There is one terminal configuration table per terminal.

In addition to the Table ID and Table Length fields, the following fields must be downloaded in the Private Use (P-60) data element, subfield 01 of the 0810 download response.

Field Name	Bytes	BASE24 Mapping
DLL Tracking	1	HPTC.CONFIGURATION-PARAMS. DLL-TRACKING-NUM
Initialization Control	1	HPTC.CONFIGURATION-PARAMS. INITIALIZATION-CNTL
Printer Baud Rate	1	HPTC.CONFIGURATION-PARAMS. PRNTR-BAUD-RATE
ECR Baud Rate	1	HPTC.CONFIGURATION-PARAMS. ECR-BAUD-RATE
Date and Time	6	YYMMDDhhmmss—Compaq NonStop system date and time. The time is adjusted according to the value set in the PTDD1. PTDD1-CORE.TIM-OFST field.
Telephone Dial Options (see “Telephone Dial Options Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. PHONE-DIAL-OPTION
Password	2	HPTC.CONFIGURATION-PARAMS. PASSWORD
Help Desk Phone	12	HPTC.CONFIGURATION-PARAMS. HELP-DESK-PHONE
Options 1 (see “Options 1 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. OPTIONS-1
Options 2 (see “Options 2 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. OPTIONS-2
Options 3 (see “Options 3 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. OPTIONS-3

Field Name	Bytes	BASE24 Mapping
Options 4 (see “Options 4 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS.OPTIONS-4
Receipt Line 2	23	PTDS1.PTDS1-CORE.TERM-NAM-LOC
Receipt Line 3	23	PTDS1.PTDS1-CORE.TERM-CITY-ST. CITY PTDS1.PTDS1-CORE.CITY-EXT PTDS1.PTDS1-CORE.TERM-CITY-ST. ST
Default Merchant Name	23	PTDS1.PTDS1-CORE.TERM-OWNER-NAM
Currency Symbol	1	HPTC.CONFIGURATION-PARAMS.CRNCY-SYMBOL
Transaction Currency Digits	1	HPTC.CONFIGURATION-PARAMS.CRNCY-DIGITS-TRANS
Currency Decimal Point Position	1	HPTC.CONFIGURATION-PARAMS.CRNCY-DEC-POSITION
Language Indicator	1	HPTC.CONFIGURATION-PARAMS.LANGUAGE-IND
Settlement Currency Digits	1	HPTC.CONFIGURATION-PARAMS.CRNCY-DIGITS-SETL
Check Service	1	HPTC.CONFIGURATION-PARAMS.CHK-SERVICE
Check Guarantee Issuer Table ID	1	HPTC.CONFIGURATION-PARAMS.ISS-TBL-ID
Check Guarantee Acquirer Table ID	1	HPTC.CONFIGURATION-PARAMS.ACQ-TBL-ID
Local Terminal Options (see “Local Terminal Options Bits,” below)	1	HPTC.CONFIGURATION-PARAMS.TERM-OPTS

Field Name	Bytes	BASE24 Mapping
Default Transaction	1	HPTC.CONFIGURATION-PARAMS. DFLT-TRAN
Debit Issuer ID	1	HPTC.CONFIGURATION-PARAMS. DEBIT-ISS-TBL-ID
Debit Acquirer ID	1	HPTC.CONFIGURATION-PARAMS. DEBIT-ACQ-TBL-ID
Phone Line Hold	1	HPTC.CONFIGURATION-PARAMS. PHONE-LINE-HLD
Reserved for future use	2	HPTC.CONFIGURATION-PARAMS. USER-FLD-1
PIN Pad Type	1	HPTC.CONFIGURATION-PARAMS. PIN-PAD-TYPE
Printer Type	1	HPTC.CONFIGURATION-PARAMS. PRINTER-TYPE
Cash Back Limit	2	HPTC.CONFIGURATION-PARAMS. CASH-BACK-LMT
NMS Call Schedule	1	HPTC.CONFIGURATION-PARAMS. NMS-CALL-SCHED
Check Prompts 1 (see “Check Prompts 1 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1
Check Prompts 2 (see “Check Prompts 2 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2
Check Prompts 3 (see “Check Prompts 3 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-3
Merchant Surcharge	2	HPTC.CONFIGURATION-PARAMS. MRCH-SURCHARGE
Options 5 (see “Options 5 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. OPTIONS-5

Field Name	Bytes	BASE24 Mapping
Reserved for future use	8	Reserved for future use
Maintenance Password	6	HPTC.CONFIGURATION-PARAMS. MAINT-PASSWORD
Reserved for future use	8	Reserved for future use
Options 7	1	HPTC.CONFIGURATION-PARAMS. OPTIONS-7
Check Services Flag	4	HPTC.CONFIGURATION-PARAMS. CHK-SRVC-FLGS
Options 8 (see “Options 8 Bits,” below)	1	HPTC.CONFIGURATION-PARAMS. OPTIONS-8
Currency Code	2	HPTC.CONFIGURATION-PARAMS. CURR-CDE
Country Code	2	HPTC.CONFIGURATION-PARAMS. CNTRY-CDE
Terminal Capabilities	3	HPTC.CONFIGURATION-PARAMS. TERM-CAP
Additional Terminal Capabilities	5	HPTC.CONFIGURATION-PARAMS. ADDNL-TERM-CAP
Reserved for future use	5	Reserved for future use

Telephone Dial Options Bits

The Telephone Dial Options field contains eight bits that identify telephone operation options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0–2	Unused	Reserved for future use

Bit	Bit Name	BASE24 Mapping
3	Auto Answer	HPTC.CONFIGURATION-PARAMS. PHONE-DIAL-OPTION.AUTO- ANSWER
4	Referral Dialing	HPTC.CONFIGURATION-PARAMS. PHONE-DIAL-OPTION.REFERRAL- DIALING
5	Memory Dialing	HPTC.CONFIGURATION-PARAMS. PHONE-DIAL-OPTION.MEMORY- DIALING
6	Toll Dialing	HPTC.CONFIGURATION-PARAMS. PHONE-DIAL-OPTION.TOLL- DIALING
7	Keyboard Dialing	HPTC.CONFIGURATION-PARAMS. PHONE-DIAL-OPTION.KBD-DIALING

Options 1 Bits

The Options 1 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Amount Dual Entry	HPTC.CONFIGURATION-PARAMS. OPTIONS-1.AMOUNT-DUAL-ENTRY
1	Display Magnetic Stripe Reader Input	HPTC.CONFIGURATION-PARAMS. OPTIONS-1.DISPLAY-MSR
2	Tip Processing	HPTC.CONFIGURATION-PARAMS. OPTIONS-1.TIP-PROCESSING
3	Cashier Processing	HPTC.CONFIGURATION-PARAMS. OPTIONS-1.CASHIER-PROCESSING
4	Lodging Processing	HPTC.CONFIGURATION-PARAMS. OPTIONS-1.LODGING-PROCESSING

Bit	Bit Name	BASE24 Mapping
5	Print Time on Receipt	HPTC.CONFIGURATION-PARAMS. OPTIONS-1.PRNT-TIM-ON-RECEIPT
6	Business Date Format	HPTC.CONFIGURATION-PARAMS. OPTIONS-1.BUS-DAT-FRMT
7	Reconciliation Method	HPTC.CONFIGURATION-PARAMS. OPTIONS-1.RECONCILE-METHOD

Options 2 Bits

The Options 2 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Reserved	Reserved for future use
1	Printer Used	HPTC.CONFIGURATION-PARAMS. OPTIONS-2.PRNTR-ATTACHED
2	Print <i>Total</i> on Receipt	HPTC.CONFIGURATION-PARAMS. OPTIONS-2.TTL-ON-RCPT
3	Till Processing	HPTC.CONFIGURATION-PARAMS. OPTIONS-2.TILL-PROCESSING
4	Refund Transactions Processed Offline	HPTC.CONFIGURATION-PARAMS. OPTIONS-2.RFND-PROCESSED-OFFL
5	Void Transactions Processed Offline	HPTC.CONFIGURATION-PARAMS. OPTIONS-2.VOID-PROCESSED-OFFL
6	Enable Receipt Preprint	HPTC.CONFIGURATION-PARAMS. OPTIONS-2.RCPT-PRE-PRNT
7	Card Verify Key or Debit Key	HPTC.CONFIGURATION-PARAMS. OPTIONS-2.CRD-VRFY-OR-DEBIT- KEY

Options 3 Bits

The Options 3 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Unused	Reserved for future use
1	Electronic Card Reader Interface	HPTC.CONFIGURATION-PARAMS.OPTIONS-3.ECR-INTERFACE
2	Amount Verify on PIN Pad	HPTC.CONFIGURATION-PARAMS.OPTIONS-3.PIN-PAD-AMT-VRFY
3	Print Card Verify Receipt	HPTC.CONFIGURATION-PARAMS.OPTIONS-3.CRD-VRFY-RCPT
4	<i>Take Imprint Prompt, Manual Entry</i>	HPTC.CONFIGURATION-PARAMS.OPTIONS-3.TAKE-IMPRINT
5	Check Reader Installed	HPTC.CONFIGURATION-PARAMS.OPTIONS-3.CHECK-READER-INSTALLED
6	Card Read from PIN Pad	HPTC.CONFIGURATION-PARAMS.OPTIONS-3.CRD-READ-FROM-PIN-PAD
7	Unused	Reserved for future use

Options 4 Bits

The Options 4 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Adjust by Cashier	HPTC.CONFIGURATION-PARAMS.OPTIONS-4.ADJ-BY-CASHIER

Bit	Bit Name	BASE24 Mapping
1	Prompt Server Not Cashier	HPTC.CONFIGURATION-PARAMS.OPTIONS-4.PROMPT-SRVR-NOT-CASHIER
2	Batch Review by Cashier	HPTC.CONFIGURATION-PARAMS.OPTIONS-4.BATCH-REVIEW-BY-CASHIER
3	Tender Cash Prompt	HPTC.CONFIGURATION-PARAMS.OPTIONS-4.TENDER-CASH-PROMPT
4-5	Reserved	Reserved for future use
6	Duplicate Processing	HPTC.CONFIGURATION-PARAMS.OPTIONS-4.DUP-PROCESSING
7	Unused	Reserved for future use

Local Terminal Options Bits

The Local Terminal Options field contains eight bits that identify options that the terminal user can change at the terminal. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Keyboard Locked	HPTC.CONFIGURATION-PARAMS.TERM-OPTS.KEYBOARD-LOCKED
1	Void Password	HPTC.CONFIGURATION-PARAMS.TERM-OPTS.VOID-PSWD-DISABLE
2	Refund Password	HPTC.CONFIGURATION-PARAMS.TERM-OPTS.RFND-PSWD-DISABLE
3	Adjustment Password	HPTC.CONFIGURATION-PARAMS.TERM-OPTS.ADJ-PSWD-DISABLE
4	Report Password	HPTC.CONFIGURATION-PARAMS.TERM-OPTS.RPT-PSWD-DISABLE
5-7	Unused	Reserved for future use

Check Prompts 1 Bits

The Check Prompts 1 field contains eight bits that identify the active prompts for the check key. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Enter Address	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1.ADDR
1	Drivers License Number	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1.LICENSE-NUM
2	Enter Check Number	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1.CHK-NUM
3	Enter Transit Number	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1.TRANSIT-NUM
4	Enter Route Number	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1.RTE-NUM
5	Enter Last Four Digits of Account	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1.LAST-4-ACCT-DIGITS
6	Enter Check Account Number	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1.CHK-ACCT-NUM
7	Enter MICR Number	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-1.MICR-NUM

Check Prompts 2 Bits

The Check Prompts 2 field contains eight bits that identify the active prompts for the check key. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Expiration Date MMY	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2.EXP-DAT

Bit	Bit Name	BASE24 Mapping
1	Idle Prompt	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2.IDLE-PROMPT
2	Enter Trans Code	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2.TXN-CDE
3	Enter ID Number	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2.ID-NUM
4	Birthdate MMDDYY	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2.BIRTH-DAT
5	Enter Phone Number	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2.PHONE-NUM
6	Enter ZIP Code	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2.POST-CDE
7	State Code	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-2.ST-CDE

Check Prompts 3 Bits

The Check Prompts 3 field contains eight bits that identify the active prompts for the check key. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0–6	Not used	Reserved for future use
7	Enter Check Number If Not Scanned	HPTC.CONFIGURATION-PARAMS. CHK-PROMPTS-3.CHK-NUM-NO-SCAN

Options 5 Bits

The Options 5 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Host Selection Type	HPTC.CONFIGURATION-PARAMS. OPTIONS-5.HOST-SELECT-TYPE
1	No Host Select Prompt at Settlement	HPTC.CONFIGURATION-PARAMS. OPTIONS-5.HOST-SELECT-PROMPT
2–7	Not used	Reserved for future use

Options 8 Bits

The Options 8 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Customer Activated	HPTC.CONFIGURATION-PARAMS. OPTIONS-8.CUST-ACTVT
1	Term Online Capable	HPTC.CONFIGURATION-PARAMS. OPTIONS-8.TERM-ONLN-CAP
2	Online PIN Verification	HPTC.CONFIGURATION-PARAMS. OPTIONS-8.ONLN-PIN-VRFY
3	Offline PIN Verification	HPTC.CONFIGURATION-PARAMS. OPTIONS-8.OFFLN-PIN-VRFY
4	Term with SCR	HPTC.CONFIGURATION-PARAMS. OPTIONS-8.TERM-SCR
5	ICC EMV Enabled	HPTC.CONFIGURATION-PARAMS. OPTIONS-8.ICC-EMV-ENBL
6	Allow PIN Bypass	HPTC.CONFIGURATION-PARAMS. OPTIONS-8.ALW-PIN-BYP

Bit	Bit Name	BASE24 Mapping
7	Early Amount (EMV)	HPTC.CONFIGURATION-PARAMS.OPTIONS-8.ERLY-AMT-EMV

Card Range Table

The card range table is used to specify up to 32 card ranges and to define their parameters.

In addition to the Table ID and Table Length fields, the following fields must be downloaded in the Private Use (P-60) data element, subfield 02 of the 0810 download response. The set of fields shown below can occur 6 times.

The field names in the BASE24 mapping column are for the first 16 card ranges. The second 16 card ranges are mapped to CARD-RANGE-2 instead of CARD-RANGE-1.

Field Name	Bytes	BASE24 Mapping
Card Range Number	1	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.CRD-RANGE-INDEX
PAN Range Low	5	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.LOW-PREFIX
PAN Range High	5	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.HIGH-PREFIX
Issuer Table ID	1	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.ISS-TBL-ID
Acquirer Table ID	1	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.ACQ-TBL-ID
PAN Length	1	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.PAN-LEN
Options 1 (see “Options 1 Bits,” below)	1	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-1
Options 2 (not used)	1	Reserved for future use
Options 3 (not used)	1	Reserved for future use

Field Name	Bytes	BASE24 Mapping
Options 4 (not used)	1	Reserved for future use
Issuer Table ID Number 2	1	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.ISS-TBL-ID-2
Acquirer Table ID Number 2	1	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.ACQ-TBL-ID-2
Reserved for future use	2	Reserved for future use

Options 1 Bits

The Options 1 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Debit Yes or No Prompt	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-1.DEBIT-PROMPT
1	Debit Key Only	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-1.DEBIT-KEY-ONLY
2	Select Debit From PIN Pad	HPTC.CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-1.DEBIT-PIN-PAD
3–7	Not used	Reserved for future use

Issuer Table

The issuer table is used to define the parameters associated with a card issuer.

In addition to the Table ID and Table Length fields, the following fields must be downloaded in the Private Use (P-60) data element, subfield 03 of the 0810 download response. The set of fields shown below can occur 3 times.

Field Name	Bytes	BASE24 Mapping
Issuer Number	1	HPTC.ISS-TABLE.ISS-TBL-DATA.ISS-TBL-INDEX
Issuer Table ID	1	HPTC.ISS-TABLE.ISS-TBL-DATA.ISS-TBL-ID
Card Name	10	HPTC.ISS-TABLE.ISS-TBL-DATA.CRD-NAM
Referral Phone Number	12	HPTC.ISS-TABLE.ISS-TBL-DATA.RFRL-PHONE-NUM
Options 1 (see “Options 1 Bits,” below)	1	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1
Options 2 (see “Options 2 Bits,” below)	1	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2
Options 3 (see “Options 3 Bits,” below)	1	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-3
Options 4 (see “Options 4 Bits,” below)	1	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4
Default Account	1	HPTC.ISS-TABLE.ISS-TBL-DATA.DEFAULT-ACCT
Reserved for future use	2	Reserved for future use
Floor Limit	2	HPTC.ISS-TABLE.ISS-TBL-DATA.FLOOR-LIMIT
Reauthorization Margin Percentage	1	HPTC.ISS-TABLE.ISS-TBL-DATA.REAUTHORIZE-MARGIN
PIN Entry Transactions (see “PIN Entry Transactions Bits,” below)	1	HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS

Field Name	Bytes	BASE24 Mapping
Additional Prompts Select (see “Additional Prompts Select Bits,” below)	1	HPTC.ISS-TABLE.ISS-TBL-DATA. ADNL-PROMPTS
Card Type	1	HPTC.ISS-TABLE.ISS-TBL-DATA. CRD-TYPE
Down Payment Indicator	1	HPTC.ISS-TABLE.ISS-TBL-DATA. DOWN-PMNT-IND
Options 5 (see “Options 5 Bits,” below)	1	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-5
Options 6 (see “Options 6 Bits,” below)	1	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-6
Reserved for future use	4	Reserved for future use

Options 1 Bits

The Options 1 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Select Account Type	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-1.ACCT-SELECT
1	PIN Entry	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-1.PIN-ENTRY
2	Manual PAN Allowed	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-1.MANUAL-PAN-ENTRY
3	Expiration Date Required	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-1.EXP-DAT-REQ
4	Offline Entry Allowed	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-1.OFFLINE-TRAN-ALLOW

Bit	Bit Name	BASE24 Mapping
5	Voice Referral Allowed	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-1.VOICE-REFERRAL
6	Descriptors Required	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-1.PRODUCT-CODES
7	Adjust Allowed	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-1.ADJUSTMENT-ALLOWED

Options 2 Bits

The Options 2 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Perform Mod-10 Check on PAN	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-2.MOD-10-CHECK
1	Invoice Number Required	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-2.INVOICE-NUM
2	Print Receipt	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-2.RECEIPT
3	Capture Transactions	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-2.CAPTURE-TRANS
4	Check Expiration Date	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-2.CHK-EXP-DAT
5	Be My Guest	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-2.BE-MY-GUEST
6	Block Refunds	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-2.BLOCK-RFND
7	Block Card Verify	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-2.BLOCK-CRD-VRFY

Options 3 Bits

The Options 3 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Block Voids	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-3.BLOCK-VOIDS
1	Cash Back	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-3.CASH-BACK
2	<i>Print Signature Not Required</i>	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-3.PRNT-SIG-NOT-RQD
3	<i>Print No Refunds Allowed</i>	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-3.PRNT-NO-RFND-ALLOWED
4	Refund Processing Offline	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-3.RFND-PROCESSED-OFFL
5	Void Processing Offline	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-3.VOID-PROCESSED-OFFL
6	Allow Cash Only Transaction	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-3.CASH-ONLY-ALLOW
7	Generate Six-Digit Auth Code	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-3.GENERATE-AUTH-CDE

Options 4 Bits

The Options 4 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Debit Transaction	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-4.DEBIT-TRAN

Bit	Bit Name	BASE24 Mapping
1	Verify Last Four PAN Digits	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.VRFY-PAN
2	Authorize Over Floor Limit	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.AUTH-OVR-FLR
3	Auto Reprint Failed Receipt	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.AUTO-REPRINT
4	Ignore Track 1 Error	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.TRK1-ERROR
5	Prompt For Tax	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.TAX-PROMPT
6	Prompt For Five-Digit Credit Plan	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.CREDIT-PLAN-PROMPT
7	Prompt For Product Codes	HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.PROD-CDE-PROMPT

PIN Entry Transactions Bits

The PIN Entry Transactions field contains eight bits that identify which transactions do not require a PIN. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	PIN Disabled For Debit	HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.DEBIT
1	PIN Disabled For Return	HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.RFND
2	PIN Disabled For Void	HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.VOID
3	PIN Disabled For Adjust	HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.ADJ

Bit	Bit Name	BASE24 Mapping
4	PIN Disabled For Cashback	HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.CASHBACK
5	PIN Disabled For Balance Inquiry	HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.BAL-INQ
6	PIN Disabled For Preauthorization	HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.PREAUTH
7	Unused	Reserved for future use

Additional Prompts Select Bits

The Additional Prompts Select field contains eight bits that identify which prompts from the additional prompts table will appear. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Additional Prompt 1	HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT1
1	Additional Prompt 2	HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT2
2	Additional Prompt 3	HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT3
3	Additional Prompt 4	HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT4
4	Additional Prompt 5	HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT5
5	Additional Prompt 6	HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT6
6	Additional Prompt 7	HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT7
7	Additional Prompt 8	HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT8

Options 5 Bits

The Options 5 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Do Not Capture Debit Transactions	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-5.DO-NOT-CAPTR-DB-TXN
1	EBT Processing	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-5.EBT-PROCESS
2	Generation Number Prompt	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-5.GEN-NUM-PROMPT
3	Additional Refund Amount Prompts	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-5.ADNL-RFND-PROMPT
4	Purchase Order Number Prompt	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-5.PO-NUM-PROMPT
5	Bar Amount Prompt	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-5.BAR-AMT-PROMPT
6	Unused	Reserved for future use
7	Disable Printing RRN	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-5.DISABLE-PRINT-RRN

Options 6 Bits

The Options 6 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	AVS Enabled on Sale	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-6.AVS-ON-SALE
1	Cardholder Address Required	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-6.CRDHLDR-ADDR-REQ

Bit	Bit Name	BASE24 Mapping
2	Cardholder ZIP Code Required	HPTC.ISS-TABLE.ISS-TBL-DATA. OPTIONS-6.CRDHLDR-ZIP-CDE-REQ
3–7	Unused	Reserved for future use

Acquirer Table

The acquirer table is used to define parameters associated with an acquirer or host.

In addition to the Table ID and Table Length fields, the following fields must be downloaded in the Private Use (P-60) data element, subfield 04 of the 0810 download response.

Field Name	Bytes	BASE24 Mapping
Acquirer Number	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. ACQ-TBL-INDEX
Acquirer ID	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. ACQ-TBL-ID
Acquirer Program Name	10	HPTC.ACQ-TABLE.ACQ-TBL-DATA. ACQ-PROG-NAM
Acquirer Name	10	HPTC.ACQ-TABLE.ACQ-TBL-DATA. ACQ-NAM
Primary Transaction Telephone Number	12	HPTC.ACQ-TABLE.ACQ-TBL-DATA. PRI-TRANS-PHONE-NUM
Primary Transaction Connection Time	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. PRI-TRANS-CNCT-TIM
Primary Transaction Dial Attempts	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. PRI-TRANS-DIAL-ATTEMPTS
Secondary Transaction Telephone Number	12	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SECNDRY-TRANS-PHONE-NUM
Secondary Transaction Connection Time	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SECNDRY-TRANS-CNCT-TIM

Field Name	Bytes	BASE24 Mapping
Secondary Transaction Dial Attempts	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SCNDRY-TRANS-DIAL-ATTEMPTS
Primary Settlement Telephone Number	12	HPTC.ACQ-TABLE.ACQ-TBL-DATA. PRI-SETL-PHONE-NUM
Primary Settlement Connection Time	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. PRI-SETL-CNCT-TIM
Primary Settlement Dial Attempts	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. PRI-SETL-DIAL-ATTEMPTS
Secondary Settlement Telephone Number	12	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SECNDRY-SETL-PHONE-NUM
Secondary Settlement Connection Time	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SECNDRY-SETL-CNCT-TIM
Secondary Settlement Dial Attempts	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SECNDRY-SETL-DIAL-ATTEMPTS
Modem Mode (Transactions)	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. MODEM-MDE
Options 1 (see “Options 1 Bits,” below)	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-1
Options 2 (see “Options 2 Bits,” below)	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-2
Options 3 (see “Options 3 Bits,” below)	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-3
Options 4 (not used)	1	Reserved for future use
Network International ID	2	HPTC.ACQ-TABLE.ACQ-TBL-DATA. NET-INTL-ID
Card Acceptor Terminal ID	8	HPTC.ACQ-TABLE.ACQ-TBL-DATA. CRD-ACCPT-TERM-ID
Card Acceptor Identification Code	15	HPTC.ACQ-TABLE.ACQ-TBL-DATA. CRD-ACCPT-ID-CDE

Field Name	Bytes	BASE24 Mapping
Time Out	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. TIM-OUT
Current Batch Number	3	PTDD1.PTDD1-CORE.BATCH-NUM
Next Batch Number	3	PTDD1.PTDD1-CORE.BATCH-NUM plus 1
Settle Time	4	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SETL-TIM
Settle Day	2	HPTC.ACQ-TABLE.ACQ-TBL-DATA. PREV-SETL-DAY
Encrypted PIN Key	8	PTDD1.PTDD1-CORE.ENCN-PIN. ENCN-KEYS.M-KEY
Master PIN Key Index	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. PIN-KEY-INDEX
Encrypted MAC Key	8	Zero filled
Encrypted MAC Key Index	1	Zero filled
Visa I Terminal ID	23	HPTC.ACQ-TABLE.ACQ-TBL-DATA. VISA-I-TERM-ID
Modem Mode (Settlement)	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SETL-MODEM-MDE
Maximum Auto- settlement Attempts	1	HPTC.ACQ-TABLE.ACQ-TBL-DATA. MAX-AUTO-SETL
Settlement Logon ID	3	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SETL-LOGON-ID
Settlement Protocol Name	10	HPTC.ACQ-TABLE.ACQ-TBL-DATA. SETL-PROTO-NAM
Acquiring Institution ID Code	6	HPTC.ACQ-TABLE.ACQ-TBL-DATA. ACQ-INST-ID-CDE

Options 1 Bits

The Options 1 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0–2	Reserved by Hypercom	Reserved for future use
3	Ignore Host Time	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-1.IGNORE-HOST-TIM
4–7	Reserved by Hypercom	Reserved for future use

Options 2 Bits

The Options 2 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0–1	Reserved by Hypercom	Reserved for future use
2	Print Visa II Merchant Number Instead of Acquirer ID	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-2.PRNT-VISA-MRCH-NUM
3	Send Current Batch Number in 01x0 and 02x0 Messages	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-2.CUR-BATCH-NUM
4	MAC Key Exchange Method	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-2.MAC-KEY-EXCHNG
5	PIN Key Exchange Method	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-2.PIN-KEY-EXCHNG
6	Disable Statistics Following Settlement	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-2.DISABLE-STATS
7	Disable NBD Header	HPTC.ACQ-TABLE.ACQ-TBL-DATA. OPTIONS-2.DISABLE-NBD-HDR

Options 3 Bits

The Options 3 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Disable 0220 Messages	HPTC.ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-3.DISABLE-220-MSG
1	ACH Totals Report on Settlement	HPTC.ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-3.ACH-TOTS-RPT
2	Non-Settlement Acquirer	HPTC.ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-3.NON-SETL-ACQ
3	HDC Mode	HPTC.ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-3.HDC-MODE
4–7	Unused	Reserved for future use

Descriptor Table

The descriptor table is used to define the parameters of a product description code.

In addition to the Table ID and Table Length fields, the following fields must be downloaded in the Private Use (P-60) data element, subfield 05 of the 0810 download response. The set of fields shown below can occur 8 times.

Field Name	Bytes	BASE24 Mapping
Descriptor Number	1	HPTC.PROD-TABLE.PROD-DATA.PROD-INDEX
Selection Key	1	HPTC.PROD-TABLE.PROD-DATA.SELCT-KEY
Transmission Code	2	HPTC.PROD-TABLE.PROD-DATA.CDE
Display/Print Text	20	HPTC.PROD-TABLE.PROD-DATA.DESCRPTION

Additional Data Prompt Table

The additional data prompt table is used to provide the ability for adding customer-defined data prompts for transaction processing.

In addition to the Table ID and Table Length fields, the following fields must be downloaded in the Private Use (P-60) data element, subfield 09 of the 0810 download response. The set of fields shown below can occur 8 times.

Field Name	Bytes	BASE24 Mapping
Prompt Number	1	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-NUM
Prompt ID	2	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-ID
Prompt Text	20	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-TEXT
Transaction Select 1 (see “Transaction Select 1 Bits,” below)	1	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1
Transaction Select 2 (see “Transaction Select 2 Bits,” below)	1	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-2
Transaction Select 3 (not used)	1	Reserved for future use
Transaction Select 4 (not used)	1	Reserved for future use
Input Length—Lower	1	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.INPUT-LGTH-MIN
Input Length—Higher	1	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.INPUT-LGTH-MAX
Data Type	1	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.DATA-TYPE
Reserved for future use	5	Reserved for future use

Transaction Select 1 Bits

The Transaction Select 1 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0	Cash	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.CASH
1	Card Verify	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.CRD-VRFY
2	Preauthorization	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.PRE-AUTH
3	Check	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.CHK
4	Debit	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.DB
5	Refund	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.RFND
6	Offline Sale	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.OFFL-SALE
7	Authorization or Sale	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.AUTH-SALE

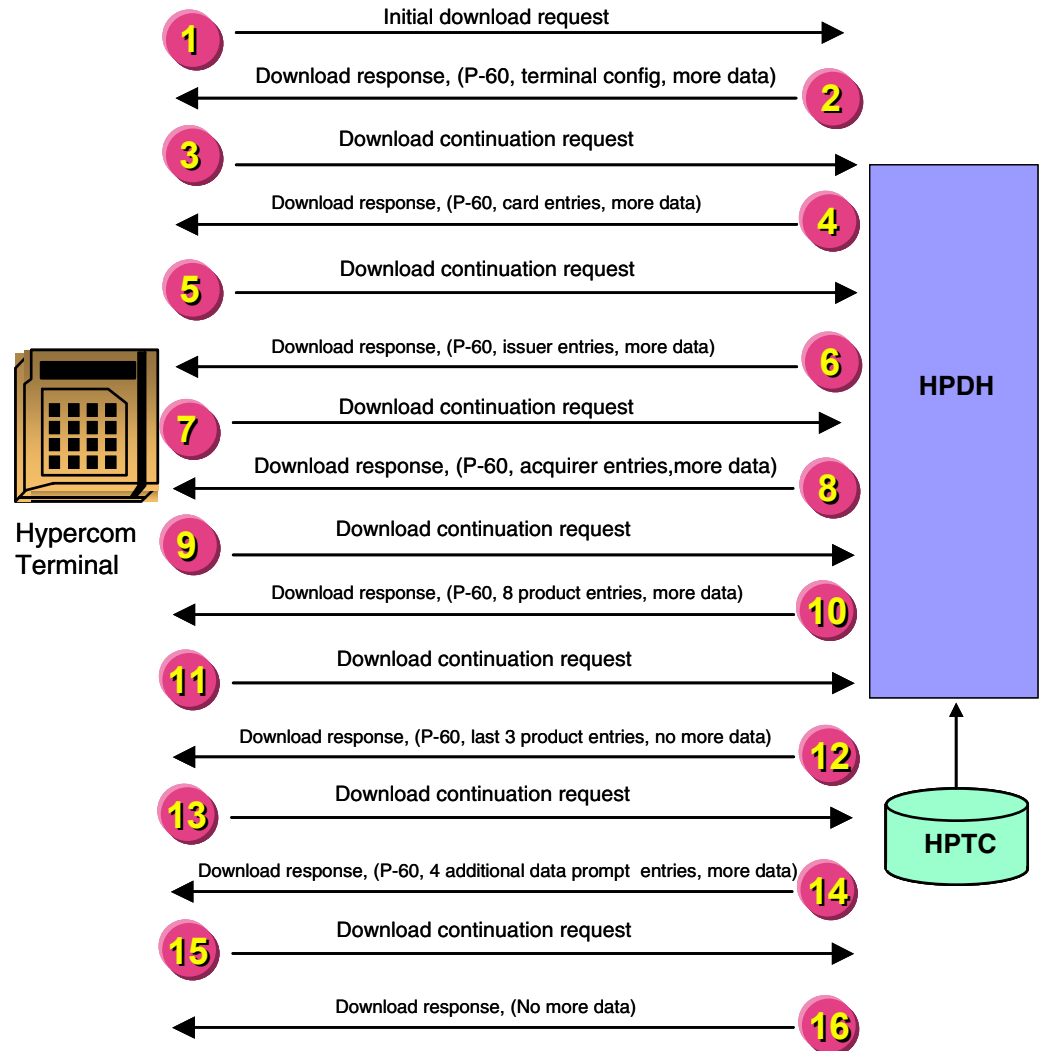
Transaction Select 2 Bits

The Transaction Select 2 field contains eight bits that identify processing options. The following table shows the description and BASE24 mapping for each bit. Bit 7 is the most significant bit.

Bit	Bit Name	BASE24 Mapping
0–6	Not used	Reserved for future use
7	Settlement	HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-2.SETL

Hypercom T7 Download Example

The following example illustrates the steps taken during a successful download to the Hypercom T7 terminal. For simplicity, message interactions with the NAC, XPNET process, and Router module are not shown.



1. The terminal formats and sends a download request to the HPDH module. The last digit of the processing code (930000) passed in the request message is 0, indicating that this is an initial request.
2. The HPDH module follows these steps to send a response to the terminal:
 - a. Resets the load index counter in device-dependent data to 0 to indicate the initial request. The counter is used to track the multiple download messages that follow.

- b. Reads the 01 HPTC record for the correct terminal group and builds a response message containing terminal configuration data.
 - c. Increases the load index by one and sets the last digit of the processing code (930001) to 1, indicating that it is expecting a download continuation message.
 - d. Sends the response to the terminal.
3. The terminal formats and sends a download continuation request message with a processing code ending in 1.
4. The HPDH module recognizes the download continuation request, checks the load index in device-dependent data, and determines that the first card range table should be read and formatted. The two card range table records contain 16 card range entries each. The HPDH module reads six entries at one time, builds a response message, and sends the message to the terminal. The processing code still ends in 1.

Note: Steps 3 and 4 are repeated until the HPDH module encounters the first entry which has zero-filled high and low prefix fields or all 32 entries have been downloaded. If a zero-filled entry occurs in the first card range table (indicating that the table is not full), the card range table processing stops, and the second card range table will be skipped in the initialization processing. For this reason, the first card range table must be filled before the second card range table can contain usable entries.

5. The terminal formats and sends a download continuation request message with a processing code ending in 1 (continuation).
6. The HPDH module recognizes the download continuation request, checks the load index in device-dependent data, and determines that the issuer table should be read and formatted. The issuer table contains up to 16 entries, which are formatted three at a time into a response message and sent to the terminal. The processing code still ends in 1.

Note: Steps 5 and 6 are repeated until the HPDH module encounters an entry with 00 in the issuer table ID field or all 16 entries in the issuer table have been downloaded.

7. The terminal formats and sends a download continuation request message with a processing code that ends in 1.
8. The HPDH module recognizes the download continuation request, checks the load index in device-dependent data, and determines that the acquirer table should be read and formatted. The acquirer table contains up to eight entries, which are formatted one at a time into a response message and sent to the terminal. The processing code still ends in 1.

Note: Steps 7 and 8 are repeated until the HPDH module encounters an entry with 00 in the acquirer table ID field or all eight entries in the acquirer table have been downloaded.

9. The terminal formats a download continuation request message with a processing code that ends in 1.
10. The HPDH module recognizes the download continuation request, checks the load index in device-dependent data, and determines that the product description table should be formatted. The product description table contains 11 entries but only 8 at one time can be formatted into the response message. The HPDH module formats the first 8 entries (even if they are blank filled) and sends a response to the device. The processing code still ends in 1.
11. The terminal formats and sends a download continuation request message that ends in 1.
12. The HPDH module recognizes the download continuation request, checks the load index in device-dependent data, and determines that the remaining entries of the product description table should be formatted. The HPDH module formats the last three entries (even if they are blank filled) and sends a response to the terminal. The processing code still ends in 1.
13. The terminal formats and sends a download continuation request message with a processing code that ends in 1.
14. The HPDH module recognizes the download continuation request, checks the load index in device-dependent data, and determines that the additional data prompt table should be read and formatted. The HPDH module reads the additional data prompt table in groups of 4 entries, formats the entries in a response entry, and sends it to the terminal. The processing code still ends in 1.

Note: Steps 13 and 14 are repeated if the additional data prompt table contains more than four entries. The additional data prompt table can contain a maximum of eight entries.

15. The terminal formats and sends a download continuation request message with a processing code that ends in 1.
16. The HPDH module recognizes the download continuation request and then checks the load index in device-dependent data and determines that no HPTC records remain to be downloaded. The HPDH module sets the last digit of the processing code to 0, indicating that this is the last message and sends a response to the terminal.

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6: HPDH Configuration Considerations

This section contains information concerning additional decisions the customer must make when configuring the Hypercom Device Handler (HPDH) module. These configuration decisions include the following:

- PIN encryption
- Derived Unique Key per Transaction (DUKPT)
- Sequence number checking
- Message authentication codes (MACs)
- Draft capture
- Transaction accumulation totals
- Handshaking

This section also includes an example configuration for a Hypercom Device Handler module.

PIN Encryption

The HPDH module supports the ANSI 9.8 PIN block that can be used with Hypercom terminals. The HPDH module supports PIN encryption with the DES algorithm performed in hardware or software. PIN encryption performed in hardware or software uses the ANSI PIN block, 12 right-most digits of the PAN excluding the check digit encryption method. The PIN encryption method is set in the PIN ENCRYPTION TYPE field on screen 7 of the BASE24-pos Terminal Data files (PTD). Whether a terminal can support PIN entry is indicated in the VALIDATE PIN field on screen 7 of the PTD. For more information on the PIN fields contained in the PTD, refer to the ***BASE24-pos Files Maintenance Manual***.

PIN encryption keys can be sent to Hypercom terminals in downloads or in the response to a Hypercom Logon request transaction. In these two cases, PIN encryption keys are sent encrypted under the terminal master key set up in the PIN MASTER KEY field on screen 7 of the PTD.

The master key is used to encode and decode PIN keys that are transmitted between Hypercom terminals and the host. The master key injected into the terminal must be the same as the master key in the PTD. In the case of Hypercom T7 terminals, the PIN master key must be adjusted for odd parity. Odd-parity adjustment on the master key means that each byte in the master key contains an odd number of ones in its binary representation. If a memory error occurs within the terminal, there is a 50 percent chance that the number of bits in any given byte will change to an even number. Using odd-parity adjustment makes it possible for the terminal to detect changes in the contents of the master key. Refer to appendix B for a description of the algorithm used for odd-parity adjustment.

In general, if a PIN is included in a request, it is verified by the HPDH module. The HPDH module does not enforce the presence of a PIN in any request. To require that a PIN be included in a particular request, enforcement must be at the terminal level or Router/Authorization module level using the Card Prefix File (CPF) configuration. For more information concerning the CPF, refer to the CPF documentation contained in the ***BASE24 Base Files Maintenance Manual***. For an in-depth discussion of PINs, refer to the ***BASE24 Transaction Security Manual***.

Derived Unique Key per Transaction (DUKPT)

POS devices attached to the HPDH module can use derived unique key per transaction (DUKPT) key management for PIN communications (session) keys. The HPDH module supports DUKPT management of single-length PIN communications keys. The Transaction Security Services process is optional for single-length PIN communications keys, and is required for longer PIN communications keys.

When using DUKPT key management, a unique base derivation key must be manually entered or injected into each POS device. The same base derivation key must be loaded into a database maintained by the host. When a POS device derives a unique key for a transaction, it must include the Key Serial Number (KSN) in the request message sent to the host.

The Derivation Key File (KEYD) is used for Hypercom device terminals that use DUKPT key management. The KEYD can contain derivation keys or key locators. Derivation keys are used by the HPDH module to translate incoming DUKPT-encrypted PIN blocks received from the terminal into a single-length Master/Session key PIN block. Key locators are nonencrypted values used by the Transaction Security Services process to locate hardware-encrypted key information maintained in the Transaction Security Services database.

Values in the PIN ENCRYPTION TYPE field on PTD screen 7 control the use of DUKPT for PIN communications keys. Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about the PTD.

For more information on DUKPT key management, refer to the ***BASE24 Transaction Security Services Processing Guide***

Sequence Number Checking

Sequence number checking is intended to ensure that BASE24-pos has received and processed every transaction only once. The Systems Trace Number (P-11) data element in the incoming ISO message is used to detect duplicate sequence numbers. The Systems Trace Number (P-11) data element is also used to match 0400 reversal messages with their corresponding 0210 response messages. Sequence numbers that are internally generated are used to match a failed message (that is, a message that the XPNET process was unable to deliver to its intended destination) with the corresponding 0210 reversal data.

In any case, where an unanticipated response is received by a Hypercom terminal, the terminal generates a reversal for the transaction in question, and continues to send the reversal until it is acknowledged by the HPDH module. The HPDH module sends the reversal to BASE24 for further processing, if possible. Otherwise the reversal is dropped.

Message Authentication Codes (MACs)

The HPDH module supports the Hypercom proprietary algorithm for software verification of message authentication codes (MACs). The Hypercom proprietary algorithm uses a form of the DES encryption algorithm to generate the MAC. Separate MAC master keys and MAC working keys are maintained by the HPDH module. The MAC working key, encrypted under the MAC master key, can be sent to terminals in downloads or in response to a Hypercom Logon request.

The Hypercom proprietary algorithm is described in the *Hypercom ECB Message Authentication Code Generation* document available from the vendor. MACs allow any data sent between the terminal and the HPDH module to be verified. The combination of the MAC and the Hypercom proprietary algorithm is designed to protect financial transaction messages against accidental or deliberate alteration. In addition, it protects against the fraudulent insertion of messages.

MACs are supported for Hypercom T4, T5, T6 and T7 terminals. For these terminals, the customer determines whether to support MACs and, if so, how they are implemented.

The HPDH module supports software MAC generation and verification only. If MACs are implemented, the MAC TYPE field on screen 7 of the PTD must be set to 01, indicating software MAC support. If the HPDH module is configured to validate MACs, a MAC is included in every request from a terminal and the MAC is validated. If the HPDH module is not configured to validate MACs, the MAC is not validated, even if it is present in the message. MACs are not supported for initialization and logon requests. If MACs are supported, the entire message is encoded.

In addition to setting the MAC TYPE field in the PTD, the MAC REQUIRED field located in the card definition table on screen 3 of the Hypercom Configuration File (HPCF) must be set up. If the PTD is set up for MAC software support (a value of 01 in the MAC TYPE field on screen 7), the MAC REQUIRED field in the HPCF must contain a value of 1, indicating that software MACs are supported for the card prefix range defined on the HPCF screen. If MACs are not supported, the MAC REQUIRED field in the HPCF must contain a value of 0, indicating that software MACs are not supported for the card prefix range defined on the HPCF screen. Up to 16 card definition table screens are contained in the HPCF record.

Hypercom terminals may be configured for software message authentication and hardware (security device) PIN encryption using Transaction Security Services (TSS). This combination of software message authentication and hardware PIN encryption is permitted only when the TERMINAL TYPE field on screen 1 of the PTD is set to the value D0, indicating a Hypercom TDS terminal.

In order to provide for this combination, the MAC TYPE field on screen 7 of the PTD must be set to the value 01, indicating software message authentication. Also, the PIN ENCRYPTION TYPE field on screen 7 of the PTD must be set to the value 03, which indicates security device and PIN/PAD PIN block, or 04, which indicates security device and ANSI standard PIN/PAN PIN block. In addition, the SECURE-DEV-TYP param in the Logical Network Configuration File (LCONF) must be set to a value of TSS. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for more information.

Setting Up MAC Use

Whenever MACs are generated at a terminal supported by the HPDH module, the Message Authentication Code (P-64) data element must be included in both the request and the response. In addition, the MAC Terminal Master Key must be manually entered or injected into the terminal. The same key also must be entered in the MAC MASTER KEY field located on screen 7 of the PTD. The other key used in generating MACs, the MAC Communications Key, is randomly generated by the HPDH module. Although this key is written to the PTD record by the HPDH module, the value is not available through BASE24 files maintenance. The MAC Communications Key generated by the HPDH module is encrypted under the MAC Terminal Master Key and downloaded to the terminal.

Generating and Verifying a MAC

When sending a request, a Hypercom terminal can calculate a MAC and place it in the Message Authentication Code (P-64) data element of the request. When the HPDH module receives a request other than an initialization or logon message, it performs message authentication based on the value in the MAC TYPE field on PTD screen 7.

Hypercom terminals support only software-generated MACs; therefore, the only valid values in the MAC TYPE field on PTD screen 7 are 00 (no message authentication) and 01 (software message authentication). If the MAC TYPE field on PTD screen 7 is set to a value other than 00 or 01, the HPDH module logs an error message and declines the request with Hypercom response code 02 (unable to process).

If the MAC TYPE field on PTD screen 7 is set to a value of 00, message authentication is not performed.

If the MAC TYPE field on PTD screen 7 is set to a value of 01, the HPDH module checks to see if the Message Authentication Code (P-64) data element has been included in the request.

- If the Message Authentication Code (P-64) data element is not present, the HPDH module logs an error message and declines the request with Hypercom response code 30 (format error).
- If the Message Authentication Code (P-64) data element is present, the HPDH module verifies the MAC generated by the terminal by generating another MAC based on the message received from the terminal and comparing the two MACs. If the two MACs do not match, the HPDH module declines the request with Hypercom response code 02 (unable to process).

Failed MAC Procedure

The HPDH module generates a new MAC communications key if it has been unable to verify the MAC after a configured number of successive requests from a single terminal. The POS-DH-CONS-MAC-ERR-LMT param in the Logical Network Configuration File (LCONF) is used to configure the number of consecutive MAC errors allowed from a single terminal before the HPDH module generates a new MAC key. The HPDH module maintains an invalid MAC counter in the device dependent data of the BASE24-pos Terminal Data Dynamic File—general data (PTDD1) for this purpose. Whenever a message is successfully verified using MACs, this counter is reset to zero.

If the number of requests from a single terminal is greater than the number of requests allowed (as configured in the POS-DH-CONS-MAC-ERR-LMT param in the LCONF), the HPDH module assumes that the HPDH module MAC keys are not the same as the MAC key for the terminal. In this case, the HPDH module sets a flag in the subsequent response to the terminal indicating that a download is required. The HPDH module then generates a new MAC key.

When a terminal is unable to verify a MAC received from the HPDH module in a transaction response with monetary impact, the terminal must generate a reversal. The reversal is not encoded using MACs. The HPDH module sends a 0400 reversal message to the authorizer of the transaction. The MAC reversal must be the next request from the terminal. If any other requests are processed before the reversal, the reversal is dropped and an error message is logged.

Draft Capture

BASE24-pos allows for processing retailer transactions as draft capture or non-draft capture. Non-draft capture transactions are transactions for which the paper receipt represents the actual settlement instrument. Paper receipts must be physically presented for deposit in order for the transaction to be settled. While the transaction could have been authorized by BASE24-pos, the BASE24-pos record cannot be used to settle the item.

Draft capture transactions are electronically captured at the time of the transaction and are sufficient in themselves to enact a funds transfer for the transaction amount. BASE24-pos identifies draft capture transactions in its files and includes draft capture totals on its retailer reports so that the retailer sponsor can settle with the retailer.

Draft capture is a settlement tool only. It has no impact on the authorization of transactions. Cardholders are still checked against the same authorization criteria without regard for whether a transaction is draft capture or non-draft capture. Thus, the retailer is not incurring additional risk of cardholder fraud by using or not using draft capture.

Draft Capture Options

Customers using the HPDH module have two draft capture options available to them. These options are authorization only with paper follow-up, and authorization and draft capture.

Different draft capture options can be selected for each terminal and card type. Customers set the TP field on PTD screens 8, 9, and 10 to indicate the draft capture option they have selected for each terminal and card type. Valid values are as follows:

- 0 = Authorization only with paper follow-up
- 1 = Authorization and draft capture

Although the value of 3 (terminal determines data capture mode for each transaction) is generally acceptable in this field, the HPDH module does not support the value.

Authorization Only with Paper Follow-up

The authorization only with paper follow-up draft capture option obtains online authorization for the amount of the customer purchase or other type of transaction, depending on the data. In an authorization only environment, the card issuer waits for paper follow-up before posting the transaction and settling with the retailer. If a card type is set up to use this option, none of the transactions for that card type from the specified terminal are draft capture.

With this option, BASE24 updates the cardholder usage and activity totals, logs transactions to the POS Transaction Log File (PTLF), and updates the service totals and authorization totals in the PTDD1.

Authorization and Draft Capture

The authorization and draft capture option is a one-step process that enables settlement to occur without paper follow-up. This option performs an authorization on the transaction and at the same time automatically captures the draft electronically. Using electronic draft capture, no paper follow-up is required by the card issuer to post the transaction and settle with the retailer. The procedure is completed in one transaction.

With this option, BASE24-pos updates the cardholder usage and activity totals, logs the transaction to the PTLF, and updates service totals, authorization totals, and draft capture totals in the PTDD1.

Any card configured on Card Definition Table screen 3 of the HPCF, as a debit card (a value of 2 in the PROCESSING TYPE field) must contain a value of 1 in the TP field on PTD screens 8, 9, and 10 for the terminal.

Setting the Draft Capture Flag in the PSTM

The draft capture flag indicates whether BASE24-pos balances should immediately reflect transaction activity. The Router module sets this draft capture flag using the TRAN-PROFILE field in the POS Standard Internal Message (PSTM).

The HPDH module uses the TP field on PTD screens 8, 9, and 10 to set the TRAN-PROFILE field in the PSTM. The Router module then moves the value in the TRAN-PROFILE field of the PSTM to the DFT-CAPTURE-FLG field in the PSTM. Valid values for the DFT-CAPTURE-FLG field are as follows:

- 0 = Authorization only with paper follow-up
- 1 = Authorization and draft capture

The transaction profile values and the corresponding draft capture flag settings are shown in the following table.

TRAN- PROFILE	DFT-CAPTURE- FLG	Transaction Type
0	0	All transactions
1	0	Preauthorization purchase, card verification, balance inquiry, and 0100 transactions
	1	Normal purchase, mail or telephone order, refund, cash advance, purchase with cash back, adjustment, and preauthorization purchase completion transactions

For request or force-post transactions (PSTM message types 0200 and 0220, respectively) from a Hypercom terminal, the TRAN-PROFILE field in the PSTM received from the terminal can be 0 or 1. For a reversal transaction (PSTM message type 0420) from a Hypercom terminal, the transaction profile must be the same as that of the original transaction.

Transaction Accumulation Totals

The BASE24-pos HPDH module supports balancing terminals by batch. The terminal forwards its totals to the HPDH module with the corresponding close request. If the terminal totals disagree with the HPDH module totals in the PTDD1, they are logged to the PTLF. If draft capture totals are out of balance, the terminal uploads the transactions. The HPDH module accumulates transactions during the upload, then logs the totals to the PTLF when the upload is complete.

PTDD1 Totals

The first step the HPDH module performs in determining if the batch is in balance is to calculate the batch totals in the PTDD1:

PTDD1 Net Total Business = Batch Debits – Batch Credits – Batch Adjustments

PTDD1 Net Draft Capture Business = Batch Draft Capture Debits – Batch Draft Capture Credits – Batch Draft Capture Adjustments

The HPDH module then logs the PTDD1 totals to the PTLF.

Terminal Totals

The second step is to calculate the terminal totals:

Terminal Net Total Business = (Draft Capture Card Debits + Debit Card Debits + Authorization Only Card Debits) – (Draft Capture Card Credits + Debit Card Credits + Authorization Only Card Credits)

Terminal Net Draft Capture Business = (Draft Capture Card Debits + Debit Card Debits) – (Draft Capture Card Credits + Debit Card Credits)

Note: If the PTDD1 net draft capture business is greater than or less than the terminal net draft capture business, then all draft capture and debit card transactions are uploaded in order to determine where the out-of-balance situation lies.

The terminal totals are mapped to the PTLF in the following manner:

Terminal Totals	PTLF Totals
Terminal Total Purchases	PTLF Batch Debits
Terminal Total Credits	PTLF Batch Credits
0	PTLF Batch Adjustments
Terminal Draft Capture Card Credits + Terminal Debit Card Credits	PTLF Batch Draft Capture Credits
Terminal Draft Capture Card Debits + Terminal Debit Card Debits	PTLF Batch Draft Capture Debits
0	PTLF Batch Draft Capture Adjustments

In addition, the terminal totals sent by the terminal are moved into the PTLF user data.

Maintaining Totals

Hypercom terminals maintain totals for draft capture credits and debits, debit account credit and debits, and authorization only credits and debits. BASE24 maintains totals for draft capture and authorization only debits, credits, and adjustments; and draft capture debits, credits, and adjustments. Although the Hypercom terminal maintains totals in a different manner than BASE24, the HPDH module is able to determine which PTDD1 totals to update when a transaction comes into the HPDH module from the Hypercom terminal. The HPDH module updates the PTDD1 totals based on transaction information included in the message it receives.

The HPDH module sends the following message responses to the terminal to indicate if terminal totals agree with HPDH module totals. Two lines of information are displayed on the terminal.

- If the terminal batch and draft capture totals agree with the HPDH module batch and draft capture totals, the HPDH module sends a “BATCH BALANCED” response to the terminal. The second line on the terminal contains a “BATCH COMPLETE” response.

- If the batch totals do not match, but the draft capture totals do match, the HPDH module sends an “OUT OF BALANCE” response to the terminal. The second line on the terminal indicates the amount the terminal is out of balance by displaying an “ERR (\$\$\$\$\$\$. \$)” message, where \$ is the monetary amount in whole and fractional currency units. An amount surrounded by parentheses indicates the terminal totals are less than the BASE24 totals. An amount without parentheses indicates the terminal totals are greater than the PTDD1 totals.
- If the batch totals match, but the draft capture totals do not match, the HPDH module sends an “UPLOAD STARTED” response to the terminal. The HPDH module accumulates totals during the upload, then logs the totals to the PTLF when the upload is complete. The second line on the terminal displays a “BATCH TOTAL OK” message.

The HPDH module always accumulates batch, shift, and day totals. Upon a batch close for the corresponding period, the HPDH module records batch totals in the PTLF and resets the batch totals. All totals (that is, batch, shift, and day) are cleared if a close transaction is entered any time within the retailer’s cutover window set in the POS Retailer Definition File (PRDF).

The BALANCE SUPPORT field on PTD screen 3 indicates to the HPDH module which totals are supported by the terminal by means of close transactions. This field must be set to 0 or 9 for Hypercom. A 0 in this field indicates that only batch totals are supported and a 9 indicates that no totals are supported by the terminal. Totals specified in the PTDD1 are allowed to accumulate until reset when a batch close transaction is entered. Other totals are reset at the logical network cutover.

The HPDH module must not clear PTDD1 batch totals prior to sending a response for a balancing request to the terminal. The HPDH sets a flag in the PTDD1 to indicate a batch close transaction was processed by the terminal and the HPDH module is waiting for confirmation that the terminal has received the response sent by the HPDH module. The WAIT-CONFIRM-FLG in the PTDD1 can be set to the following values for the HPDH module:

- 0 = Not waiting for confirmation
- 1 = Waiting for confirmation, batch close performed

When a batch close transaction is processed, the HPDH module sets the WAIT-CONFIRM-FLG to 1. This indicates the HPDH module does not clear batch totals in the PTDD1. When the HPDH module receives the next transaction that it determines not to be a duplicate administrative transaction, the batch totals are cleared and the WAIT-CONFIRM-FLG is reset to 0. If the HPDH module determines the next transaction to be a duplicate administrative transaction (by

comparing this transaction to the previous transaction) the batch totals are not cleared. This scenario continues until the HPDH module determines a transaction not to be a duplicate.

Handshaking

Handshaking is an 0800 test message sent to the terminal by the HPDH module. The terminal responds by sending the appropriate response message. Handshaking allows the HPDH module to ensure that the lines of communication with the terminal are open.

The handshake request contains the following data elements:

- Message Type ID
- Bit Map
- Processing Code (P-3)
- Transaction Time (P-12)
- Transaction Date (P-13)
- Network International ID (P-24)
- Card Acceptor Terminal ID (P-41)
- Card Acceptor ID (P-42)

Passthrough Configuration Example

The Hypercom Device Handler (HPDH) module allows customers the flexibility of configuring their systems in various ways in order to suit their individual needs. Although some hardware limitations do exist, customers using BASE24 as a passthrough to send and receive messages define their own configuration parameters using the BASE24 files described earlier in this section. The following pages contain an example that describes one way in which a system might be configured using the HPDH module and BASE24 as a passthrough for network management system (NMS) processing.

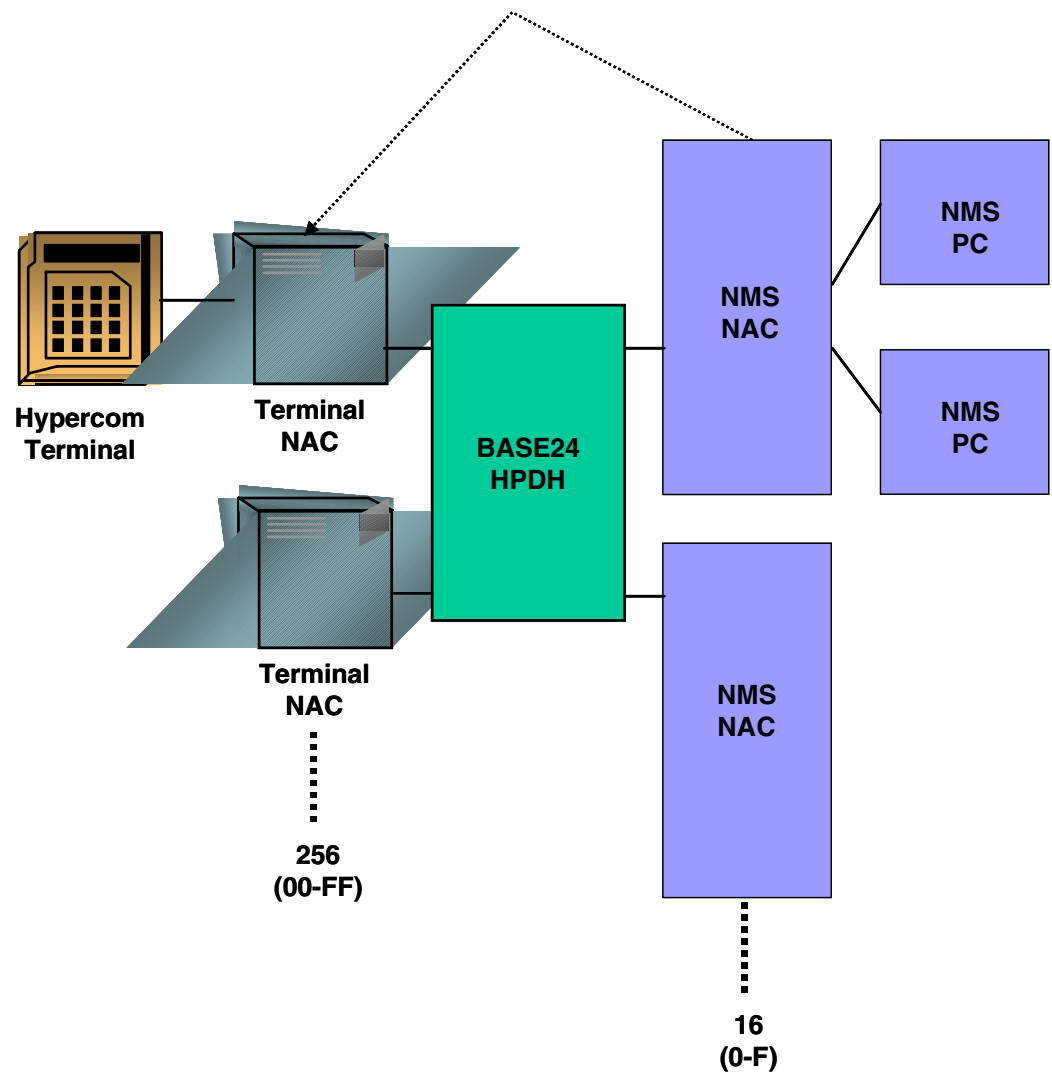
HPDH Module Configuration

The configuration of a system using an HPDH module can be split into two major portions. Although these major portions are interrelated, they are handled separately in this subsection to enhance understanding.

The first major portion of the HPDH module configuration is the terminal side. The terminal side processes routine messages originating at or being sent to a terminal network access controller (NAC). This portion is illustrated on the left-hand side of the diagram shown below.

The second major portion of the HPDH module configuration is the network management system (NMS) side. The NMS side processes diagnostic messages originating at or being sent to an NMS NAC. This portion is illustrated on the right-hand side of the following diagram.

The diagram illustrates an example of a complete configuration, containing both the terminal and NMS portions, for a system using the HPDH module. The dashed line in the diagram illustrates a configuration where BASE24 is not used and the terminal NAC and NMS NAC are directly connected. The remainder of this subsection describes both the terminal side and the NMS side.



Terminal Configuration

Terminals are connected to the HPDH module using terminal NACs. The terminal NAC is an interface between the terminal and the HPDH module. Messages being sent from the terminal to the HPDH module or to the terminal from the HPDH module must pass through the terminal NAC before they reach their destination. BASE24 allows for support of up to 256 terminal NACs. The number of terminals that can be attached to a single terminal NAC is dependent on the hardware limitations of that terminal NAC.

Each terminal NAC is identified in a POS-HPDH-NAC-ID*identifier* param located in the LCONF, where *identifier* is a two-character hexadecimal value in the range of 00 through FF. These LCONF params contain the terminal NAC IDs and their corresponding station names. Upon initialization, the HPDH module uses these LCONF params to build an internal table containing all of the terminal NAC IDs and their associated station names. This LCONF param needs to be set up only if network management system (NMS) processing is being used.

Network Management System (NMS) Configuration

Network management system (NMS) PCs are connected to the HPDH module using NMS NACs. The NMS NAC is an interface between the NMS PC and the HPDH module. Network management messages sent from the NMS PC to the HPDH module or to the terminal from the HPDH module must pass through the NMS NAC. BASE24 allows for support of up to 16 NMS NACs. The number of NMS PCs that can be attached to a single NMS NAC is dependent on the hardware limitations of that NMS NAC.

Each NMS NAC is identified in a POS-HPDH-NMS-ID*identifier* param located in the LCONF, where *identifier* is a hexadecimal value in the range of 0 through F. These LCONF params contain the NMS NAC IDs and their corresponding station names. Upon initialization, the HPDH module uses these LCONF params to build an internal table containing all of the NMS NAC IDs and their associated station names. This LCONF param needs to be set up only if NMS processing is being used.

7: Transaction Message Flows

When a Hypercom terminal sends a message to BASE24, it is received by the XPNET process, which forwards the message to the Hypercom Device Handler (HPDH) module.

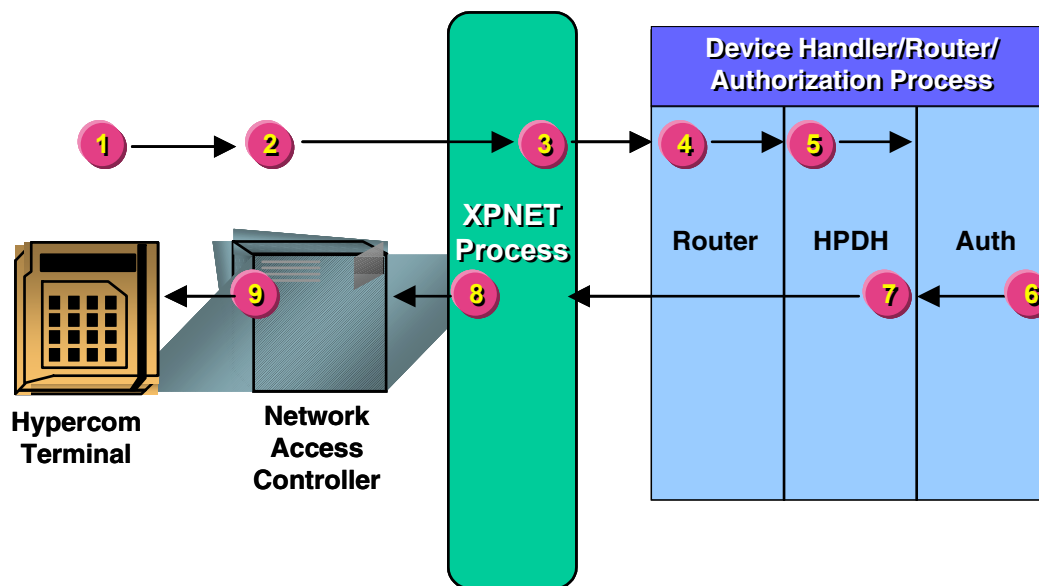
The HPDH module is bound into a BASE24-pos Router/Authorization process. Any messages coming into the bound process are first received by the Router module, which then routes the message to the Device Handler module.

This section contains examples of the message flows listed below. Each example includes a diagram illustrating the message flow, followed by a description of each step:

- Normal transaction with response
- Normal transaction without response—terminal times out
- Normal transaction—unable to deliver response to terminal
- Normal transaction—NAC unable to deliver response to terminal
- NMS passthrough completes successfully
- NMS unable to deliver message to NAC
- Settlement transaction—batch in balance
- Settlement transaction—draft capture totals in balance and total business out of balance
- Settlement transaction—draft capture totals out of balance

Normal Transaction with Response

The diagram below illustrates the transaction message flow in a scenario containing a normal transaction with a response.



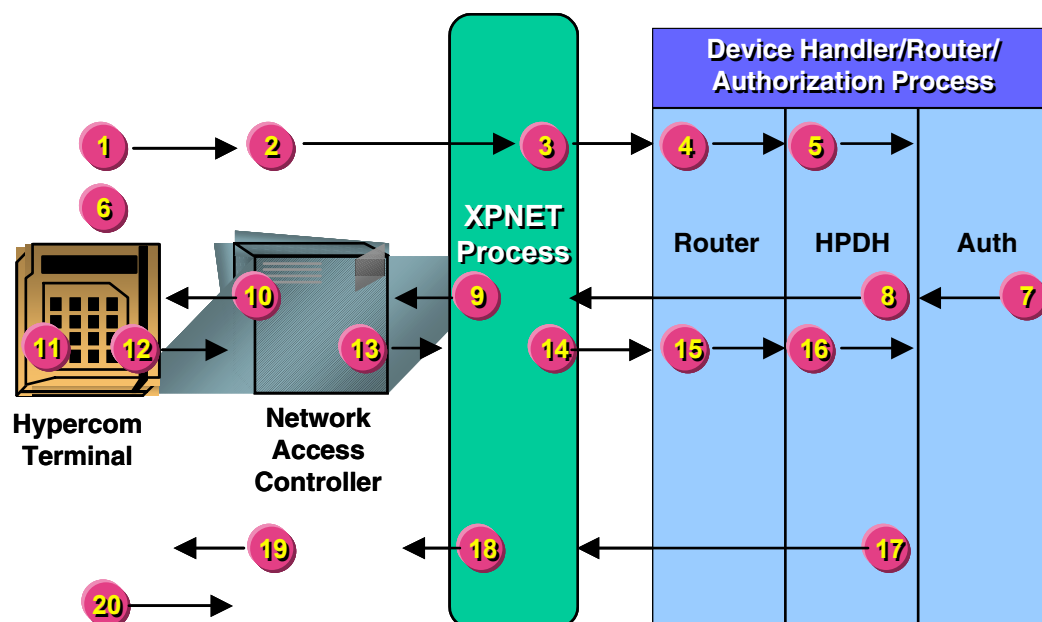
Steps

Processing

- | | |
|------------|--|
| 1, 2, 3, 4 | The terminal transmits a 0200 request message to the HPDH module. This message travels through the network access controller (NAC), the XPNET process and the Router module. |
| 5 | The HPDH module sends the 0200 request message to the Authorization module. |
| 6 | The Authorization module sends the HPDH module a 0210 response. |
| 7, 8, 9 | The HPDH module sends the 0210 response to the terminal. This message travels through the XPNET process and the NAC. |

Normal Transaction without Response—Terminal Times Out

The diagram below illustrates the transaction message flow in a scenario containing a normal transaction without a response.



Steps

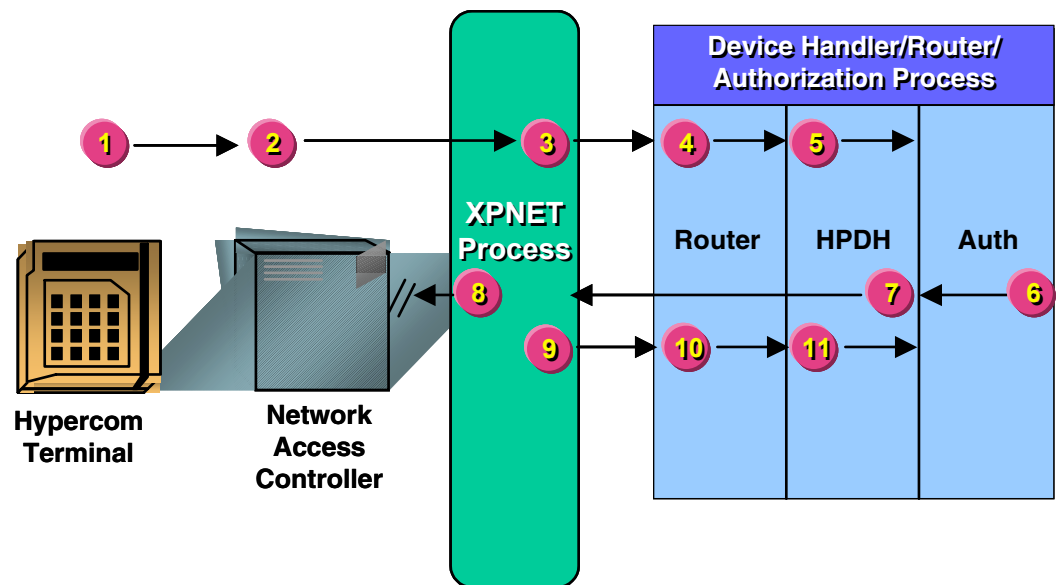
Processing

- | | |
|------------|---|
| 1, 2, 3, 4 | The terminal transmits a 0200 request message to the HPDH module. This message travels through the network access controller (NAC), the XPNET process, and the Router module. |
| 5 | The HPDH module sends the 0200 request message to the Authorization module. |
| 6 | The terminal times out and stores a 0400 reversal transaction in the batch. |
| 7 | The Authorization module sends the HPDH module a 0210 response. |
| 8, 9, 10 | The HPDH module sends the 0210 response to the terminal. This message travels through the XPNET process and the NAC. |
| 11 | The clerk initiates another transaction. |

Steps	Processing
12, 13, 14, 15	The terminal releases the 0400 message from the batch and sends it to the HPDH module before it processes the requested transaction. The message travels through the NAC, the XPNET process, and the Router module.
16	The HPDH module sends the 0400 message to the Authorization module.
17, 18, 19	The HPDH module sends a 0410 reversal response message to the terminal. This message travels through the XPNET process, and the NAC.
20	The terminal sends the transaction that was requested in step 11.

Normal Transaction—Unable to Deliver Response to Terminal

The diagram below illustrates the transaction message flow in a scenario containing a normal transaction where the XPNET process is unable to deliver a response to the terminal.



Steps

Processing

- | | |
|------------|---|
| 1, 2, 3, 4 | The terminal transmits a 0200 request message to the HPDH module. This message travels through the network access controller (NAC), the XPNET process, and the Router module. |
| 5 | The HPDH module sends the 0200 request message to the Authorization module. |
| 6 | The Authorization module sends a 0210 response to the HPDH module. |
| 7, 8 | The HPDH module sends the response to the terminal. This message travels through the XPNET process but fails to reach its destination. |
| 9, 10 | The XPNET process returns the 0210 response to the HPDH module as undeliverable. This message travels through the Router module. |

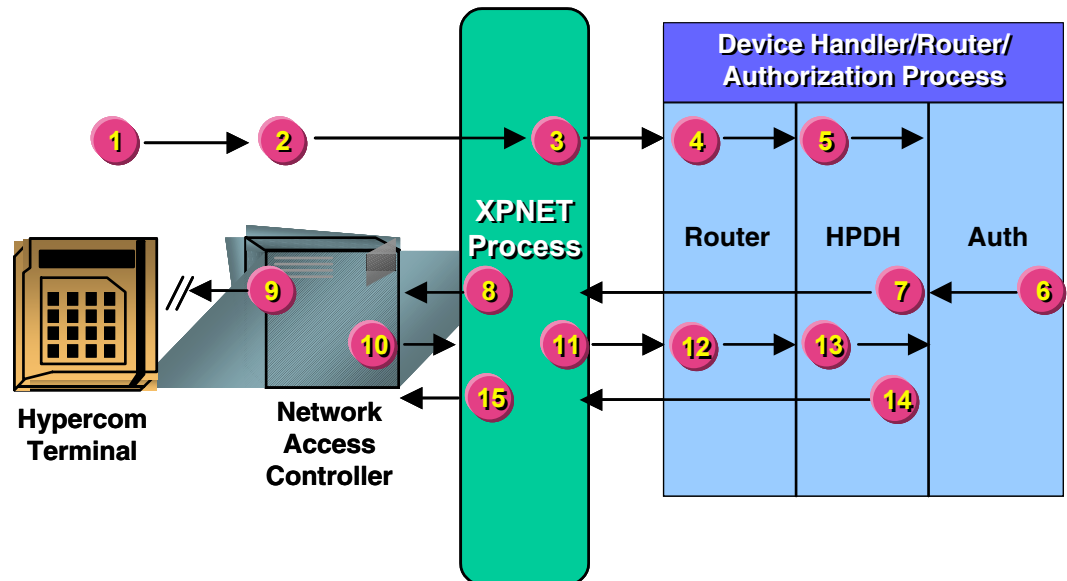
Steps

Processing

- | | |
|----|---|
| 11 | The HPDH module recognizes a failed message, generates a 0400 reversal message if the transaction was approved, and sends it to the Authorization module. |
|----|---|

Normal Transaction - NAC Unable to Deliver Response to Terminal

The diagram below illustrates the transaction message flow in a scenario containing a normal transaction where the network access controller (NAC) is unable to deliver a response to the terminal.



Steps

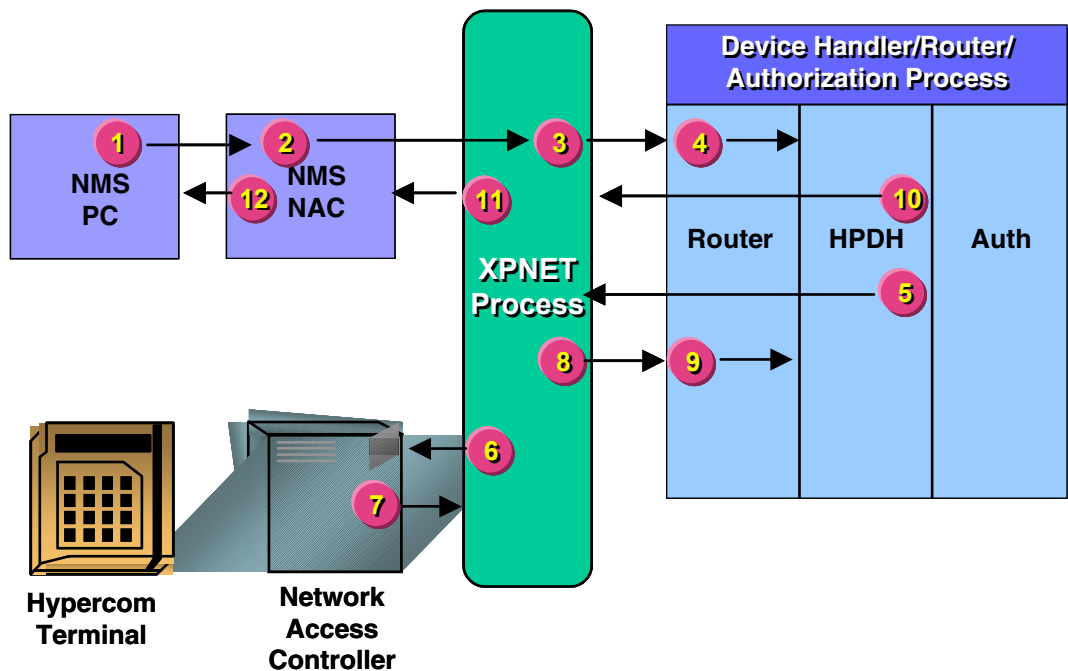
Processing

- | | |
|------------|---|
| 1, 2, 3, 4 | The terminal transmits a 0200 request message to the HPDH module. This message travels through the NAC, the XPNET process, and the Router module. |
| 5 | The HPDH module sends the 0200 request message to the Authorization module. |
| 6 | The Authorization module sends a 0210 response to the HPDH module. |
| 7, 8, 9 | The HPDH module sends the 0210 response to the terminal. This message travels through the XPNET process and NAC but is unable to reach its destination. |

Steps	Processing
10, 11, 12	The NAC generates a 0400 reversal message and sends it to the HPDH module. This message travels through the XPNET process and the Router module.
13	The HPDH module reverses the transaction if the transaction was approved.
14, 15	Regardless of whether the transaction was approved, the HPDH module sends a 0410 response to the NAC. This message travels through the XPNET process.

NMS Passthrough Completes Successfully

The Network Management System (NMS) PC can communicate with the XPNET process. BASE24 acts as a passthrough for NMS messages between the terminal network access controller (NAC) and the NMS NAC. The diagram below illustrates the transaction message flow in a scenario containing an NMS transaction where NMS passthrough completes successfully.



Steps

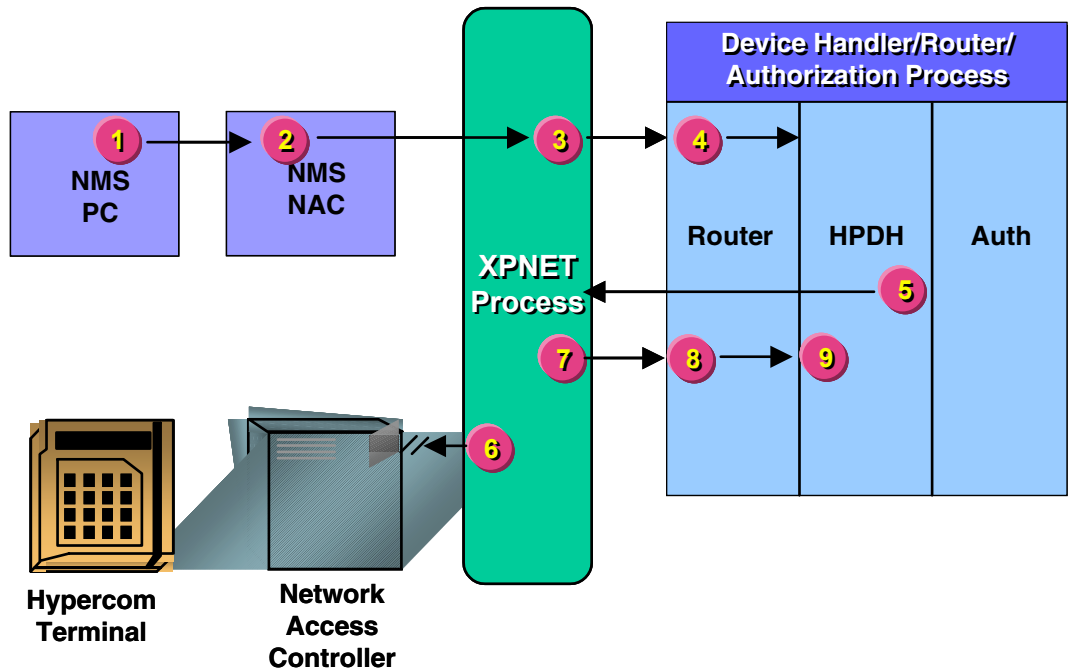
Processing

- | | |
|---------|--|
| 1 | The operator at the NMS PC requests Network Management System (NMS) statistics. The NMS PC sends the request to the NMS NAC. |
| 2, 3, 4 | The NMS NAC formats an NMS request and sends it to the HPDH module. This message travels through the XPNET process and the Router module. |
| 5, 6 | The HPDH module, using information contained in the Transport Protocol Data Unit (TPDU) message header, determines the appropriate terminal NAC to which the NMS message should be routed. The HPDH module then formats the message and sends it to the NAC. This message travels through the XPNET process. |

Steps	Processing
7, 8, 9	The NAC formats an NMS response and transmits it to the HPDH module. This message travels through the XPNET process and the Router module.
10, 11	The HPDH module, using information contained in the Transport Protocol Data Unit (TPDU) message header, determines the appropriate NMS NAC to which the NMS message should be routed. The HPDH module then formats the message to the NMS NAC. This message travels through the XPNET process.
12	The NMS NAC routes the message to the appropriate NMS PC.

NMS Unable to Deliver Message to NAC

The Network Management System (NMS) PC can communicate with the XPNET process. BASE24 acts as a passthrough for NMS messages between the terminal network access controller (NAC) and the NMS NAC. The diagram below illustrates the transaction message flow in a scenario where an NMS is unable to deliver a message to a NAC.



Steps

Processing

- | | |
|---------|--|
| 1 | The operator at the NMS PC requests Network Management System (NMS) statistics. The NMS PC sends the request to the NMS NAC. |
| 2, 3, 4 | The NMS NAC formats the NMS request and sends it to the HPDH module. This message travels through the XPNET process and the Router module. |

Steps	Processing
5, 6	The HPDH module, using information contained in the Transport Protocol Data Unit (TPDU) message header, determines the appropriate terminal NAC to which the NMS message should be routed. The HPDH module then formats a message and routes it to the XPNET process. The XPNET process sends the message to the terminal NAC, but the message does not reach its destination.
7, 8	The XPNET process returns the request to the HPDH module as undeliverable. This message travels through the Router module.
9	The HPDH module recognizes a failed message for an NMS and drops the message.

The diagram illustrates the XPNET Process flow, showing the sequence of steps from the Hypercom Terminal to the Network Access Controller and then to the XPNET Process and the Device Handler/Router/Authorization Process.

Hypercom Terminal (represented by a terminal icon) and **Network Access Controller** (represented by a server rack icon) are connected to the **XPNET Process** (represented by a green vertical bar).

The **XPNET Process** is connected to the **Device Handler/Router/Authorization Process** (represented by a blue table structure).

The flow is as follows:

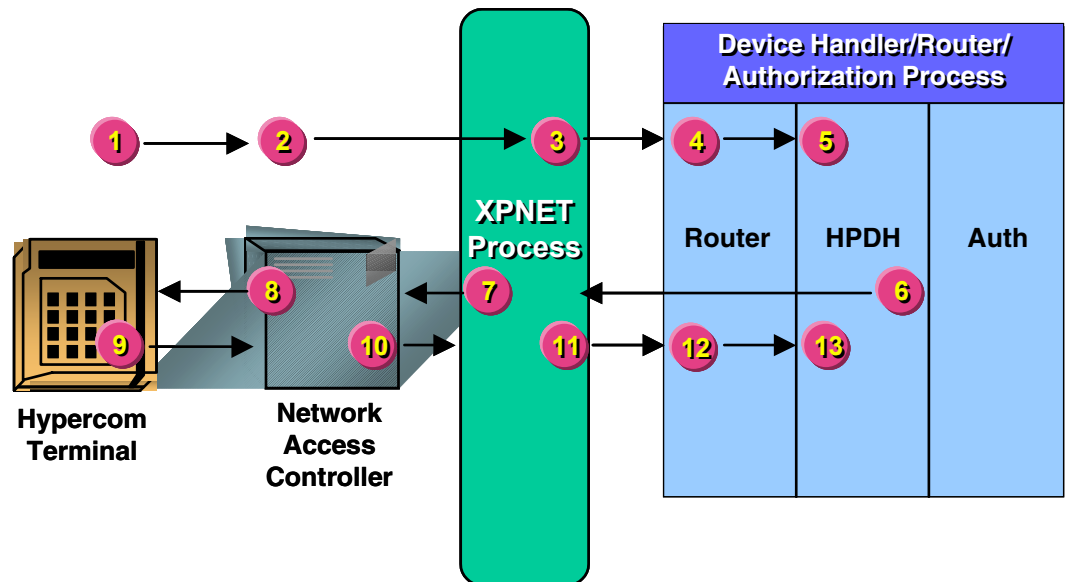
- Step 1: Hypercom Terminal sends data to Step 2.
- Step 2: Network Access Controller sends data to Step 3.
- Step 3: XPNET Process sends data to Step 4.
- Step 4: Router sends data to Step 5.
- Step 5: HPDH sends data to Step 6.
- Step 6: Auth sends data to Step 7.
- Step 7: XPNET Process sends data to Step 8.
- Step 8: Network Access Controller sends data to Step 9.
- Step 9: Hypercom Terminal sends data to Step 10.
- Step 10: Network Access Controller sends data to Step 11.
- Step 11: XPNET Process sends data to Step 12.
- Step 12: Router sends data to Step 13.
- Step 13: HPDH sends data to Step 14.

7-13

Steps	Processing
6, 7, 8	The HPDH module formats a 0510 batch close response and sends it to the terminal with a response code of 00, indicating the transaction is complete. This message travels through the XPNET process and the NAC.
9, 10, 11, 12	After a successful batch close, the terminal sends a terminal statistics message (message type 0800, process code 91). This message travels through the NAC, the XPNET process, and the Router module.
13	The HPDH module logs the terminal statistics to the PTLF.

Settlement Transaction—Draft Capture Totals In Balance and Total Business Out of Balance

The diagram below illustrates the transaction message flow in a scenario containing a Hypercom batch close transaction where the draft capture totals are in balance, but the total business is out of balance.



Steps

1, 2, 3, 4

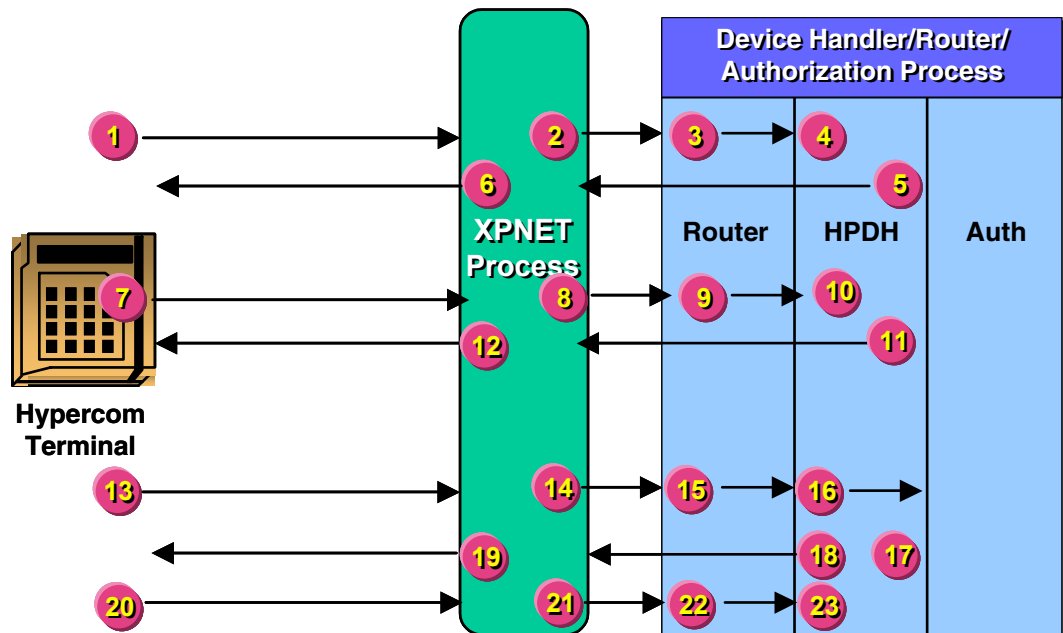
Processing

The terminal sends a 0500 batch close request message to the HPDH module. The second digit of the processing code contained in the request message is a 2 (920000), indicating this is an initial request. This message travels through the network access controller (NAC), the XPNET process, and the Router module.

Steps	Processing
5	The HPDH module determines whether the terminal should be cut over to the next posting date based on the cutover parameters set in the POS Retailer Definition File (PRDF). The HPDH module logs PTD totals by card type (service totals) to the POS Transaction Log File (PTLF). The HPDH module compares terminal totals to PTD totals and determines that the draft capture total business is in balance, but the total business (including authorization only transactions) is out of balance. In this case, PTD batch totals are logged to the PTLF with the out-of-balance flag set. Terminal batch totals are logged to the PTLF with the out-of-balance flag set. If the HPDH module is cutting the terminal over, network totals are logged to the PTLF and the posting date is increased by one.
6, 7, 8	The HPDH module formats a 0510 batch close response and sends it to the terminal with a response code of 00, indicating the transaction is complete. This message travels through the XPNET process and the NAC.
9, 10, 11, 12	After a successful batch close, the terminal sends a terminal statistics message (message type 0800, process code 91). This message travels through the NAC, the XPNET process, and the Router module.
13	The HPDH module logs the terminal statistics to the PTLF.

Settlement Transaction—Draft Capture Totals Out of Balance

The diagram below illustrates the transaction message flow in a scenario containing a Hypercom batch close transaction where the draft capture totals are out of balance.



Steps

1, 2, 3

Processing

The terminal sends a 0500 batch close request message to the HPDH module. The second digit of the processing code contained in the request message is a 2 (920000), indicating this is an initial request. This message travels through the XPNET process and the Router module.

Steps	Processing
4	The HPDH module determines whether the terminal should be cut over to the next posting date based on the cutover parameters set in the POS Retailer Definition File (PRDF). The HPDH module logs PTD totals by card type (service totals) to the POS Transaction Log File (PTLF). The HPDH module compares terminal totals to PTD totals and determines that the draft capture total business is out of balance. In this case, PTD batch totals are logged to the PTLF with the out-of-balance flag set. Terminal batch totals are logged to the PTLF with the out-of-balance flag set.
5, 6	The HPDH module formats and sends a 0510 batch close response to the terminal with a response code of 95, indicating the terminal should start a batch upload. This message travels through the XPNET process.
7, 8, 9	The terminal formats and sends a 0320 upload request message with the first transaction in its batch to the HPDH module. This message travels through the XPNET process and the Router module.
10	The HPDH module formats the transaction and logs it to the PTLF.
11, 12	The HPDH module formats and sends a 0330 acknowledgment response to the terminal. This message travels through the XPNET process. Steps 7–12 are repeated for each transaction in the terminal batch.
13, 14, 15	The terminal sends a 0500 batch close request message to the HPDH module. The second digit of the processing code contained in the request message is a 6 (960000), indicating this is the final request. This message travels through the XPNET process and the Router module.
16	The HPDH module formats the transaction as a batch totals record and sends it to the Authorization module for logging to the PTLF.
17	If the HPDH module is cutting the terminal over, network totals are logged to the PTLF and the posting date is increased by one.

Steps	Processing
18, 19	The HPDH module formats a 0510 batch close response and sends it to the terminal with a response code of 00, indicating the transaction is complete. This message travels through the XPNET process.
20, 21, 22	After a successful batch close, the terminal sends a terminal statistics message (message type 0800, process code 91). This message travels through the XPNET process and the Router module.
23	The HPDH module logs the terminal statistics to the PTLF.

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8: Hypercom Configuration File (HPCF)

The Hypercom Device Handler (HPDH) module is the exclusive reader of the Hypercom Configuration File (HPCF). The HPCF is used to configure Hypercom T4, T5, T6 and T7 terminals. This file must contain a record for every terminal group in the network using the HPDH module with a Hypercom device. The HPCF is accessed using BASE24-pos files maintenance. If an HPCF record is changed, the HPDH module must be warmbooted in order for the changes to take effect.

BASE24-pos files maintenance manipulates records through the use of formatted screens. This section describes the formatted screens used to access the HPCF.

The HPCF is set up in the same manner as most other files in the BASE24-pos system. Refer to the *BASE24 Base Files Maintenance Manual*, the *BASE24-pos Files Maintenance Manual*, and the *BASE24 CRT Access Manual* for more information concerning files maintenance screens. The information provided in this section, however, should be sufficient to set up the necessary records in the HPCF for persons familiar with BASE24 CRT access and files maintenance.

Refer to section 9 for information about configuring Hypercom T7 terminals.

Hypercom T4, T5, and T6 Terminal Configuration Data

The Hypercom Configuration File (HPCF) contains configuration data used for downloading to a Hypercom T4, T5, or T6 terminal. The data contained in this file is divided into the following categories:

Category	Function
Configuration parameters	Contains information that applies to the terminal operation, such as initialization information and dialing options. There is one set of configuration parameters for each terminal group.
Product descriptions	Defines the parameters associated with a product description code. There can be up to 9 product descriptions for each terminal group.
Card definitions	Defines issuer and acquirer information for card prefix ranges. There can be up to 16 card definitions for each terminal group.
Range definitions	Defines additional card prefix ranges that can use the card definitions. There can be up to 100 range definitions for each terminal group for a total of 116 card prefix ranges that can be associated with the 16 card definitions.

The categories above form a single physical HPCF record. When adding or updating records, you either can enter information on all screens before pressing the **F3** or **F5** key, or you can enter information on one screen, press the **F3** or **F5** key, and continue to the next screen. Refer to the ***BASE24 CRT Access Manual*** for more information about BASE24 file maintenance function keys.

Configuration Parameters Screen

The Configuration Parameters screen enables the entry of configuration parameters necessary during processing. One configuration record must exist for each terminal. This screen is shown below, followed by descriptions of its fields.

```

BASE24-POS  HYPERCOM CONFIG PARAMS  LLLL      YY/MM/DD  HH:MM  01 OF 04
      TERM GROUP

      INITIALIZATION CONTROL  0 (NO RESET)      INITIAL PASSWORD 0000
      PRINTER CONTROL  0 (NO PRINTER)          HELP DESK PHONE
      ECR BAUD RATE  0 (NO ECR)                DLL REVISION      # 00
      CHECK GUARANTEE  0 (NONE)                REAUTHORIZE MARGIN % 00
      DEFAULT TRAN  0 (SALE)
      SUPER PASSWORD  0 (NOT USED)
      PIN PAD CONTROL  0 (NO ENCRYPTION)

      TERMINAL OPTIONS : 0) DISABLE, 1) ENABLE
      0 REFERRAL DIALING      0 KEYBOARD LOCKED      0 RESTAURANT TIP
      0 MEMORY DIALING      0 VOID PASSWORD DISABLE  0 BALANCE INQUIRY
      0 TOLL DIALING      0 REFUND PASSWORD DISABLE  0 AMT DUAL ENTRY
      0 DIALING      0 TRAN SRCH PWD DISABLE  0 DISP MAG STRIPE
      0 PRINT 'TOTAL' ON RCPT 0 VAL PAN LGTH OFFL DISABLE 0 REFUNDS OFFLINE
      0 0-9 CUST # ON RECEIPT      0 VOIDS OFFLINE

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

TERM GROUP — Identifies the terminal group associated with this terminal. This field is the key to the HPCF and must match a terminal group in the BASE24-pos Terminal Data files (PTD).

Field Length: 1–4 alphanumeric characters
 Required Field: Yes
 Default Value: No default value
 Data Name: HPCF.TERM-GRP

INITIALIZATION CONTROL — A code indicating whether the terminal resets and requests a partial batch download from the host after initialization. This field is currently not supported by BASE24-pos. Valid values are as follows:

0 = No, terminal does not reset after initialization.
 1 = Yes, terminal does reset after initialization.

Field Length: 1–2 numeric characters
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.INITIALIZATION-
CNTL

INITIAL PASSWORD — The initial password downloaded to a terminal from the HPCF.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000
Data Name: HPCF.CONFIGURATION-PARAMS.PASSWORD

PRINTER CONTROL — A code indicating the type of printer that is attached to the specific terminal. Valid values are as follows:

0 = No printer
1 = P4 preprinted
2 = P4 not preprinted
3 = CAT055
4 = CAT053
5 = Citizen ID-560RS

Field Length: 1–2 numeric characters
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.PRINTER-
CONTROL

HELP DESK PHONE — The telephone number of the help desk associated with this terminal. The help desk is used when a problem occurs with the system or terminal. This field is left-justified and blank filled.

Example: 4023907600
Field Length: 1–14 numeric characters
Required Field: No
Default Value; No default value
Data Name: HPCF.CONFIGURATION-PARAMS.HELP-DESK-
PHONE

ECR BAUD RATE — A code indicating the baud rate for an electronic cash register (ECR) attached to the terminal. Valid values are as follows:

- 0 = No ECR
- 1 = 1200 bps
- 2 = 2400 bps
- 3 = 4800 bps

Field Length: 1–2 numeric characters
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.ECR-BAUD-RATE

DLL REVISION # — A code indicating the download revision number. This field should match the download revision number contained in the download request.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPCF.CONFIGURATION-PARAMS.DLL-REVISION-
NUM

CHECK GUARANTEE — A code indicating the check guarantee option supported at the terminal. Valid values are as follows:

- 0 = None
- 1 = Telecredit with check MICR data
- 2 = TeleCheck with drivers license
- 3 = TeleCheck with account number
- 4 = JBS with check MICR data
- 5 = Sight draft
- 8 = Telecredit with account number

Field Length: 1–2 numeric characters
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.CHECK-
GUARANTEE-OPTIONS

REAUTHORIZE MARGIN % — A code indicating the percentage of reauthorization margin. The range of valid values is 00–99. This field is currently not supported by BASE24-pos.

Field Length: 2 numeric characters
Required Field: No
Default Value: 00
Data Name: HPCF.CONFIGURATION-PARAMS.REAUTHORIZE-MARGIN

DEFAULT TRAN — A code indicating the default transaction code used by the terminal. Valid values are as follows:

0 = Sale
1 = Cash
2 = Balance inquiry

Field Length: 1–2 numeric characters
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.DEFAULT-TRAN-CODE

SUPER PASSWORD — A code indicating how a supervisor password is being used. The supervisor password is downloaded to the terminal in the INITIAL PASSWORD field. Valid values are as follows:

0 = Password not used
1 = Used as local password
2 = Used as PIN (not supported by Hypercom)
3 = Used as PIN with administration card (not supported by Hypercom)

Field Length: 1–2 numeric characters
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.SUPERVISOR-PASSWORD

PIN PAD CONTROL — A code identifying the type of PIN pad connected to the terminal. Valid values are as follows:

- 0 = No encryption
- 1 = Encryption in terminal
- 2 = CAT033
- 3 = S4
- 4 = S4B
- 5 = S4C

Field Length: 1–2 numeric characters
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.PIN-PAD-CNTL

TERMINAL OPTIONS

The remaining fields on screen 1 are used to define parameters associated with the terminal's functionality.

REFERRAL DIALING — A code indicating whether the terminal dials a referral telephone number on referral responses. The referral telephone number is taken from the REFERRAL PHONE # field on screen 3 of the HPCF. Valid values are as follows:

- 0 = No, the terminal does not dial a referral telephone number.
- 1 = Yes, the terminal dials a referral telephone number.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.PHONE-DIAL-
OPTION.REFERRAL-DIALING

KEYBOARD LOCKED — A code indicating whether the terminal keyboard is locked after a download and requires a password to unlock it. If locked, the password set in the INITIAL PASSWORD field must be entered in order to unlock the keyboard after the download is completed. Valid values are as follows:

- 0 = No, the keyboard is not locked.
- 1 = Yes, the keyboard is locked.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.LOCAL-TERM-
OPTIONS.KEYBOARD-LOCKED

RESTAURANT TIP — A code indicating whether the terminal prompts a clerk to enter an additional amount (tip amount) on purchase transactions. Valid values are as follows:

- 0 = No, the terminal does not prompt a clerk to enter an additional amount on purchase transactions.
- 1 = Yes, the terminal prompts a clerk to enter an additional amount on purchase transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.TERM-OPTIONS.
RESTAURANT-TIP

MEMORY DIALING — A code indicating whether the terminal supports memory dialing. Valid values are as follows:

- 0 = No, the terminal does not support memory dialing.
- 1 = Yes, the terminal supports memory dialing.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.PHONE-DIAL-
OPTION.MEMORY-DIALING

VOID PASSWORD DISABLE — A code indicating whether a password is required for void transactions. Valid values are as follows:

- 0 = No, do not disable the password requirement (that is, require a password).
- 1 = Yes, disable the password requirement (that is, do not require a password).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.LOCAL-TERM-
OPTIONS.VOID-PASSWORD-DISABLE

BALANCE INQUIRY — A code indicating whether a terminal supports balance inquiry transactions. Valid values are as follows:

- 0 = No, the terminal does not support Balance Inquiry transactions.
- 1 = Yes, the terminal supports Balance Inquiry transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.TERM-OPTIONS.
BALANCE-INQUIRY

TOLL DIALING — A code indicating whether a terminal supports toll dialing. Valid values are as follows:

- 0 = No, the terminal does not support toll dialing.
- 1 = Yes, the terminal supports toll dialing.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.PHONE-DIAL-
OPTION.TOLL-DIALING

REFUND PASSWORD DISABLE — A code indicating whether a password is required for refund transactions. Valid values are as follows

- 0 = No, do not disable the password requirement (that is, require a password).
- 1 = Yes, disable the password requirement (that is, do not require a password).

Field Length: 1 numeric character

Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.LOCAL-TERM-
OPTIONS.REFUND-PASSWORD-DISABLE

AMT DUAL ENTRY — A code indicating whether the terminal prompts a clerk to verify (reenter) transaction amounts. Valid values are as follows:

0 = No, the terminal does not prompt the clerk to reenter transaction amounts.
1 = Yes, the terminal prompts the clerk to reenter transaction amounts.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.OPTIONS-1.
AMOUNT-DUAL-ENTRY

DIALING — A code indicating whether a terminal supports the use of a telephone. Valid values are as follows:

0 = No, the terminal does not support the use of a telephone.
1 = Yes, the terminal supports the use of a telephone.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.PHONE-DIAL-
OPTION.DIALING

TRAN SRCH PWD DISABLE — A code indicating whether a password is required to search a batch of transactions. Valid values are as follows:

0 = No, do not disable the password requirement (that is, require a password).
1 = Yes, disable the password requirement (that is, do not require a password).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.LOCAL-TERM-
OPTIONS.TRAN-SEARCH-PASSWORD-DISABLE

DISP MAG STRIPE — A code indicating whether the terminal displays Track 2 data when a card is read by the device. Valid values are as follows:

- 0 = No, the terminal does not display Track 2 data when a card is read by the device.
- 1 = Yes, the terminal displays Track 2 data when a card is read by the device.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.DISPLAY-MAG-
STRIPE

PRINT 'TOTAL' ON RCPT — A code indicating whether the word *Total* is printed on the customer receipt. Valid values are as follows:

- 0 = No, do not print *Total* on the customer receipt.
- 1 = Yes, print *Total* on the customer receipt.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.TERM-OPTIONS.
PRINT-TOTAL

VAL PAN LGTH OFFL DISABLE — A code indicating whether the PAN length should be validated on offline transactions. Valid values are as follows:

- 0 = No, do not disable PAN length validation (that is, PAN length validation remains enabled).
- 1 = Yes, disable PAN length validation (that is, PAN length validation is disabled).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.TERM-OPTIONS.
VAL-PAN-LGTH-OFFLINE-DISABLE

REFUNDS OFFLINE — A code indicating whether refund processing is allowed offline. Valid values are as follows:

- 0 = No, refund processing is not allowed offline.
- 1 = Yes, refund processing is allowed offline.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.TERM-OPTIONS.
REFUNDS-OFFLINE

0-9 CUST # ON RECEIPT — A code indicating whether the clerk is prompted for a 0–9 digit customer number to be printed on the receipt. Valid values are as follows:

- 0 = No, do not prompt the clerk for a customer number.
- 1 = Yes, prompt the clerk for a customer number.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.TERM-OPTIONS.
CUSTOMER-NUMBER-ON-RECEIPT

VOIDS OFFLINE — A code indicating whether void processing is allowed offline. Valid values are as follows:

- 0 = No, void processing is not allowed offline.
- 1 = Yes, void processing is allowed offline.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CONFIGURATION-PARAMS.TERM-OPTIONS.
VOIDS-OFFLINE

Product Data Screen

Use the Product Data screen to define up to nine product codes and the keys with which they are associated. Product data is defined by the customer and is downloaded to the terminal, although the data cannot be displayed on the terminal by an operator.

You can use product codes to distinguish between the goods and services offered by a retailer. For example, a service station can choose one product code to track petroleum transactions, another to track food transactions, and another to track mechanical services. Retailers can then separate their transactions according to the product code categories they set up on the Hypercom Product Data screen.

The Hypercom Product Data screen is shown below, followed by descriptions of its fields.

BASE24-POS

HYPERCOM PRODUCT DATA

LLLL

YY/MM/DD

HH:MM

02 OF 04

TERM GROUP

SELECT KEY

PRODUCT CODE

PRODUCT DESCRIPTION

01

02

03

04

05

06

07

08

09

***** BASE24 *****

NEW PAGE:

FILE DESTINATION:

NEW LOGICAL NETWORK ID:

F12-HELP

SELECT KEY — A number indicating the terminal function key associated with the product code being defined. The range of valid values is 01–09.

Field Length: System-protected
Data Name: Not applicable

PRODUCT CODE — A user-defined code identifying a retailer's product or line of business. This code is sent by the terminal in requests if the product code is configured in the card definition table. This field can occur up to nine times. The range of valid values is 01–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPCF.PRODUCT-CODE.CDE

PRODUCT DESCRIPTION — The product description associated with the PRODUCT CODE defined in the previous field. This field is user-defined and can occur up to nine times.

Field Length: 1–20 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: HPCF.PRODUCT-CODE.DESCRPTION

Card Definition Table Screen Function Keys

The use of two function keys on the Card Definition Table screen varies from the standard function keys explained in the *BASE24 CRT Access Manual*. The use of these function keys is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F14	Forward — Moves forward to the next card definition table entry. For example, if you are currently working on table entry 1, pressing the F14 key accesses table entry 2. If you are working on table entry 16, pressing the F14 key accesses table entry 1.
F15	Back — Moves back to the previous card definition entry. For example, if you are currently working on table entry 2, pressing the F15 key accesses table entry 1. If you are working on table entry 1, pressing the F15 key accesses table entry 16.

Card Definition Table Screen

Use the Card Definition Table screen to define the characteristics of a card range. You can define up to 16 card ranges. The parameters set up on this screen apply to transactions initiated with cards within the defined prefix range. Hypercom Card Definition Table screen 3 is shown below, followed by descriptions of its fields.

```

BASE24-POS    HYPERCOM CARD DEF TABLE LLLL          YY/MM/DD  HH:MM  03 OF 04
TERM GROUP          TABLE ENTRY 01 OF 16

PREFIX (LOW)  00000000          PRIMARY PHONE  #
PREFIX (HIGH) 00000000          BACKUP PHONE   #
CARD NAME                                REFERRAL PHONE #
DEFAULT ACCOUNT 0 (DEFAULT)      FLOOR LIMIT    0000
PROCESSING TYPE 0 (CREDIT AUTH ONLY)

TRACK SENT 0 (TRACK II)          NETWORK INT'L ID 000
TERMINAL TIMEOUT 00 SECONDS      CARD ACCEPTOR ID

CUSTOMER ACCT          0          PRODUCT CODES          0
PIN REQUIRED            0          ADJUSTMENTS ALLOWED 0
MANUAL PAN ENTRY ALLOWED 0        MOD 10 CHECK          0
EXPIRATION DATE REQ'D  0          INVOICE # REQUIRED    0
OFFLINE TRANS ALLOWED  0          PURCHASE CASH BACK 0
VOICE REFERRALS        0          MAC REQUIRED           0
PRINT RECEIPT          0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                                  F12-HELP

```

TABLE ENTRY 01 OF 16 — A number indicating which occurrence of this screen is being displayed on the terminal. Up to 16 screens defining 16 prefix ranges and their associated parameters can exist. This field acts as an index into the card definition table. The range of valid values is 01–16. You can change the entry number by pressing the **F14** key to go forward and the **F15** key to go backward through the screens.

Field Length: System-protected
Data Name: Not applicable

PREFIX (LOW) — A number identifying the lowest value within the card prefix range defined. Any card prefixes that are equal to or greater than this value and equal to or less than the value in the PREFIX (HIGH) field are included within the range. All cards falling within the range defined use the processing options defined on this screen.

Field Length: 8 numeric characters
Required Field: Yes
Default Value: 00000000
Data Name: HPCF.CARD-DEFINITION-TBL.LOW-PREFIX

PRIMARY PHONE # — The telephone number of the primary network for transactions initiated with cards within the defined prefix range.

Example: 4023907600
Field Length: 1–14 numeric characters
Required Field: No
Default Value: No default value
Data Name: HPCF.CARD-DEFINITION-TBL.PRIMARY-PHONE-
NUM

PREFIX (HIGH) — A number identifying the highest value within the card prefix range defined. Any card prefixes that are equal to or less than this value and equal to or greater than the value in the PREFIX (LOW) field are included within the range. All cards falling within the range defined use the processing options specified on this screen.

Field Length: 8 numeric characters
Required Field: Yes
Default Value: 00000000
Data Name: HPCF.CARD-DEFINITION-TBL.HIGH-PREFIX

BACKUP PHONE # — The telephone number of the backup network for transactions initiated with cards within the defined prefix range.

Example: 4023907600
Field Length: 1–14 numeric characters
Required Field: No
Default Value: No default value
Data Name: HPCF.CARD-DEFINITION-TBL.BACKUP-PHONE-NUM

CARD NAME — The card name displayed on the terminal and receipts for transactions initiated with cards within the defined prefix range.

Field Length: 1–10 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: HPCF.CARD-DEFINITION-TBL.CARD-NAM

REFERRAL PHONE # — The telephone number of the network to receive referrals concerning transactions initiated with cards within the defined prefix range.

Example: 4023907600
Field Length: 1–14 numeric characters
Required Field: No
Default Value: No default value
Data Name: HPCF.CARD-DEFINITION-TBL.REFERRAL-PHONE-
NUM

DEFAULT ACCOUNT — A code identifying the account type the terminal uses during processing for cards within the specified prefix range. This field is used by the terminal if the value in the CUSTOMER ACCT SELECT field on this screen is set to 0. Otherwise, the customer decides the account type to be sent by the terminal during processing. Valid values are as follows:

0 = Default
1 = Savings
2 = Checking
3 = Credit
4 = Universal

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.DEFAULT-ACCT

FLOOR LIMIT — The maximum amount, in whole currency units, allowed per transaction that are entered offline and initiated with cards within the defined card prefix range.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000
Data Name: HPCF.CARD-DEFINITION-TBL.FLOOR-LIMIT

PROCESSING TYPE — A code indicating the type of card processing supported for cards within the defined prefix range. Valid values are as follows:

- 0 = Credit authorization only
- 1 = Credit draft capture
- 2 = Debit
- 3 = Visa I

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.CARD-PROC-TYP

TRACK SENT — A code indicating which card track data the terminal sends to BASE24 during processing for cards within the defined prefix range. Valid values are as follows:

- 0 = Track 2
- 1 = Track 1 (not supported by BASE24)
- 2 = Track 1 and Track 2 (not supported by BASE24)

Field Length: 1–2 numeric characters
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.TRACK-SENT

NETWORK INT'L ID — Not currently supported by BASE24-pos. However, a value must be entered in this field.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000
Data Name: HPCF.CARD-DEFINITION-TBL.NETWORK-
INTERNATIONAL-ID

TERMINAL TIMEOUT — The number of seconds a terminal waits before a request is timed out for cards within the defined prefix range.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 30
Data Name: HPCF.CARD-DEFINITION-TBL.TERMINAL-TIMEOUT

CARD ACCEPTOR ID — A code identifying the card acceptor for this transaction, if the transaction originated from an acquirer host. This is only used in cases where the acquirer is not the actual card acceptor.

Field Length: 1–15 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: HPCF.CARD-DEFINITION-TBL.CRD-ACCEPTOR-ID

CUSTOMER ACCT — A code indicating whether the terminal is to use the value contained in the DEFAULT ACCOUNT field during processing, or if the customer is to decide the account type the terminal uses during processing for cards within the defined prefix range. Valid values are as follows:

- 0 = No, the customer does not select the account type to use during processing for the defined prefix range. The terminal uses the default account number.
- 1 = Yes, the customer selects the account type to use during processing for the defined prefix range.

If this field is set up so the customer is determining the account type (a value of 1), a PIN pad must be available for entering the account type, since the terminal prompts the customer to enter the account type using the PIN pad.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.ACCT-
SELECT

PRODUCT CODES — A code indicating whether cards within the specified prefix range require entry of a product code. Product codes are used to identify a customer's line of business. Valid values are as follows:

- 0 = No, the terminal does not prompt for product code entry for the defined prefix range.
- 1 = Yes, the terminal prompts for product code entry for the defined prefix range.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.
PRODUCT-CODES

PIN REQUIRED — A code indicating whether cards within the defined prefix range require PINs. Valid values are as follows:

- 0 = No, PINs are not required for the defined prefix range.
- 1 = Yes, PINs are required for the defined prefix range.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.PIN-
ENTRY

ADJUSTMENTS ALLOWED — A code indicating whether adjustments can be performed for cards within the defined prefix range.

Adjustments can only be performed if the PROCESSING TYPE field on this screen is set to 1 (authorize and capture). Valid values are as follows:

- 0 = No, adjustments are not allowed for cards within the defined prefix range.
- 1 = Yes, adjustments are allowed for cards within the defined prefix range.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.
ADJUSTMENT-ALLOWED

MANUAL PAN ENTRY ALLOWED — A code indicating whether the operator can manually enter the primary account number (PAN) for cards within the defined prefix range. Valid values are as follows:

- 0 = No, manual PAN entry is not allowed for cards within the defined prefix range.
- 1 = Yes, manual PAN entry is allowed for cards within the defined prefix range.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.
MANUAL-PAN-ENTRY

MOD 10 CHECK — A code indicating whether MOD 10 checks are used for cards within the defined prefix range. Valid values are as follows:

- 0 = No, terminal MOD 10 checks are not used for the cards within the defined prefix range.
- 1 = Yes, terminal MOD 10 checks are used for the cards within the defined prefix range.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.MOD-
10-CHECK

EXPIRATION DATE REQ'D — A code indicating whether a customer is required to enter an expiration date when the MANUAL PAN ENTRY ALLOWED field is entered. Valid values are as follows:

- 0 = No, the terminal does not prompt for an expiration date on manual PAN entry.
- 1 = Yes, the terminal prompts for an expiration date on manual PAN entry.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.
MANUAL-EXP-DAT

INVOICE # REQUIRED — A code indicating whether the terminal prompts a customer for the invoice number for cards within the defined prefix range. Valid values are as follows:

- 0 = No, the terminal does not prompt for an invoice number for cards within the defined prefix range.
- 1 = Yes, the terminal prompts for an invoice number for cards within the defined prefix range.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.
INVOICE-NUM

OFFLINE TRANS ALLOWED — A code indicating whether offline transactions can be performed for cards within the defined prefix range. Valid values are as follows:

- 0 = No, offline transactions cannot be performed for cards within the defined prefix range.
- 1 = Yes, offline transactions can be performed for cards within the defined prefix range.

Field Length: 1 numeric character

Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.
OFFLINE-TRAN-ALLOW

PURCHASE CASH BACK — A code indicating whether purchase with cash back transactions are allowed for cards within the defined prefix range.

This field is used only if the PROCESSING TYPE field is set to 2. Valid values are as follows:

- 0 = No, purchase with cash back transactions are not allowed for cards within the defined prefix range.
- 1 = Yes, purchase with cash back transactions are allowed for cards within the defined prefix range.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.CASH-
BACK-ALLOWED

VOICE REFERRALS — A code indicating whether the terminal automatically dials the referral telephone number on referral responses for cards within the defined prefix range. The referral telephone number is taken from the REFERRAL PHONE # field. Valid values are as follows:

- 0 = No, voice referrals are not allowed for cards within the defined prefix range.
- 1 = Yes, voice referrals are allowed for cards within the defined prefix range.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.
VOICE-REFERRAL

MAC REQUIRED — A code indicating whether the terminal calculates a message authentication code (MAC) and sends it to BASE24 for validation during processing for cards within the defined prefix range. Valid values are as follows:

0 = No, do not calculate and send MACs for cards within the defined prefix range.

1 = Yes, calculate and send MACs for cards within the defined prefix range.

Note: This field must be the same for all 16 Hypercom Card Definition Table screens.

Field Length: 1 numeric character

Required Field: Yes

Default Value: 0

Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.MAC

PRINT RECEIPT — A code indicating whether a receipt is printed for transactions being performed within the defined prefix range. Valid values are as follows:

0 = No, do not print a receipt for transactions being performed.

1 = Yes, print a receipt for transactions being performed.

Field Length: 1 numeric character

Required Field: Yes

Default Value: 0

Data Name: HPCF.CARD-DEFINITION-TBL.PROC-OPTIONS.
RECEIPT

Range Definition Table Screen Function Keys

The use of two function keys on the Range Definition Table screen varies from the standard function keys explained in the *BASE24 CRT Access Manual*. The uses of these functions keys are explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F14	Forward — Moves forward to the next group of range definition table entries. For example, if you are currently working on entries 001–010, pressing the F14 key accesses entries 011–020. If you are working on entries 091–100, pressing the F14 key accesses entries 001–010.
F15	Back — Moves back to the previous group of range definition table entries. For example, if you are currently working on entries 011–020, pressing the F15 key accesses entries 001–010. If you are working on entries 001–010, pressing the F15 key accesses entries 091–100.

ENTRIES — The range of table entries currently being displayed. This information can be changed by pressing the **F14** key to go forward or the **F15** key to go backward through the screens.

Field Length: System-protected
Data Name: Not applicable

TABLE ENTRY — A value identifying the entry in the range definition table.

Field Length: System-protected
Data Name: Not applicable

PREFIX (LOW) — A number identifying the lowest value within the card prefix range defined in this record. Any card prefixes that are equal to or greater than this value and equal to or less than the value in the **PREFIX (HIGH)** field are included within the range.

Field Length: 8 numeric characters
Required Field: Yes
Default Value: 00000000
Data Name: HPCF.RDT.RANGE-DEFINITION-TBL.LOW-PREFIX

PREFIX (HIGH) — A number identifying the highest value within the card prefix range defined in this record. Any card prefixes that are equal to or less than this value and equal to or greater than the value in the **PREFIX (LOW)** field are included within the range.

Field Length: 8 numeric characters
Required Field: Yes
Default Value: 00000000
Data Name: HPCF.RDT.RANGE-DEFINITION-TBL.HIGH-PREFIX

CARD DEFINITION TABLE INDEX — A number indicating which card definition table on screen 3 contains the configuration data used for the prefix range defined. This field is the index into the card definition table. Valid values are 01 through 16.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPCF.RDT.RANGE-DEFINITION-TBL.CARD-
DEFINITION-TBL-INDEX

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9: Hypercom T7 Configuration File (HPTC)

The Hypercom Device Handler (HPDH) module is the exclusive reader of the Hypercom T7 Configuration File (HPTC). The HPTC is used to configure Hypercom T7 terminals. This file must contain a record for every terminal group in the network using the HPDH module with a Hypercom T7 device. The HPTC is accessed using BASE24-pos files maintenance. If an HPTC record is changed, the HPDH module must be warmbooted and the device initialized in order for the changes to take effect.

BASE24-pos files maintenance manipulates records through the use of formatted screens. This section describes the formatted screens used to access the HPTC.

The HPTC is set up in the same manner as most other files in the BASE24-pos system. Refer to the *BASE24 Base Files Maintenance Manual*, the *BASE24-pos Files Maintenance Manual*, and the *BASE24 CRT Access Manual* for more information concerning files maintenance screens. The information provided in this section, however, should be sufficient to set up the necessary records in the HPTC for persons familiar with BASE24 CRT access and files maintenance.

Refer to section 8 for information about configuring Hypercom T4, T5, and T6 terminals.

Hypercom T7 Terminal Configuration Data

The Hypercom T7 Configuration File (HPTC) contains configuration data used for downloading to a Hypercom T7 terminal. The data contained in this file is divided into the following categories:

Category	Function
Configuration parameters	Contains information that applies to the terminal operation, such as initialization information and dialing options. There is one configuration parameters table for each terminal group.
Card ranges	Specifies a range of card prefixes accepted by terminals in the terminal group. Identifies the issuers table entries and the acquirers table entries used with the card prefix range. There can be up to 32 card ranges for each terminal group.
Issuers	Defines the parameters associated with a card issuer. There can be up to 16 sets of issuer parameters for each terminal group.
Acquirers	Defines the parameters associated with an acquirer or host. There can be up to 8 sets of acquirer parameters for each terminal group.
Product descriptions	Defines the parameters associated with a product description code. There can be up to 11 product description codes for each terminal group.
Additional data prompts	Defines additional data prompts that can be displayed during transaction processing. There can be up to 8 additional data prompts for each terminal group.

The HPTC contains a table for each data category. While all of these tables are contained in the HPTC, each table is an individual physical record. When adding or updating the HPTC, you must press the **F3** or **F5** key each time you enter data on a screen. For more information about BASE24 file maintenance function keys, refer to the ***BASE24 CRT Access Manual***.

Screen 1 Function Keys

The use of one function key on HPTC screen 1 varies from the standard function keys explained in the *BASE24 CRT Access Manual*. The use of this function key is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F9	Next Screen — Displays screen 2 for the specified record type. For example, if you enter 01 in the RECORD TYPE field and press the F9 key, the system displays Hypercom T7 Configuration Parameters screen 2. If you enter 02 in the RECORD TYPE field and press the F9 key, the system displays Hypercom T7 Card Table screen 2.

Screen 1

Hypercom T7 Configuration File (HPTC) screen 1 serves as a menu to the HPTC. This screen enables you to access the screens associated with each HPTC table. These screens are arranged by record type. For example, record type 01 contains the screens for the configuration parameters table. HPTC screen 1 fields are described below.

TERMINAL GROUP — The group to which the terminals belong. This value corresponds to the TERM GROUP field on BASE24-pos Terminal Data files (PTD) screen 1. The TERMINAL GROUP field combined with the RECORD TYPE field are the key to the HPTC.

Field Length: 1–4 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: HPTC.KEY-S.TERM-GRP

RECORD TYPE — A code indicating the table to which you want access. The RECORD TYPE field combined with the TERMINAL GROUP field are the key to the HPTC. Valid values are as follows:

01 = Configuration parameters table
02 = Card range table (first 16 entries)
03 = Card range table (second 16 entries)
04 = Issuer table
05 = Acquirer table
06 = Product description table
07 = Additional data prompts table
0C = ICC table ID
0D = CHIPDATA table ID
0E = KEYDATA table ID
0F = REVKEY table ID

Field Length: 2 numeric characters
Required Field: Yes
Default Value: No default value
Data Name: HPTC.KEY-S.REC-TYPE

Configuration Parameters Table Screen 2

Hypercom T7 Configuration Parameters Table screen 2 contains parameters for configuring all terminals in a terminal group. One configuration parameters record exists for each terminal group.

You can display Hypercom T7 Configuration Parameters Table screen 2 by entering a valid terminal group in the **TERMINAL GROUP** field and **01** in the **RECORD TYPE** field of HPTC screen 1 and pressing the **F9** key.

Hypercom T7 Configuration Parameters Table screen 2 is shown below, followed by descriptions of its fields.

```
BASE24-POS  HYPERCOM T7 CONFIG PARAM LLLL      YY/MM/DD  HH:MM  02 OF 04
              TERMINAL GROUP:                RECORD TYPE:  01
```

```

INITIALIZATION CONTROL  00
PRINTER BAUD RATE      00
ECR BAUD RATE          00
CHECK GUARANTEE/SERVICE 00
LANGUAGE               00
HELP DESK PHONE #      000000000000000000000000
PASSWORD               0000
DOWN LINE LOAD NUMBER  00
ISSUER TABLE ID:      CHECK GUARANTEE/SERVICE 00  DEBIT 00
ACQUIRER TABLE ID:    CHECK GUARANTEE/SERVICE 00  DEBIT 00
AMOUNT DIGITS:         TRANSACTION 00              SETTLEMENT 00
CURRENCY SYMBOL        0
DECIMAL POINT POSITION  00
TELEPHONE LINE HOLD    00
DEFAULT TRANSACTION    00
```

```

***** BASE24 *****
FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F12-HELP
```

INITIALIZATION CONTROL — A code indicating whether the terminal is to clear all transactions from its batch storage. This is used to ensure that the terminal being used as a replacement terminal has no old transactions remaining in the batch. Valid values are as follows:

- 00 = No, do not clear transactions from batch storage or request a batch download.
- 01 = Yes, clear tables and request a batch download.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.INITIALIZATION-CNTL

PRINTER BAUD RATE — A code indicating the baud rate for the attached printer. Valid values are as follows:

00 = 4,800 bits per second
01 = 2,400 bits per second
02 = 1,200 bits per second

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.PRNTR-BAUD-RATE

ECR BAUD RATE — A code indicating the baud rate for an attached electronic cash register. Valid values are as follows:

00 = 4,800 bits per second
01 = 2,400 bits per second
02 = 1,200 bits per second

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.ECR-BAUD-RATE

CHECK GUARANTEE/SERVICE — A code indicating which check service is to be used or the type of identification needed to complete the transaction. Valid values are as follows:

00 = No service
01 = Telecredit with check MICR data
02 = TeleCheck with drivers license
03 = TeleCheck with account number
04 = JBS with check MICR data
07 = Bank card with account number

- 08 = Telecredit with drivers license
- 09 = Full MICR, including account number and routing number
- 11 = Generic identification
- 12 = Checking account with check MICR data
- 13 = Drivers license number
- 14 = ZIP code and telephone number
- 15 = Last four digits of MICR number and address
- 16 = Drivers license number and address
- 17 = Address and telephone number
- 18 = Credit card account number and expiration date
- 20 = NCN proprietary check reader with check MICR data
- 21 = NDCCHECK proprietary check reader with check MICR data
- 22 = NDCCHECK proprietary check reader with drivers license

Values 11–22 can be entered in this field; however, BASE24 does not use them in processing.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.CHK-SERVICE

LANGUAGE — A code indicating the language to be used by the terminal to display messages. Valid values are as follows:

- 00 = English
- 01 = Spanish
- 02 = Portuguese
- 03 = Chinese
- 04 = Thai

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.LANGUAGE-IND

HELP DESK PHONE # — The telephone number of the help desk that services the terminal group. The value in this field is left-justified and blank filled.

Field Length: 1–24 numeric characters

Required Field: Yes
Default Value: 000000000000000000000000
Data Name: HPTC.CONFIGURATION-PARAMS.HELP-DESK-
PHONE

PASSWORD — The password used when performing certain transactions at this terminal. This password is included in a terminal download and can be changed by the terminal operator. The range of valid values is 0000–9999.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000
Data Name: HPTC.CONFIGURATION-PARAMS.PASSWORD

DOWN LINE LOAD NUMBER — The value used by the terminal for tracking downline loads. The terminal sends this value in statistics messages.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.DLL-TRACKING-
NUM

ISSUER TABLE ID

The next two fields contain issuer table IDs for check service and debit card transactions.

CHECK GUARANTEE/SERVICE — The issuer table ID used for check service transactions. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.ISS-TBL-ID

DEBIT — The issuer table ID used for debit card transactions. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.DEBIT-ISS-TBL-ID

ACQUIRER TABLE ID

The next two fields contain acquirer table IDs for check service and debit card transactions.

CHECK GUARANTEE/SERVICE — The acquirer table ID used for check service transactions. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.ACQ-TBL-ID

DEBIT — The acquirer table ID used for debit card transactions. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.DEBIT-ACQ-TBL-ID

AMOUNT DIGITS

The next two fields contain the number of digits to be used for transaction and settlement amounts.

TRANSACTION — The number of digits allowed in the transaction amount entry or display. The range of valid values is 00–12.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.CRNCY-DIGITS-TRANS

SETTLEMENT — The number of digits allowed in the settlement amount entry or display. The range of valid values is 00–12.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.CRNCY-DIGITS-SETL

CURRENCY SYMBOL — The leading character used in the amount entry or display.

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.CRNCY-SYMBOL

DECIMAL POINT POSITION — The position (from the right) of the decimal point in the amount entry or display. The range of valid values is 00–03. The value 00 indicates no decimal point. Amounts with 2 places after the decimal point use a value of 03.

The value in this field must correspond to the value in the CURRENCY CODE field on BASE24-pos Terminal Data files (PTD) screen 2. For example, a code of 840 in the CURRENCY CODE field identifies the United States currency, which has 2 places after the decimal point. The corresponding value in the DECIMAL POINT POSITION field is 03.

Field Length: 2 numeric characters

Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.CRNCY-DEC-
POSITION

TELEPHONE LINE HOLD — A code indicating the number of seconds the phone line should be held open after a transaction. Valid values are as follows:

00 = 2 seconds
01 = 5 seconds
02 = 10 seconds
03 = 15 seconds
04 = 20 seconds
05 = 30 seconds
06 = 45 seconds
07 = 60 seconds
08 = 75 seconds
09 = 90 seconds
10 = 120 seconds
11 = 150 seconds
12 = 300 seconds
13 = 600 seconds
14 = 900 seconds
15 = Indefinitely

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.PHONE-LINE-HLD

DEFAULT TRANSACTION — A code indicating the terminal default transaction type. Valid values are as follows:

00 = Normal sale
01 = Cash
02 = Balance inquiry
03 = Offline

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.DFLT-TRAN

Configuration Parameters Table Screen 3

Hypercom T7 Configuration Parameters Table screen 3 enables you to set the terminal options for each terminal group. You can display this screen by pressing the **F9** key from Hypercom T7 Configuration Parameters Table screen 2. Pressing the **F9** key from this screen displays HPTC screen 4.

Hypercom T7 Configuration Parameters Table screen 3 is shown below, followed by descriptions of its fields.

```

BASE24-POS    HYPERCOM T7 CONFIG PARAM LLLL    YY/MM/DD  HH:MM  03 OF 04
      TERMINAL GROUP:                RECORD TYPE:  01

      TERMINAL OPTIONS : 0) DISABLE, 1) ENABLE
0 AUTO ANSWER          0 REFERRAL DIALING        0 MEMORY DIALING
0 TOLL DIALING          0 KEYBOARD DIALING        0 AMOUNT DUAL ENTRY
0 DISPLAY TRACK2 DATA  0 TIP PROCESSING         0 CASHIER PROCESSING
0 LODGING PROCESSING    0 TIME ON RECEIPT         0 PRINTER USED
0 KEYBOARD LOCKED       0 VOID PASSWORD DISABLE  0 REFUND PASSWORD DISABLE
0 ADJUST PASSWORD DISABLE 0 REPORT PASSWORD DISABLE 0 TOTAL ON RECEIPT
0 REFUND PROCESSED OFFLIN 0 VOID PROCESSED OFFLINE 0 RECEIPT PRE-PRINT
0 CHECK READER USED     0 ADJUST BY CASHIER    0 PROMPT SRVR NOT CASHIER
0 BATCH REVIEW BY CASHIER 0 TILL PROCESSING        0 CARD READ FROM PIN PAD
0 TAKE IMPRINT, KEY ENTRY 0 PRINT CARD VERIFY RCPT  0 AMOUNT VRFY ON PIN PAD
0 ECR INTERFACE         0 TENDER/CASH PROMPT
CARD VERIFY/DEBIT KEY  0                      RECON METHOD 0
BUSINESS DATE FORMAT   0
PRINTER TYPE    00                      PIN PAD TYPE: 00

***** BASE24 *****
      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
      F12-HELP

```

AUTO ANSWER — A code indicating whether the terminal supports automatic answering. Valid values are as follows:

- 0 = No, automatic answering is not allowed.
- 1 = Yes, automatic answering is allowed.

Field Length: 1 numeric character
 Required Field: Yes
 Default Value: 0
 Data Name: HPTC.CONFIGURATION-PARAMS.PHONE-DIAL-
 OPTION.AUTO-ANSWER

REFERRAL DIALING — A code indicating whether the terminal supports the dialing of the referral phone number with a referral response. Valid values are as follows:

- 0 = No, referral dialing is not allowed.
- 1 = Yes, referral dialing is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.PHONE-DIAL-
OPTION.REFERRAL-DIALING

MEMORY DIALING — A code indicating whether the terminal supports memory dialing. Valid values are as follows:

- 0 = No, memory dialing is not allowed.
- 1 = Yes, memory dialing is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.PHONE-DIAL-
OPTION.MEMORY-DIALING

TOLL DIALING — A code indicating whether the terminal supports toll dialing. Valid values are as follows:

- 0 = No, toll dialing is not allowed.
- 1 = Yes, toll dialing is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.PHONE-DIAL-
OPTION.TOLL-DIALING

KEYBOARD DIALING — A code indicating whether the terminal supports keyboard dialing. Valid values are as follows:

- 0 = No, keyboard dialing is not allowed.
- 1 = Yes, keyboard dialing is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.PHONE-DIAL-
OPTION.KBD-DIALING

AMOUNT DUAL ENTRY — A code indicating whether the terminal prompts the clerk to verify the transaction amounts by reentering the amount. Valid values are as follows:

- 0 = No, amount dual entry is not required.
- 1 = Yes, amount dual entry is required.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-1.
AMOUNT-DUAL-ENTRY

DISPLAY TRACK2 DATA — A code indicating whether the terminal displays Track 2 information when a card is passed through the card reader. Valid values are as follows:

- 0 = No, do not display Track 2 data.
- 1 = Yes, display Track 2 data.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-1.
DISPLAY-MSR

TIP PROCESSING — A code indicating whether the clerk is prompted to enter an additional amount (tip amount) on purchase transactions. Valid values are as follows:

- 0 = No, do not prompt for a tip amount.
- 1 = Yes, prompt for a tip amount.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-1.TIP-PROCESSING

CASHIER PROCESSING — A code indicating whether cashier processing is allowed by the terminal. Valid values are as follows:

- 0 = No, cashier processing is not allowed.
- 1 = Yes, cashier processing is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-1.CASHIER-PROCESSING

LODGING PROCESSING — A code indicating whether lodging processing is allowed at the terminal. Valid values are as follows:

- 0 = No, lodging processing is not allowed.
- 1 = Yes, lodging processing is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-1.LODGING-PROCESSING

TIME ON RECEIPT — A code indicating whether the time is to be printed on receipts. Valid values are as follows:

- 0 = No, the time is not to be printed on receipts.
- 1 = Yes, the time is to be printed on receipts.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-1.PRNT-TIM-ON-RECEIPT

PRINTER USED — A code indicating whether a printer is attached to the terminal. Valid values are as follows:

- 0 = No, a printer is not attached to the terminal.
- 1 = Yes, a printer is attached to the terminal.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-2.PRNTR-ATTACHED

KEYBOARD LOCKED — A code indicating whether the terminal keyboard is locked after a download and requires a password to unlock it. Valid values are as follows:

- 0 = No, the keyboard is not locked after a download.
- 1 = Yes, the keyboard is locked after a download.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.TERM-OPTS.KEYBOARD-LOCKED

VOID PASSWORD DISABLE — A code indicating whether a password is required for void transactions. Valid values are as follows:

- 0 = No, do not disable the password requirement (that is, require a password).
- 1 = Yes, disable the password requirement (that is, do not require a password).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.TERM-OPTS.VOID-PSWD-DIASBLE

REFUND PASSWORD DISABLE — A code indicating whether a password is required for refund transactions. Valid values are as follows:

- 0 = No, do not disable the password requirement (that is, require a password).
- 1 = Yes, disable the password requirement (that is, do not require a password).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.TERM-OPTS.RFND-PSWD-DISABLE

ADJUST PASSWORD DISABLE — A code indicating whether a password is required for adjustment transactions. Valid values are as follows:

- 0 = No, do not disable the password requirement (that is, require a password).
- 1 = Yes, disable the password requirement (that is, do not require a password).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.TERM-OPTS.ADJ-PSWD-DISABLE

REPORT PASSWORD DISABLE — A code indicating whether a password is required for report transactions. Valid values are as follows:

- 0 = No, do not disable the password requirement (that is, require a password).
- 1 = Yes, disable the password requirement (that is, do not require a password).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.TERM-OPTS.RPT-
PSWD-DISABLE

TOTAL ON RECEIPT — A code indicating whether the word *Total* is to be printed on the receipt. Valid values are as follows:

- 0 = No, do not print *Total* on the receipt.
- 1 = Yes, print *Total* on the receipt.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-2.TTL-
ON-RCPT

REFUND PROCESSED OFFLIN — A code indicating whether refunds can be processed offline. Valid values are as follows:

- 0 = No, refunds cannot be processed offline.
- 1 = Yes, refunds can be processed offline.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-2.RFND-
PROCESSED-OFFL

VOID PROCESSED OFFLINE — A code indicating whether voids can be processed offline. Valid values are as follows:

- 0 = No, voids cannot be processed offline.
- 1 = Yes, voids can be processed offline.

Field Length: 1 numeric character

Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-2.VOID-
PROCESSED-OFFL

RECEIPT PRE-PRINT — A code indicating whether receipt preprinting is allowed at the terminal. Valid values are as follows:

0 = No, receipt preprinting is not allowed.
1 = Yes, receipt preprinting is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAM

MS.OPTIONS-2.RCPT-PRE-PRNT

CHECK READER USED — A code indicating whether a check reader is installed. Valid values are as follows:

0 = No, a check reader is not installed.
1 = Yes, a check reader is installed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-3.
CHECK-READER-INSTALLED

ADJUST BY CASHIER — A code indicating whether the cashier can perform adjustments. Valid values are as follows:

0 = No, a cashier cannot perform adjustments.
1 = Yes, a cashier can perform adjustments.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-4.ADJ-
BY-CASHIER

PROMPT SRVR NOT CASHIER — A code indicating whether the server should be prompted instead of the cashier. Valid values are as follows:

- 0 = No, do not prompt the server, prompt the cashier.
- 1 = Yes, prompt the server, do not prompt the cashier.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-4.
PROMPT-SRVR-NOT-CASHIER

BATCH REVIEW BY CASHIER — A code indicating whether a cashier can perform a batch review. Valid values are as follows:

- 0 = No, a cashier cannot perform a batch review.
- 1 = Yes, a cashier can perform a batch review.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-4.BATCH-
REVIEW-BY-CASHIER

TILL PROCESSING — A code indicating whether a cashier is tied to a specific cash drawer (till). Valid values are as follows:

- 0 = No, a cashier is not tied to a specific cash drawer.
- 1 = Yes, a cashier is tied to a specific cash drawer.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-2.TILL-
PROCESSING

CARD READ FROM PIN PAD — A code indicating whether the card is read from the PIN pad. Valid values are as follows:

- 0 = No, the card is not read from the PIN pad.
- 1 = Yes, the card is read from the PIN pad.

The PIN PAD TYPE field on this screen must contain a valid value for this option to operate.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-3.CRD-
READ-FROM-PIN-PAD

TAKE IMPRINT, KEY ENTRY — A code indicating whether a card imprint is to be taken on a transaction that is entered manually. Valid values are as follows:

0 = No, do not take a card imprint.
1 = Yes, take a card imprint.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-3.TAKE-
IMPRINT

PRINT CARD VERIFY RCPT — A code indicating whether a receipt is to be printed for a card verify transaction. Valid values are as follows:

0 = No, do not print a receipt.
1 = Yes, print a receipt.

The following fields must contain specific values for this terminal option to operate:

- The PRINTER USED field on this screen must contain a value of 1.
- The PRINTER TYPE field on this screen must contain a valid value.
- The PRINT RECEIPT field on Issuer Table screen 2 must contain a value of 1.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-3.CRD-
VRFY-RCPT

AMOUNT VRFY ON PIN PAD — A code indicating whether the amount is to be verified on transactions when a PIN pad is used. Valid values are as follows:

- 0 = No, do not verify the amount.
- 1 = Yes, verify the amount.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-3.PIN-PAD-AMT-VRFY

ECR INTERFACE — A code indicating whether an electronic cash register interface is supported. Valid values are as follows:

- 0 = No, an electronic cash register interface is not supported.
- 1 = Yes, an electronic cash register interface is supported.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-3.ECR-INTERFACE

TENDER/CASH PROMPT — A code indicating whether the tender/cash prompt is supported. Valid values are as follows:

- 0 = No, the tender/cash prompt is not supported.
- 1 = Yes, the tender/cash prompt is supported.

The following fields must contain specific values for this terminal option to operate:

- The DEBIT TRANSACTION field on Issuer Table screen 3 must contain a value of 1.
- The CASH BACK field on Issuer Table screen 3 must contain a value of 1.

Field Length: 1 numeric character

Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-4.
TENDER-CASH-PROMPT

CARD VERIFY/DEBIT KEY — A code indicating whether the card verify key or the debit key is enabled. Valid values are as follows:

0 = Card verify key is enabled
1 = Debit key is enabled

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-2.CRD-
VRFY-OR-DEBIT-KEY

RECON METHOD — A code indicating the reconciliation method used by the terminal group. Valid values are as follows:

0 = Entered totals
1 = Confirmed totals

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-1.
RECONCILE-METHOD

BUSINESS DATE FORMAT — A code indicating the format used for business dates. Valid values are as follows:

0 = Month before day (MMDD)
1 = Day before month (DDMM)

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-1.BUS-
DAT-FRMT

PRINTER TYPE — A code indicating the type of printer used. Valid values are as follows:

00 = P4
01 = P7
02 = Roll printer
03 = VFI 250
04 = T7P 40 columns
05 = T7P 23 columns
06 = P8F
07 = P8S

Values 08 through 99 are reserved for future Hypercom support or custom support.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.PRINTER-TYPE

PIN PAD TYPE — A code indicating the type of PIN pad used. Valid values are as follows:

00 = S7 or S8
01 = Atalla version 1
02 = Atalla version 2
03 = Intellicut 6515
04 = Application 2000
05 = S7CR
06 = S8D

Values 07 through 99 are reserved for future Hypercom support or custom support.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.PIN-PAD-TYPE

Configuration Parameters Table Screen 4

Hypercom T7 Configuration Parameters Table screen 4 enables you to set the terminal options for each terminal group. You can display this screen by pressing the **F9** key from Hypercom T7 Configuration Parameters Table screen 3. Hypercom T7 Configuration Parameters Table screen 4 is the last screen for record type 01. Pressing the **F9** key from this screen displays HPTC screen 1.

Hypercom T7 Configuration Parameters Table screen 4 is shown below followed by descriptions of its fields.

```

BASE24-POS  HYPERCOM T7 CONFIG PARAM LLLL      YY/MM/DD  HH:MM  04 OF 04
      TERMINAL GROUP:                RECORD TYPE:  01

CASH BACK LIMIT  0000  MAINTENANCE PASSWORD  000000
NMS CALL SCHEDULE  00

___ EARLY AMT (EMV)      ___ OFFLINE PIN VERIFY
___ ALLOW PIN BYPASS    ___ ONLINE PIN VERIFY
___ ICC EMV ENABLED     ___ TERMINAL ONLINE CAPABLE
___ TERM WITH SCR       ___ CUSTOMER ACTIVATED

CURRENCY CODE ____  COUNTRY CODE ____  TERM CAPABILITIES ____
ADDNL TERM CAPABILITIES _____

***** BASE24 *****
FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F12-HELP

```

CASH BACK LIMIT — The maximum amount of cash back permitted for a transaction. The range of valid values is 0000–9999.

The following fields must contain specific values for this terminal option to operate:

- The TENDER/CASH PROMPT field on Configuration Parameters Table screen 3 must contain a value of 0.
- The DEBIT TRANSACTION field on Issuer Table screen 3 must contain a value of 1.

- The CASH BACK field on Issuer Table screen 3 must contain a value of 1.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000
Data Name: HPTC.CONFIGURATION-PARAMS.CASH-BACK-LMT

MAINTENANCE PASSWORD — The password used with terminal configuration and download transactions.

At least one of the first two characters entered in this field must be a nonzero value for the password to be valid.

Field Length: 6 numeric characters
Required Field: Yes
Default Value: 000000
Data Name: HPTC.CONFIGURATION-PARAMS.MAINT-PASSWORD

NMS CALL SCHEDULE — The number of days for the terminal to dial into a network management system (e.g., Hypercom Term-Master program), for scheduling purposes. The range of valid values is 00–19. A value of 00 indicates that call scheduling is disabled.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CONFIGURATION-PARAMS.NMS-CALL-SCHED

EARLY AMOUNT (EMV) — A code indicating whether there is an early amount entry for EMV transactions. Valid values are as follows:

- 0 = No early amount entry for EMV transactions.
1 = Yes, there is an early amount entry for EMV transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.ERLY-AMT-EMV

ALLOW PIN BYPASS — A code indicating whether PIN bypasses are allowed from the terminal. Valid values are as follows:

- 0 = Do not allow PIN bypass.
- 1 = Allow PIN bypass.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.ALW-PIN-BYP

ICC EMV ENABLED — A code indicating whether the terminal is ICC EMV enabled. Valid values are as follows:

- 0 = ICC EMV is disabled.
- 1 = ICC EMV is enabled

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.ICC-EMV-ENBL

TERM WITH SCR — A code indicating whether the terminal is SCR enabled. Valid values are as follows:

- 0 = Terminal is not SCR enabled.
- 1 = Terminal is SCR enabled.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.TRM-SCR

OFFLINE PIN VERIFY — A code indicating whether there is offline PIN verification. Valid values are as follows:

- 0 = Offline PIN verification is not allowed.
- 1 = Offline PIN verification is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.OFFLN-PIN-VRFY

ONLINE PIN VERIFY — A code indicating whether there is online PIN verification. Valid values are as follows:

- 0 = Online PIN verification is not allowed.
- 1 = Online PIN verification is allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.ONLN-PIN-VRFY

TERMINAL ONLINE CAPABLE — A code indicating whether the terminal is online capable. Valid values are as follows:

- 0 = The terminal is not online capable.
- 1 = The terminal is online capable.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.TERM-ONLN-CAP

CUSTOMER ACTIVATED — A code indicating whether the terminal is customer activated. Valid values are as follows:

- 0 = The terminal is not customer activated.
- 1 = The terminal is customer activated.

Field Length: 1 numeric character

Required Field: Yes
Default Value: 0
Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.CUST-ACTVT

CURRENCY CODE — The local currency used at the terminal. Refer to the vendor documentation for a complete set of valid values.

Field Length: 4 alphanumeric characters
Required Field: Yes
Default Value: 0000
Data Name: HPTC.CONFIGURATION-PARAMS.CURR-CDE

COUNTRY CODE — The location of the terminal. Refer to the vendor documentation for a complete set of valid values.

Field Length: 4 alphanumeric characters
Required Field: Yes
Default Value: 0000
Data Name: HPTC.CONFIGURATION-PARAMS.CNTRY-CDE

TERM CAPABILITIES — The standard capabilities of the terminal. Refer to the vendor documentation for a complete set of valid values.

Field Length: 6 alphanumeric characters
Required Field: Yes
Default Value: 000000
Data Name: HPTC.CONFIGURATION-PARAMS.TERM-CAP

ADDNL TERM CAPABILITIES — The additional capabilities of the terminal. Refer to the vendor documentation for a complete set of valid values.

Field Length: 10 alphanumeric characters
Required Field: Yes
Default Value: 0000000000
Data Name: HPTC.CONFIGURATION-PARAMS.ADDNL-TERM-CAP

Card Range Table Screen 2 Function Keys

The use of two function keys on Card Range Table screen 2 varies from the standard function keys explained in the ***BASE24 CRT Access Manual***. The use of these function keys is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F14	Forward — Moves forward to the next card range table entry. For example, if you are currently working on table entry 1, pressing the F14 key accesses table entry 2. If you are working on table entry 16, pressing the F14 key accesses table entry 1.
F15	Back — Moves back to the previous card range table entry. For example, if you are currently working on table entry 2, pressing the F15 key accesses table entry 1. If you are working on table entry 1, pressing the F15 key accesses table entry 16.

Card Range Table Screen 2

Hypercom T7 Card Range Table screen 2 enables you to configure the card range table for the terminal group. You can define up to 16 card prefix ranges.

You can display Hypercom T7 Card Range Table screen 2 by entering 02 or 03 in the RECORD TYPE field and pressing the **F2** key. Record type 02 is for the first 16 entries in the card range table; record type 03 is for the second 16 entries.

Hypercom T7 Card Range Table screen 2 is shown below, followed by descriptions of its fields.

```

BASE24-POS  HYPERCOM T7 CARD TABLE  LLLL      YY/MM/DD  HH:MM  02 OF 03
      TERMINAL GROUP:                RECORD TYPE:  02
      TABLE ENTRY:      1  OF  16

```

```

      PREFIX (LOW)      0000000000
      PREFIX (HIGH)     0000000000
      ISSUER TABLE ID   00
      ACQUIRER TABLE ID 00
      ISSUER TABLE ID #2 00
      ACQUIRER TABLE ID #2 00
      PAN LENGTH         00

```

```

***** BASE24 *****
      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
      F12-HELP

```

TABLE ENTRY — The position in the card range table. The range of valid values is 1–16. This field is controlled by the **F14** and **F15** keys. To advance to the next table entry (for example, move to table entry 2 from table entry 1), press the **F14** key. To go back to the previous table entry, press the **F15** key.

Field Length: System-protected

Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.CRD-RANGE-INDEX (when working with record type 02)

 HPTC.CARD-RANGE-2.CRD-RANGE-TBL.CRD-RANGE-INDEX (when working with record type 03)

PREFIX (LOW) — The lower limit of the card prefix range.

Field Length: 10 numeric characters
Required Field: Yes
Default Value: 0000000000
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.LOW-PREFIX (when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.LOW-PREFIX (when working with record type 03)

PREFIX (HIGH) — The upper limit of the card prefix range.

Field Length: 10 numeric characters
Required Field: Yes
Default Value: 0000000000
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.HIGH-PREFIX (when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.HIGH-PREFIX (when working with record type 03)

ISSUER TABLE ID — The issuer table entry to be used when selecting credit cards within the card prefix range. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.ISS-TBL-ID (when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.ISS-TBL-ID (when working with record type 03)

ACQUIRER TABLE ID — The acquirer table entry to be used when selecting credit cards within the card prefix range. The range of valid values is 00–99

Field Length: 2 numeric characters
Required Field: Yes

Default Value: 00
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.ACQ-TBL-ID
(when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.ACQ-TBL-ID
(when working with record type 03)

ISSUER TABLE ID # 2 — The debit issuer table entry to be used for dual (credit and debit) cards in the card prefix range. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.ISS-TBL-ID-2 (when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.ISS-TBL-ID-2 (when working with record type 03)

ACQUIRER TABLE ID # 2 — The debit acquirer table entry to be used for dual (credit and debit) cards in the card prefix range. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.ACQ-TBL-ID-2 (when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.ACQ-TBL-ID-2 (when working with record type 03)

PAN LENGTH — The length of the PAN for the card prefix range. The range of valid values is 00–19. A value of 00 indicates that the PAN length is not checked.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.PAN-LEN
(when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.PAN-LEN
(when working with record type 03)

Card Range Table Screen 3 Function Keys

The use of two function keys on Card Range Table screen 3 varies from the standard function keys explained in the BASE24 CRT Access Manual. The use of these keys is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F14	Forward — Moves forward to the next card range table entry. For example, if you are currently working on table entry 1, pressing the F14 key accesses table entry 2. If you are working on table entry 16, pressing the F14 key accesses table entry 1.
F15	Back — Moves back to the previous card range table entry. For example, if you are currently working on table entry 2, pressing the F15 key accesses table entry 1. If you are working on table entry 1, pressing the F15 key accesses table entry 16.

DEBIT Y/N PROMPT — A code indicating whether a prompt is displayed to identify a debit card transaction. Valid values are as follows:

- 0 = No, do not display a prompt for identifying debit card transactions.
- 1 = Yes, display a prompt for identifying debit card transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-1.
DEBIT-PROMPT (when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.OPTIONS-1.
DEBIT-PROMPT (when working with record type 03)

DEBIT KEY ONLY — A code indicating whether debit-key-only processing is supported. When the terminal uses debit-key-only processing, a clerk or cardholder can initiate a debit card transaction only by pressing the Debit key on the Hypercom terminal. Valid values are as follows:

- 0 = No, do not support debit-key-only processing.
- 1 = Yes, support debit-key-only processing.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-1.
DEBIT-KEY-ONLY (when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.OPTIONS-1.
DEBIT-KEY-ONLY (when working with record type 03)

PIN PAD DEBIT SELECT — A code indicating whether the customer can use the PIN pad to select a debit card account. Valid values are as follows:

- 0 = No, do not permit the customer to use the PIN pad to select a debit card account.
- 1 = Yes, permit the customer to use the PIN pad to select a debit card account.

Field Length: 1 numeric character
Required Field: Yes

Default Value: 0
Data Name: HPTC.CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-1.
DEBIT-PIN-PAD (when working with record type 02)
HPTC.CARD-RANGE-2.CRD-RANGE-TBL.OPTIONS-1.
DEBIT-PIN-PAD (when working with record type 03)

Issuer Table Screen 2 Function Keys

The use of two function keys on Issuer Table screen 2 varies from the standard function keys explained in the ***BASE24 CRT Access Manual***. The use of these function keys is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F14	Forward — Moves forward to the next issuer table entry. For example, if you are currently working on table entry 1, pressing the F14 key accesses table entry 2. If you are working on table entry 16, pressing the F14 key accesses table entry 1.
F15	Back — Moves back to the previous issuer table entry. For example, if you are currently working on table entry 2, pressing the F15 key accesses table entry 1. If you are working on table entry 1, pressing the F15 key accesses table entry 16.

Issuer Table Screen 2

Hypercom T7 Issuer Table screen 2 enables you to maintain issuer parameters of the issuer table for a terminal group. You can display Hypercom T7 Issuer Table screen 2 by entering 04 in the RECORD TYPE field and pressing the **F2** key.

Hypercom T7 Issuer Table screen 2 is shown below, followed by descriptions of its fields.

```

BASE24-POS  HYPERCOM T7 ISSUER TABLE LLLL      YY/MM/DD  HH:MM  02 OF 03
      TERMINAL GROUP:                RECORD TYPE:  04
      TABLE ENTRY:                   1  OF  16

ISSUER TABLE ID      00
CARD NAME
REFERRAL PHONE #
DEFAULT ACCOUNT TYPE  00
FLOOR LIMIT           0000
REAUTHORIZE MARGIN %  00
OPTIONS : 0) DISABLE, 1) ENABLE
0 CUSTOMER ACCT SELECTION 0  ADJUSTMENTS ALLOWED      0  BLOCK REFUNDS
0 PIN ENTRY REQUIRED      0  MOD 10 CHECK ON PAN        0  BLOCK CARD VERIFY
0 MANUAL PAN ENTRY ALLOWED 0  INVOICE # REQUIRED      0  PROMPT FOR TAX
0 EXPIRATION DATE REQUIRED 0  PRINT RECEIPT          0  IGNORE TRACK 1 ERROR
0 OFFLINE TRANS ALLOWED  0  CAPTURE TRANSACTION      0  AUTOREPRNT FAILED RCPT
0 VOICE REFERRAL ALLOWED  0  CHECK EXPIRATION DATE
0 PRODUCT CODES REQUIRED  0  BE MY GUEST

***** BASE24 *****
      FILE DESTINATION:                NEW LOGICAL NETWORK ID:
      F12-HELP

```

TABLE ENTRY — The position in the issuer table. The range of valid values is 1–16. This field is controlled by the **F14** and **F15** keys. To advance to the next table entry (for example, move to table entry 2 from table entry 1), press the **F14** key. To go back to the previous table entry, press the **F15** key.

Field Length: System-protected
 Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ISS-TBL-INDEX

ISSUER TABLE ID — The ID of the issuer table entry. This value must match the issuer table ID in the card range table. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ISS-TBL-ID

CARD NAME — The card name that is displayed to the terminal operator and printed on the receipt.

Field Length: 1–10 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.CRD-NAM

REFERRAL PHONE # — The telephone number used when a referral response code is sent to the terminal.

Field Length: 1–24 numeric characters with no embedded spaces
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.RFRL-PHONE-NUM

DEFAULT ACCOUNT TYPE — The data that is placed in the third position of the processing code for all financial transactions. This value is overwritten if an account is selected during transaction entry. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.DEFAULT-ACCT

FLOOR LIMIT — The floor limit for the terminal group. If the transaction amount is below the amount in this field, the terminal will approve the transaction locally and assign an approval code. If the transaction amount is more than the value in this field, the terminal sends the transaction to the host for authorization.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.FLOOR-LIMIT

REAUTHORIZE MARGIN% — The percentage used for reauthorizing a transaction after it has been adjusted. This field is reserved for future use.

Field Length: 2 numeric characters
Required Field: No
Default Value: 00
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.REAUTHORIZE-MARGIN

OPTIONS

The following fields are processing options that determine the interaction allowed or required between the Hypercom terminal and the terminal operator for the issuer table entry.

CUSTOMER ACCT SELECTION — A code indicating whether the terminal is to send the default account or is to prompt the customer to enter an account number at the PIN pad. Valid values are as follows:

- 0 = The terminal sends the default account.
- 1 = The customer enters the account number at PIN pad.

Note: A PIN pad is required if this field is set to 1.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.ACCT-SELECT

ADJUSTMENTS ALLOWED — A code indicating whether adjustments are allowed for authorization and capture transactions (processing type 1) only. Valid values are as follows:

- 0 = No, adjustments are not allowed.
- 1 = Yes, adjustments are allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.
ADJUSTMENT-ALLOWED

BLOCK REFUNDS — A code indicating whether refund transactions can be blocked. Valid values are as follows:

- 0 = No, do not block refund transactions.
- 1 = Yes, block refund transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.BLOCK-
RFND

PIN ENTRY REQUIRED — A code indicating whether a PIN entry is required. Valid values are as follows:

- 0 = No, a PIN entry is not required.
- 1 = Yes, a PIN entry is required.

The following fields must contain specific values for this terminal option to operate:

- The DEBIT TRANSACTION field on Issuer Table screen 3 must contain a value of 1.
- At least one of the PIN ENTRY NOT REQUIRED fields on Issuer Table screen 3 must contain a value of 0. The field for each transaction that requires a PIN must contain a value of 0.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.PIN-ENTRY

MOD 10 CHECK ON PAN — A code indicating whether the terminal should perform a MOD 10 check on the PAN. Valid values are as follows:

- 0 = No, do not perform a MOD 10 check on the PAN.
- 1 = Yes, perform a MOD 10 check on the PAN.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.MOD-10-CHECK

BLOCK CARD VERIFY — A code indicating whether card verification transactions can be blocked. Valid values are as follows:

- 0 = No, do not block card verification transactions.
- 1 = Yes, block card verification transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.BLOCK-CRD-VRFY

MANUAL PAN ENTRY ALLOWED — A code indicating whether manual PAN entry is allowed. Valid values are as follows:

- 0 = No, manual PAN entry is not allowed.
- 1 = Yes, manual PAN entry is allowed.

Field Length: 1 numeric character

Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.
MANUAL-PAN-ENTRY

INVOICE # REQUIRED — A code indicating whether the terminal prompts for an invoice number. Valid values are as follows:

0 = No, the terminal does not prompt for an invoice number.
1 = Yes, the terminal prompts for an invoice number.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.
INVOICE-NUM

PROMPT FOR TAX — A code indicating whether the terminal prompts for a tax amount. Valid values are as follows:

0 = No, the terminal does not prompt for a tax amount.
1 = Yes, the terminal prompts for a tax amount.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.TAX-
PROMPT

EXPIRATION DATE REQUIRED — A code indicating whether a card expiration date is required. The only valid value is 1 (expiration date required).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.EXP-DAT-
REQ

PRINT RECEIPT — A code indicating whether a receipt is printed. Valid values are as follows:

- 0 = No, do not print a receipt.
- 1 = Yes, print a receipt.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.RECEIPT

IGNORE TRACK 1 ERROR — A code indicating whether a Track 1 error should be ignored. Valid values are as follows:

- 0 = No, do not ignore a Track 1 error.
- 1 = Yes, ignore a Track 1 error.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.TRK1-
ERROR

OFFLINE TRANS ALLOWED — A code indicating whether offline transactions are allowed. Valid values are as follows:

- 0 = No, offline transactions are not allowed.
- 1 = Yes, offline transactions are allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.
OFFLINE-TRAN-ALLOW

CAPTURE TRANSACTION — A code indicating whether transactions are to be captured. Valid values are as follows:

- 0 = No, do not capture the transactions.
- 1 = Yes, capture the transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.
CAPTURE-TRANS

AUTOREPRINT FAILED RCPT — A code indicating whether a receipt that does not print properly should be reprinted automatically. Valid values are as follows:

- 0 = No, do not automatically reprint a receipt that does not print properly.
- 1 = Yes, automatically reprint a receipt that does not print properly.

The following fields must contain specific values for this terminal option to operate:

- The PRINTER USED field on Configuration Parameters Table screen 3 must contain a value of 1.
- The PRINTER TYPE field on Configuration Parameters Table screen 3 must contain a valid value.
- The PRINT RECEIPT field on this screen must contain a value of 1.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.AUTO-
REPRINT

VOICE REFERRAL ALLOWED — A code indicating whether a voice referral is allowed. Valid values are as follows:

- 0 = No, voice referral is not allowed.
- 1 = Yes, voice referral is allowed. The terminal dials the referral telephone number on a referral response.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.VOICE
REFERRAL

CHECK EXPIRATION DATE — A code indicating whether the terminal should check the expiration date. Valid values are as follows:

- 0 = No, do not check the expiration date.
- 1 = Yes, check the expiration date.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.CHK-EXP-DAT

PRODUCT CODES REQUIRED — A code indicating whether product code entry is required. Valid values are as follows:

- 0 = No, product code entry is not required.
- 1 = Yes, product code entry is required. The terminal prompts for a product code.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.PRODUCT-CODES

BE MY GUEST — A code indicating whether Be-My-Guest processing is allowed. Valid values are as follows:

- 0 = No, do not allow Be-My-Guest processing.
- 1 = Yes, allow Be-My-Guest processing.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.BE-MY-GUEST

Issuer Table Screen 3 Function Keys

The use of two function keys on Issuer Table screen 3 differ from the standard function keys explained in the *BASE24 CRT Access Manual*. The use of these keys is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F14	Forward — Moves forward to the next issuer table entry. For example, if you are currently working on table entry 1, pressing the F14 key accesses table entry 2. If you are working on table entry 16, pressing the F14 key accesses table entry 1.
F15	Back — Moves back to the previous issuer table entry. For example, if you are currently working on table entry 2, pressing the F15 key accesses table entry 1. If you are working on table entry 1, pressing the F15 key accesses table entry 16.

Issuer Table Screen 3

Hypercom T7 Issuer Table screen 3 contains processing options for the issuer table. You can display this screen by pressing the **F9** key from Hypercom T7 Issuer Table screen 2.

Hypercom T7 Issuer Table screen 3 is shown below, followed by descriptions of its fields.

```

BASE24-POS  HYPERCOM T7 ISSUER TABLE LLLL      YY/MM/DD  HH:MM  03 OF 03
      TERMINAL GROUP:                RECORD TYPE:  04
      TABLE ENTRY:                   1  OF  16

OPTIONS : 0) DISABLE, 1) ENABLE
0 BLOCK VOIDS                0 DEBIT TRANSACTION      0 AVS ENABLED ON SALE
0 CASH BACK                  0 VRFY LAST 4 PAN DGTS    0 CARDHOLDER ADDRESS
0 PRINT SIG NOT REQUIRED      0 AUTH OVER FLOOR LMT  0 CARDHOLDER ZIP CODE
0 PRINT NO RFNDS ALLOWED    0 PROMPT FOR CUST INFO
0 RFND PROCESSED OFFLINE    0 PROMPT FOR BAR AMOUNTS
0 VOID PROCESSED OFFLINE    0 ADNL RFND AMT PROMPTS
0 CASH ONLY TRAN ALLOWED

PIN ENTRY NOT REQUIRED                ADDITIONAL PROMPTS
0 DEBIT      0 CASHBACK              0 PROMPT 1    0 PROMPT 2
0 RETURN     0 BAL INQUIRY           0 PROMPT 3    0 PROMPT 4
0 VOID       0 PREAUTH               0 PROMPT 5    0 PROMPT 6
0 ADJUST                    0 PROMPT 7    0 PROMPT 8

***** BASE24 *****
      FILE DESTINATION:                NEW LOGICAL NETWORK ID:
      F12-HELP

```

TABLE ENTRY — The position in the issuer table. The range of valid values is 1–16. This field is controlled by the **F14** and **F15** keys. To advance to the next table entry (for example, move to table entry 2 from table entry 1), press the **F14** key. To go back to the previous table entry, press the **F15** key.

Field Length: System-protected
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ISS-TBL-INDEX

OPTIONS

The next 16 fields are processing options that determine the interaction allowed or required between the Hypercom terminal and the terminal operator for the issuer table entry.

BLOCK VOIDS — A code indicating whether void transactions are to be blocked. Valid values are as follows:

- 0 = No, do not block void transactions.
- 1 = Yes, block void transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.BLOCK-VOIDS

DEBIT TRANSACTION — A code indicating whether debit transactions are allowed. Valid values are as follows:

- 0 = No, do not allow debit transactions.
- 1 = Yes, allow debit transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.DEBIT-TRAN

AVS ENABLED ON SALE — A code indicating whether address verification is performed on a sale transaction. Valid values are as follows:

- 0 = No, address verification is not enabled.
- 1 = Yes, address verification is enabled.

The CARDHOLDER ADDRESS field on this screen, the CARDHOLDER ZIP CODE field on this screen, or both fields must contain a value of 1 for this terminal option to operate.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-6.AVS-ON-SALE

CASH BACK — A code indicating whether cash back transactions are allowed. Valid values are as follows:

- 0 = No, do not allow cash back transactions.
- 1 = Yes, allow cash back transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.CASH-
BACK

VRFY LAST 4 PAN DGTS — A code indicating whether the terminal is to perform PAN verification on the last four PAN digits. Valid values are as follows:

- 0 = No, do not perform PAN verification.
- 1 = Yes, perform PAN verification on the last four digits of the PAN.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.VRFY-
PAN

CARDHOLDER ADDRESS — A code indicating whether the cardholder's address is required to complete a transaction. Valid values are as follows:

- 0 = No, the cardholder's address is not required.
- 1 = Yes, the cardholder's address is required.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-6.
CRDHLDR-ADDR-REQ

PRINT SIG NOT REQUIRED — A code indicating whether to print the phrase *Signature not required* on the receipt. Valid values are as follows:

- 0 = No, do not print *Signature not required* on the receipt.
- 1 = Yes, print *Signature not required* on the receipt.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.PRNT-
SIG-NOT-RQD

AUTH OVER FLOOR LIMIT — A code indicating whether transactions that are over the floor limit can be authorized. Valid values are as follows:

- 0 = No, do not authorize transactions that are over the floor limit.
- 1 = Yes, authorize transactions that are over the floor limit.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.AUTH-
OVR-FLR

CARDHOLDER ZIP CODE — A code indicating whether the cardholder's ZIP Code is required to complete a transaction. Valid values are as follows:

- 0 = No, the cardholder's ZIP Code is not required.
- 1 = Yes, the cardholder's ZIP Code is required.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-6.
CRDHLDR-ZIP-CDE-REQ

PRINT NO RFNDS ALLOWED — A code indicating whether the message ****No refunds allowed* is to be printed on the receipt. Valid values are as follows:

- 0 = No, do not print ****No refunds allowed* on the receipt.
- 1 = Yes, print ****No refunds allowed* on the receipt.

Field Length: 1 numeric character

Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.PRNT-
NO-RFND-ALLOWED

PROMPT FOR CUST INFO — A code indicating whether a prompt for purchase order information is displayed for purchase card transactions. Valid values are as follows:

- 0 = No, do not display a prompt for purchase order information.
- 1 = Yes, display a prompt for purchase order information.

The DEBIT TRANSACTION field on this screen must contain a value of 1 for this terminal option to operate when the Mail or Telephone Order transaction is performed.

The DEBIT TRANSACTION field on this screen must contain a value of 0 for this terminal option to operate when a transaction other than the Mail or Telephone Order transaction is performed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-5.PO-NUM-
PROMPT

RFND PROCESSED OFFLINE — A code indicating whether refunds can be processed offline. Valid values are as follows:

- 0 = No, do not allow refunds to be processed offline.
- 1 = Yes, allow refunds to be processed offline.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.RFND-
PROCESSED-OFFL

PROMPT FOR BAR AMOUNTS — A code indicating whether a prompt for amounts purchased from the bar is displayed in the transaction sequence. Valid values are as follows:

- 0 = No, do not display a prompt for bar amounts.
- 1 = Yes, display a prompt for bar amounts.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-5.BAR-AMT-PROMPT

VOID PROCESSED OFFLINE — A code indicating whether void transactions can be processed offline. Valid values are as follows:

- 0 = No, do not allow void transactions to be processed offline.
- 1 = Yes, allow void transactions to be processed offline.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.VOID-PROCESSED-OFFL

ADNL RFND AMT PROMPTS — A code indicating whether an additional refund amount prompt is displayed. Valid values are as follows:

- 0 = No, do not display an additional refund amount prompt.
- 1 = Yes, display an additional refund amount prompt.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-5.ADNL-RFND-PROMPT

CASH ONLY TRAN ALLOWED — A code indicating whether cash-only transactions are allowed at the terminal. Valid values are as follows:

- 0 = No, cash-only transactions are not allowed.
- 1 = Yes, cash-only transactions are allowed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.CASH-ONLY-ALLOW

PIN ENTRY NOT REQUIRED

The following fields pertain to PIN entry for various transactions.

DEBIT — A code indicating whether not entering a PIN on debit card transactions is permitted. Valid values are as follows:

- 0 = No, do not permit debit card transactions without PIN entry (that is, PIN entry is required for debit card transactions).
- 1 = Yes, permit debit card transactions without PIN entry (that is, PIN entry is not required for debit card transactions).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.DEBIT

CASHBACK — A code indicating whether not entering a PIN on cash-back transactions is permitted. Valid values are as follows:

- 0 = No, do not permit cash-back transactions without PIN entry (that is, PIN entry is required for cash-back transactions).
- 1 = Yes, permit cash-back transactions without PIN entry (that is, PIN entry is not required for cash-back transactions).

Field Length: 1 numeric character

Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.
CASHBACK

RETURN — A code indicating whether not entering a PIN on return transactions is permitted. Valid values are as follows:

- 0 = No, do not permit return transactions without PIN entry (that is, PIN entry is required for return transactions).
- 1 = Yes, permit return transactions without PIN entry (that is, PIN entry is not required for return transactions).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.
RFND

BAL INQUIRY — A code indicating whether not entering a PIN on balance inquiry transactions is permitted. Valid values are as follows:

- 0 = No, do not permit balance inquiry transactions without PIN entry (that is, PIN entry is required for balance inquiry transactions).
- 1 = Yes, permit balance inquiry transactions without PIN entry (that is, PIN entry is not required for balance inquiry transactions).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.
BAL-INQ

VOID — A code indicating whether not entering a PIN on void transactions is permitted. Valid values are as follows:

- 0 = No, do not permit void transactions without PIN entry (that is, PIN entry is required for void transactions).
- 1 = Yes, permit void transactions without PIN entry (that is, PIN entry is not required for void transactions).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.
VOID

PREAUTH — A code indicating whether not entering a PIN on preauthorization transactions is permitted. Valid values are as follows:

- 0 = No, do not permit preauthorization transactions without PIN entry (that is, PIN entry is required for preauthorization transactions).
- 1 = Yes, permit preauthorization transactions without PIN entry (that is, PIN entry is not required for preauthorization transactions).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.
PREAUTH

ADJUST — A code indicating whether not entering a PIN on adjustment transactions is permitted. Valid values are as follows:

- 0 = No, do not permit adjustment transactions without PIN entry (that is, PIN entry is required for adjustment transactions).
- 1 = Yes, permit adjustment transactions without PIN entry (that is, PIN entry is not required for adjustment transactions).

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.
ADJ

ADDITIONAL PROMPTS

The following fields identify which prompts from Additional Prompt screen 2 appear for this issuer. Transactions in which the prompts appear are defined on Additional Prompt screen 2.

Note: The HPDH module downloads additional prompt configuration information to the Hypercom T7 terminal. However, the BASE24-pos standard product does not use the data entered by a terminal operator in response to an additional prompt.

PROMPT 1 — A code indicating whether prompt 1 from Additional Prompt screen 2 appears during transactions for this issuer. Valid values are as follows:

- 0 = No, prompt 1 does not appear.
- 1 = Yes, prompt 1 appears.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.
PROMPT1

PROMPT 2 — A code indicating whether prompt 2 from Additional Prompt screen 2 appears during transactions for this issuer. Valid values are as follows:

- 0 = No, prompt 2 does not appear.
- 1 = Yes, prompt 2 appears.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.
PROMPT2

PROMPT 3 — A code indicating whether prompt 3 from Additional Prompt screen 2 appears during transactions for this issuer. Valid values are as follows:

- 0 = No, prompt 3 does not appear.
- 1 = Yes, prompt 3 appears.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.
PROMPT3

PROMPT 4 — A code indicating whether prompt 4 from Additional Prompt screen 2 appears for this issuer. Valid values are as follows:

- 0 = No, prompt 4 does not appear.
- 1 = Yes, prompt 4 appears.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.
PROMPT4

PROMPT 5 — A code indicating whether prompt 5 from Additional Prompt screen 2 appears for this issuer. Valid values are as follows:

- 0 = No, prompt 5 does not appear.
- 1 = Yes, prompt 5 appears.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.
PROMPT5

PROMPT 6 — A code indicating whether prompt 6 from Additional Prompt screen 2 appears for this issuer. Valid values are as follows:

- 0 = No, prompt 6 does not appear.
- 1 = Yes, prompt 6 appears.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.
PROMPT6

PROMPT 7 — A code indicating whether prompt 7 from Additional Prompt screen 2 appears for this issuer. Valid values are as follows:

- 0 = No, prompt 7 does not appear.
- 1 = Yes, prompt 7 appears.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.
PROMPT7

PROMPT 8 — A code indicating whether prompt 8 from Additional Prompt screen 2 appears for this issuer. Valid values are as follows:

0 = No, prompt 8 does not appear.
1 = Yes, prompt 8 appears.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.
PROMPT8

Acquirer Table Screen 2 Function Keys

The use of two function keys on Acquirer Table screen 2 varies from the standard function keys explained in the *BASE24 CRT Access Manual*. The use of these keys is explained below.

The first column of information shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F14	Forward — Moves forward to the next acquirer table entry. For example, if you are currently working on table entry 1, pressing the F14 key accesses table entry 2. If you are working on table entry 8, pressing the F14 key accesses table entry 1.
F15	Back — Moves back to the previous acquirer table entry. For example, if you are currently working on table entry 2, pressing the F15 key accesses table entry 1. If you are working on table entry 1, pressing the F15 key accesses table entry 8.

Acquirer Table Screen 2

Hypercom T7 Acquirer Table screen 2 enables you to maintain the acquirer table for a card range. Acquirer table entries are downloaded to the terminal. You can display this screen by entering a valid terminal group in the **TERMINAL GROUP** field and 05 in the **RECORD TYPE** field and pressing the **F2** key.

Hypercom T7 Acquirer Table screen 2 is shown below, followed by a description of its fields.

```

BASE24-POS  HYPERCOM T7 ACQUIRER TBL LLLL      YY/MM/DD  HH:MM  02 OF 03
      TERMINAL GROUP:                RECORD TYPE:  05
      TABLE ENTRY:          1  OF  8
ACQUIRER TABLE ID              00
ACQUIRER PROGRAM NAME
ACQUIRER NAME
PRIMARY PHONE #
PRIMARY CONNECT TIME            00 (SECS)  PRIMARY DIAL ATTEMPTS            00
SECONDARY PHONE #
SECONDARY CONNECT TIME          00 (SECS)  SECONDARY DIAL ATTEMPTS            00
PRIMARY SETL PHONE #
PRIMARY SETL CONNECT TIME        00 (SECS)  PRIMARY SETL DIAL ATTEMPTS        00
SECONDARY SETL PHONE #
SECONDARY SETL CONNECT TIME      00 (SECS)  SECONDARY SETL DIAL ATTEMPTS      00
TERMINAL TIMEOUT                 00 (SECS)
NETWORK INTERNATIONAL ID         0000
MODEM MODE                       00      (BELL 212A 1200 BPS SDLC)
CARD ACCEPTOR ID CODE            CARD ACCEPTOR TERMINAL ID

***** BASE24 *****
      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
      F12-HELP

```

TABLE ENTRY — The position this entry holds in the acquirer table. The range of valid values is 1–8. This field is controlled by the **F14** and **F15** keys. To advance to the next table entry (for example, move to table entry 2 from table entry 1), press the **F14** key. To go back to the previous table entry, press the **F15** key.

Field Length: System-protected

Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.ACQ-TBL-INDEX

ACQUIRER TABLE ID — The ID that matches the acquirer ID in the card range table. The range of valid values is 00–99.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.ACQ-TBL-ID

ACQUIRER PROGRAM NAME — The program responsible for the communications with this acquirer. The only valid value is ACI.

Field Length: 1–10 alphabetic or numeric characters
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.ACQ-PROG-NAM

ACQUIRER NAME — The name used for display and printing purposes for this acquirer.

Field Length: 1–10 alphabetic or numeric characters
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.ACQ-NAM

PRIMARY PHONE # — The first telephone number the terminal dials for transactions. If no connection is made using this telephone number, the terminal uses the secondary telephone number.

Field Length: 1–24 numeric characters with no embedded spaces
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.PRI-TRANS-
PHONE-NUM

PRIMARY CONNECT TIME — The amount of time in seconds to wait for a connection after dialing the primary telephone number.

Field Length: 2 numeric characters

Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.PRI-TRANS-CNCT-TIM

PRIMARY DIAL ATTEMPTS — The number of times the terminal dials the primary telephone number before dialing the secondary telephone number.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.PRI-TRANS-DIAL-ATTEMPTS

SECONDARY PHONE # — The backup telephone number the terminal dials for transactions. The terminal dials this telephone number if there is no connection made with the primary telephone number after the specified number of dial attempts. If the terminal cannot get a connection with the secondary telephone number, the transaction is aborted.

Field Length: 1–24 numeric characters with no embedded spaces
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-TRANS-PHONE-NUM

SECONDARY CONNECT TIME — The amount of time in seconds to wait for a connection after dialing the secondary telephone number.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-TRANS-CNCT-TIM

SECONDARY DIAL ATTEMPTS — The number of times the terminal dials the secondary telephone number before aborting the transaction with a communications error.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-
TRANS-DIAL-ATTEMPTS

PRIMARY SETL PHONE # — The first telephone number the terminal dials for settlement. If no connection is made by the terminal, the secondary telephone number is dialed.

Field Length: 1–24 numeric characters with no embedded spaces
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.PRI-SETL-PHONE-
NUM

PRIMARY SETL CONNECT TIME — The amount of time in seconds to wait for a connection after dialing the primary settlement telephone number.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.PRI-SETL-CNCT-
TIM

PRIMARY SETL DIAL ATTEMPTS — The number of times the terminal dials the primary settlement telephone number before dialing the secondary settlement telephone number.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.PRI-SETL-DIAL-
ATTEMPTS

SECONDARY SETL PHONE # — The backup telephone number the terminal dials for settlement. The terminal dials this number if there is no connection made with the primary settlement telephone number after the specified number of dial attempts. If the terminal cannot make a connection with the secondary settlement telephone number, settlement is aborted.

Field Length: 1–24 numeric characters with no embedded spaces
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-SETL-
PHONE-NUM

SECONDARY SETL CONNECT TIME — The amount of time in seconds to wait for a connection after dialing the secondary settlement telephone number.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-SETL-
CNCT-TIM

SECONDARY SETL DIAL ATTEMPTS — The number of times the terminal dials the secondary settlement telephone number before aborting settlement.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-SETL-
DIAL-ATTEMPTS

TERMINAL TIMEOUT — The number of seconds in the host response timeout. If the terminal does not receive a response from the host in the number of seconds specified in this field, the transaction is aborted.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.TIM-OUT

NETWORK INTERNATIONAL ID — The value the terminal sends in the Network International ID (P-24) data element of the Hypercom External Message.

BASE24-pos does not use this value in processing. However, Hypercom devices require the Network International ID (P-24) data element in the Hypercom External Message.

Field Length: 4 numeric characters
 Required Field: Yes
 Default Value: 0000
 Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.NET-INTL-ID

MODEM MODE — A code identifying the mode the modem is utilizing. Refer to the *Hypercom T7 Standard Software Initialization Parameters* for additional information about these modem modes. Valid values are as follows:

00 = Bell 212A 1200 bps SDLC
 01 = Bell 212A 1200 bps Async (7 bits, 1 stop, E parity)
 02 = CCITT V.22 1200 bps SDLC
 03 = CCITT V.22 1200 bps Async (7 bits, 1 stop, E parity)
 04 = Bell 103 300 bps Async (7 bits, 1 stop, E parity)
 05 = CCITT V.21 300 bps Async (7 bits, 1 stop, E parity)
 06 = CCITT V.22bis 2400 bps SDLC
 07 = CCITT V.22bis 2400 bps Async (7 bits, 1 stop, E parity)
 08 = Bell V.22 BIS 2400 bps SDLC
 09 = Bell V.22 BIS 2400 bps Async (7 bits, 1 stop, E parity)
 10 = Bell 103 300 bps Async (7 bits, 1 stop, E parity)
 11 = Bell 103 300 bps Envoy
 12 = Bell 103 300 bps Visa II
 13 = CCITT V.21 300 bps Visa II
 14 = Bell 212A 1200 bps Visa II
 15 = CCITT V.22 1200 bps Visa II
 16 = Bell V.22 BIS 2400 bps Visa II
 17 = CCITT V.22 BIS 2400 bps Visa II
 18 = Bell 103 300 bps Envoy
 19 = Bell 103 300 bps Envoy
 20 = CCITT V.22 1200 bps SDLC
 21 = Bell 103 300 bps FDR
 22 = Bell 212A 1200 bps FDR
 23 = Bell 103 300 bps FDR
 24 = Bell 103 1200 bps Nabanco
 25 = Bell 212A 1200 bps Nabanco
 26 = Bell V.22 BIS 2400 bps Nabanco

27 = Bell 103 300 bps Nabanco
28 = Bell 212A 1200 bps Nabanco
29 = Bell V.22 BIS 2400 bps Nabanco
30 = Bell 212A 1200 bps NBD
31 = CCITT V.22 1200 bps SDLC
32 = Bell 212A 1200 bps SDLC (7 bits, 1 stop, E parity)
33 = Bell 212A 1200 bps Async (7 bits, 1 stop, E parity)
34 = Bell 212A 1200 bps FDR
35 = Bell 212A 1200 bps Envoy
36 = Bell 212A 1200 bps Async (7 bits, 1 stop, E parity)
37 = Bell 103 300 bps Advantis
38 = Bell 212A 1200 bps Advantis
39 = V.22 BIS 2400 bps Advantis
40 = V.22 1200 bps Epsco
41 = V.21 300 bps Visa/EMEA
42 = V.22 1200 bps Visa/EMEA
43 = Bell 212A 1200 bps MagCard
44 = V.23 1200 bps Cafis
45 = Bell 103/212A 300/1200 bps FDC
46 = Bell 212A 1200 bps Geisco
47 = Bell 212A 1200 bps Envoy
48 = V.22 1200 bps Envoy
49 = V.22 1200 bps Visa I (7 bits, 1 stop, E parity)

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.MODEM-MDE

CARD ACCEPTOR ID CODE — The card acceptor identification code the terminal sends in the Card Acceptor ID (P-42) data element of the Hypercom External Message.

Field Length: 1–15 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.CRD-ACCPT-ID-CDE

CARD ACCEPTOR TERMINAL ID — The card acceptor terminal identification the terminal sends in the Card Acceptor Terminal ID (P-41) data element of the Hypercom External Message.

Field Length:	1–8 alphanumeric characters
Required Field:	No
Default Value:	No default value
Data Name:	HPTC.ACQ-TABLE.ACQ-TBL-DATA.CRD-ACCPT-TERM-ID

Acquirer Table Screen 3 Function Keys

The use of two function keys on Acquirer Table screen 3 varies from the standard function keys explained in the *BASE24 CRT Access Manual*. The use of these function keys is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this key.

Key	Description
F14	Forward — Moves forward to the next acquirer table entry. For example, if you are currently working on table entry 1, pressing the F14 key accesses table entry 2. If you are working on table entry 8, pressing the F14 key accesses table entry 1.
F15	Back — Moves back to the previous acquirer table entry. For example, if you are currently working on table entry 2, pressing the F15 key accesses table entry 1. If you are working on table entry 1, pressing the F15 key accesses table entry 8.

Acquirer Table Screen 3

Hypercom T7 Acquirer Table screen 3 enables you to set processing options for the acquirer table. You can display Hypercom T7 Acquirer Table screen 3 by pressing the **F9** key from Hypercom T7 Acquirer Table screen 2.

Hypercom T7 Acquirer Table screen 3 is shown below, followed by descriptions of its fields.

```

BASE24-POS  HYPERCOM T7 ACQUIRER TBL LLLL      YY/MM/DD  HH:MM  03 OF 03
      TERMINAL GROUP:                RECORD TYPE:  05
      TABLE ENTRY:                   1  OF  8

OPTIONS : 0) DISABLE, 1) ENABLE
0 IGNORE HOST TIME

PREVIOUS SETTLE DAY    00  (DD)
SETTLE TIME            0000 (HHMM)
VISA I TERMINAL ID
SETL MODEM MODE       00 (BELL 212A 1200 BPS SDLC)
MAX AUTO-SETL ATTEMPTS 00
MASTER PIN KEY INDEX
MASTER MAC KEY INDEX
PIN KEY EXCHANGE METHOD 0 (TERM-MASTER)
MAC KEY EXCHANGE METHOD 0 (TERM-MASTER)

***** BASE24 *****
      FILE DESTINATION:                NEW LOGICAL NETWORK ID:
      F12-HELP

```

TABLE ENTRY — The position this entry holds in the Acquirer table. The range of valid values is 1–8. This field is controlled by the **F14** and **F15** keys. To advance to the next table entry (for example, move to table entry 2 from table entry 1), press the **F14** key. To go back to the previous table entry, press the **F15** key.

Field Length: System-protected
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.ACQ-TBL-INDEX

OPTIONS

The following field is a processing option in the acquirer table.

IGNORE HOST TIME — A code indicating whether to ignore the host time.
Valid values are as follows:

- 0 = No, do not ignore host time.
- 1 = Yes, ignore host time.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-1.
IGNORE-HOST-TIM

PREVIOUS SETTLE DAY — The day of the month when settlement previously occurred. The range of valid values is 00–31.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.PREV-SETL-DAY

SETTLE TIME — The time (hhmm) when settlement automatically occurs. The time 0000 disables settlement.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.SETL-TIM

VISA I TERMINAL ID — The merchant or terminal ID used for this acquirer when the host uses a Visa I message format. This field is not supported.

Field Length: 1–23 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.VISA-I-TERM-ID

SETL MODEM MODE — A code identifying the mode the modem is using for settlement. Refer to the *Hypercom T7 Standard Software Initialization Parameters* for additional information about these modem modes. Valid values are as follows:

- 00 = Bell 212A 1200 bps SDLC
- 01 = Bell 212A 1200 bps Async (7 bits, 1 stop, E parity)
- 02 = CCITT V.22 1200 bps SDLC
- 03 = CCITT V.22 1200 bps Async (7 bits, 1 stop, E parity)
- 04 = Bell 103 300 bps Async (7 bits, 1 stop, E parity)
- 05 = CCITT V.21 300 bps Async (7 bits, 1 stop, E parity)
- 06 = CCITT V.22bis 2400 bps SDLC
- 07 = CCITT V.22bis 2400 bps Async (7 bits, 1 stop, E parity)
- 08 = Bell V.22 BIS 2400 bps SDLC
- 09 = Bell V.22 BIS 2400 bps Async
- 10 = Bell 103 300 bps Async
- 11 = Bell 103 300 bps Envoy
- 12 = Bell 103 300 bps Visa II
- 13 = CCITT V.21 300 bps Visa II
- 14 = Bell 212A 1200 bps Visa II
- 15 = CCITT V.22 1200 bps Visa II
- 16 = Bell V.22 BIS 2400 bps Visa II
- 17 = CCITT V.22 BIS 2400 bps Visa II
- 18 = Bell 103 300 bps Envoy
- 19 = Bell 103 300 bps Envoy
- 20 = CCITT V.22 1200 bps SDLC
- 21 = Bell 103 300 bps FDR
- 22 = Bell 212A 1200 bps FDR
- 23 = Bell 103 300 bps FDR
- 24 = Bell 103 1200 bps Nabanco
- 25 = Bell 212A 1200 bps Nabanco
- 26 = Bell V.22 BIS 2400 bps Nabanco
- 27 = Bell 103 300 bps Nabanco
- 28 = Bell 212A 1200 bps Nabanco
- 29 = Bell V.22 BIS 2400 bps Nabanco
- 30 = Bell 212A 1200 bps NBD
- 31 = CCITT V.22 1200 bps SDLC
- 32 = Bell 212A 1200 bps SDLC
- 33 = Bell 212A 1200 bps Async
- 34 = Bell 212A 1200 bps FDR
- 35 = Bell 212A 1200 bps Envoy
- 36 = Bell 212A 1200 bps Async
- 37 = Bell 103 300 bps Advantis
- 38 = Bell 212A 1200 bps Advantis

39 = V.22 BIS 2400 bps Advantis
40 = V.22 1200 bps Epsco
41 = V.21 300 bps Visa/EMEA
42 = V.22 1200 bps Visa/EMEA
43 = Bell 212A 1200 bps MagCard
44 = V.23 1200 bps Cafis
45 = Bell 103/212A 300/1200 bps FDC
46 = Bell 212A 1200 bps Geisco
47 = Bell 212A 1200 bps Envoy
48 = V.22 1200 bps Envoy
49 = V.22 1200 bps Visa I

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.SETL-MODEM-MDE

MAX AUTO-SETL ATTEMPTS — The number of times the terminal attempts to settle automatically. Attempts are made every five minutes.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.MAX-AUTO-SETL

MASTER PIN KEY INDEX — The index of the master PIN key used to encrypt the PIN key. BASE24 supports the use of only one PIN key. The range of valid values is 1–9 and A–C.

Field Length: 1 alphabetic or numeric character
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.PIN-KEY-INDEX

MASTER MAC KEY INDEX — The index of the master MAC key used to encrypt the MAC key. BASE24 supports the use of only one MAC key. The range of valid values is 1–9 and A–C.

Field Length: 1 alphabetic or numeric character
Required Field: Yes
Default Value: No default value
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.MAC-KEY-INDEX

PIN KEY EXCHANGE METHOD — A code indicating where the master PIN key originates. Valid values are as follows:

0 = Hypercom Term-Master program
1 = Host (BASE24 is treated as a host by the Hypercom T7 device)

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-2.PIN-KEY-EXCHNG

MAC KEY EXCHANGE METHOD — A code indicating where the master MAC key originates. Valid values are as follows:

0 = Hypercom Term-Master program
1 = Host (BASE24 is treated as a host by the Hypercom T7 device)

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: HPTC.ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-2.MAC-KEY-EXCHNG

Product Data Screen 2

Hypercom T7 Product Data screen 2 enables you to define and maintain the product codes stored in the description table. Product data is defined by the customer and is downloaded to the terminal.

You can use the product codes to distinguish between the goods and services offered by a retailer. For example, a service station can choose one product code to track petroleum transactions, another to track food transactions, and another to track mechanical services. Retailers can then separate their transactions according to the product code categories they set up on Hypercom T7 Product Data screen 2.

You can display Hypercom T7 Product Data screen 2 by entering a valid terminal group in the **TERMINAL GROUP** field and 06 in the **RECORD TYPE** field and pressing the **F2** key.

Hypercom T7 Product Data screen 2 is shown below, followed by descriptions of its fields.

```
BASE24-POS  HYPERCOM T7 PRODUCT DATA LLLL      YY/MM/DD  HH:MM  02 OF 02
              TERMINAL GROUP:                RECORD TYPE:  06
```

SELECT KEY	PRODUCT CODE	PRODUCT DESCRIPTION
-----	-----	-----
00	00	00000000000000000000
01	00	00000000000000000000
02	00	00000000000000000000
03	00	00000000000000000000
04	00	00000000000000000000
05	00	00000000000000000000
06	00	00000000000000000000
07	00	00000000000000000000
08	00	00000000000000000000
09	00	00000000000000000000
B	00	00000000000000000000

```
***** BASE24 *****
              FILE DESTINATION:          NEW LOGICAL NETWORK ID:
              F12-HELP
```


SELECT KEY — The terminal key associated with the table. The range of valid values is 00–09 and the letter B.

Field Length: System-protected
Occurs: 11 times
Data Name: HPTC.PROD-TABLE.PROD-DATA.SELCT-KEY

PRODUCT CODE — A user-defined code identifying a retailer's product or line of business. This code is sent by the terminal in requests if the product code is configured in the issuer table. The range of valid values is 00–99.

Field Length: 2 numeric characters
Occurs: 11 times
Required Field: No
Default Value: 00
Data Name: HPTC.PROD-TABLE.PROD-DATA.CDE

PRODUCT DESCRIPTION — The product description associated with the product code.

Field Length: 1–20 alphanumeric characters
Occurs: 11 times
Required Field: No
Default Value: 00000000000000000000
Data Name: HPTC.PROD-TABLE.PROD-DATA.DESCRPTION

Additional Prompt Screen 2

Hypercom T7 Additional Prompt screen 2 enables you to define and maintain additional data prompts for transaction processing.

Note: The HPDH module downloads additional prompt configuration information to the Hypercom T7 terminal. However, the BASE24-pos standard product does not use the data entered by a terminal operator in response to an additional prompt.

You can display Hypercom T7 Additional Prompt screen 2 by entering a valid terminal group in the **TERMINAL GROUP** field and 07 in the **RECORD TYPE** field and pressing the **F2** key.

Hypercom T7 Additional Prompt screen 2 is shown below, followed by descriptions of its fields.

```

BASE24-POS  HYPERCOM T7 ADDL PROMPT  LLLL      YY/MM/DD  HH:MM  02 OF 02
          TERMINAL GROUP:                RECORD TYPE:  07

          TERMINAL OPTIONS : 0) DISABLE, 1) ENABLE

          C  V  P          R  O  A  S
          A  R  R  C  D  F  F  U  E
PROMPT      PROMPT      DATA  LGTH  LGTH  S  F  E  H  E  N  F  T  T
#   ID      TEXT IF ENABLED  TYPE  LOW  HIGH  H  Y  A  K  B  D  L  H  L
1              00  00  00  0  0  0  0  0  0  0  0  0  0  0
2              00  00  00  0  0  0  0  0  0  0  0  0  0  0
3              00  00  00  0  0  0  0  0  0  0  0  0  0  0
4              00  00  00  0  0  0  0  0  0  0  0  0  0  0
5              00  00  00  0  0  0  0  0  0  0  0  0  0  0
6              00  00  00  0  0  0  0  0  0  0  0  0  0  0
7              00  00  00  0  0  0  0  0  0  0  0  0  0  0
8              00  00  00  0  0  0  0  0  0  0  0  0  0  0

***** BASE24 *****
          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
          F12-HELP

```

TERMINAL OPTIONS

The following fields define additional data prompts for transaction processing.

PROMPT # — The number that identifies the prompt information. The range of valid values is 1–8.

The corresponding ADDITIONAL PROMPTS field on Issuer Table screen 3 must contain a value of 1 for this terminal option to operate. Each value in the PROMPT # field has a separate ADDITIONAL PROMPTS field on Issuer Table screen 3.

Field Length: System-protected
Occurs: 8 times
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-
NUM

PROMPT ID — The value that is sent in the transaction detail to define the prompt. This value is sent instead of the prompt text.

Field Length: 2 alphanumeric characters
Occurs: 8 times
Required Field: No
Default Value: No default value
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-
ID

PROMPT TEXT IF ENABLED — The text that is displayed as a prompt at the terminal.

Field Length: 1–20 alphanumeric characters
Occurs: 8 times
Required Field: No
Default Value: No default value
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-
TEXT

DATA TYPE — A code that identifies the type of data input accepted in response to the prompt. Valid values are as follows:

00 = All input is accepted
01 = Only numbers are accepted
02 = Alphanumeric characters are accepted

Field Length: 2 numeric characters
Occurs: 8 times

Required Field: Yes
Default Value: 00
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.DATA-
TYPE

INPT LGTH LOW — The minimum length of the data input accepted in response to the prompt. The range of valid values is 00–30.

Field Length: 2 numeric characters
Occurs: 8 times
Required Field: Yes
Default Value: 00
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.INPUT-
LGTH-MIN

INPT LGTH HIGH — The maximum length of the data input accepted in response to the prompt. The range of valid values is 00–30.

Field Length: 2 numeric characters
Occurs: 8 times
Required Field: Yes
Default Value: 00
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.INPUT-
LGTH-MAX

CASH — A code that identifies whether the prompt should be displayed with a cash transaction. Valid values are as follows:

0 = No, do not display the prompt with a cash transaction.
1 = Yes, display the prompt with a cash transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-1.CASH

VRFY — A code that identifies whether the prompt should be displayed with a card verification transaction. Valid values are as follows:

- 0 = No, do not display the prompt with a card verification transaction.
- 1 = Yes, display the prompt with a card verification transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-1.CRD-VRFY

PREA — A code that identifies whether the prompt should be displayed with a preauthorization transaction. Valid values are as follows:

- 0 = No, do not display the prompt with a preauthorization transaction.
- 1 = Yes, display the prompt with a preauthorization transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-1.PRE-AUTH

CHK — A code that identifies whether the prompt should be displayed with a check transaction. Valid values are as follows:

- 0 = No, do not display the prompt with a check transaction.
- 1 = Yes, display the prompt with a check transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-1.CHK

DEB — A code that identifies whether the prompt should be displayed with a debit transaction. Valid values are as follows:

- 0 = No, do not display the prompt with a debit transaction.
- 1 = Yes, display the prompt with a debit transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-1.DB

RFND — A code that identifies whether the prompt should be displayed with a refund transaction. Valid values are as follows:

- 0 = No, do not display the prompt with a refund transaction.
- 1 = Yes, display the prompt with a refund transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-1.RFND

OFFL — A code that identifies whether the prompt should be displayed with an offline sale transaction. Valid values are as follows:

- 0 = No, do not display the prompt with an offline sale transaction.
- 1 = Yes, display the prompt with an offline sale transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-1.OFFL-SALE

AUTH — A code that identifies whether the prompt should be displayed with a sale transaction. Valid values are as follows:

- 0 = No, do not display the prompt with a sale transaction.
- 1 = Yes, display the prompt with a sale transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-1.AUTH-SALE

SETL — A code that identifies whether the prompt should be displayed with a settlement transaction. Valid values are as follows:

- 0 = No, do not display the prompt with a settlement transaction.
- 1 = Yes, display the prompt with a settlement transaction.

Field Length: 1 numeric character
Occurs: 8 times
Required Field: Yes
Default Value: 0
Data Name: HPTC.ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-
SELECT-2.SETL

ACI Worldwide, Inc.

A: American Express Standard Industry Formats

The AMEX Data Collection option enables customers to perform draft capture on transactions originating from American Express cardholders. Generally, American Express has defined categories under which transactions are placed. The categories supported by the HPDH module are as follows:

- Lodging
- Restaurant
- General retail

For each of these categories, American Express requires different data to be sent from the device. Therefore, American Express has set forth standard industry formats for each category to ensure the data they require is captured by the device and sent to BASE24.

American Express also supports a Descriptive Billing File (DBF) that is used to submit charges to American Express without submitting paper. The format of the DBF conforms to American Express standard. The HPDH module is able to provide some of the information required in these standard industry formats and, subsequently, capture and process transactions sent using them. The HPDH module fills the fields for which it cannot provide information with blanks. It is then the responsibility of the host to determine if additional information is required to build the DBF.

In conjunction with this file, the customer must set up Standard Industrial Classification (SIC) Codes or Merchant Category Codes in the BASE24-pos Terminal Data files (PTD). SIC and Merchant Category Codes identify the merchant's line of business. Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about the PTD.

Lodging

The following format is used for the lodging industry.

Field	Length	Description
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 11 for lodging transactions.
reference-cde	9(9)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. The range of valid values is 000000000–999999999.
charge-type	9(1)	A code that indicates the type of charge made. Valid values are as follows: 1 = Lodging 2 = Restaurant 3 = Gift shop 0, 4–9 = Reserved
tab-roc-id	X(10)	The original record of charge (ROC), invoice, or another number used to identify the original transaction. This is an internal American Express code.
tax-amt	9(18)	The amount of sales tax charged for the transaction. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
arrival-dat	9(6)	The scheduled arrival date (MMDDYY).
depart-dat	9(6)	The scheduled departure date (MMDDYY).
room-rate	9(18)	The per diem rate charged for the customer's stay at the establishment. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.

Field	Length	Description
program-cde	9(1)	A code indicating the reason for the charge. If there are no special circumstances that require a code, a value of 1 is placed in this field. Valid values are as follows: 1 = Used if no other code pertains 2 = Assured reservation—no show 3 = CARDeposit 4 = Delayed charge 5 = Express service 6 = Assured reservation 0, 7–9 = Reserved

Restaurant

The following format is used by the restaurant industry.

Field	Length	Description
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 12 for restaurant transactions.
reference-cde	9(9)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. The range of valid values is 000000000–999999999.
charge-cde1	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
tab-roc-id	X(10)	The original record of charge (ROC), invoice, or another number used to identify the original transaction. This is an internal American Express code.
description-cde	X(1)	A code that describes the nature of the transaction. Valid values are as follows: 0 = Food 1 = Food and beverage 2 = Gift certificate 3–9 = User-defined A–Z = User-defined
tax-amt	9(18)	The amount of sales tax charged for the transaction. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.

Field	Length	Description
food-amt	9(18)	The total cost of food, or food and beverage if these charges are combined on the original record of charge (ROC). This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
beverage-amt	9(18)	The total cost of beverages, if itemized in the DESCRIPTION-CDE field. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
tip1-cde	X(1)	A code mapped by the host to a description that designates which employee received a tip.
tip2-cde	X(1)	A code mapped by the host to a description that designates which employee received a second tip itemized on the original record of charge (ROC).
tip1-amt	9(18)	The total amount of the tip described in the TIP1-CDE field. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
tip2-amt	9(18)	The total amount of the tip described in the TIP2-CDE field. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.

General Retail

The following format is used by the general retail industry.

Field	Length	Description
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 20 for general retail transactions.
tax-amt	9(18)	The amount of sales tax charged for the transaction. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
reference-cde	9(9)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. The range of valid values is 000000000–999999999.
charge-cde1	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
tab-roc-id	X(10)	The original record of charge (ROC), invoice, or another number used to identify the original transaction. This is an internal American Express code.
charge-cde2	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
charge-cde3	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.

Field	Length	Description
charge-cde4	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
charge-cde5	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.

ACI Worldwide, Inc.

B: Odd-Parity Adjustment Algorithm

Use the following procedures to perform odd-parity adjustment on the PIN master key.

1. Select a master key of 16 hexadecimal characters. The example 1D2F45B708C31A62 is used throughout this appendix.
2. Use the following table to convert the hexadecimal characters to Binary Coded Decimal:

Hex Value	Binary Code Value	Hex Value	Binary Code Value
0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111

The example 1D2F45B708C31A62 would be converted as follows:

1	D	2	F	4	5	B	7	0	8	C	3	1	A	6	2
0001	1101	0010	1111	0100	0101	1011	0111	0000	1000	1100	0011	0001	1010	0110	0010

3. Combine the four-bit strings into eight-bit blocks. Eight bits equal 1 byte. The bits in the previous example would produce the following combinations:

00011101 00101111 01000101 10110111 00001000 11000011 00011010 01100010

4. Drop the leftmost bit of each byte.
5. Count the number of 1s in the seven-bit blocks. The number of 1s in the previous example is as follows:

_0011101	_0101111	_1000101	_0110111	_0001000	_1000011	_0011010	_1100010
4	5	3	5	1	3	3	3

6. If the number of 1s in the block is even, put a 1 in the leftmost bit. If the number of 1s is odd, put a 0 in the leftmost bit. The first bits for the previous example would be as follows:

_0011101	_0101111	_1000101	_0110111	_0001000	_1000011	_0011010	_1100010
4	5	3	5	1	3	3	3
even	odd	odd	odd	odd	odd	odd	odd
10011101	00101111	01000101	00110111	00001000	01000011	00011010	01100010

7. Break the eight-bit blocks into four-bit blocks.
8. Translate the resulting four character blocks to hexadecimal characters using the table in step 2. The translation for the blocks in the previous example would be as follows:

1001	1101	0010	1111	0100	0101	0011	0111	0000	1000	0100	0011	0001	1010	0110	0010
9	D	2	F	4	5	3	7	0	8	4	3	1	A	6	2

The odd-parity adjusted master key is 9D2F453708431A62.

Index

A

- Acquirer Processing Code File (APCF)
 - description of, [2-2](#)
 - file usage, [3-7](#)
 - routing, [3-3](#)
- Acquirer table, T7 terminal, [5-43](#)
- Additional data prompt table, T7 terminal, [5-48](#)
- Address verification
 - overview, [1-8](#)
 - response data, [4-80](#)
- Allowed transaction checking, [3-2](#)
- ALTERATTRIBUTE command, [3-11](#)
- American Express data collection
 - external message mapping, [4-83](#)
 - formats, [A-1](#)
 - introduction to, [1-7](#)
- American Express standard industry formats, [A-1](#)
- APCF
 - see* Acquirer Processing Code File (APCF)

B

- Balancing Hypercom terminals, [6-11](#)
- BASE24 transaction security options, [1-10](#)
- BASE24-pos Terminal Data Dynamic File—general data (PTDD1), [2-4](#)
- BASE24-pos Terminal Data Dynamic File—scratch pad (PTDD2), [2-4](#)
- BASE24-pos Terminal Data files (PTD)
 - description of, [2-4](#)
 - file usage, [3-8](#)
 - transaction support, [1-14](#)
- BASE24-pos Terminal Data Static File—general data (PTDS1), [2-4](#)
- Bit map, [4-12](#)

C

- Card range table, T7 terminal, [5-34](#)
- Check verification methods, [4-86](#)

- Configuration files
 - maintaining the Hypercom Configuration File (HPCF), [8-1](#)
 - maintaining the Hypercom T7 Configuration File (HPTC), [9-1](#)
 - overview, [2-2](#)
- Cutover, [1-9](#)

D

- Data element, [4-13](#)
- DEACTIVATE FLAG command, [3-11](#)
- Derived unique key per transaction (DUKPT), [6-3](#)
- Descriptive Billing File (DBF)
 - data source for, [4-83](#)
 - description of, [A-1](#)
- Descriptor table, T7 terminal, [5-47](#)
- Download data, [5-1](#)
- Download examples
 - T4, T5, and T6 terminals, [5-7](#)
 - T7 terminals, [5-51](#)
- Download options, [1-7](#)
- Download response messages
 - acquirer table, [5-43](#)
 - additional data prompt table, [5-48](#)
 - Card Acceptor Name (P43) data element, [5-2](#)
 - card range table, [5-34](#)
 - descriptor table, [5-47](#)
 - issuer table, [5-35](#)
 - Private Use (P-60) data element, [5-2](#)
 - Private Use (P-61) data element, [5-4](#)
 - Private Use (P-62) data element, [5-5](#)
 - Private Use (P-63) data element, [5-6](#)
 - T7 message format, [5-14](#)
- Download support, [1-7](#)
- Draft capture options
 - Authorization and draft capture, [6-9](#)
 - Authorization only with paper follow-up, [6-9](#)
 - description of, [1-8](#)
- Draft capture options, configuration considerations, [6-8](#)

E

Event messages
as a feature, [1-11](#)
generation of, [3-14](#)

H

Handshaking, [6-15](#)
HPCF
see Hypercom Configuration File (HPCF)
HPDH
see Hypercom Device Handler (HPDH) module
HPTC
see Hypercom T7 Configuration File (HPTC)
Hypercom Configuration File (HPCF)
Card Definition Table screen, [8-16](#)
Configuration Parameters screen, [8-3](#)
file usage, [3-7](#)
overview, [2-2](#)
Product Data screen, [8-13](#)
Range Definition Table screen, [8-27](#)
Hypercom Device Handler (HPDH) module
process overview, [3-2](#)
supported protocols, [1-4](#)
supported terminals, [1-3](#)
transaction support, [1-14](#)
Hypercom External Message
see also Message structures
see also Message text format
Hypercom External Message overview, [4-2](#)
Hypercom Request/Response Codes, [4-28](#)
Hypercom T7 Configuration File (HPTC)
Acquirer Table screen 2, [9-62](#)
Acquirer Table screen 3, [9-71](#)
Additional Prompt screen 2, [9-78](#)
Card Range Table screen 2, [9-31](#)
Card Range Table screen 3, [9-35](#)
Configuration Parameters Table screen 2, [9-5](#)
Configuration Parameters Table screen 3, [9-12](#)
download data, [9-2](#)
file usage, [3-7](#)
Issuer Table screen 2, [9-39](#)
Issuer Table screen 3, [9-49](#)
overview, [2-2](#)
Product Data screen 2, [9-76](#)
screen 1, [9-4](#)
Hypercom terminals
configuring T4, T5, and T6 terminals, [8-1](#)
configuring T7 terminals, [9-1](#)
supported terminals, [1-3](#)

I

IDF
see Institution Definition File (IDF)
Initialization, [3-9](#)

Institution Definition File (IDF)
description of, [2-3](#)
file usage, [3-7](#)
Integration with BASE24-pos, [1-19](#)
Issuer table, T7 terminal, [5-35](#)

L

LOADFLAGOFF command, [3-13](#)
LOADFLAGON command, [3-12](#)
Log Message Configuration File (LMCF)
description of, [2-3](#)
file usage, [3-7](#)
Logical Network Configuration File (LCONF)
assigns, [2-6](#)
file usage, [3-7](#)
params, [2-7](#)
Logical Network Configuration File (LCONF) assigns
APCF, [2-6](#)
KEYD, [2-6](#)
POS-DH-PTD-EXTMEM-SWAPVOL, [2-7](#)
POS-HPDH-HPCF, [2-6](#)
POS-HPDH-HPTC, [2-6](#)
POS-PTD-DYN-GNRL, [2-7](#)
POS-PTD-DYN-SCRATCH-PAD, [2-7](#)
POS-PTD-STATIC-GNRL, [2-7](#)
Logical Network Configuration File (LCONF) params
POS-DH-DUKPT-UPDATE-METHOD, [2-7](#)
POS-DH-KEYD-READ-FROM-DISK, [2-7](#)
POS-DH-KSN-APPRV-OPT, [2-7](#)
POS-DH-KSN-CNTR-APPRV-OPT, [2-8](#)
POS-DH-KSN-DESCR, [2-8](#)
POS-DH-KSN-MAX-DIFFERENCE, [2-8](#)
POS-DH-MAX-TERMS, [2-8](#)
POS-DH-MAX-TERMS-IN-DYN-TBL, [2-8](#)
POS-DH-MAX-TERMS-IN-STATIC-TBL, [2-9](#)
POS-HPDH-AUTH-RESP-TIMEOUT, [2-10](#)
POS-HPDH-LOG-TERM-STATS, [2-10](#)
POS-HPDH-MAX-HPCFS, [2-10](#)
POS-HPDH-MAX-HPTCS, [2-10](#)
POS-HPDH-NAC-IDentifier, [2-10](#)
POS-HPDH-NMS-IDentifier, [2-10](#)
POS-HPDH-OLDMSG-TIMEOUT, [2-10](#)

M

Message authentication codes (MACs)
failed MAC procedure, [6-7](#)
generating a new MAC, [6-6](#)
introduction, [3-4](#)
setting up message authentication, [6-6](#)
Message sequencing, [1-10](#)
Message structures
adjustment sale transaction, [4-55](#)
authorization transaction, [4-29](#)
balance inquiry transaction, [4-33](#)
batch upline load transaction, [4-60](#)
card verification transaction, [4-35](#)
cash transaction, [4-39](#)

Message structures *continued*

- check guarantee transaction, [4-57](#)
- check verification transaction, [4-59](#)
- guarantee late arrival transaction, [4-47](#)
- logon transaction, [4-67](#)
- offline cash transaction, [4-51](#)
- offline sale transaction, [4-49](#)
- parameter download transaction—T4, T5, and T6 terminals, [4-69](#)
- parameter download transaction—T7 terminals, [4-70](#)
- preauthorization transaction, [4-31](#)
- refund transaction, [4-45](#)
- reversal transaction, [4-63](#)
- sale and cash back transaction, [4-41](#)
- sale completion transaction, [4-53](#)
- sale transaction, [4-37](#)
- settlement transaction, [4-64](#)
- table headings used in descriptions, [4-27](#)
- test transaction, [4-66](#)
- void sale transaction, [4-43](#)

Message text format

- authorization messages, [4-11](#)
- bit map, [4-12](#)
- file update or transfer messages, [4-11](#)
- financial transaction messages, [4-11](#)
- headings used in descriptions, [4-14](#)
- message type identifier, [4-10](#)
- NMS messages, [4-12](#)
- primary data elements, [4-13](#)
- reconciliation messages, [4-12](#)
- reversal messages, [4-12](#)

Messages specific to Hypercom, [1-18](#)

Multiple language support, [1-6](#)

Multiple terminal support, [1-6](#)

N

Network Access Controller (NAC)

- introduction to, [1-2](#)
- link to HPDH, [1-20](#)

Network Management System (NMS)

- configuration, [6-18](#)
- messages, [4-5](#)
- overview, [1-2](#)

Network text commands, [3-11](#)

O

Odd-parity adjustment algorithm, [B-1](#)

P

Parametric authorization, [1-19](#)

Passthrough configuration example, [6-16](#)

PIN encryption, [6-2](#)

POS Retailer Definition File (PRDF), [2-3](#)

POS Standard Internal Message (PSTM)

- external message mapping to, [4-2](#)
- in data element descriptions, [4-29](#)

POS Transaction Log File (PTLF)

- description of, [2-5](#)
- file usage, [3-8](#)

PRDF

- see* POS Retailer Definition File (PRDF)

Primary data element, [4-13](#)

Primary data field

- see* Primary data element

Protocols, supported, [1-4](#)

PTD

- see* BASE24-pos Terminal Data files (PTD)

PTDD1

- see* BASE24-pos Terminal Data Dynamic File—general data (PTDD1)

PTDD2

- see* BASE24-pos Terminal Data Dynamic File—scratch pad (PTDD2)

PTDS1

- see* BASE24-pos Terminal Data Static File—general data (PTDS1)

PTLF

- see* POS Transaction Log File (PTLF)

R

RESETSTATUS command, [3-13](#)

Reversal processing, [3-5](#)

Routing transactions, APCF routing, [3-3](#)

S

Sequence number checking, [6-4](#)

Settlement, [1-9](#)

STATUSEXTMEM command, [3-13](#)

T

Table IDs

- address verification request data (54), [4-79](#)
- address verification response data (55), [4-80](#)
- cash back amount (41), [4-76](#)
- CPS 2000 data (30), [4-75](#)
- customer ID (68), [4-82](#)
- CVV2 processing data (16), [4-74](#)
- description of, [4-72](#)
- EBT response data (48), [4-78](#)
- EBT terminal data (47), [4-78](#)
- level II commercial card data subfield (73), [4-82](#)
- level II commercial card indicator (15), [4-73](#)
- lodging details (11), [4-73](#)
- lodging program indicator (13), [4-73](#)
- miscellaneous amount (53), [4-79](#)
- payment plan (45), [4-77](#)
- payment services (PS2000) data (20), [4-74](#)

Table IDs *continued*

- PS2000 terminal data (21), [4-75](#)
- purchase order number (68), [4-82](#)
- surcharge amount (44), [4-77](#)
- tax amount 1 (39), [4-76](#)

Terminal balancing, [1-9](#)

Terminal configuration table, T7 terminal, [5-23](#)

TLF token, [3-5](#)

Token handling, retrieving tokens, [3-5](#)

Tokens included in PSTM mapping, [4-29](#)

Transaction accumulation totals, [6-11](#)

Transaction flow

- HPDH versions, [1-22](#)
- Router/Authorization process, [1-19](#)

Transaction message flows

- NMS passthrough completes successfully, [7-9](#)
- NMS unable to deliver message to NAC, [7-11](#)
- normal transaction with response, [7-2](#)
- normal transaction without response—terminal times out, [7-3](#)
- normal transaction—NAC unable to deliver response to terminal, [7-7](#)
- normal transaction—unable to deliver response to terminal, [7-5](#)
- settlement—batch in balance, [7-13](#)
- settlement—draft capture in balance and business out of balance, [7-15](#)
- settlement—draft capture totals out of balance, [7-17](#)

Transaction support

- administrative transactions, [1-16](#)
- expanded transaction set, [1-6](#)
- financial transactions, [1-14](#)

Transport protocol data unit (TPDU), [4-3](#)

W

Warmboot, [3-10](#)

WARMBOOT PTD command, [3-11](#)

Warmboot PTD processing, [3-10](#)

Field Name Index

Numerics

0-9 CUST # ON RECEIPT, [8-12](#)

A

ACQUIRER NAME, [9-63](#)

ACQUIRER PROGRAM NAME, [9-63](#)

ACQUIRER TABLE ID

Acquirer Table screen 2, [9-63](#)

Card Range Table screen 2, [9-32](#)

Configuration Parameters Table screen 2, [9-9](#)

ACQUIRER TABLE ID # 2, [9-33](#)

ADDITIONAL PROMPTS, [9-57](#)

ADJUST, [9-57](#)

ADJUST BY CASHIER, [9-19](#)

ADJUST PASSWORD DISABLE, [9-17](#)

ADJUSTMENTS ALLOWED

Card Definition Table screen, [8-21](#)

Issuer Table screen 2, [9-42](#)

ADNL RFND AMT PROMPTS, [9-54](#)

AMOUNT DIGITS, [9-9](#)

AMOUNT DUAL ENTRY, [9-14](#)

AMOUNT VRFY ON PIN PAD, [9-22](#)

AMT DUAL ENTRY, [8-10](#)

AUTH, [9-83](#)

AUTH OVER FLOOR LIMIT, [9-52](#)

AUTO ANSWER, [9-12](#)

AUTOREPRNT FAILED RCPT, [9-46](#)

AVS ENABLED ON SALE, [9-50](#)

B

BACKUP PHONE #, [8-17](#)

BAL INQUIRY, [9-56](#)

BALANCE INQUIRY, [8-9](#)

BATCH REVIEW BY CASHIER, [9-20](#)

BE MY GUEST, [9-47](#)

BLOCK CARD VERIFY, [9-43](#)

BLOCK REFUNDS, [9-42](#)

BLOCK VOIDS, [9-50](#)

BUSINESS DATE FORMAT, [9-23](#)

C

CAPTURE TRANSACTION, [9-45](#)

CARD ACCEPTOR ID, [8-20](#)

CARD ACCEPTOR ID CODE, [9-68](#)

CARD ACCEPTOR TERMINAL ID, [9-69](#)

CARD DEFINITION TABLE INDEX, [8-29](#)

CARD NAME, [8-18](#), [9-40](#)

CARD READ FROM PIN PAD, [9-20](#)

CARD VERIFY/DEBIT KEY, [9-23](#)

CARDHOLDER ADDRESS, [9-51](#)

CARDHOLDER ZIP CODE, [9-52](#)

CASH, [9-80](#)

CASH BACK, [9-51](#)

CASH BACK LIMIT, [9-25](#)

CASH ONLY TRAN ALLOWED, [9-55](#)

CASHBACK, [9-55](#)

CASHIER PROCESSING, [9-15](#)

CHECK EXPIRATION DATE, [9-47](#)

CHECK GUARANTEE

Configuration Parameters screen, [8-5](#)

Configuration Parameters Table screen 2, [9-6](#), [9-8](#), [9-9](#)

CHECK READER USED, [9-19](#)

CHK, [9-81](#)

CURRENCY SYMBOL, [9-10](#)

CUSTOMER ACCT, [8-20](#)

CUSTOMER ACCT SELECTION, [9-41](#)

D

DATA TYPE, [9-79](#)

DEB, [9-82](#)

DEBIT

Configuration Paramters Table screen 2, Acquirer Table ID, [9-9](#)

Configuration Paramters Table screen 2, Issuer Table ID, [9-9](#)

Issuer Table screen 3, [9-55](#)

DEBIT KEY ONLY, [9-36](#)

DEBIT TRANSACTION, [9-50](#)

DEBIT Y/N PROMPT, [9-36](#)

DECIMAL POINT POSITION, [9-10](#)

DEFAULT ACCOUNT, [8-18](#)
DEFAULT ACCOUNT TYPE, [9-40](#)
DEFAULT TRAN, [8-6](#)
DEFAULT TRANSACTION, [9-11](#)
DIALING, [8-10](#)
DISP MAG STRIPE, [8-11](#)
DISPLAY TRACK2 DATA, [9-14](#)
DLL REVISION #, [8-5](#)
DOWN LINE LOAD NUMBER, [9-8](#)

E

ECR BAUD RATE
 Configuration Parameters screen, [8-5](#)
 Configuration Parameters Table screen 2, [9-6](#)
ECR INTERFACE, [9-22](#)
ENTRIES, [8-28](#)
EXPIRATION DATE REQ'D, [8-23](#)
EXPIRATION DATE REQUIRED, [9-44](#)

F

FLOOR LIMIT
 Card Definition Table screen, [8-19](#)
 Issuer Table screen 2, [9-41](#)

H

HELP DESK PHONE, [8-4](#)
HELP DESK PHONE #, [9-7](#)

I

IGNORE HOST TIME, [9-72](#)
IGNORE TRACK 1 ERROR, [9-45](#)
INITIAL PASSWORD, [8-4](#)
INITIALIZATION CONTROL
 Configuration Parameters screen, [8-3](#)
 Configuration Parameters Table screen 2, [9-5](#)
INPT LGTH HIGH, [9-80](#)
INPT LGTH LOW, [9-80](#)
INVOICE # REQUIRED
 Card Definition Table screen, [8-23](#)
 Issuer Table screen 2, [9-44](#)
ISSUER TABLE ID
 Card Range Table screen 2, [9-32](#)
 Configuration Parameters Table screen 2, [9-8](#)
 Issuer Table screen 2, [9-40](#)
ISSUER TABLE ID # 2, [9-33](#)

K

KEYBOARD DIALING, [9-14](#)

KEYBOARD LOCKED
 Configuration Parameters screen, [8-7](#)
 Configuration Parameters Table screen 3, [9-16](#)

L

LANGUAGE, [9-7](#)
LODGING PROCESSING, [9-15](#)

M

MAC KEY EXCHANGE METHOD, [9-75](#)
MAC REQUIRED, [8-25](#)
MAINTENANCE PASSWORD, [9-26](#)
MANUAL PAN ENTRY ALLOWED
 Card Definition Table screen, [8-22](#)
 Issuer Table screen 2, [9-43](#)
MASTER MAC KEY INDEX, [9-75](#)
MASTER PIN KEY INDEX, [9-74](#)
MAX AUTO-SETL ATTEMPTS, [9-74](#)
MEMORY DIALING
 Configuration Parameters screen, [8-8](#)
 Configuration Parameters Table screen 3, [9-13](#)
MOD 10 CHECK, [8-22](#)
MOD 10 CHECK ON PAN, [9-43](#)
MODEM MODE, [9-67](#)

N

NETWORK INT'L ID, [8-20](#)
NETWORK INTERNATIONAL ID, [9-67](#)
NMS CALL SCHEDULE, [9-26](#)

O

OFFL, [9-82](#)
OFFLINE TRANS ALLOWED
 Card Definition Table screen, [8-23](#)
 Issuer Table screen 2, [9-45](#)
OPTIONS
 Acquirer Table screen 3, [9-71](#)
 Issuer Table screen 2, [9-41](#)
 Issuer Table screen 3, [9-49](#)

P

PAN LENGTH, [9-33](#)
PASSWORD, [9-8](#)
PIN ENTRY NOT REQUIRED, [9-55](#)
PIN ENTRY REQUIRED, [9-42](#)
PIN KEY EXCHANGE METHOD, [9-75](#)
PIN PAD CONTROL, [8-7](#)
PIN PAD DEBIT SELECT, [9-36](#)
PIN PAD TYPE, [9-24](#)

PIN REQUIRED, [8-21](#)
PREA, [9-81](#)
PREAUTH, [9-57](#)
PREFIX (HIGH)
 Card Definition Table screen, [8-17](#)
 Card Range Table screen 2, [9-32](#)
 Range Definition Table screen, [8-28](#)
PREFIX (LOW)
 Card Definition Table screen, [8-17](#)
 Card Range Table screen 2, [9-32](#)
 Range Definition Table screen, [8-28](#)
PREVIOUS SETTLE DAY, [9-72](#)
PRIMARY CONNECT TIME, [9-63](#)
PRIMARY DIAL ATTEMPTS, [9-64](#)
PRIMARY PHONE #
 Acquirer Table screen 2, [9-63](#)
 Card Definition Table screen, [8-17](#)
PRIMARY SETL CONNECT TIME, [9-65](#)
PRIMARY SETL DIAL ATTEMPTS, [9-65](#)
PRIMARY SETL PHONE #, [9-65](#)
PRINT 'TOTAL' ON RCPT, [8-11](#)
PRINT CARD VERIFY RCPT, [9-21](#)
PRINT NO RFNDS ALLOWED, [9-52](#)
PRINT RECEIPT
 Card Definition Table screen, [8-25](#)
 Issuer Table screen 2, [9-45](#)
PRINT SIG NOT REQUIRED, [9-51](#)
PRINTER BAUD RATE, [9-6](#)
PRINTER CONTROL, [8-4](#)
PRINTER TYPE, [9-24](#)
PRINTER USED, [9-16](#)
PROCESSING TYPE, [8-19](#)
PRODUCT CODE
 Product Data screen, [8-14](#)
 Product Data screen 2, [9-77](#)
PRODUCT CODES, [8-21](#)
PRODUCT CODES REQUIRED, [9-47](#)
PRODUCT DESCRIPTION
 Product Data screen, [8-14](#)
 Product Data screen 2, [9-77](#)
PROMPT #, [9-78](#)
PROMPT 1, [9-58](#)
PROMPT 2, [9-58](#)
PROMPT 3, [9-58](#)
PROMPT 4, [9-59](#)
PROMPT 5, [9-59](#)
PROMPT 6, [9-59](#)
PROMPT 7, [9-59](#)
PROMPT 8, [9-60](#)
PROMPT FOR BAR AMOUNTS, [9-54](#)
PROMPT FOR CUST INFO, [9-53](#)

PROMPT FOR TAX, [9-44](#)
PROMPT ID, [9-79](#)
PROMPT SRVR NOT CASHIER, [9-20](#)
PROMPT TEXT IF ENABLED, [9-79](#)
PURCHASE CASH BACK, [8-24](#)

R

REAUTHORIZE MARGIN %
 Configuration Parameters screen, [8-6](#)
 Issuer Table screen 2, [9-41](#)
RECEIPT PRE-PRINT, [9-19](#)
RECONCILIATION METHOD, [9-23](#), [9-26](#)
RECORD TYPE, [9-4](#)
REFERRAL DIALING
 Configuration Parameters screen, [8-7](#)
 Configuration Parameters Table screen 3, [9-13](#)
REFERRAL PHONE #
 Card Definition Table screen, [8-18](#)
 Issuer Table screen 2, [9-40](#)
REFUND PASSWORD DISABLE
 Configuration Parameters screen, [8-9](#)
 Configuration Parameters Table screen 3, [9-17](#)
REFUND PROCESSED OFFLIN, [9-18](#)
REFUNDS OFFLINE, [8-12](#)
REPORT PASSWORD DISABLE, [9-17](#)
RESTAURANT TIP, [8-8](#)
RETURN, [9-56](#)
RFND, [9-82](#)
RFND PROCESSED OFFLINE, [9-53](#)

S

SECONDARY CONNECT TIME, [9-64](#)
SECONDARY DIAL ATTEMPTS, [9-65](#)
SECONDARY PHONE #, [9-64](#)
SECONDARY SETL CONNECT TIME, [9-66](#)
SECONDARY SETL DIAL ATTEMPTS, [9-66](#)
SECONDARY SETL PHONE #, [9-66](#)
SELECT KEY
 Product Data screen, [8-13](#)
 Product Data screen 2, [9-77](#)
SETL, [9-83](#)
SETL MODEM MODE, [9-73](#)
SETTLE TIME, [9-72](#)
SETTLEMENT, [9-10](#)
SUPER PASSWORD, [8-6](#)

T

TABLE ENTRY

- Acquirer Table screen 2, [9-62](#)
- Acquirer Table screen 3, [9-71](#)
- Card Range Table screen 2, [9-31](#)
- Card Range Table screen 3, [9-35](#)
- Issuer Table screen 2, [9-39](#)
- Issuer Table screen 3, [9-49](#)
- Range Definition Table screen, [8-28](#)

TABLE ENTRY 01 OF 16, [8-16](#)

TAKE IMPRINT, KEY ENTRY, [9-21](#)

TELEPHONE LINE HOLD, [9-11](#)

TENDER/CASH PROMPT, [9-22](#)

TERM GROUP, [8-3](#)

TERMINAL GROUP, [9-4](#)

TERMINAL OPTIONS

- Additional Prompt screen 2, [9-78](#)
- Configuration Parameters screen, [8-7](#)

TERMINAL TIMEOUT

- Acquirer Table screen 2, [9-66](#)
- Card Definition Table screen, [8-20](#)

TILL PROCESSING, [9-20](#)

TIME ON RECEIPT, [9-16](#)

TIP PROCESSING, [9-15](#)

TOLL DIALING

- Configuration Parameters screen, [8-9](#)
- Configuration Parameters Table screen 3, [9-13](#)

TOTAL ON RECEIPT, [9-18](#)

TRACK SENT, [8-19](#)

TRAN SRCH PWD DISABLE, [8-10](#)

TRANSACTION, [9-10](#)

V

VAL PAN LGTH OFFL DISABLE, [8-11](#)

VISA I TERMINAL ID, [9-72](#)

VOICE REFERRAL ALLOWED, [9-46](#)

VOICE REFERRALS, [8-24](#)

VOID, [9-56](#)

VOID PASSWORD DISABLE

- Configuration Parameters screen, [8-8](#)
- Configuration Parameters Table screen 3, [9-17](#)

VOID PROCESSED OFFLINE

- Configuration Parameters Table screen 3, [9-18](#)
- Issuer Table screen 3, [9-54](#)

VOIDS OFFLINE, [8-12](#)

VRFY, [9-81](#)

VRFY LAST 4 PAN DGTS, [9-51](#)

Data Name Index

A

ACQ-TABLE.ACQ-TBL-DATA.ACQ-NAM, [9-63](#)
ACQ-TABLE.ACQ-TBL-DATA.ACQ-PROG-NAM, [9-63](#)
ACQ-TABLE.ACQ-TBL-DATA.ACQ-TBL-ID, [9-63](#)
ACQ-TABLE.ACQ-TBL-DATA.ACQ-TBL-INDEX, [9-62](#), [9-71](#)
ACQ-TABLE.ACQ-TBL-DATA.CRD-ACCPT-ID-CDE, [9-68](#)
ACQ-TABLE.ACQ-TBL-DATA.CRD-ACCPT-TERM-ID, [9-69](#)
ACQ-TABLE.ACQ-TBL-DATA.MAC-KEY-INDEX, [9-75](#)
ACQ-TABLE.ACQ-TBL-DATA.MAX-AUTO-SETL, [9-74](#)
ACQ-TABLE.ACQ-TBL-DATA.MODEM-MDE, [9-68](#)
ACQ-TABLE.ACQ-TBL-DATA.NET-INTL-ID, [9-67](#)
ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-1.IGNORE-HOST-TIM, [9-72](#)
ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-2.MAC-KEY-EXCHNG, [9-75](#)
ACQ-TABLE.ACQ-TBL-DATA.OPTIONS-2.PIN-KEY-EXCHNG, [9-75](#)
ACQ-TABLE.ACQ-TBL-DATA.PIN-KEY-INDEX, [9-74](#)
ACQ-TABLE.ACQ-TBL-DATA.PREV-SETL-DAY, [9-72](#)
ACQ-TABLE.ACQ-TBL-DATA.PRI-SETL-CNCT-TIM, [9-65](#)
ACQ-TABLE.ACQ-TBL-DATA.PRI-SETL-DIAL-ATTEMPTS, [9-65](#)
ACQ-TABLE.ACQ-TBL-DATA.PRI-SETL-PHONE-NUM, [9-65](#)
ACQ-TABLE.ACQ-TBL-DATA.PRI-TRANS-CNCT-TIM, [9-64](#)
ACQ-TABLE.ACQ-TBL-DATA.PRI-TRANS-DIAL-ATTEMPTS, [9-64](#)
ACQ-TABLE.ACQ-TBL-DATA.PRI-TRANS-PHONE-NUM, [9-63](#)
ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-SETL-CNCT-TIM, [9-66](#)
ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-SETL-DIAL-ATTEMPTS, [9-66](#)
ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-SETL-PHONE-NUM, [9-66](#)
ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-TRANS-CNCT-TIM, [9-64](#)
ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-TRANS-DIAL-ATTEMPTS, [9-65](#)
ACQ-TABLE.ACQ-TBL-DATA.SECNDRY-TRANS-PHONE-NUM, [9-64](#)
ACQ-TABLE.ACQ-TBL-DATA.SETL-MODEM-MDE, [9-74](#)
ACQ-TABLE.ACQ-TBL-DATA.SETL-TIM, [9-72](#)
ACQ-TABLE.ACQ-TBL-DATA.TIM-OUT, [9-66](#)
ACQ-TABLE.ACQ-TBL-DATA.VISA-I-TERM, [9-72](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.DATA-TYPE, [9-80](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.INPUT-LGTH-MAX, [9-80](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.INPUT-LGTH-MIN, [9-80](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-ID, [9-79](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-NUM, [9-79](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.PROMPT-TEXT, [9-79](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.AUTH-SALE, [9-83](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.CASH, [9-80](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.CHK, [9-81](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.CRD-VRFY, [9-81](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.DB, [9-82](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.OFFL-SALE, [9-82](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.PRE-AUTH, [9-81](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-1.RFND, [9-82](#)
ADNL-PROMPT.ADNL-PROMPT-DATA.TRAN-SELECT-2.SETL, [9-83](#)

C

CARD-DEFINITION-TBL.BACKUP-PHONE-
NUM, [8-17](#)

CARD-DEFINITION-TBL.CARD-NAM, [8-18](#)

CARD-DEFINITION-TBL.CRD-ACCEPTOR-
ID, [8-20](#)

CARD-DEFINITION-TBL.DEFAULT-ACCT, [8-18](#)

CARD-DEFINITION-TBL.FLOOR-LIMIT, [8-19](#)

CARD-DEFINITION-TBL.HIGH-PREFIX, [8-17](#)

CARD-DEFINITION-TBL.LOW-PREFIX, [8-17](#)

CARD-DEFINITION-TBL.NETWORK-
INTERNATIONAL-ID, [8-20](#)

CARD-DEFINITION-TBL.PRIMARY-PHONE-
NUM, [8-17](#)

CARD-DEFINITION-TBL.PROC-OPTIONS.ACCT-
SELCT, [8-21](#)

CARD-DEFINITION-TBL.PROC-
OPTIONS.ADJUSTMENT-ALLOWED, [8-22](#)

CARD-DEFINITION-TBL.PROC-OPTIONS.CARD-
PROC-TYP, [8-19](#)

CARD-DEFINITION-TBL.PROC-OPTIONS.CASH-
BACK-ALLOWED, [8-24](#)

CARD-DEFINITION-TBL.PROC-
OPTIONS.INVOICE-NUM, [8-23](#)

CARD-DEFINITION-TBL.PROC-
OPTIONS.MAC, [8-25](#)

CARD-DEFINITION-TBL.PROC-
OPTIONS.MANUAL-EXP-DAT, [8-23](#)

CARD-DEFINITION-TBL.PROC-
OPTIONS.MANUAL-PAN-ENTRY, [8-22](#)

CARD-DEFINITION-TBL.PROC-OPTIONS.MOD-10-
CHECK, [8-22](#)

CARD-DEFINITION-TBL.PROC-
OPTIONS.OFFLINE-TRAN-ALLOW, [8-24](#)

CARD-DEFINITION-TBL.PROC-OPTIONS.PIN-
ENTRY, [8-21](#)

CARD-DEFINITION-TBL.PROC-
OPTIONS.PRODUCT-CODES, [8-21](#)

CARD-DEFINITION-TBL.PROC-
OPTIONS.RECEIPT, [8-25](#)

CARD-DEFINITION-TBL.PROC-OPTIONS.VOICE-
REFERRAL, [8-24](#)

CARD-DEFINITION-TBL.REFERRAL-PHONE-
NUM, [8-18](#)

CARD-DEFINITION-TBL.TERMINAL-
TIMEOUT, [8-20](#)

CARD-DEFINITION-TBL.TRACK-SENT, [8-19](#)

CARD-RANGE-1.CRD-RANGE-TBL.ACQ-TBL-
ID, [9-33](#)

CARD-RANGE-1.CRD-RANGE-TBL.ACQ-TBI-ID-
2, [9-33](#)

CARD-RANGE-1.CRD-RANGE-TBL.CRD-RANGE-
INDEX, [9-31](#), [9-35](#)

CARD-RANGE-1.CRD-RANGE-TBL.HIGH-
PREFIX, [9-32](#)

CARD-RANGE-1.CRD-RANGE-TBL.ISS-TBL-
ID, [9-32](#)

CARD-RANGE-1.CRD-RANGE-TBL.ISS-TBL-ID-
2, [9-33](#)

CARD-RANGE-1.CRD-RANGE-TBL.LOW-
PREFIX, [9-32](#)

CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-
1.DEBIT-KEY-ONLY, [9-36](#)

CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-
1.DEBIT-PIN-PAD, [9-37](#)

CARD-RANGE-1.CRD-RANGE-TBL.OPTIONS-
1.DEBIT-PROMPT, [9-36](#)

CARD-RANGE-1.CRD-RANGE-TBL.PAN-
LEN, [9-33](#)

CARD-RANGE-2.CRD-RANGE-TBL.ACQ-TBL-ID-
2, [9-33](#)

CARD-RANGE-2.CRD-RANGE-TBL.ACQ-TBL-
ID, [9-33](#)

CARD-RANGE-2.CRD-RANGE-TBL.CRD-RANGE-
INDEX, [9-31](#), [9-35](#)

CARD-RANGE-2.CRD-RANGE-TBL.HIGH-
PREFIX, [9-32](#)

CARD-RANGE-2.CRD-RANGE-TBL.ISS.TBL-ID-
2, [9-33](#)

CARD-RANGE-2.CRD-RANGE-TBL.ISS-TBL-
ID, [9-32](#)

CARD-RANGE-2.CRD-RANGE-TBL.LOW-
PREFIX, [9-32](#)

CARD-RANGE-2.CRD-RANGE-TBL.OPTIONS-
1.DEBIT-KEY-ONLY, [9-36](#)

CARD-RANGE-2.CRD-RANGE-TBL.OPTIONS-
1.DEBIT-PIN-PAD, [9-37](#)

CARD-RANGE-2.CRD-RANGE-TBL.OPTIONS-
1.DEBIT-PROMPT, [9-36](#)

CARD-RANGE-2.CRD-RANGE-TBL.PAN-
LEN, [9-33](#)

CONFIGURATION-PARAMS.ACQ-TBL-ID, [9-9](#)

CONFIGURATION-PARAMS.CASH-BACK-
LMT, [9-26](#)

CONFIGURATION-PARAMS.CHECK-
GUARANTEE-OPTIONS, [8-5](#)

CONFIGURATION-PARAMS.CHK-SERVICE, [9-7](#)

CONFIGURATION-PARAMS.CRNCY-DEC-
POSITION, [9-11](#)

CONFIGURATION-PARAMS.CRNCY-DIGITS-
SETL, [9-10](#)

CONFIGURATION-PARAMS.CRNCY-DIGITS-
TRANS, [9-10](#)

CONFIGURATION-PARAMS.CRNCY-
SYMBOL, [9-10](#)

CONFIGURATION-PARAMS.DEBIT-ACQ-TBL-
ID, [9-9](#)

CONFIGURATION-PARAMS.DEBIT-ISS-TBL-ID, 9-9
 CONFIGURATION-PARAMS.DEFAULT-TRAN-CODE, 8-6
 CONFIGURATION-PARAMS.DFLT-TRAN, 9-11
 CONFIGURATION-PARAMS.DISPLAY-MAG-STRIPE, 8-11
 CONFIGURATION-PARAMS.DLL-REVISION-NUM, 8-5
 CONFIGURATION-PARAMS.DLL-TRACKING-NUM, 9-8
 CONFIGURATION-PARAMS.ECR-BAUD-RATE, 8-5, 9-6
 CONFIGURATION-PARAMS.HELP-DESK-PHONE, 8-4, 9-8
 CONFIGURATION-PARAMS.INITIALIZATION-CNTL, 8-4, 9-6
 CONFIGURATION-PARAMS.ISS-TBL-ID, 9-8
 CONFIGURATION-PARAMS.LANGUAGE-IND, 9-7
 CONFIGURATION-PARAMS.LOCAL-TERM-OPTIONS.KEYBOARD-LOCKED, 8-8
 CONFIGURATION-PARAMS.LOCAL-TERM-OPTIONS.REFUND-PASSWORD-DISABLE, 8-10
 CONFIGURATION-PARAMS.LOCAL-TERM-OPTIONS.TRAN-SEARCH-PASSWORD-DISABLE, 8-10
 CONFIGURATION-PARAMS.LOCAL-TERM-OPTIONS.VOID-PASSWORD-DISABLE, 8-9
 CONFIGURATION-PARAMS.MAINT-PASSWORD, 9-26
 CONFIGURATION-PARAMS.NMS-CALL-SCHED, 9-26
 CONFIGURATION-PARAMS.OPTIONS-1.AMOUNT-DUAL-ENTRY, 8-10, 9-14
 CONFIGURATION-PARAMS.OPTIONS-1.BUS-DAT-FRMT, 9-23
 CONFIGURATION-PARAMS.OPTIONS-1.CASHIER-PROCESSING, 9-15
 CONFIGURATION-PARAMS.OPTIONS-1.DISPLAY-MSR, 9-14
 CONFIGURATION-PARAMS.OPTIONS-1.LODGING-PROCESSING, 9-15
 CONFIGURATION-PARAMS.OPTIONS-1.PRNT-TIM-ON-RECEIPT, 9-16
 CONFIGURATION-PARAMS.OPTIONS-1.RECONCILE-METHOD, 9-23, 9-26, 9-27, 9-28, 9-29
 CONFIGURATION-PARAMS.OPTIONS-1.TIP-PROCESSING, 9-15
 CONFIGURATION-PARAMS.OPTIONS-2.CRD-VRFY-OR-DEBIT-KEY, 9-23
 CONFIGURATION-PARAMS.OPTIONS-2.PRNTR-ATTACHED, 9-16
 CONFIGURATION-PARAMS.OPTIONS-2.RCPT-PRE-PRNT, 9-19
 CONFIGURATION-PARAMS.OPTIONS-2.RFND-PROCESSED-OFFL, 9-18
 CONFIGURATION-PARAMS.OPTIONS-2.TILL-PROCESSING, 9-20
 CONFIGURATION-PARAMS.OPTIONS-2.TTL-ON-RCPT, 9-18
 CONFIGURATION-PARAMS.OPTIONS-2.VOID-PROCESSED-OFFL, 9-19
 CONFIGURATION-PARAMS.OPTIONS-3.CHECK-READER-INSTALLED, 9-19
 CONFIGURATION-PARAMS.OPTIONS-3.CRD-READ-FROM-PIN-PAD, 9-21
 CONFIGURATION-PARAMS.OPTIONS-3.CRD-VRFY-RCPT, 9-21
 CONFIGURATION-PARAMS.OPTIONS-3.ECR-INTERFACE, 9-22
 CONFIGURATION-PARAMS.OPTIONS-3.PIN-PAD-AMT-VRFY, 9-22
 CONFIGURATION-PARAMS.OPTIONS-3.TAKE-IMPRINT, 9-21
 CONFIGURATION-PARAMS.OPTIONS-4.ADJ-BY-CASHIER, 9-19
 CONFIGURATION-PARAMS.OPTIONS-4.BATCH-REVIEW-BY-CASHIER, 9-20
 CONFIGURATION-PARAMS.OPTIONS-4.PROMPT-SRVR-NOT-CASHIER, 9-20
 CONFIGURATION-PARAMS.OPTIONS-4.TENDER-CASH-PROMPT, 9-23
 CONFIGURATION-PARAMS.PASSWORD, 8-4, 9-8
 CONFIGURATION-PARAMS.PHONE-DIAL-OPTION.AUTO-ANSWER, 9-12
 CONFIGURATION-PARAMS.PHONE-DIAL-OPTION.DIALING, 8-10
 CONFIGURATION-PARAMS.PHONE-DIAL-OPTION.KBD-DIALING, 9-14
 CONFIGURATION-PARAMS.PHONE-DIAL-OPTION.MEMORY-DIALING, 8-8, 9-13
 CONFIGURATION-PARAMS.PHONE-DIAL-OPTION.REFERRAL-DIALING, 8-7, 9-13
 CONFIGURATION-PARAMS.PHONE-DIAL-OPTION.TOLL-DIALING, 8-9, 9-13
 CONFIGURATION-PARAMS.PHONE-LINE-HLD, 9-11
 CONFIGURATION-PARAMS.PIN-PAD-CNTL, 8-7
 CONFIGURATION-PARAMS.PIN-PAD-TYPE, 9-24
 CONFIGURATION-PARAMS.PRINTER-CONTROL, 8-4
 CONFIGURATION-PARAMS.PRINTER-TYPE, 9-24
 CONFIGURATION-PARAMS.PRNTR-BAUD-RATE, 9-6
 CONFIGURATION-PARAMS.REAUTHORIZE-MARGIN, 8-6

CONFIGURATION-PARAMS.SUPERVISOR-PASSWORD, [8-6](#)
CONFIGURATION-PARAMS.TERM-OPTIONS.BALANCE-INQUIRY, [8-9](#)
CONFIGURATION-PARAMS.TERM-OPTIONS.CUSTOMER-NUMBER-ON-RECEIPT, [8-12](#)
CONFIGURATION-PARAMS.TERM-OPTIONS.PRINT-TOTAL, [8-11](#)
CONFIGURATION-PARAMS.TERM-OPTIONS.REFUNDS-OFFLINE, [8-12](#)
CONFIGURATION-PARAMS.TERM-OPTIONS.RESTAURANT-TIP, [8-8](#)
CONFIGURATION-PARAMS.TERM-OPTIONS.VAL-PAN-LGTH-OFFLINE-DISABLE, [8-11](#)
CONFIGURATION-PARAMS.TERM-OPTIONS.VOIDS-OFFLINE, [8-12](#)
CONFIGURATION-PARAMS.TERM-OPTS.ADJ-PSWD-DISABLE, [9-17](#)
CONFIGURATION-PARAMS.TERM-OPTS.KEYBOARD-LOCKED, [9-16](#)
CONFIGURATION-PARAMS.TERM-OPTS.RFND-PSWD-DISABLE, [9-17](#)
CONFIGURATION-PARAMS.TERM-OPTS.RPT-PSWD-DISABLE, [9-18](#)
CONFIGURATION-PARAMS.TERM-OPTS.VOID-PSWD-DIASBLE, [9-17](#)

I

ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT1, [9-58](#)
ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT2, [9-58](#)
ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT3, [9-58](#)
ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT4, [9-59](#)
ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT5, [9-59](#)
ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT6, [9-59](#)
ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT7, [9-60](#)
ISS-TABLE.ISS-TBL-DATA.ADNL-PROMPTS.PROMPT8, [9-60](#)
ISS-TABLE.ISS-TBL-DATA.CRD-NAM, [9-40](#)
ISS-TABLE.ISS-TBL-DATA.DEFAULT-ACCT, [9-40](#)
ISS-TABLE.ISS-TBL-DATA.FLOOR-LIMIT, [9-41](#)
ISS-TABLE.ISS-TBL-DATA.ISS-TBL-ID, [9-40](#)
ISS-TABLE.ISS-TBL-DATA.ISS-TBL-INDEX, [9-39](#), [9-49](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.ACCT-SELECT, [9-41](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.ADJUSTMENT-ALLOWED, [9-42](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.EXP-DAT-REQ, [9-44](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.MANUAL-PAN-ENTRY, [9-44](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.OFFLINE-TRAN-ALLOW, [9-45](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.PIN-ENTRY, [9-43](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.PRODUCT-CODES, [9-47](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-1.VOICE-REFERRAL, [9-46](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.BE-MY-GUEST, [9-47](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.BLOCK-CRD-VRFY, [9-43](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.BLOCK-RFND, [9-42](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.CAPTURE-TRANS, [9-46](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.CHK-EXP-DAT, [9-47](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.INVOICE-NUM, [9-44](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.MOD-10-CHECK, [9-43](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-2.RECEIPT, [9-45](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.BLOCK-VOIDS, [9-50](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.CASH-BACK, [9-51](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.CASH-ONLY-ALLOW, [9-55](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.PRNT-NO-RFND-ALLOWED, [9-53](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.PRNT-SIG-NOT-RQD, [9-52](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.RFND-PROCESSED-OFFL, [9-53](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-3.VOID-PROCESSED-OFFL, [9-54](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.AUTH-OVR-FLR, [9-52](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.AUTO-REPRINT, [9-46](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.DEBIT-TRAN, [9-50](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.TAX-PROMPT, [9-44](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.TRK1-ERROR, [9-45](#)

ISS-TABLE.ISS-TBL-DATA.OPTIONS-4.VRFY-PAN, [9-51](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-5.ADNL-RFND-PROMPT, [9-54](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-5.BAR-AMT-PROMPT, [9-54](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-5.PO-NUM-PROMPT, [9-53](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-6.AVS-ON-SALE, [9-50](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-6.CRDHLDR-ADDR-REQ, [9-51](#)
ISS-TABLE.ISS-TBL-DATA.OPTIONS-6.CRDHLDR-ZIP-CDE-REQ, [9-52](#)
ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.ADJ, [9-57](#)
ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.BAL-INQ, [9-56](#)
ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.CASHBACK, [9-56](#)
ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.DEBIT, [9-55](#)
ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.PREAUTH, [9-57](#)
ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.RFND, [9-56](#)
ISS-TABLE.ISS-TBL-DATA.PIN-ENTRY-TRANS.VOID, [9-57](#)
ISS-TABLE.ISS-TBL-DATA.REAUTHORIZE-MARGIN, [9-41](#)
ISS-TABLE.ISS-TBL-DATA.RFRL-PHONE-NUM, [9-40](#)

K

KEY-S.REC-TYPE, [9-4](#)
KEY-S.TERM-GRP, [9-4](#)

P

PROD-TABLE.PROD-DATA.CDE, [9-77](#)
PROD-TABLE.PROD-DATA.DESCRPTION, [9-77](#)
PROD-TABLE.PROD-DATA.SELCT-KEY, [9-77](#)
PRODUCT-CODE.CDE, [8-14](#)
PRODUCT-CODE.DESCRPTION, [8-14](#)

R

RDT.RANGE-DEFINITION-TBL.CARD-DEFINITION-TBL-INDEX, [8-29](#)
RDT.RANGE-DEFINITION-TBL.HIGH-PREFIX, [8-28](#)
RDT.RANGE-DEFINITION-TBL.LOW-PREFIX, [8-28](#)

T

TERM-GRP, [8-3](#)

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